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Drugeon et al.

(10) **Patent No.: US 12,401,825 B2**

(45) **Date of Patent: Aug. 26, 2025**

(54) **ENCODER, DECODER, ENCODING METHOD, AND DECODING METHOD**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Panasonic Intellectual Property Corporation of America**, Torrance, CA (US)

2023/0082783 A1* 3/2023 Wang H04N 19/188
375/240.02
2023/0156201 A1* 5/2023 Hendry H04N 19/187
375/240.02

(72) Inventors: **Virginie Drugeon**, Darmstadt (DE);
Tadamasa Toma, Osaka (JP); **Takahiro Nishi**, Nara (JP); **Kiyofumi Abe**, Osaka (JP)

FOREIGN PATENT DOCUMENTS

WO WO-2021235327 A1 * 11/2021 H04N 19/31

(73) Assignee: **PANASONIC INTELLECTUAL PROPERTY CORPORATION OF AMERICA**, Torrance, CA (US)

OTHER PUBLICATIONS

Bross et al: "Versatile Video Coding (Draft 7)", 16th Meeting: Geneva, CH, Oct. 1-11, 2019 (The Joint Video Exploration Team of ISO/IEC JTC1/SC29/WG11 and ITU-T SG. 16), No. JVET-Q0046; m51618 Dec. 18, 2019 (Dec. 18, 2019), XP03022421.*

(Continued)

(21) Appl. No.: **17/977,184**

Primary Examiner — Tsion B Owens

(22) Filed: **Oct. 31, 2022**

(74) *Attorney, Agent, or Firm* — Wenderoth, Lind & Ponack, L.L.P.

(65) **Prior Publication Data**

US 2023/0059074 A1 Feb. 23, 2023

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2021/018301, filed on May 13, 2021.

(60) Provisional application No. 63/026,978, filed on May 19, 2020.

(51) **Int. Cl.**
H04N 19/31 (2014.01)
H04N 19/70 (2014.01)

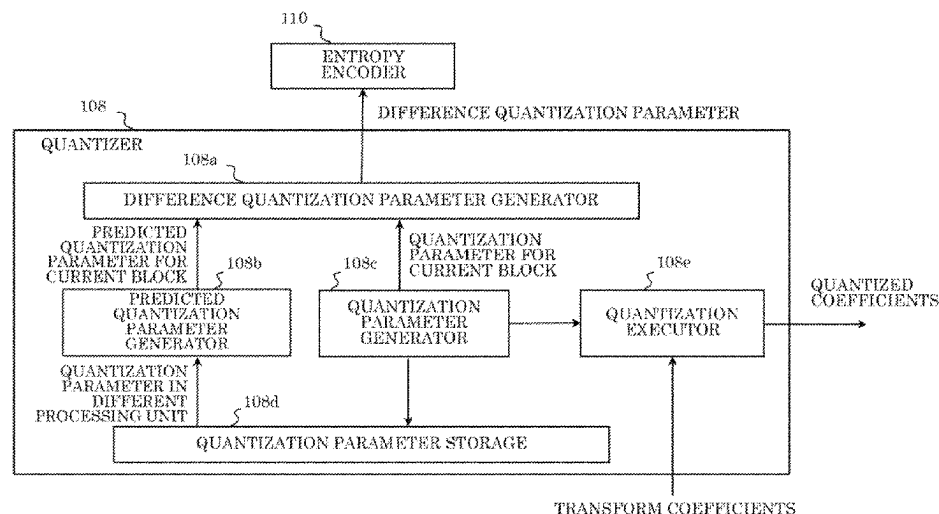
(52) **U.S. Cl.**
CPC **H04N 19/70** (2014.11); **H04N 19/31** (2014.11)

(58) **Field of Classification Search**
CPC H04N 19/70; H04N 19/31
See application file for complete search history.

(57) **ABSTRACT**

An encoder includes circuitry and memory coupled to the circuitry. In operation, in the case where a video parameter set is used for encoding a video, the circuitry writes a first parameter into the video parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The circuitry writes a second parameter into a sequence parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. In the case where the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value.

12 Claims, 123 Drawing Sheets



(56)

References Cited

OTHER PUBLICATIONS

International Search Report (ISR) issued on Jul. 13, 2021 in International (PCT) Application No. PCT/JP2021/018301.
H.265 (ISO/IEC 23008-2 HEVC (High Efficiency Video Coding)), Dec. 1, 2013.
Benjamin Bross et al., “Versatile Video Coding (Draft 9)”, Joint Video Experts Team (JVET) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, Document: JVET-R2001-vA, Apr. 2020, pp. 38-45, 94-113.
Virginie Drugeon et al., “AHG8/AHG9: Clarifications to HRD specification for single-layer and multi-layers bitstreams”, Joint Video Experts Team (JVET) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, Document: JVET-S0097, May 2020, pp. 1-3.
Ye-Kui Wang et al., “AHG9/AHG8/AHG12: On parameter sets and picture header”, Joint Video Experts Team (JVET) of ITU-T SG16

WP3 and ISO/IEC JTC1/SC29/WG11, Document: JVET-S0049-v1, May 2020, pp. 1-6.
Extended European Search Report issued Jul. 24, 2023 in corresponding European Patent Application No. 21807709.7.
Bross et al: “Versatile Video Coding (Draft 7)”, 16th Meeting: Geneva, CH, Oct. 1-11, 2019 (The Joint Video Exploration Team of ISO/IEC JTC1/SC29/WG11 and ITU-T SG.16) No. JVET-Q0046; m51618, Dec. 18, 2019 (Dec. 18, 2019), XP030222421.
Bross B; Chen J; Liu S; Y-K Wang: “Versatile Video Coding (Draft 9)”, 130. MPEG meeting; Apr. 20, 2020-Apr. 24, 2020; Alpbach (Motion Picture Expert Group or ISO/IEC JTC1/SC29/WG 11), No. JVET-R2001.vA, May 15, 2020 (May 15, 2020), pp. 1-524, XP030287936.
Bross, et al., “Versatile Video Coding (Draft 7)”, Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 16th Meeting: Geneva, CH, Oct. 1-11, 2019, Document: JVET-P2001-vE.

* cited by examiner

FIG. 1

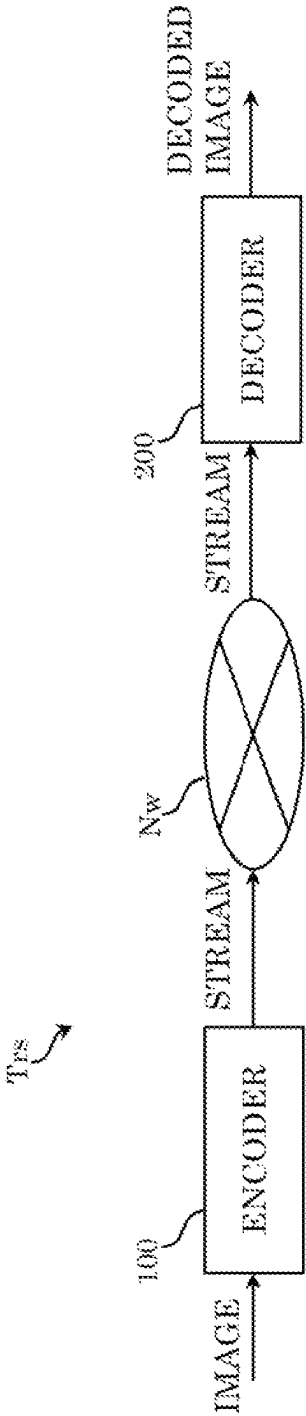


FIG. 2

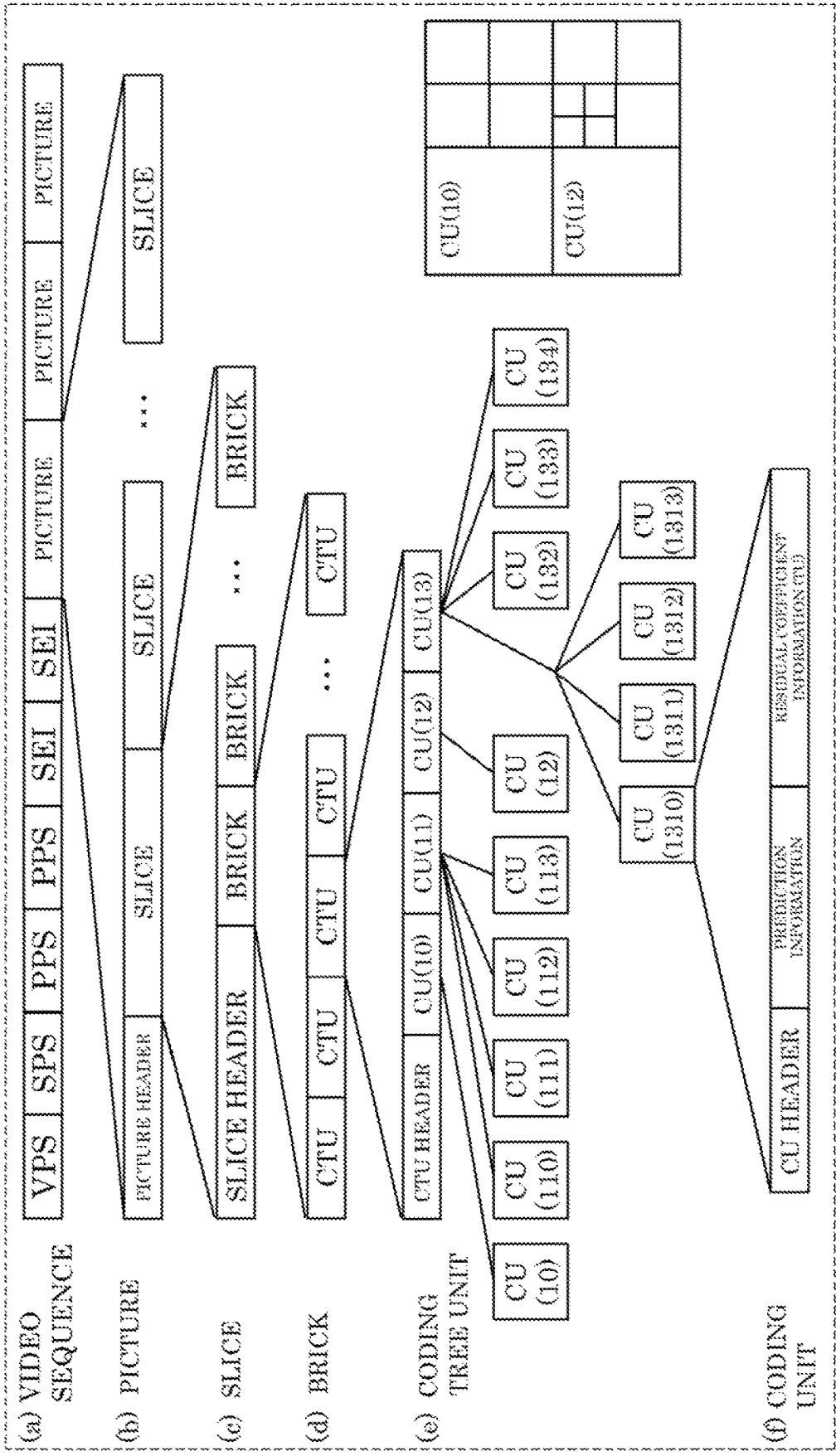


FIG. 3

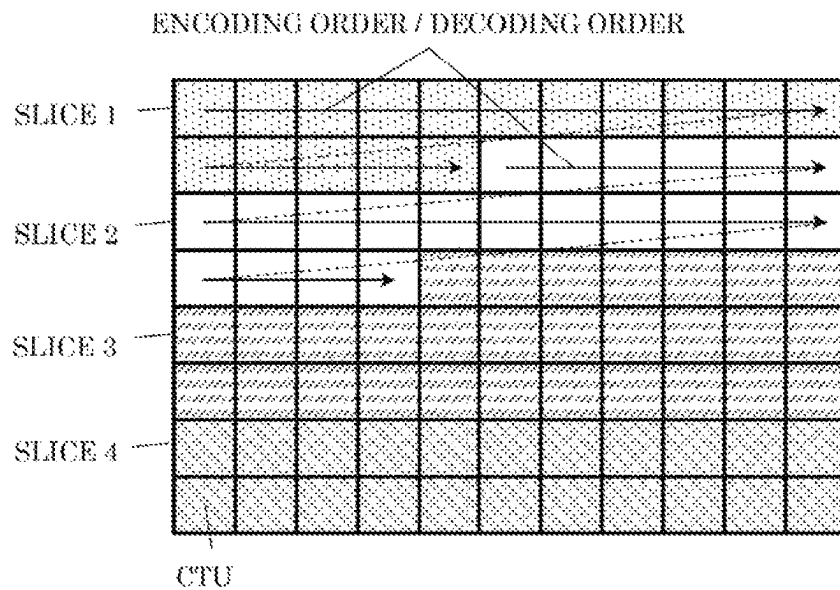
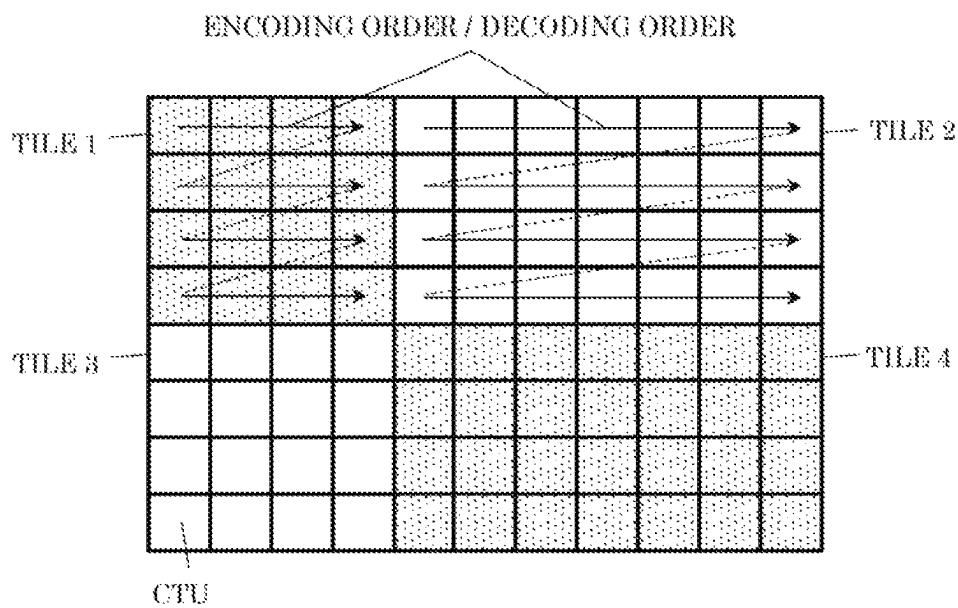
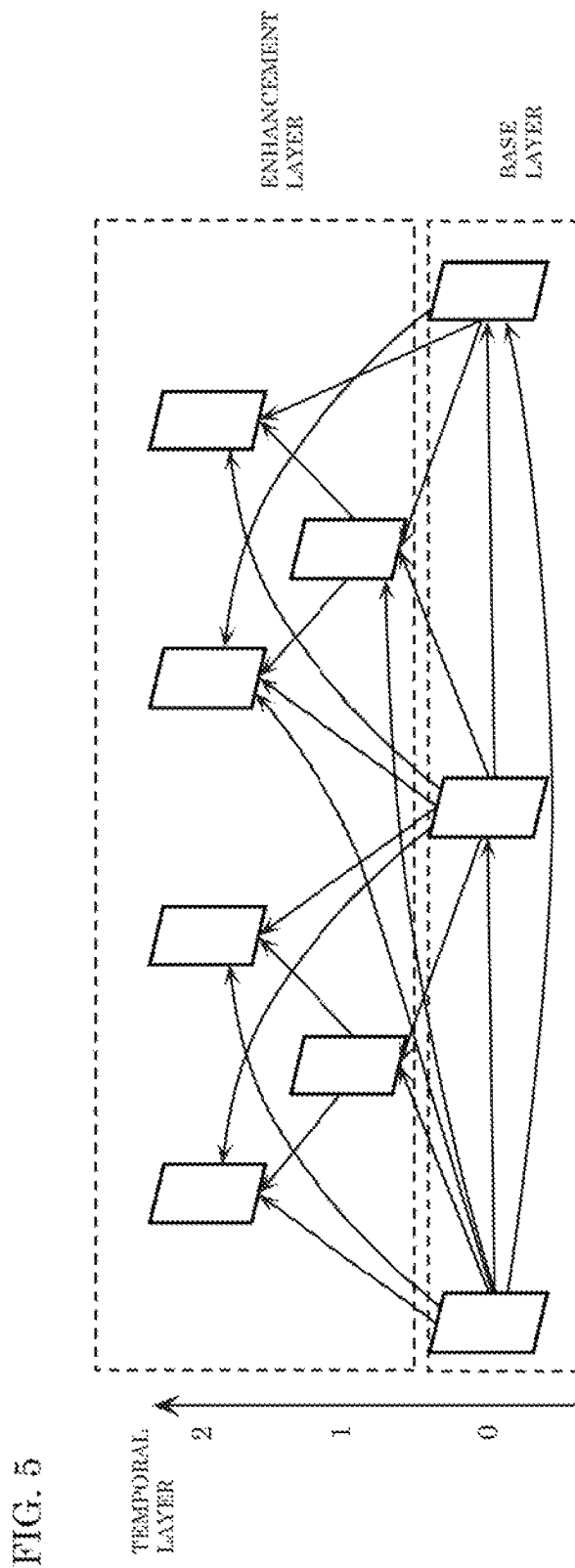


FIG. 4





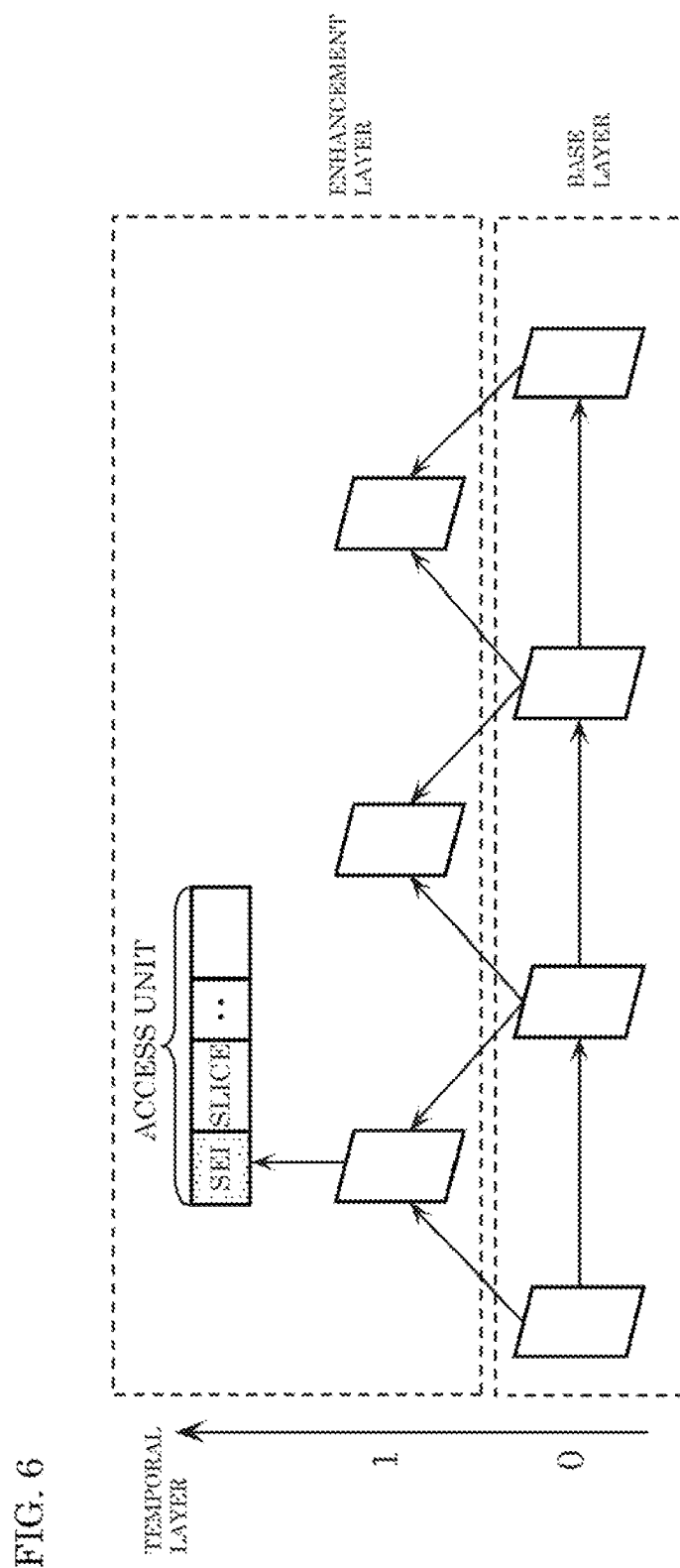


FIG. 7

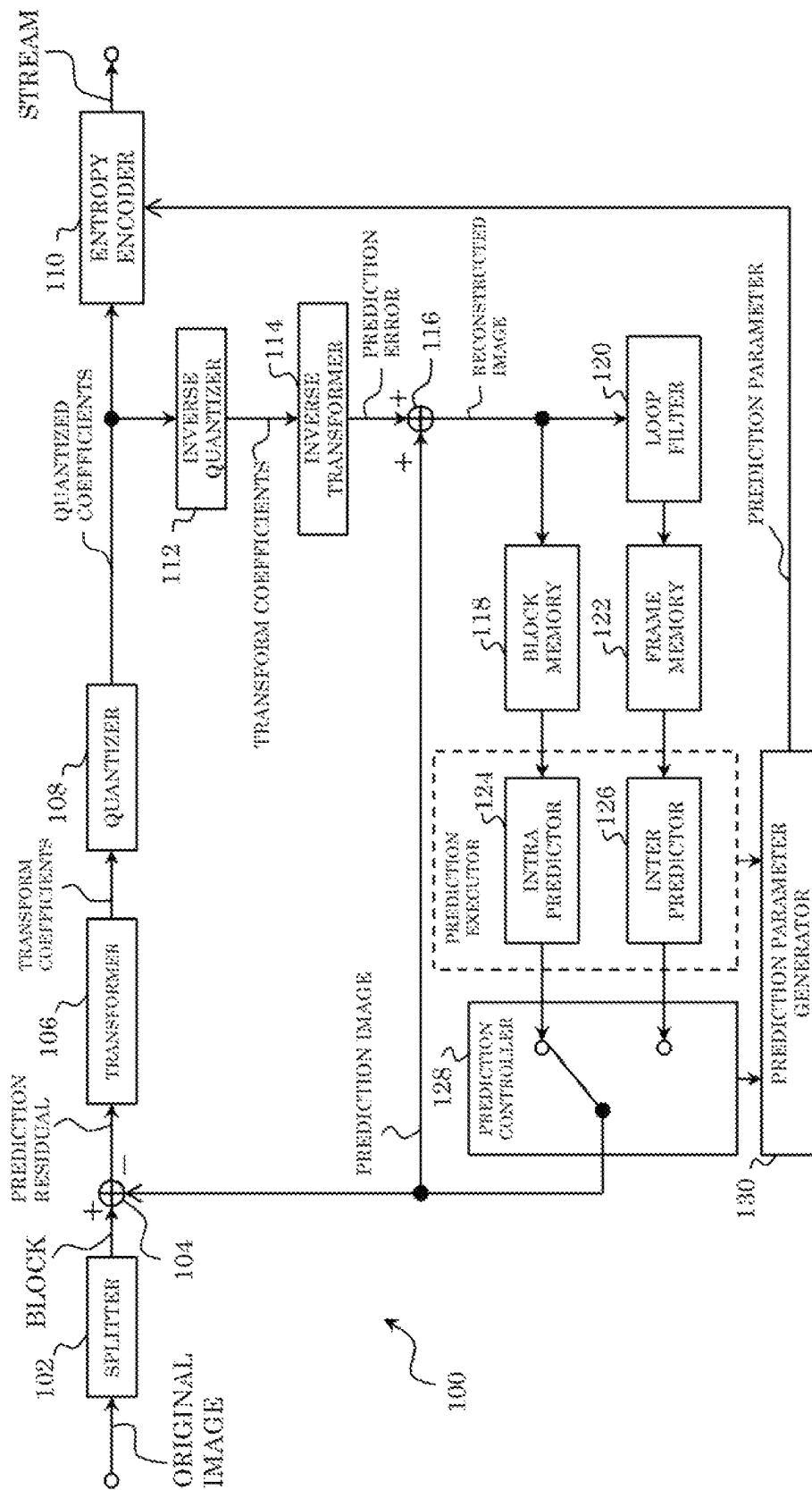


FIG. 8

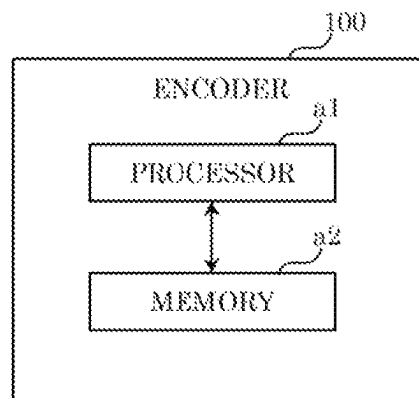


FIG. 9

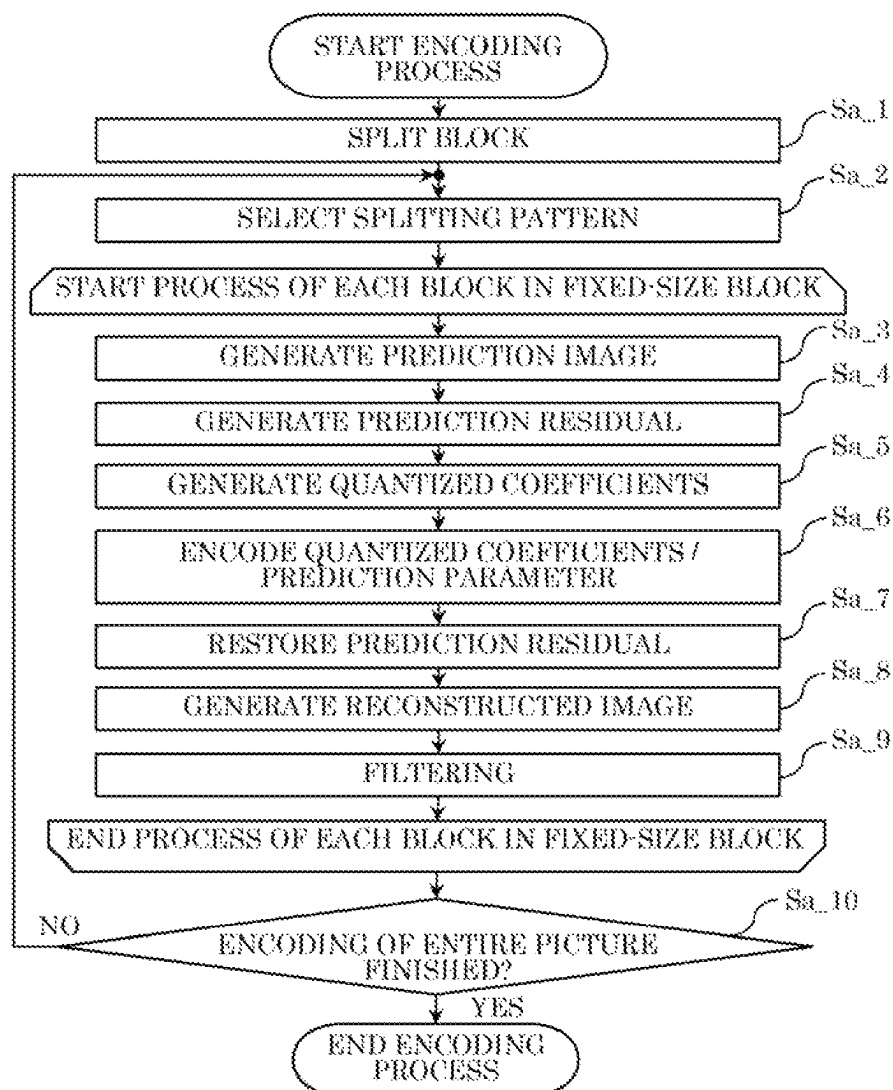


FIG. 10

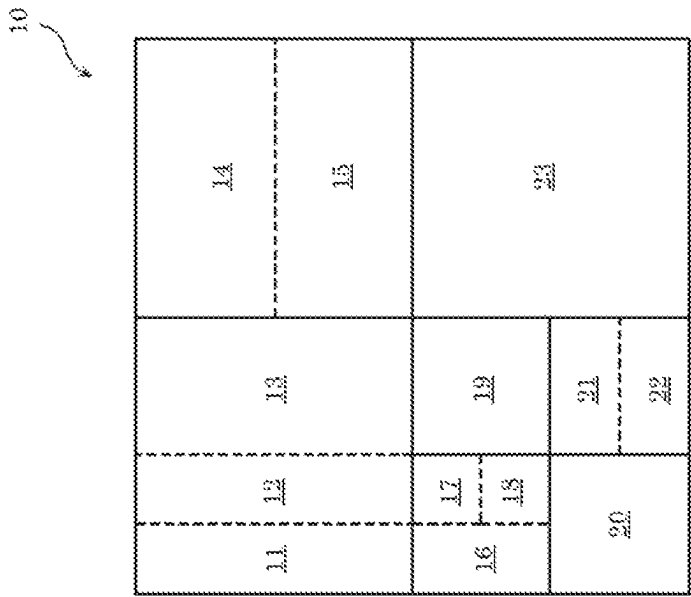


FIG. 11

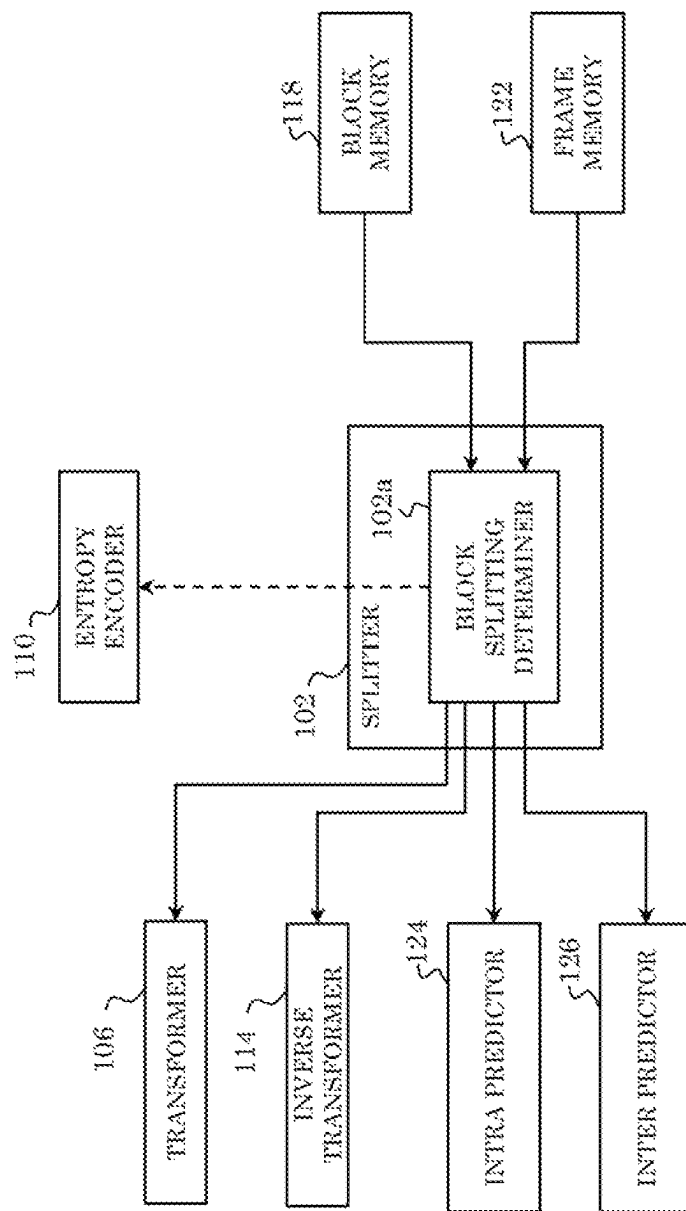


FIG. 12

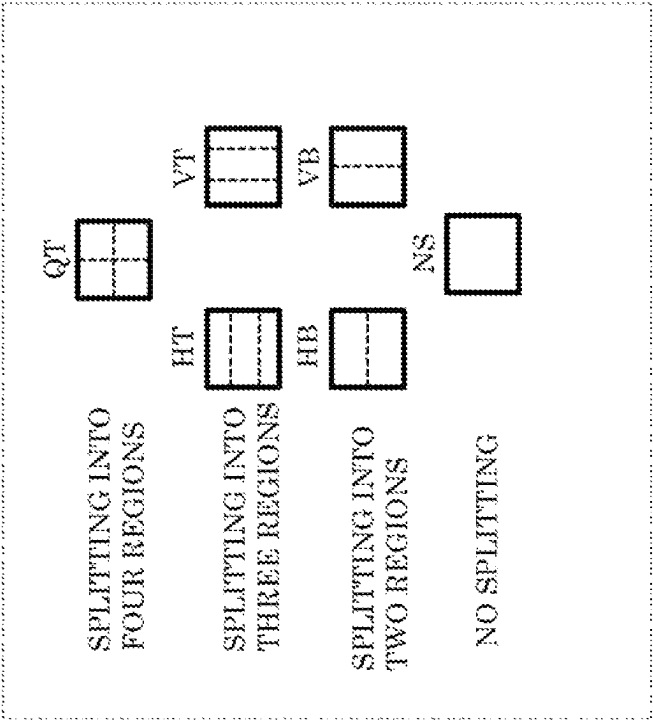


FIG. 13A

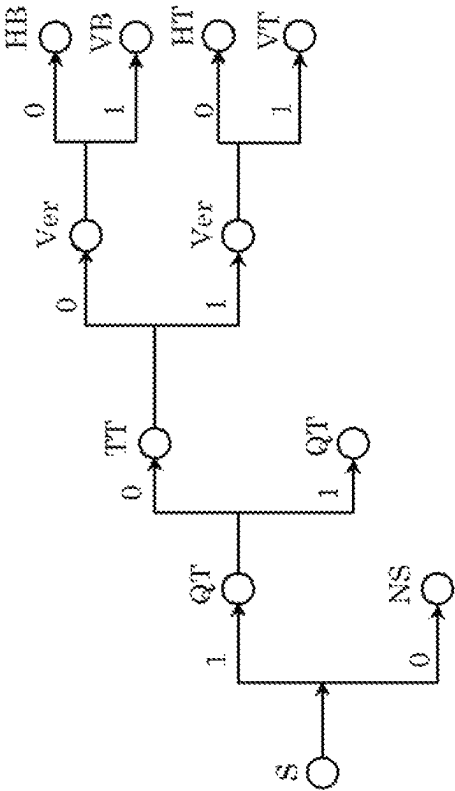


FIG. 13B

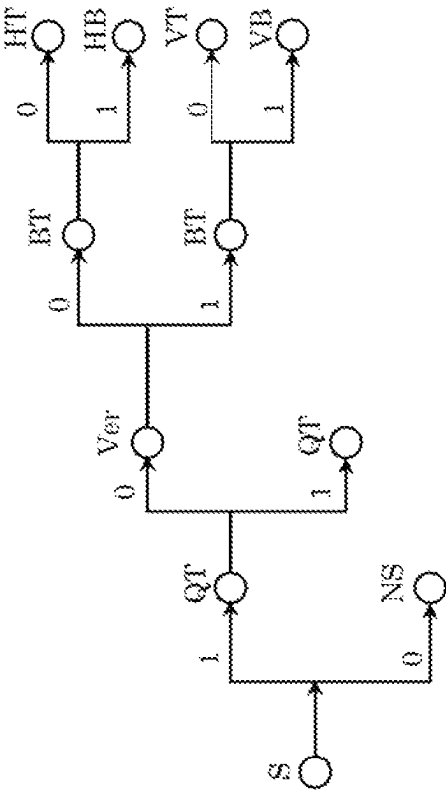


FIG. 14

TRANSFORM TYPE	BASIS FUNCTION $T_i(j)$, $i, j = 0, 1, \dots, N-1$
DCT-II	$T_i(j) = \omega_0 \cdot \sqrt{\frac{2}{N}} \cdot \cos\left(\frac{\pi \cdot i \cdot (2j+1)}{2N}\right)$ $\text{WHERE } \omega_0 = \begin{cases} \sqrt{\frac{2}{N}} & i=0 \\ 1 & i \neq 0 \end{cases}$
DCT-V	$T_i(j) = \omega_0 \cdot \omega_1 \cdot \sqrt{\frac{2}{2N-1}} \cdot \cos\left(\frac{2\pi \cdot i \cdot j}{2N-1}\right)$ $\text{WHERE } \omega_0 = \begin{cases} \sqrt{\frac{2}{N}} & i=0 \\ 1 & i \neq 0 \end{cases}, \omega_1 = \begin{cases} \sqrt{\frac{2}{N}} & j=0 \\ 1 & j \neq 0 \end{cases}$
DCT-VIII	$T_i(j) = \sqrt{\frac{4}{2N+1}} \cdot \cos\left(\frac{\pi \cdot (2i+1) \cdot (2j+1)}{4N+2}\right)$
DST-I	$T_i(j) = \sqrt{\frac{2}{N+1}} \cdot \sin\left(\frac{\pi \cdot (i+1) \cdot (j+1)}{N+1}\right)$
DST-VII	$T_i(j) = \sqrt{\frac{4}{2N+1}} \cdot \sin\left(\frac{\pi \cdot (2i+1) \cdot (2j+1)}{2N+1}\right)$

FIG. 15

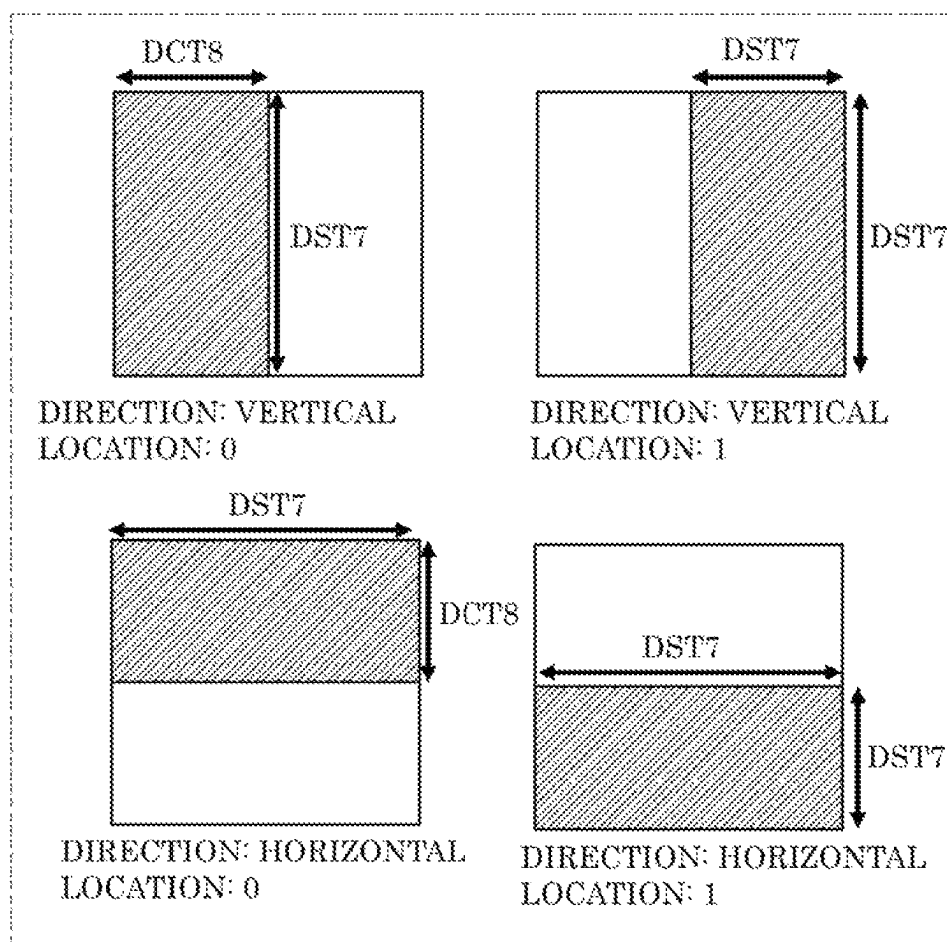


FIG. 16

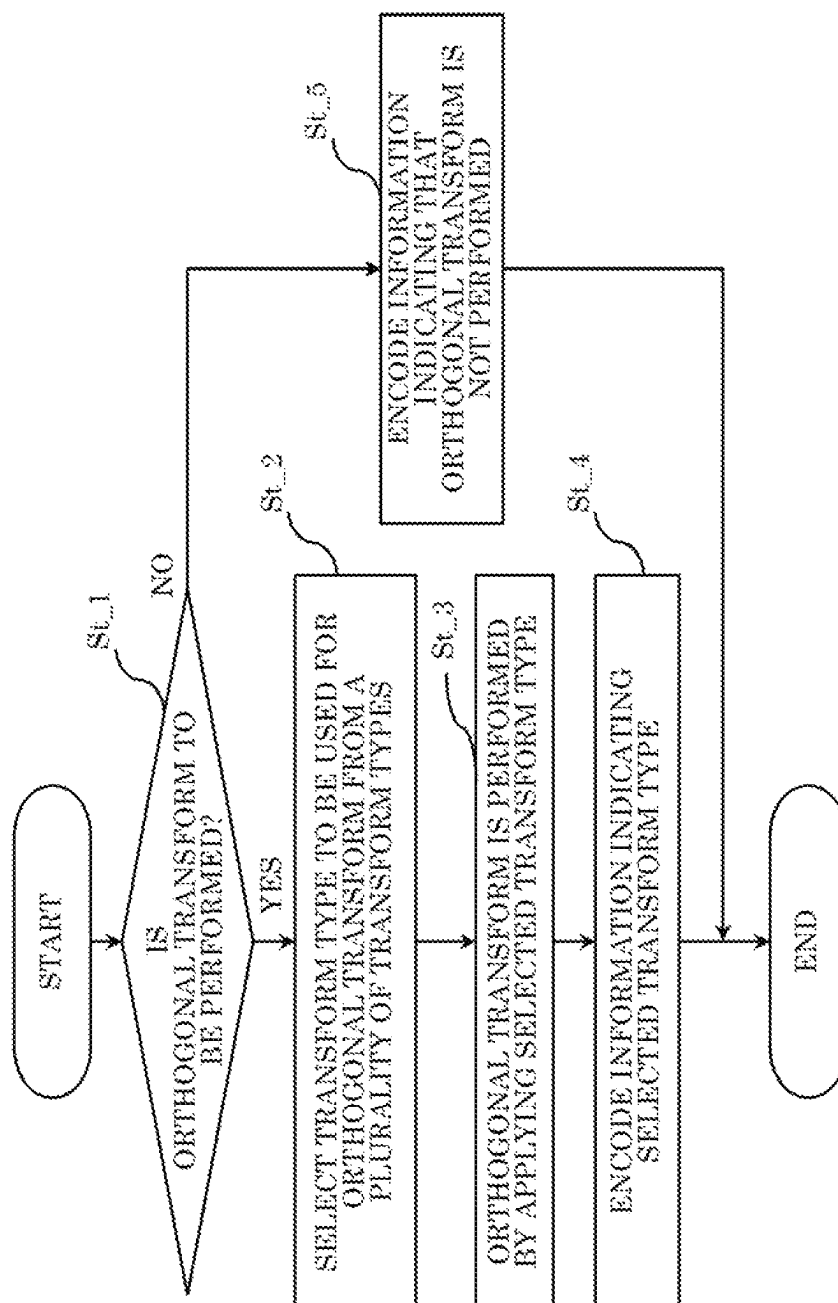


FIG. 17

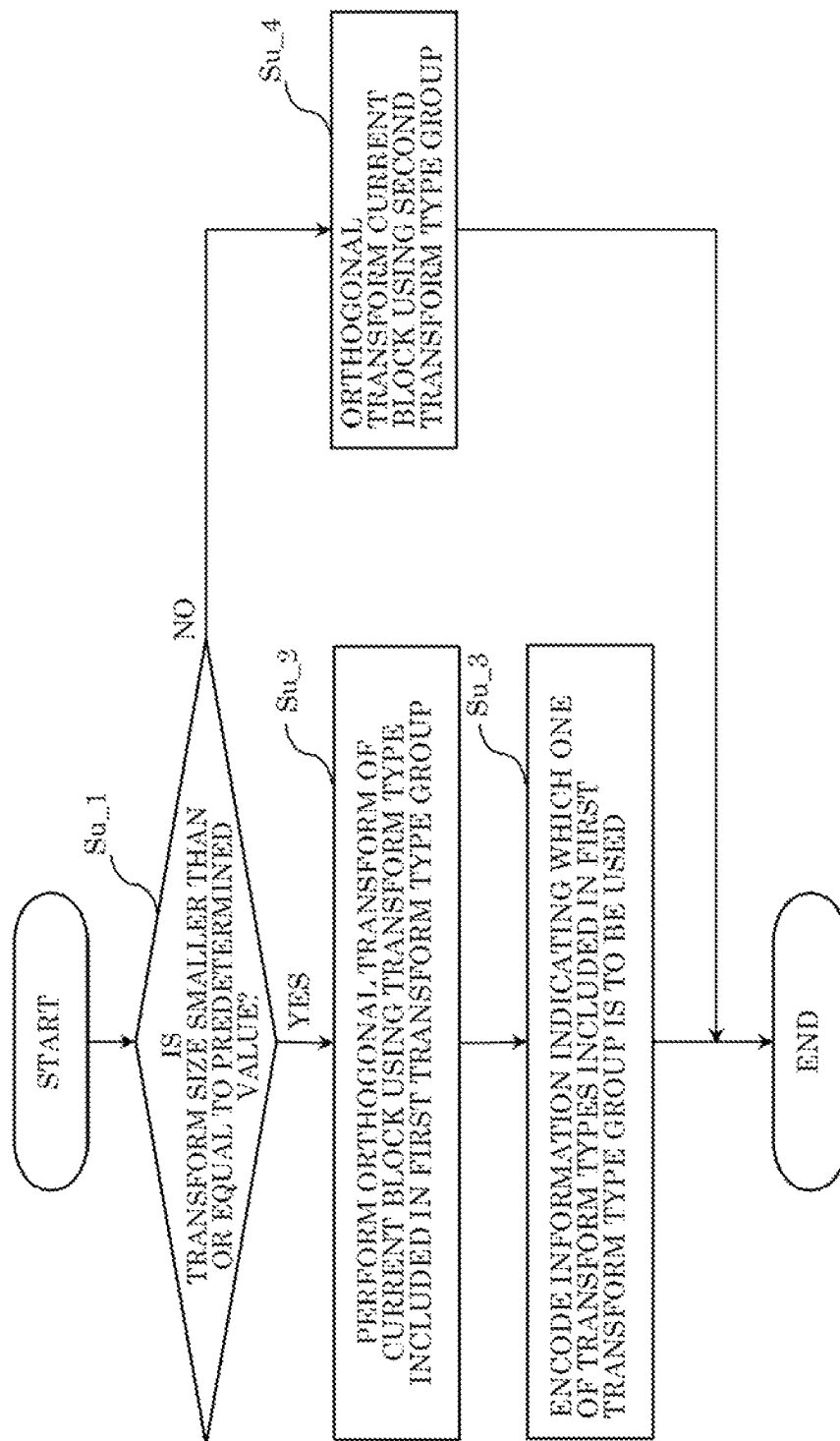


FIG. 18

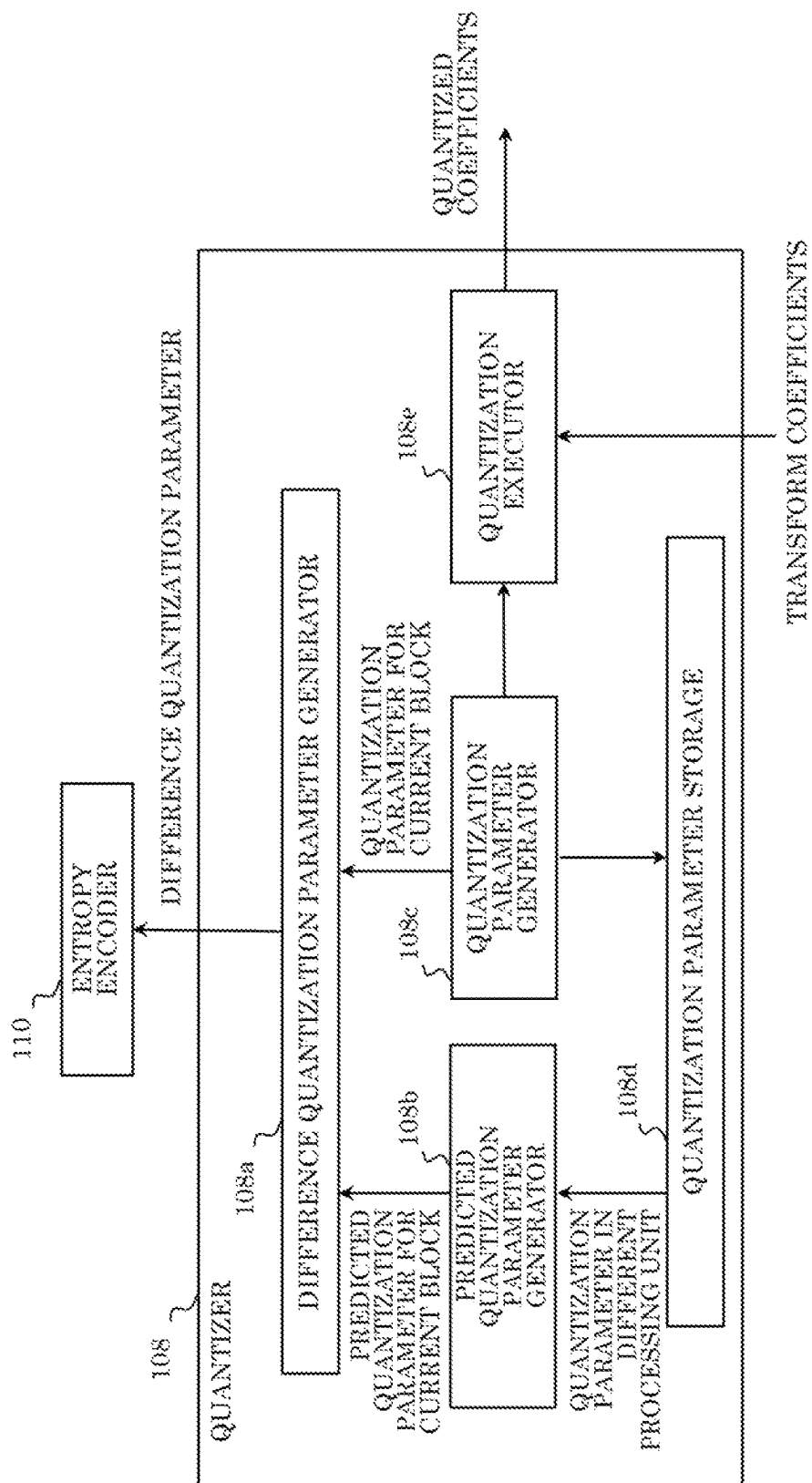
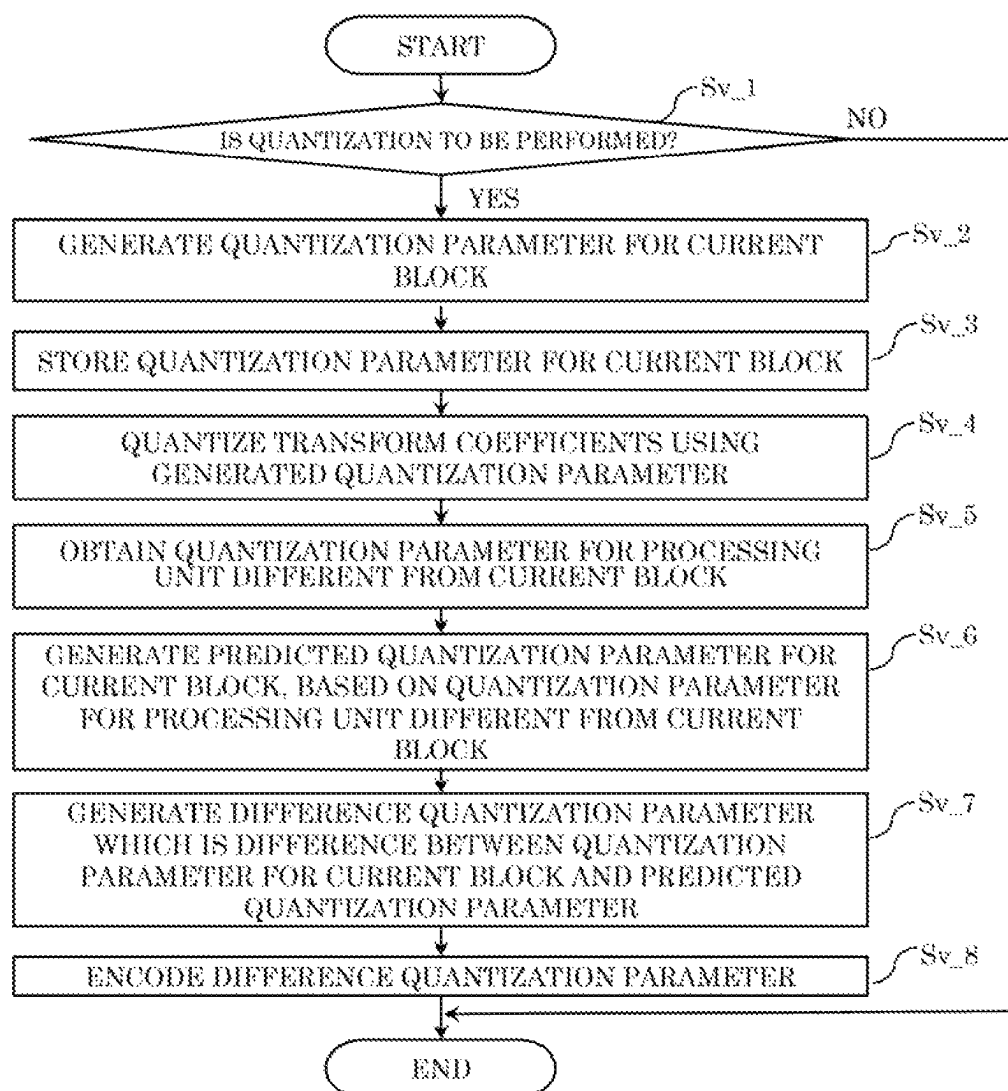


FIG. 19



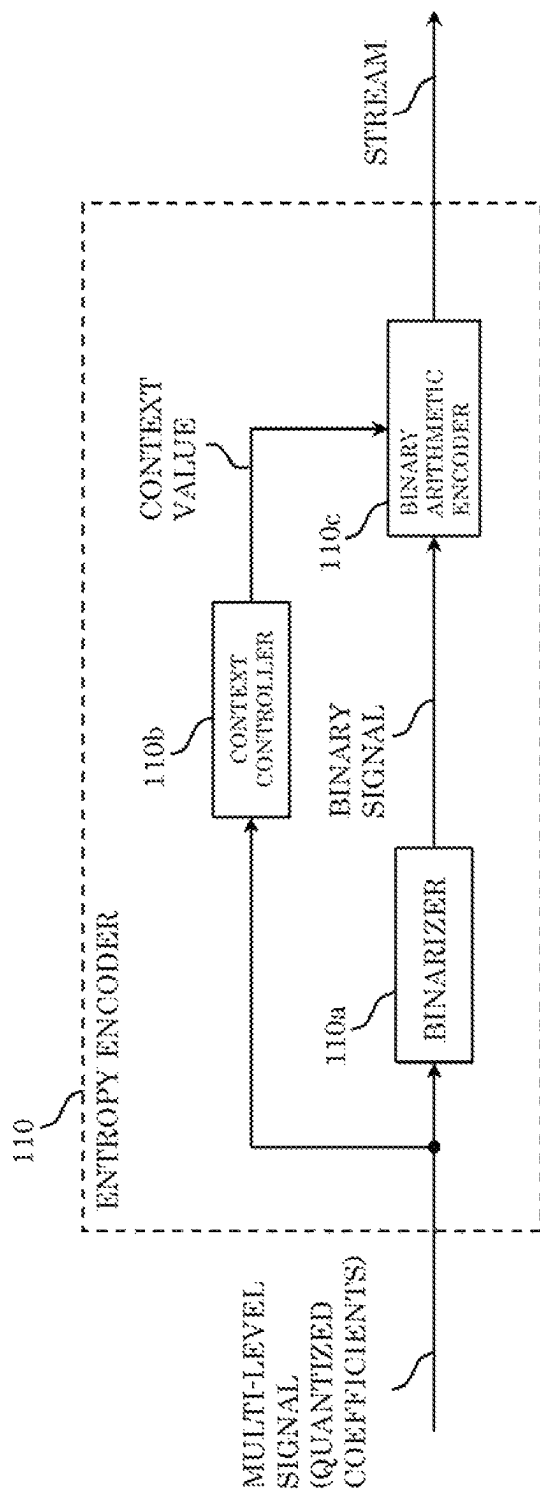


FIG. 20

FIG. 21

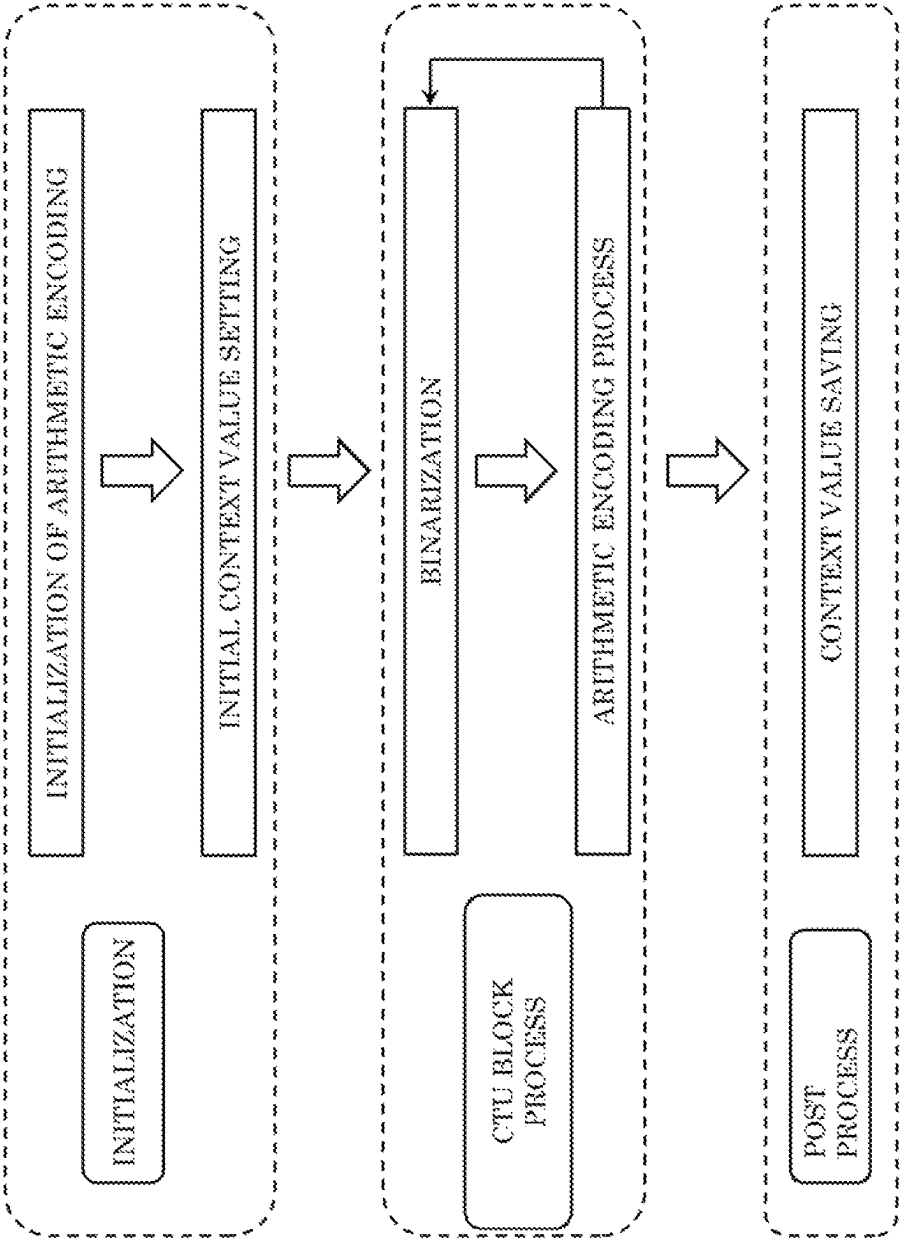


FIG. 22

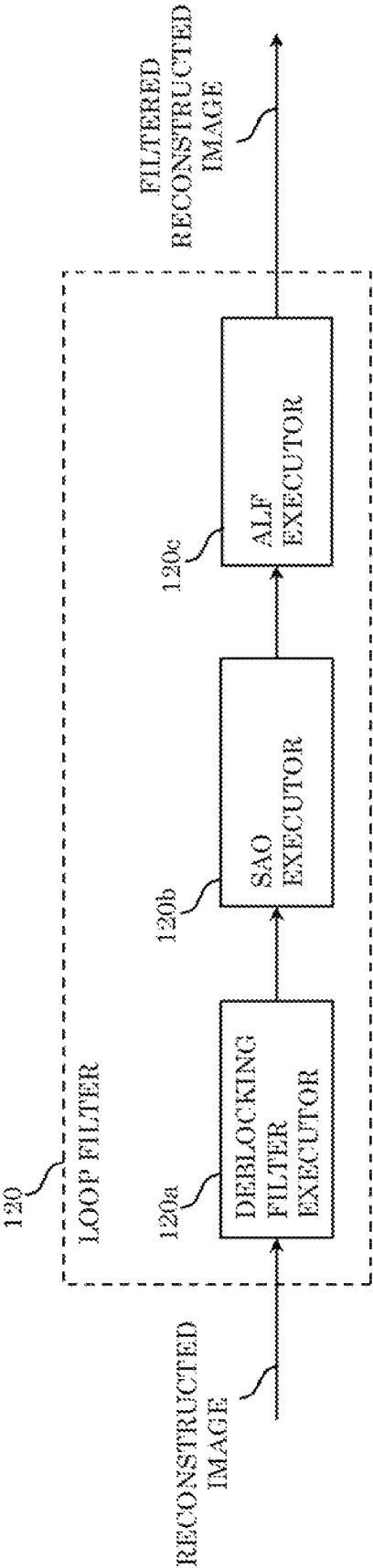


FIG. 23A

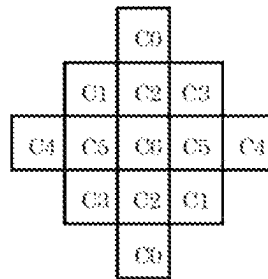


FIG. 23B

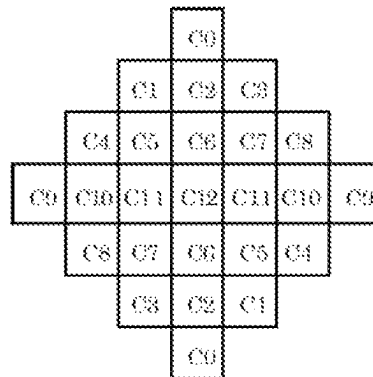


FIG. 23C

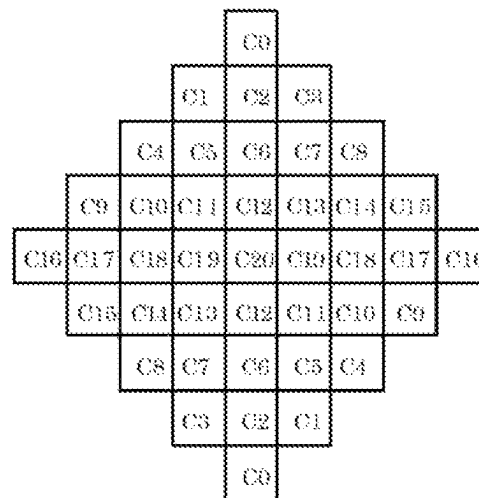


FIG. 23D

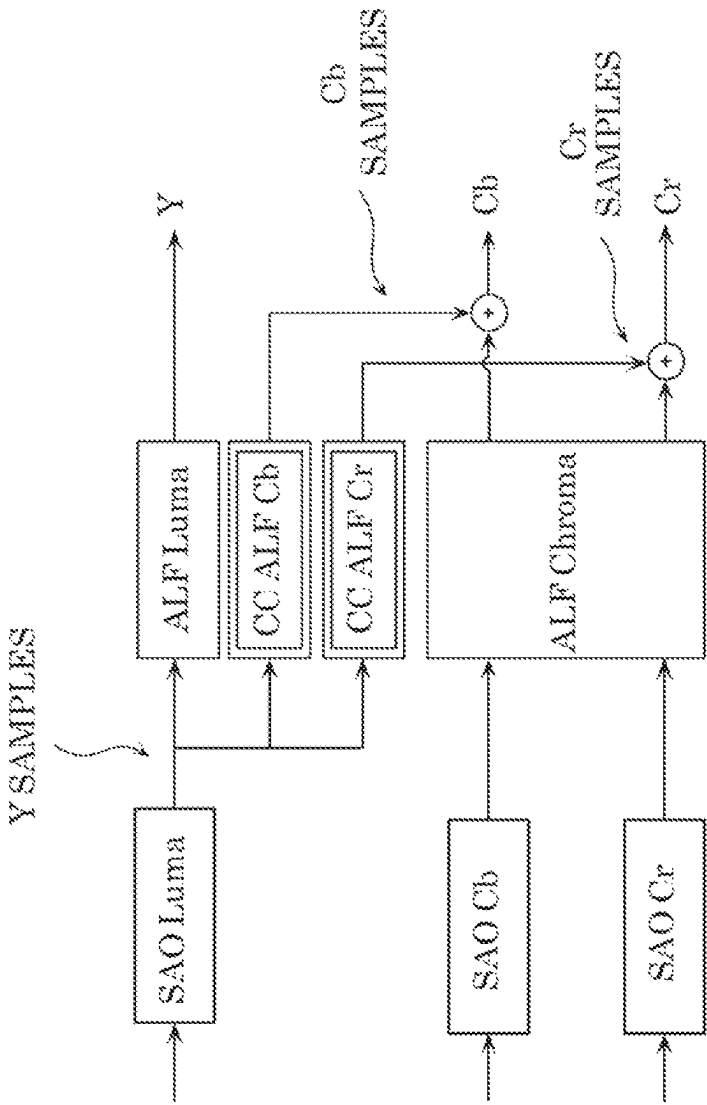
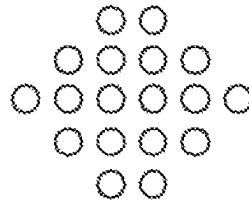


FIG. 28E



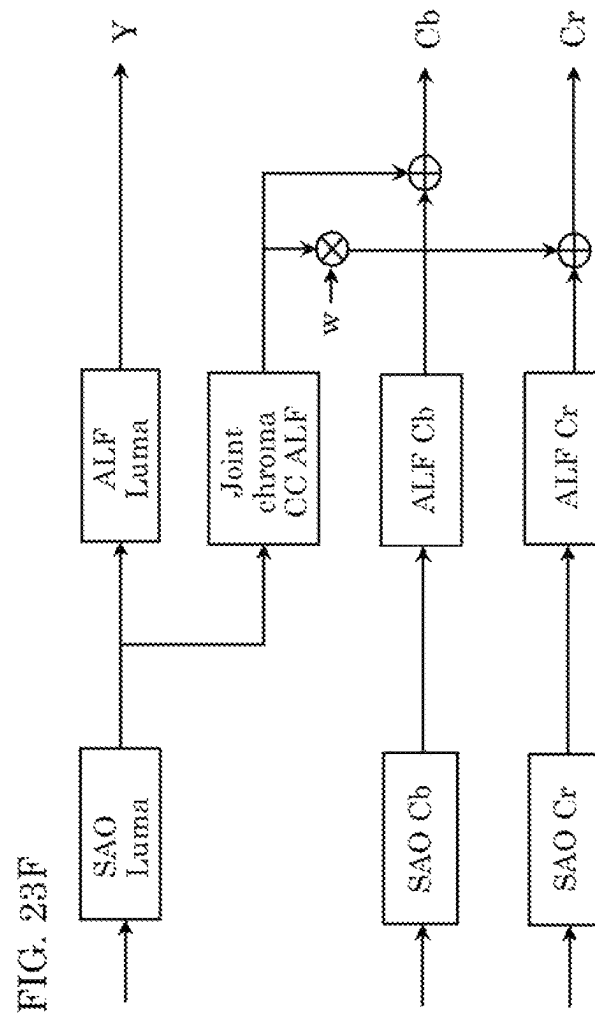


FIG. 23G

Index	1	2	3	4	5	6	7
w	1/4	1/2	3/4	1	4	2	4/3

FIG. 24

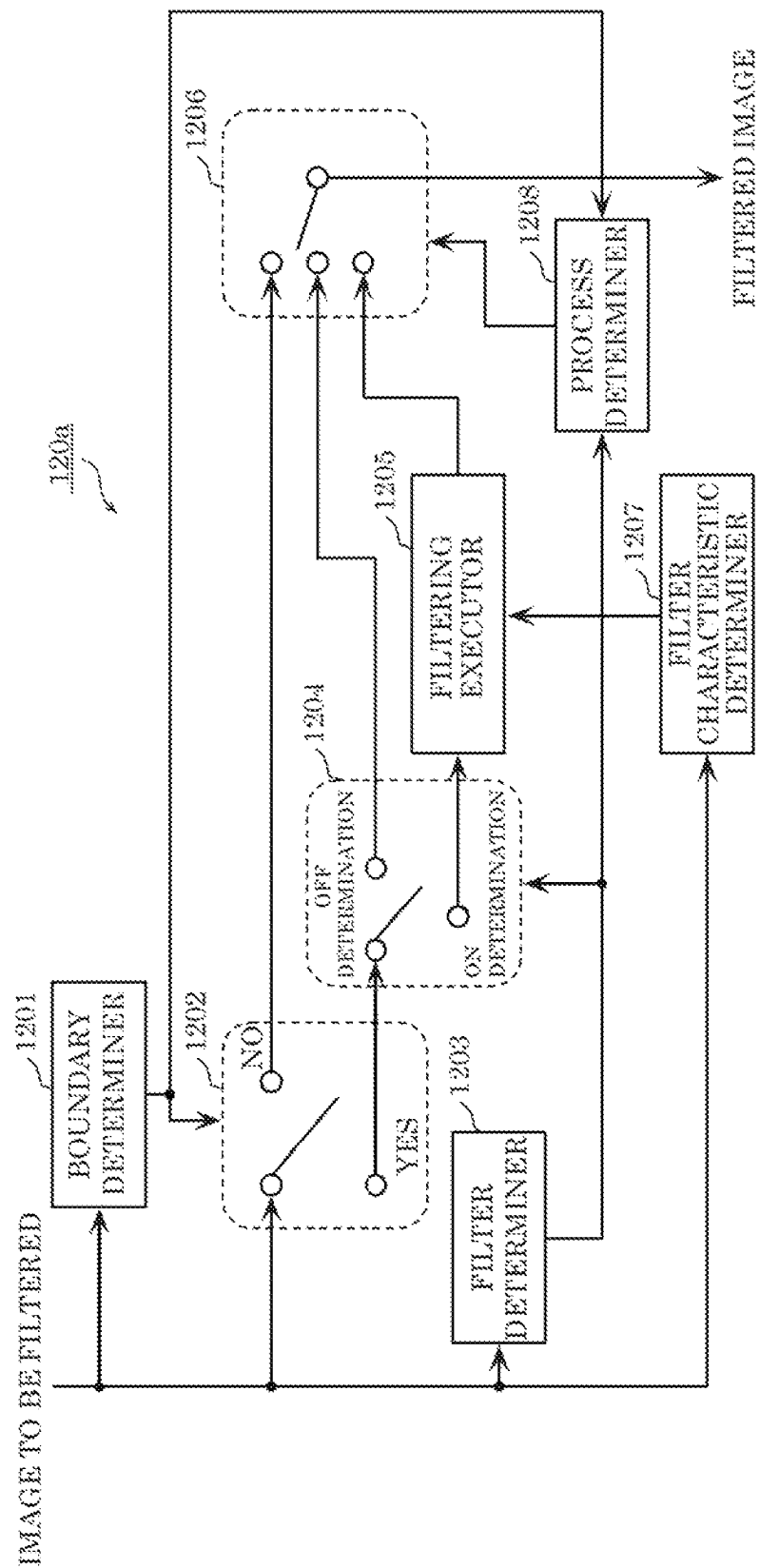


FIG. 25

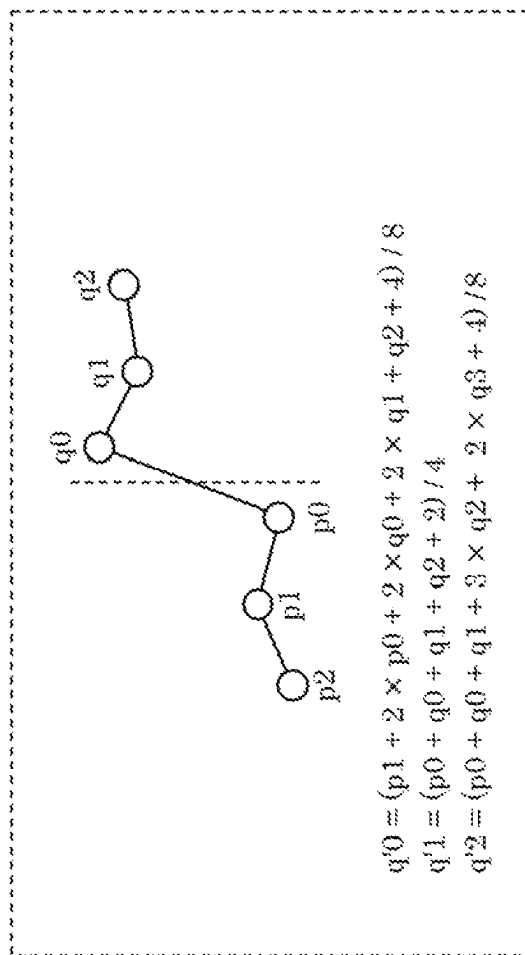


FIG. 26

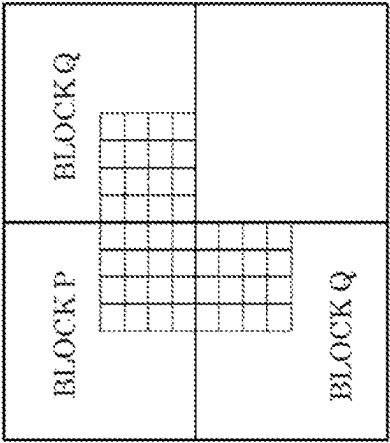


FIG. 27

CONDITIONS FOR BLOCKS P AND Q LOCATED ACROSS BOUNDARY FOR WHICH B _s VALUE IS TO BE DETERMINED	B _s VALUES
- AT LEAST ONE OF THE BLOCKS IS INTRA PREDICTION BLOCK	2
- AT LEAST ONE OF THE BLOCKS INCLUDES NON-ZERO TRANSFORM COEFFICIENT AND HAS A SIDE WHICH COINCIDES WITH BOUNDARY BETWEEN ORTHOGONAL TRANSFORM BLOCKS	1
- ABSOLUTE VALUE OF DIFFERENCE BETWEEN MOTION VECTORS OF THE TWO BLOCKS ACROSS BOUNDARY CORRESPONDS TO ONE OR MORE PIXELS	1
- REFERENCE IMAGES FOR MOTION COMPENSATION OF THE TWO BLOCKS ACROSS BOUNDARY OR THE NUMBERS OF MOTION VECTORS OF THE TWO BLOCKS ARE DIFFERENT	1
- THE OTHERS	0

FIG. 28

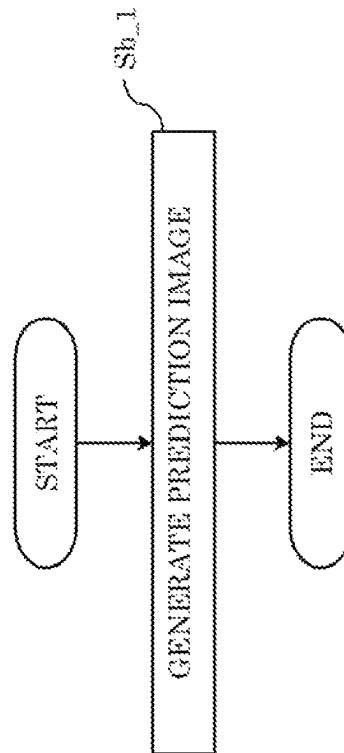


FIG. 29

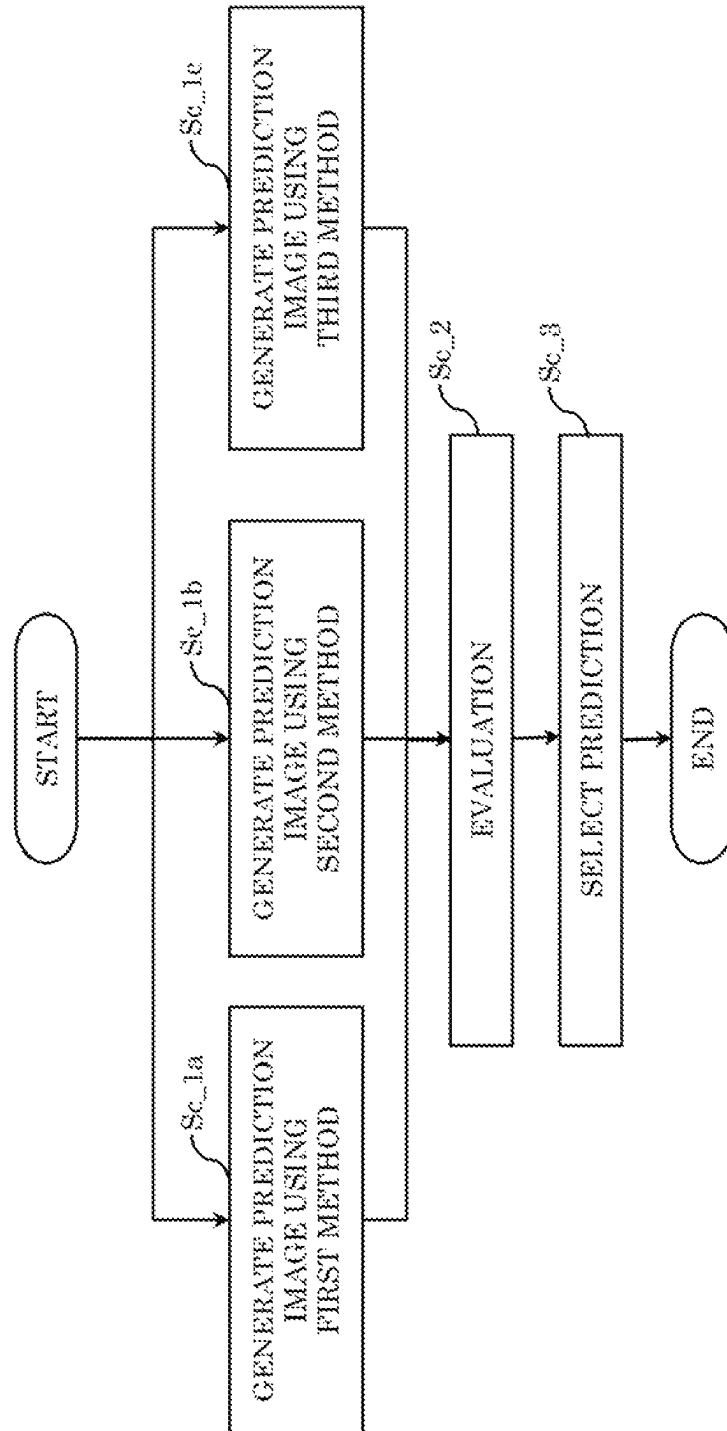


FIG. 30

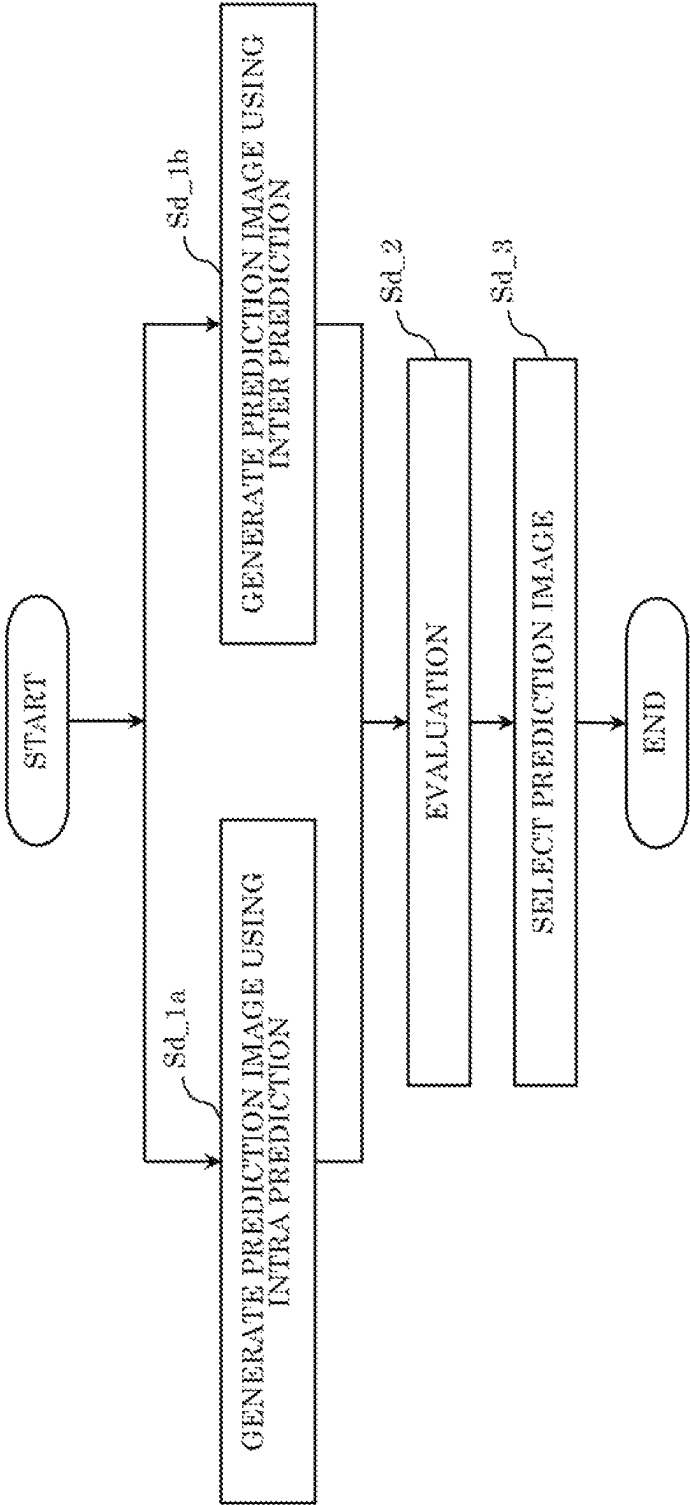


FIG. 31

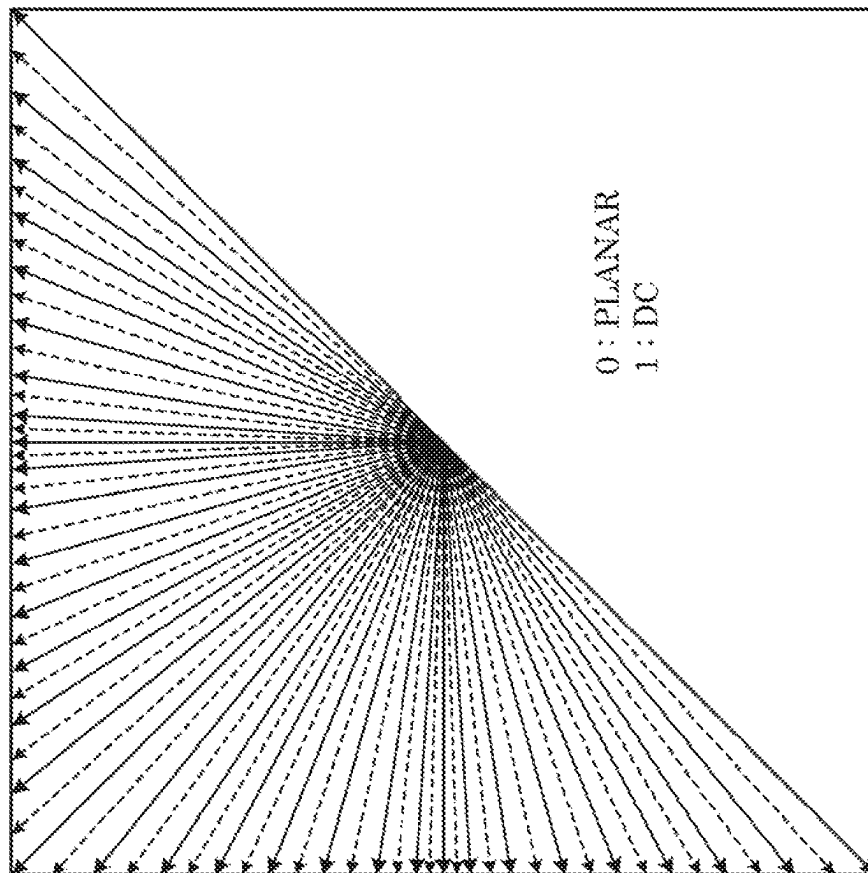


FIG. 32

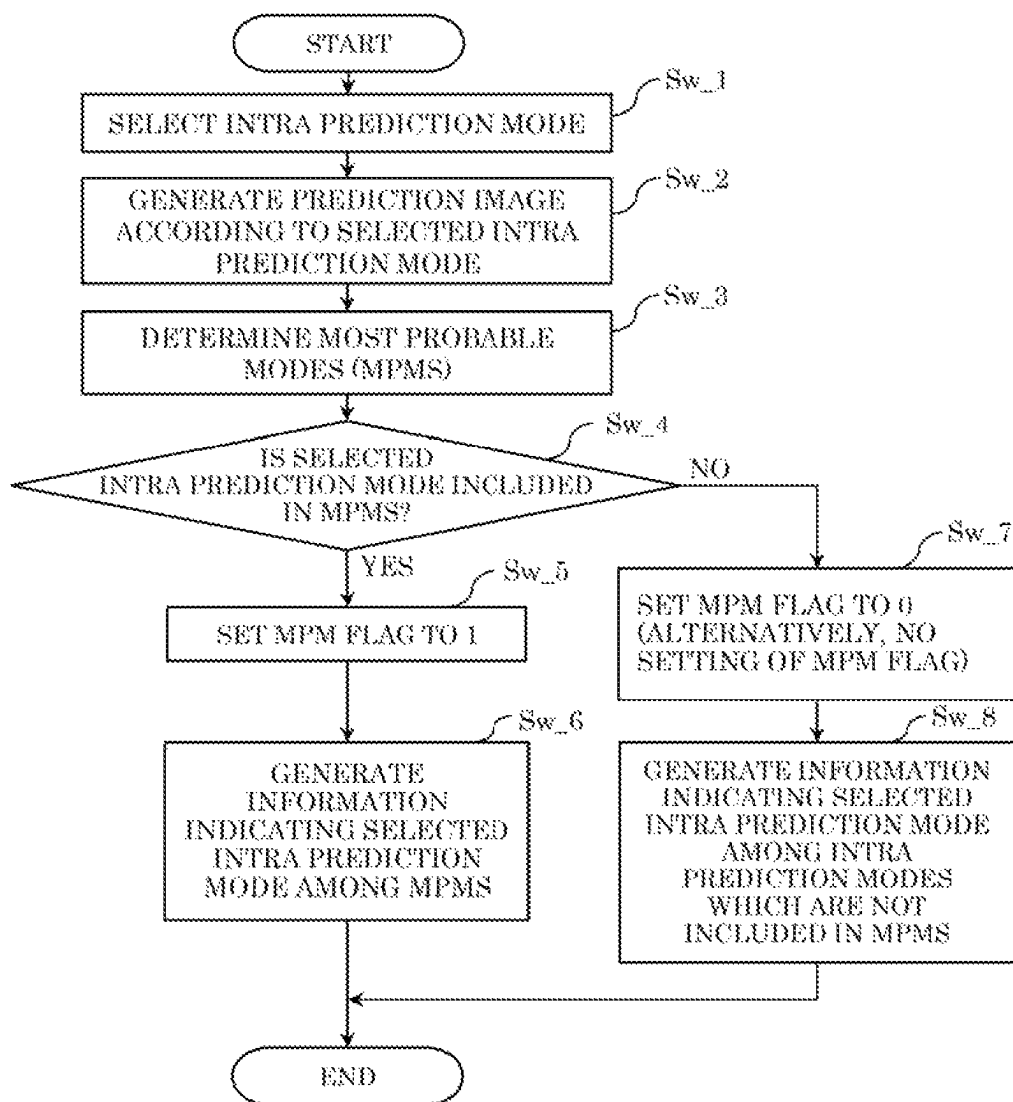
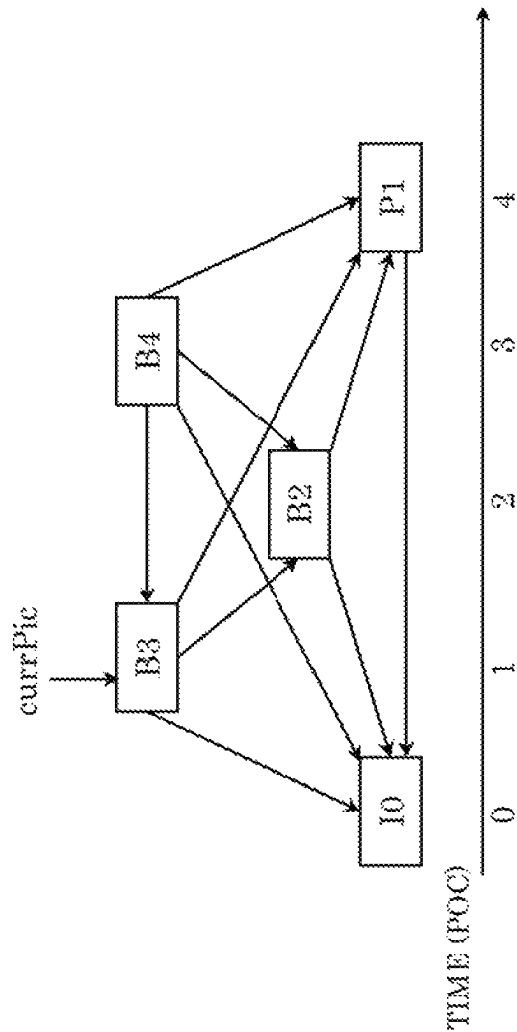


FIG. 33



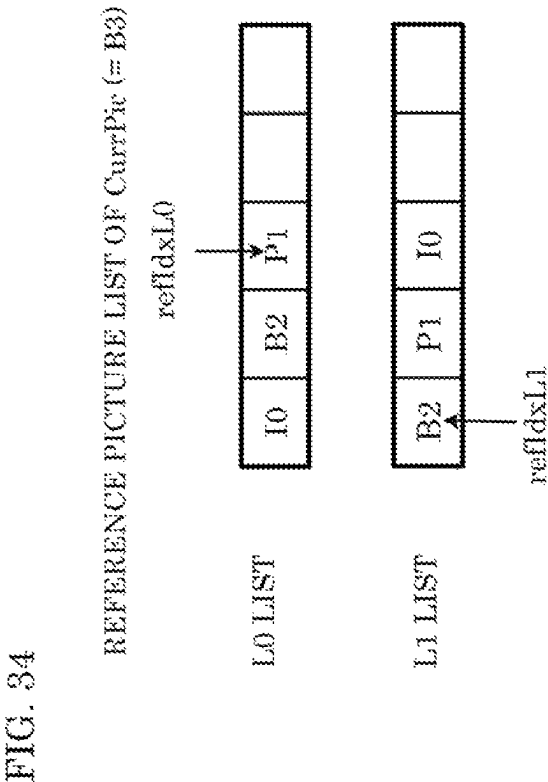


FIG. 35

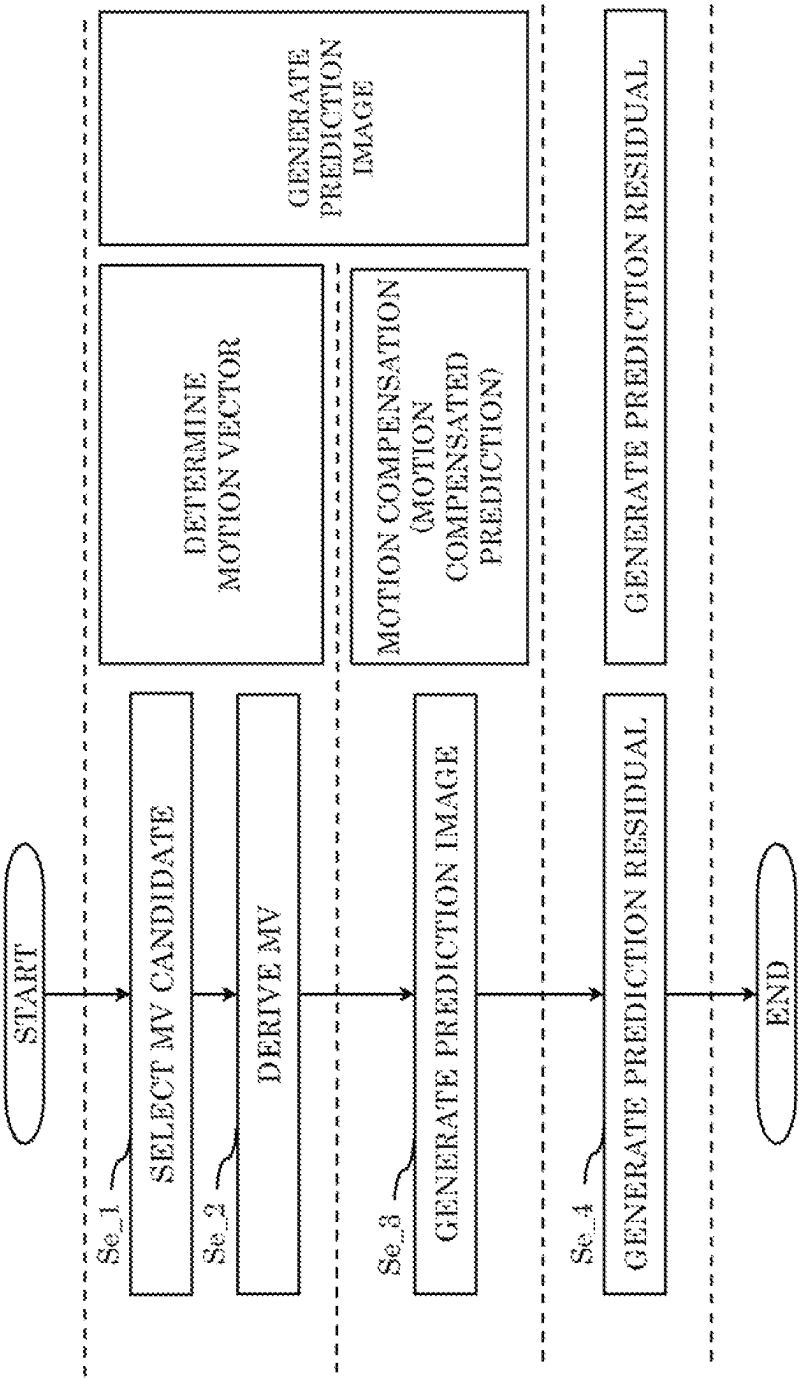


FIG. 36

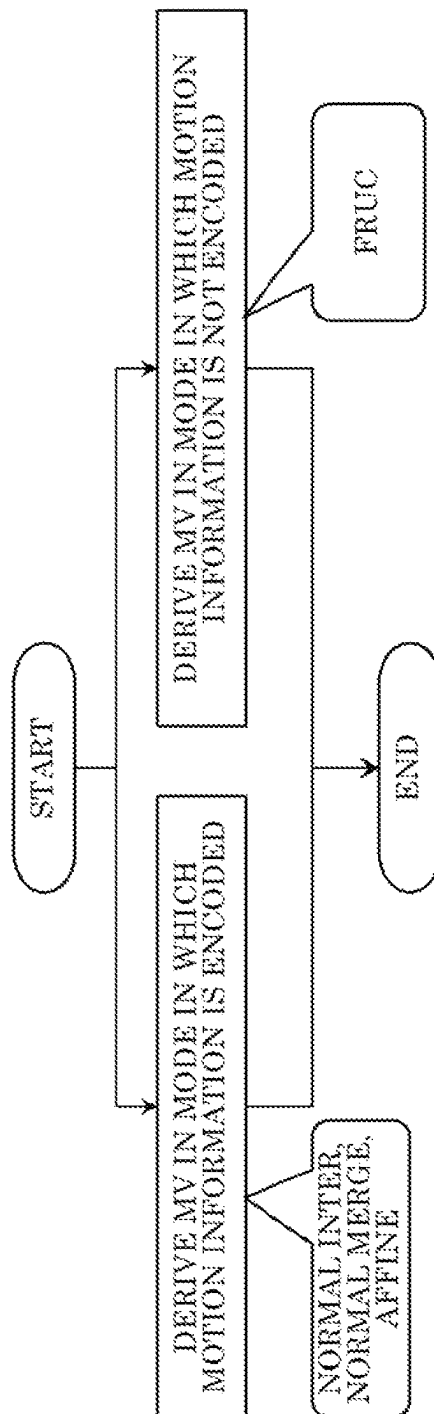


FIG. 37

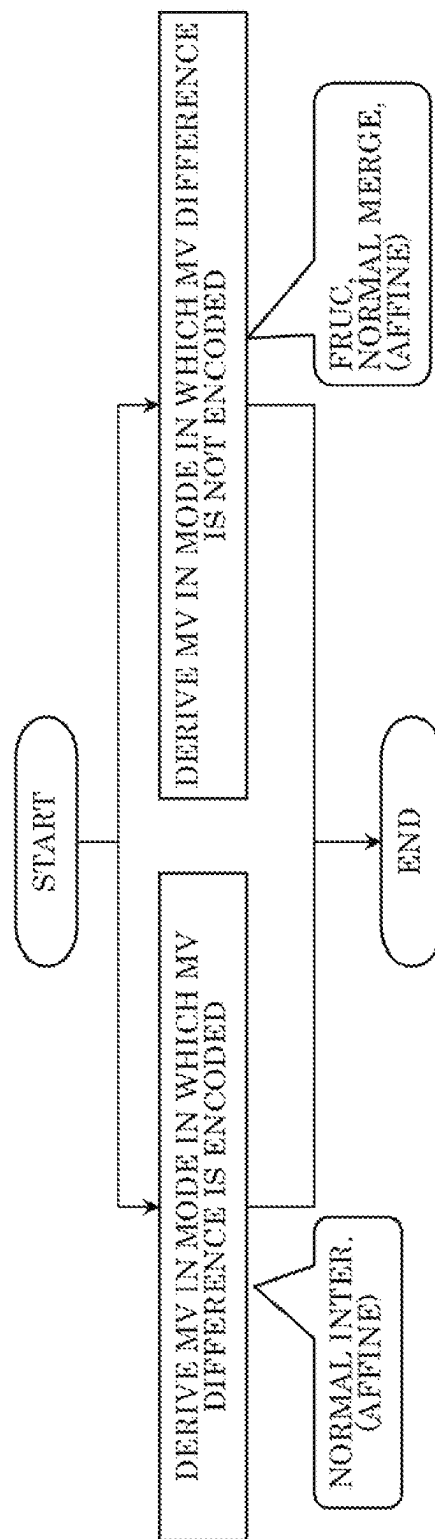


FIG. 38A

MV DERIVATION MODE		MOTION INFORMATION	MV DIFFERENCE
MERGE MODE	INTER MODE	○	○
	OTHER THAN MMVD MODE	○	×
	MMVD MODE	○	△
FRUC MODE		×	×

FIG. 38B

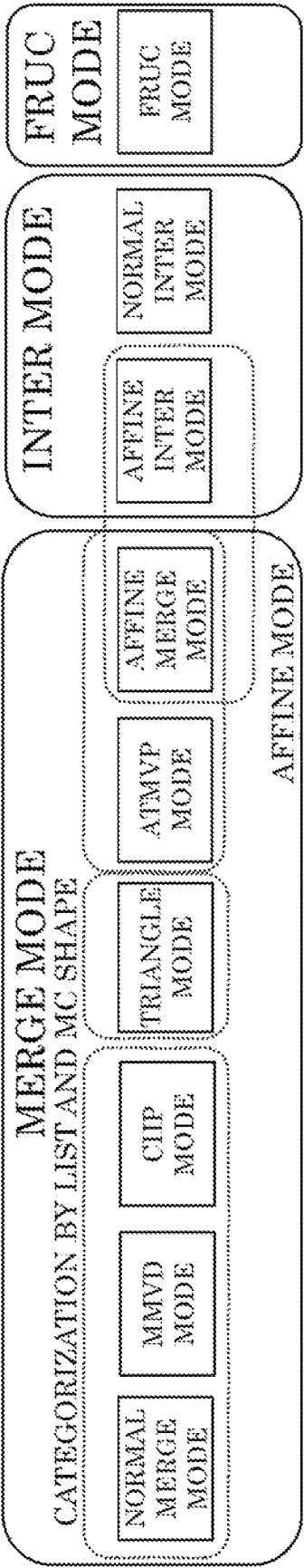


FIG. 39

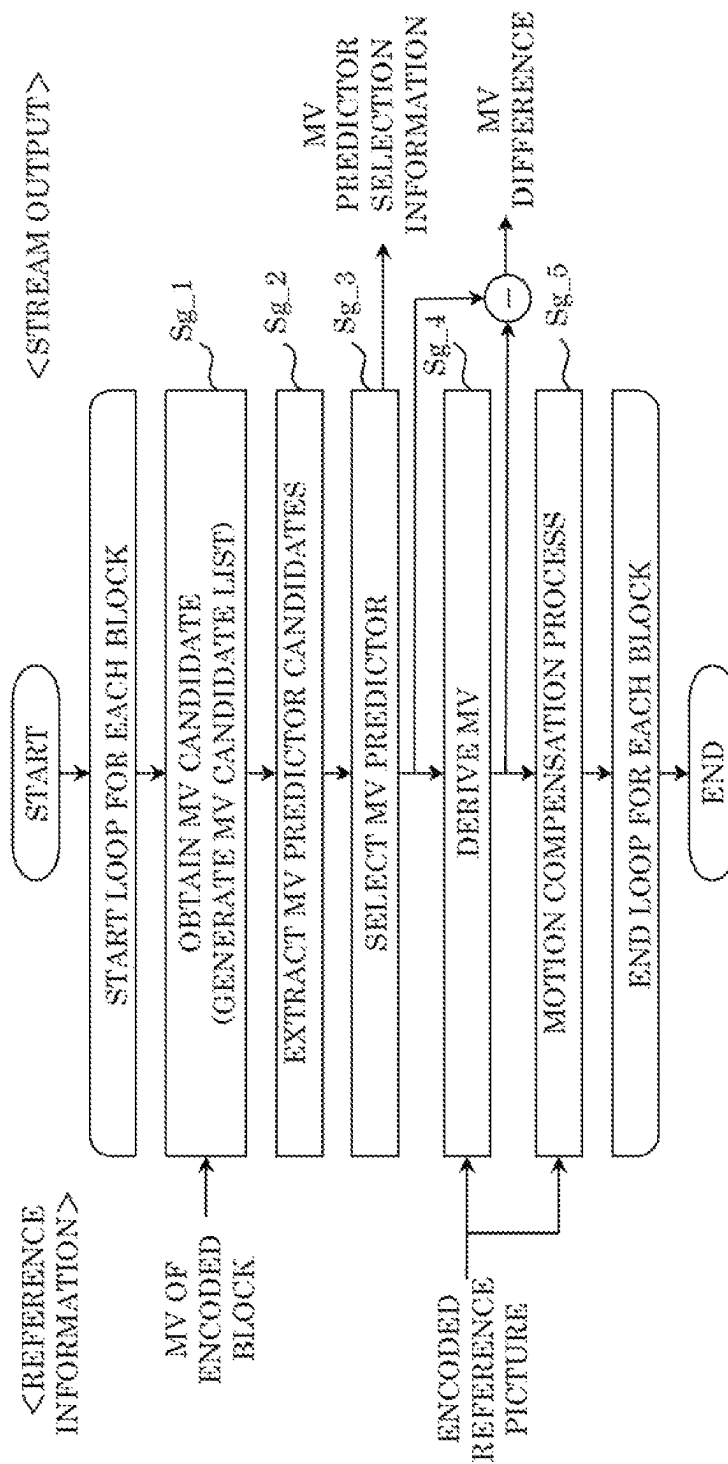


FIG. 40

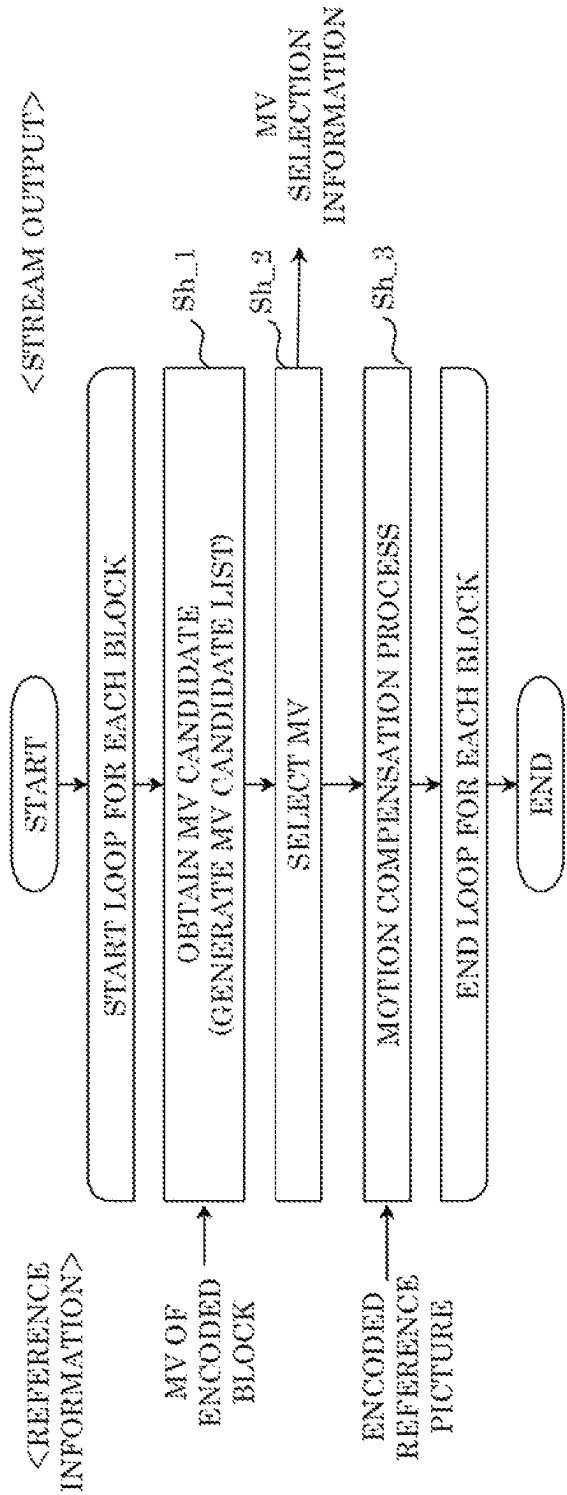


FIG. 41

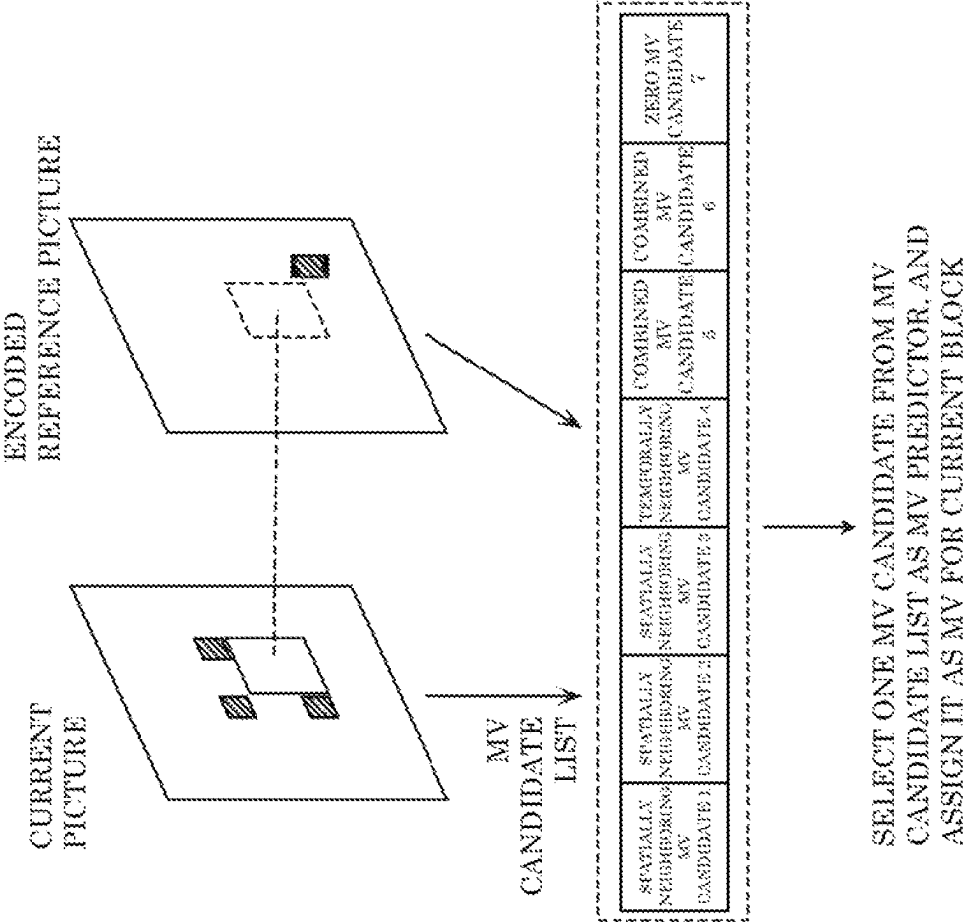


FIG. 42

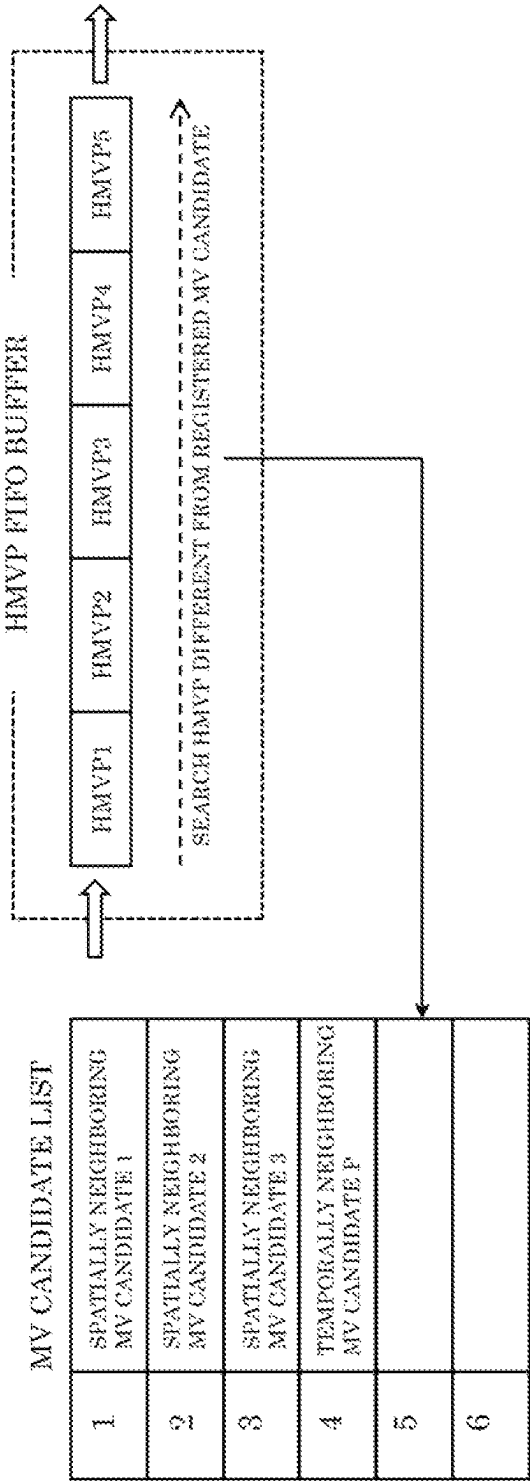


FIG. 43

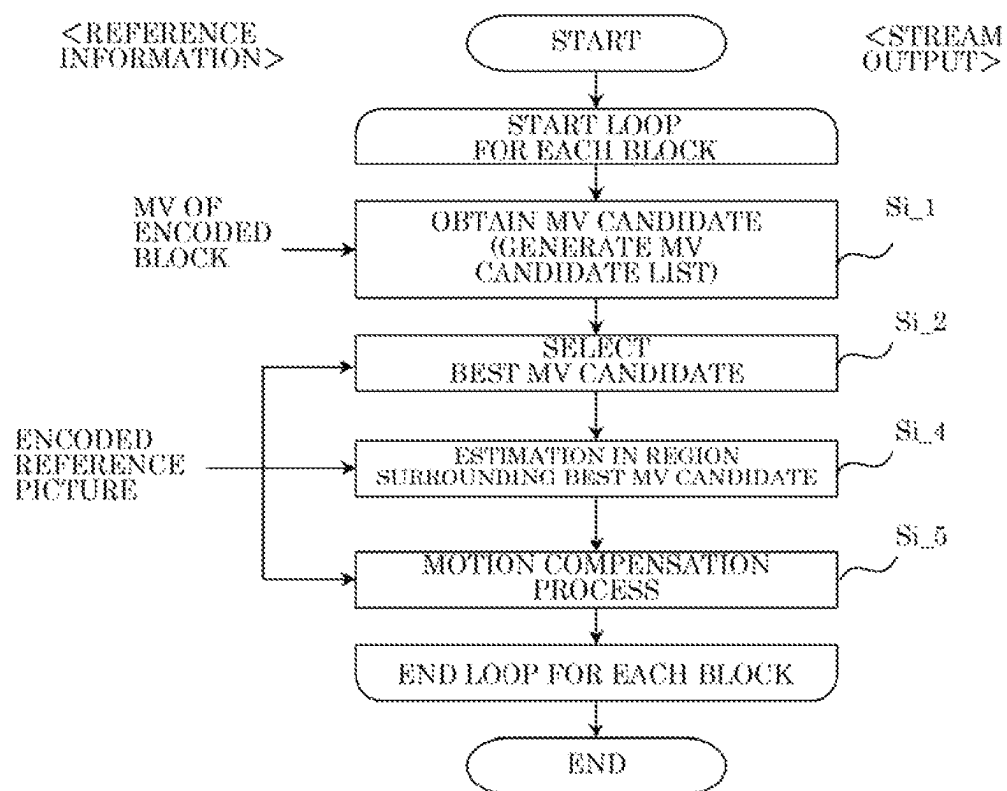


FIG. 44

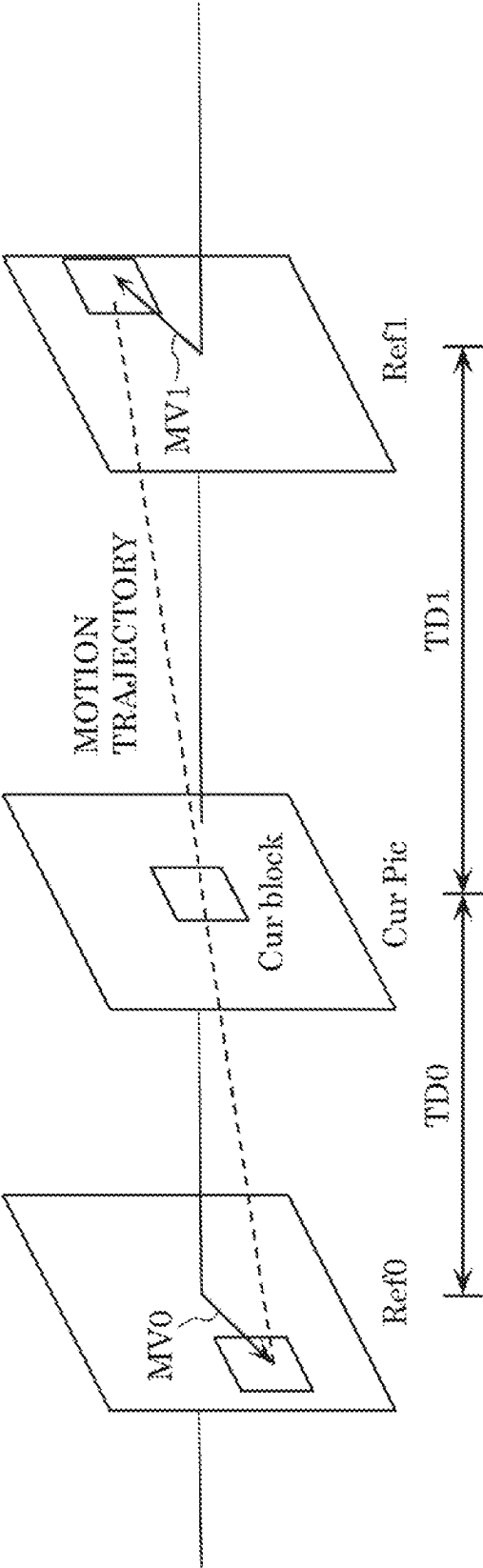


FIG. 45

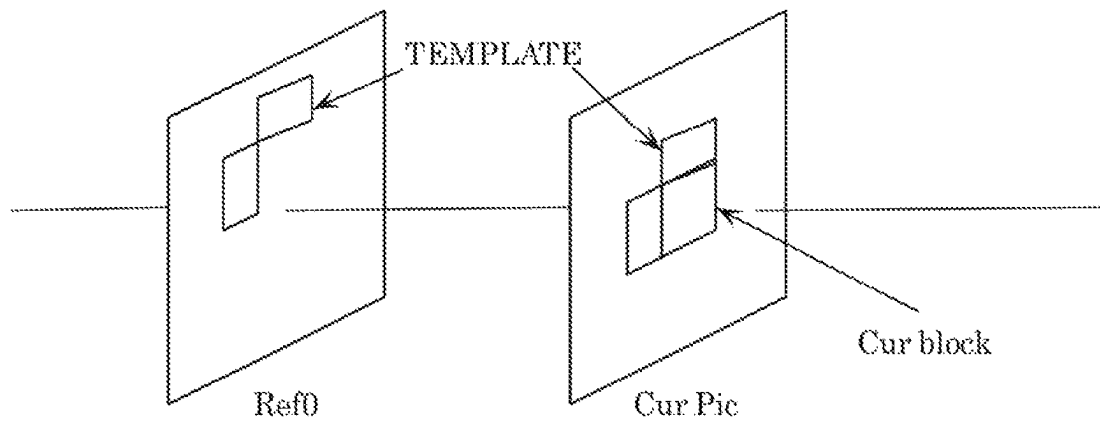


FIG. 46A

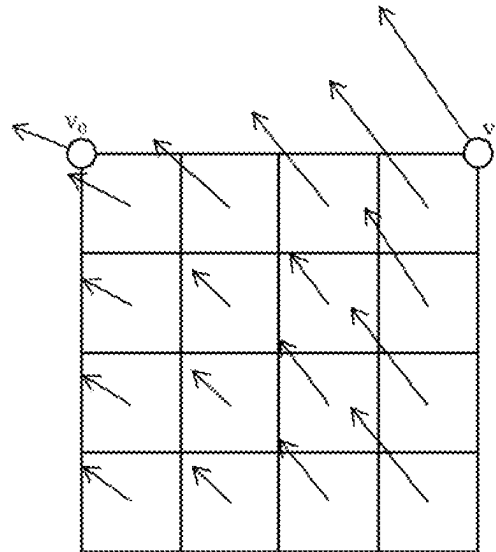


FIG. 46B

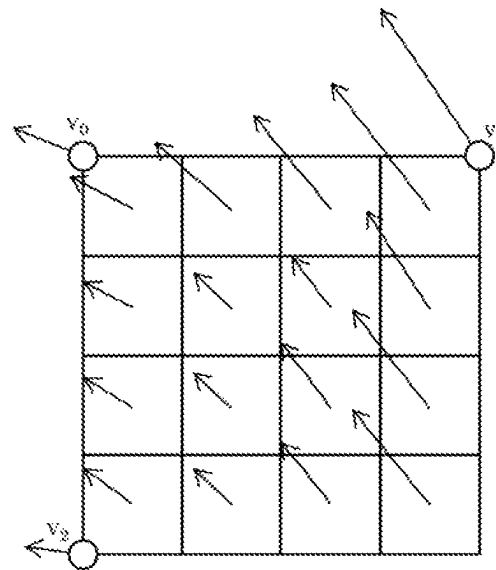


FIG. 47A

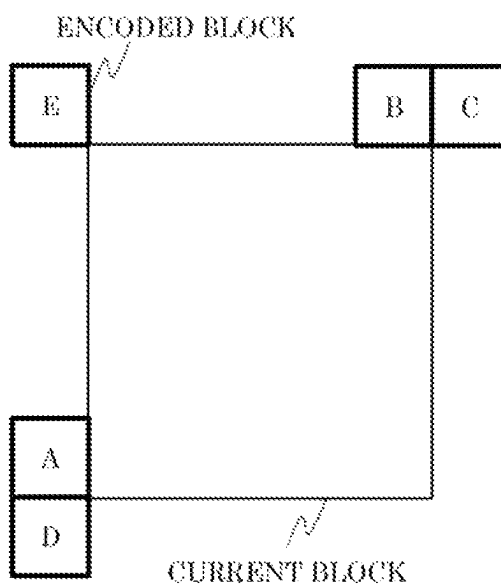


FIG. 47B

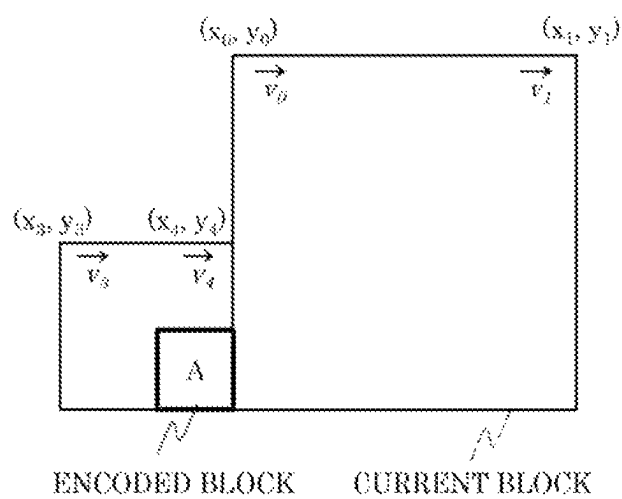


FIG. 47C

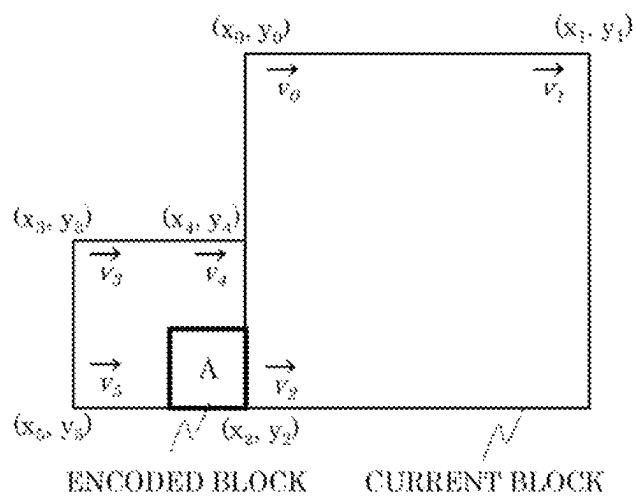


FIG. 48A

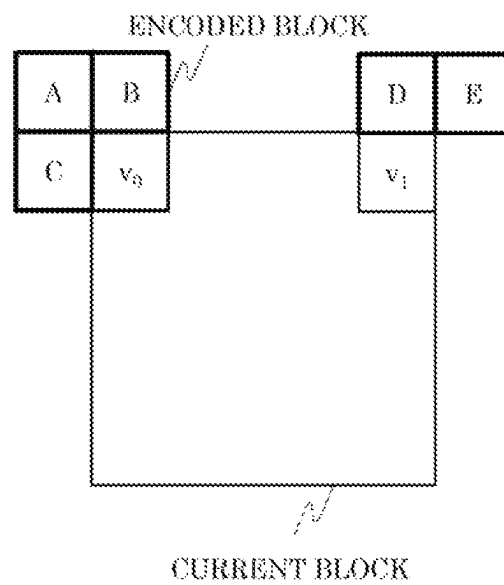


FIG. 48B

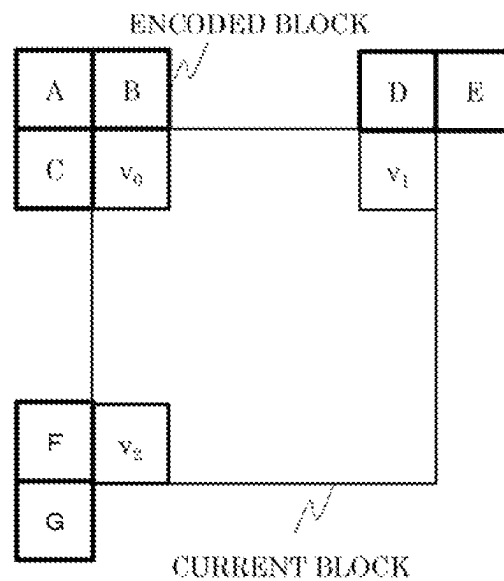


FIG. 49A

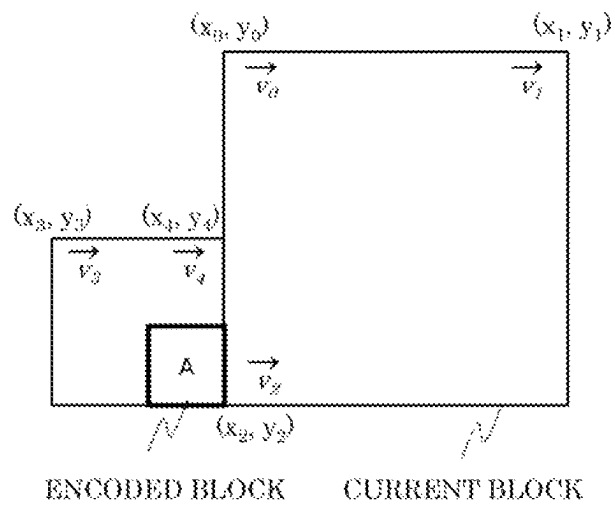
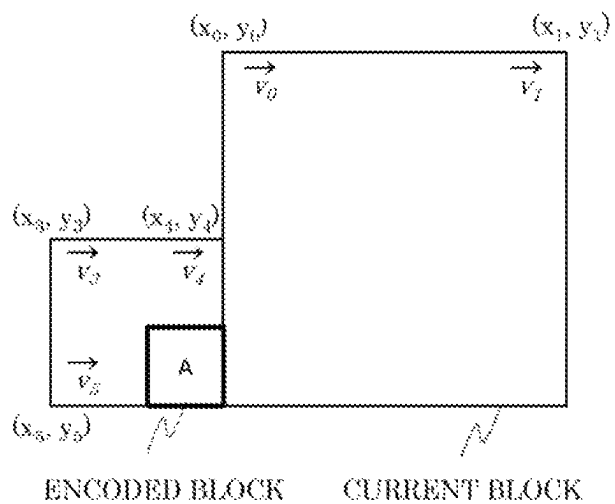


FIG. 49B



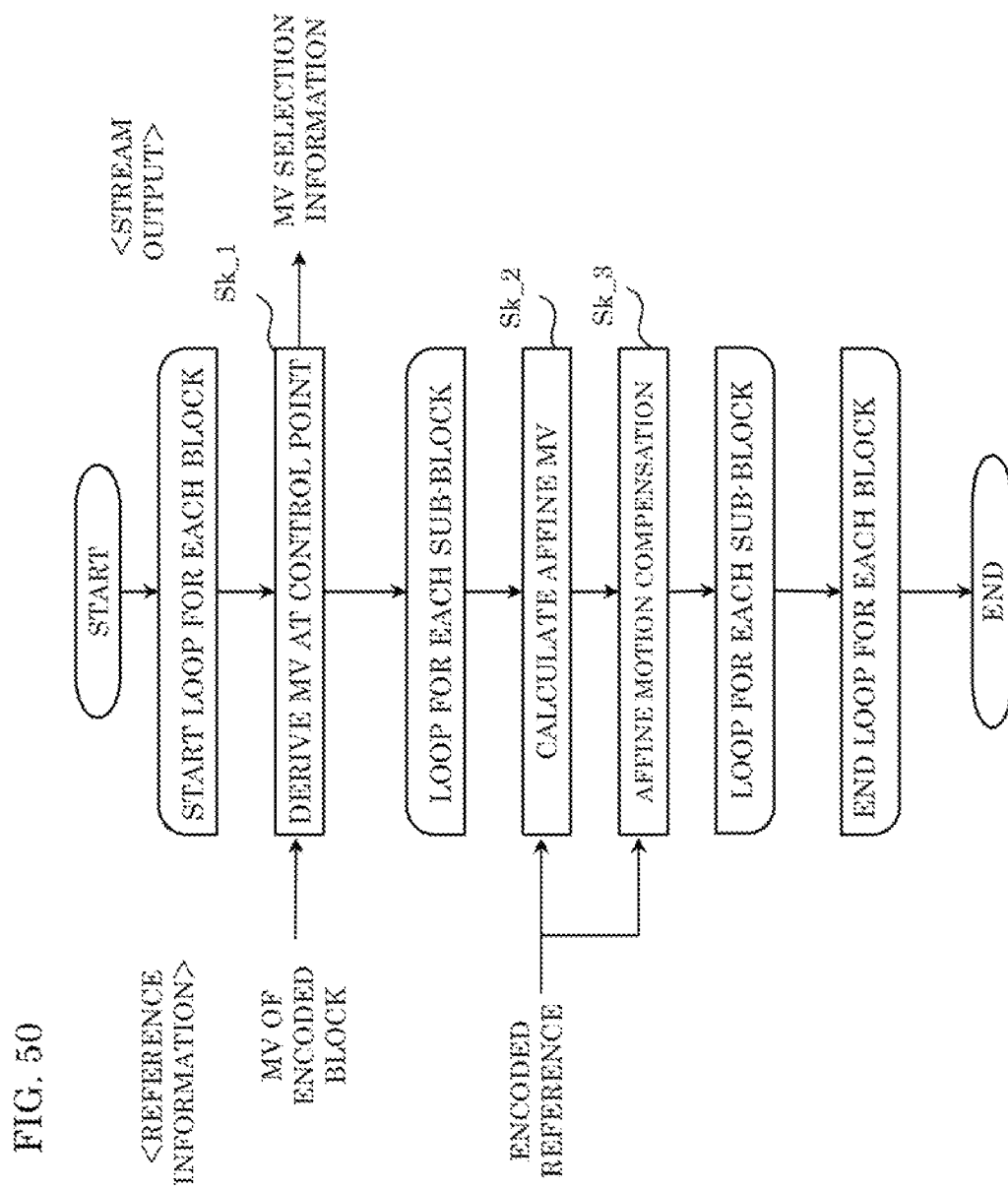


FIG. 51

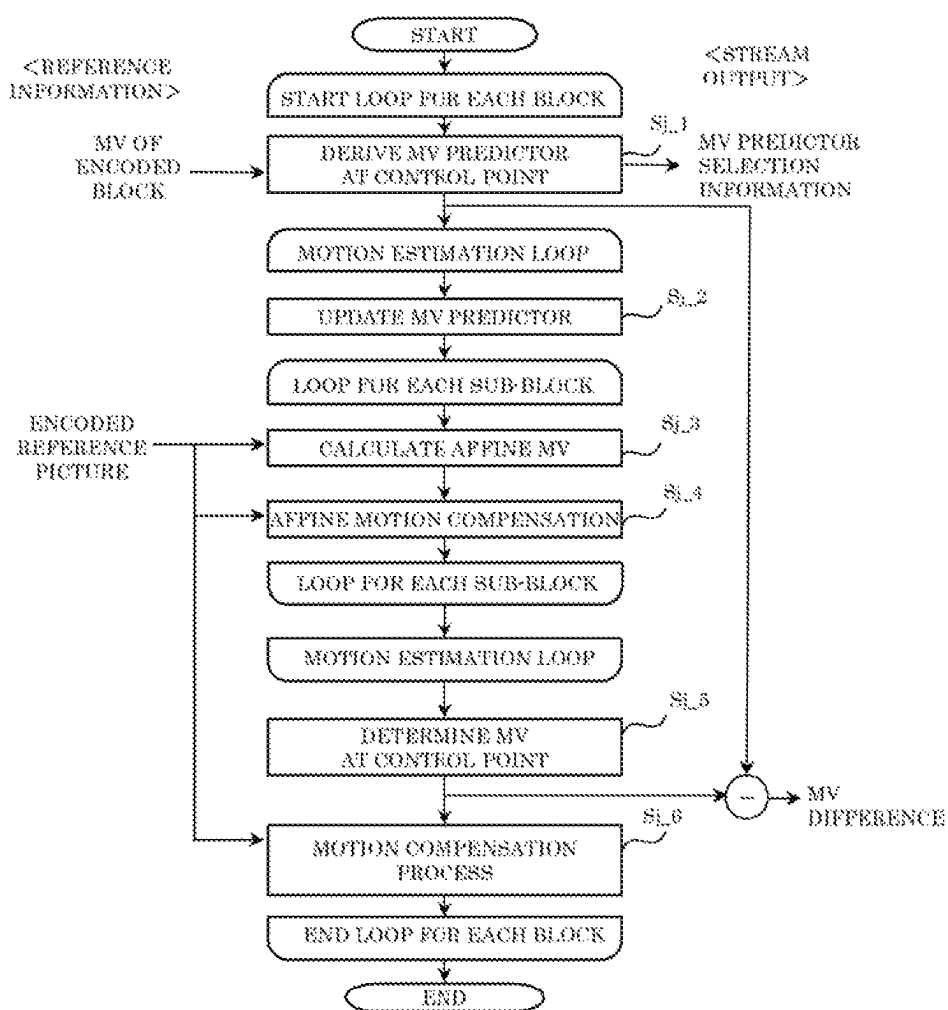


FIG. 52A

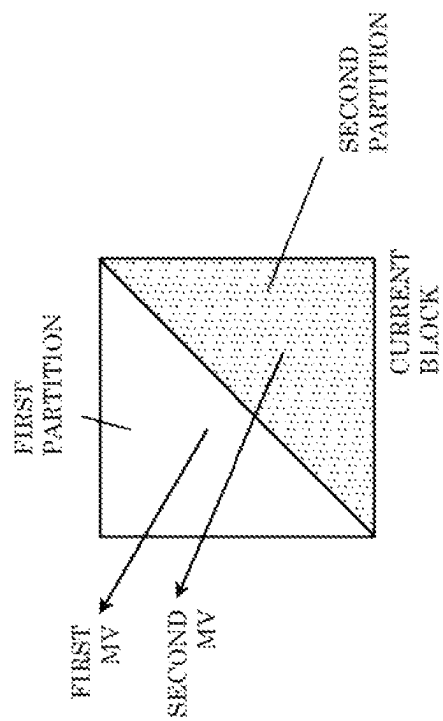


FIG. 52B

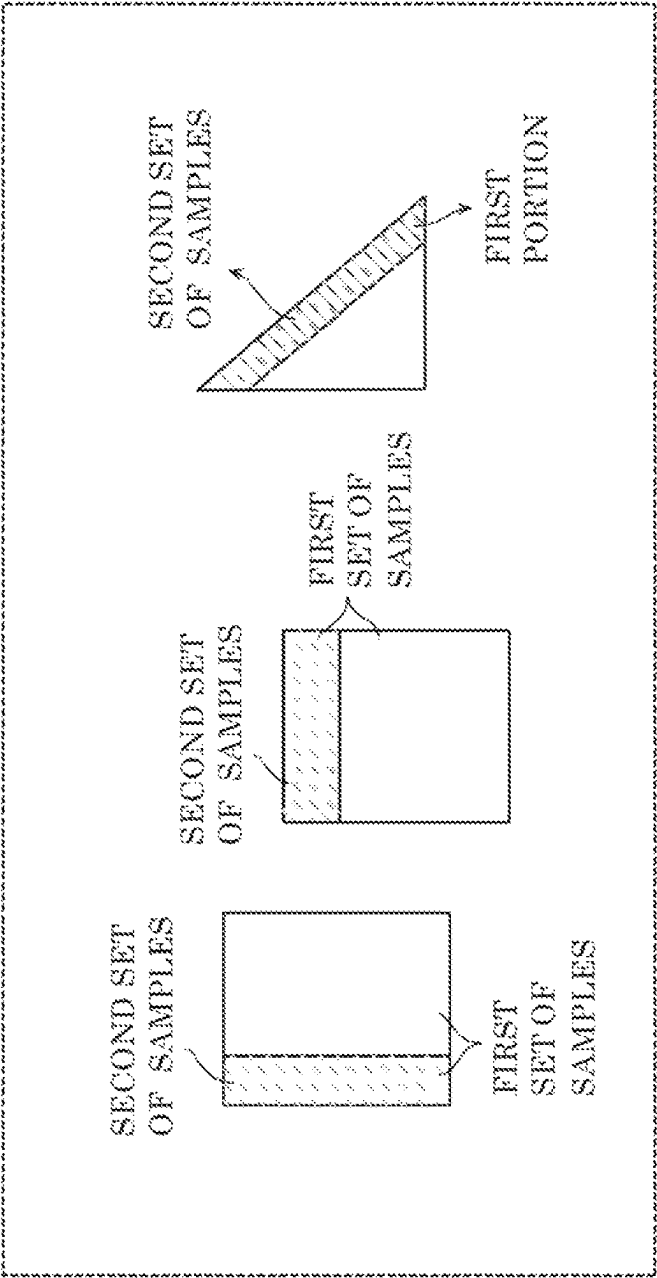


FIG. 52C

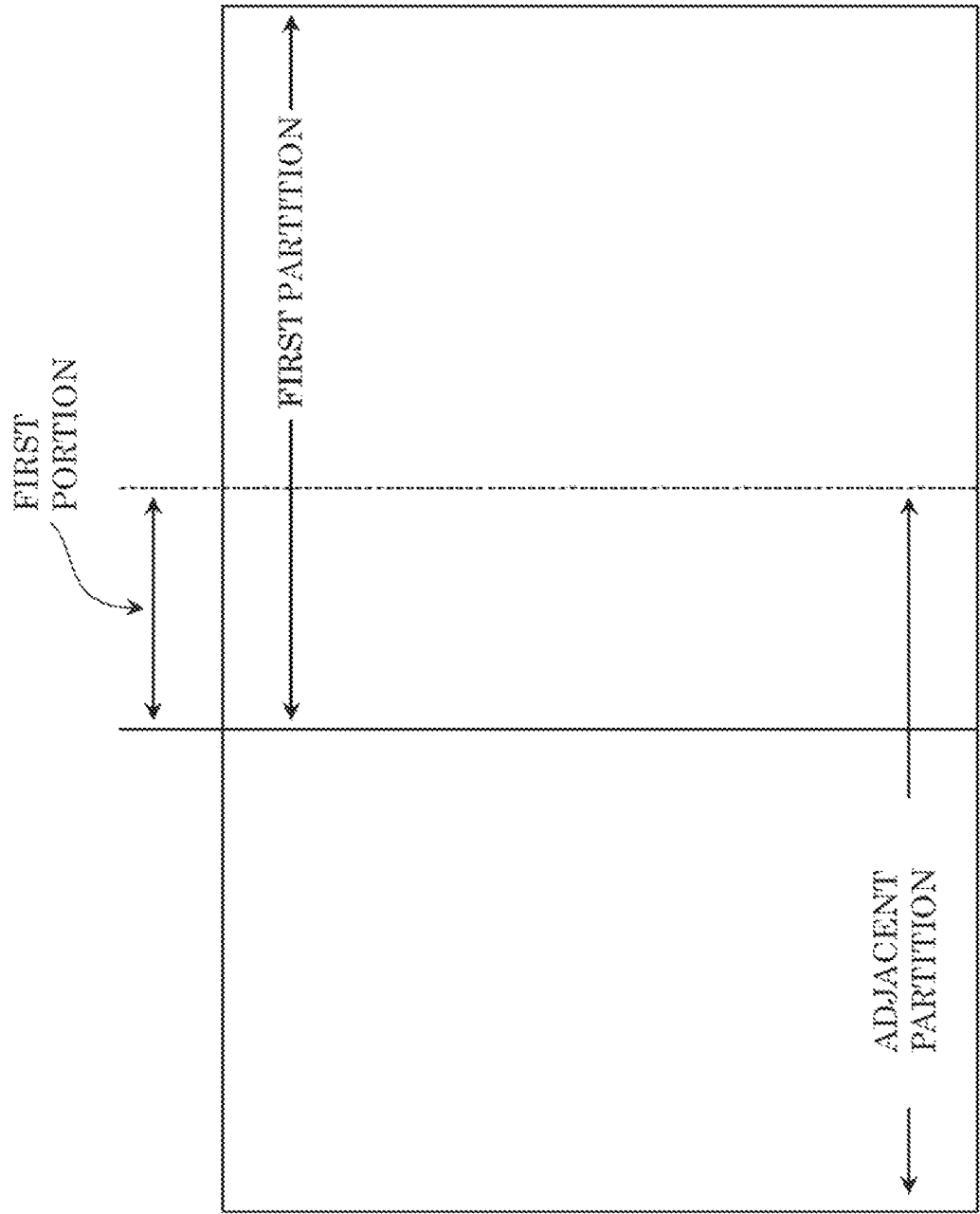


FIG. 53

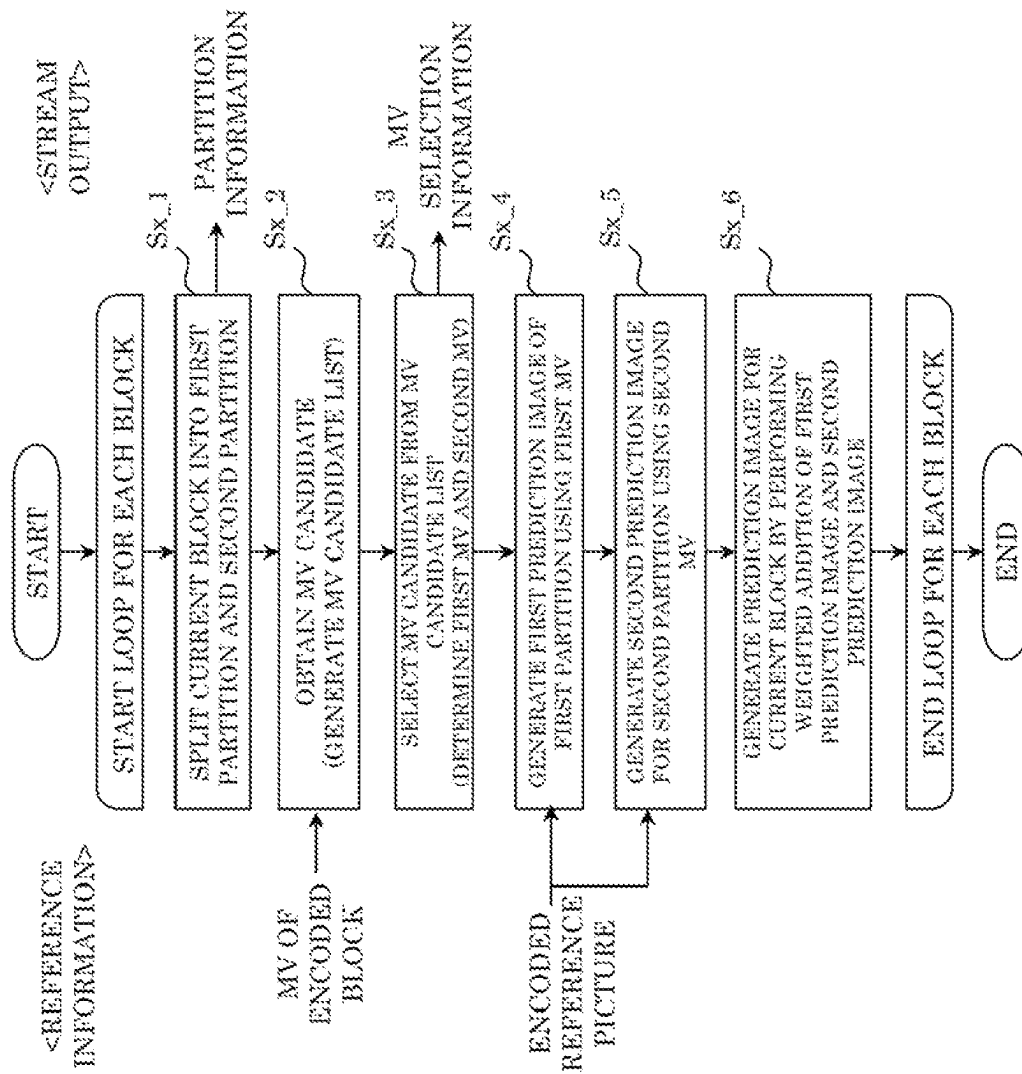


FIG. 54

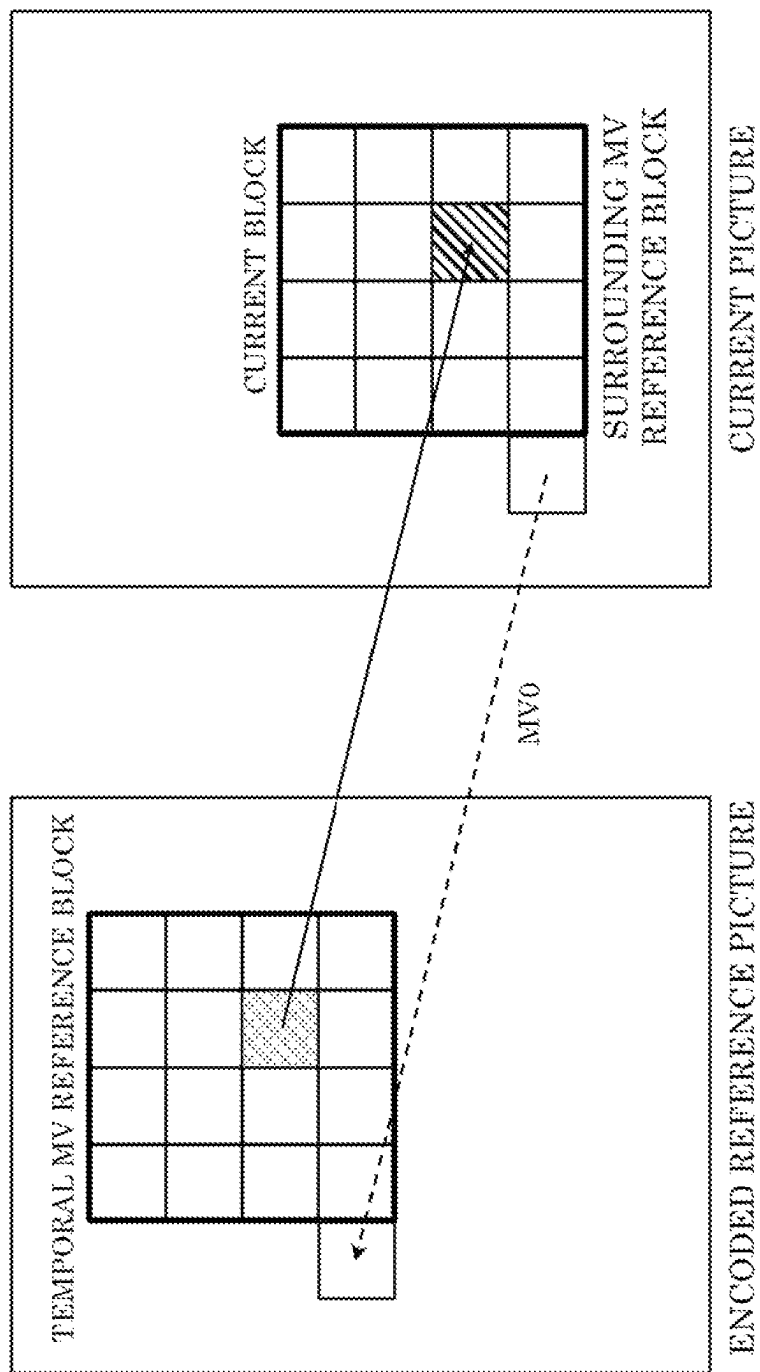


FIG. 55

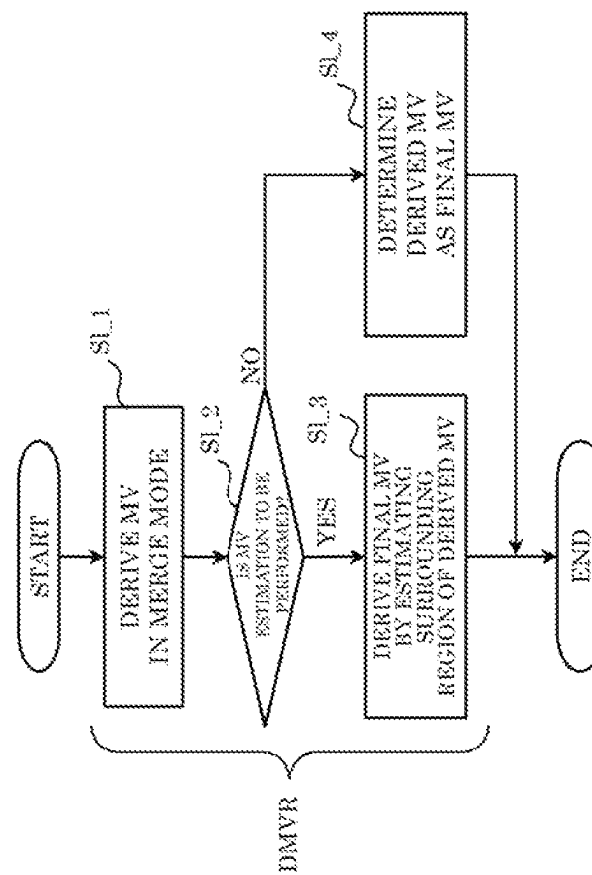


FIG. 56

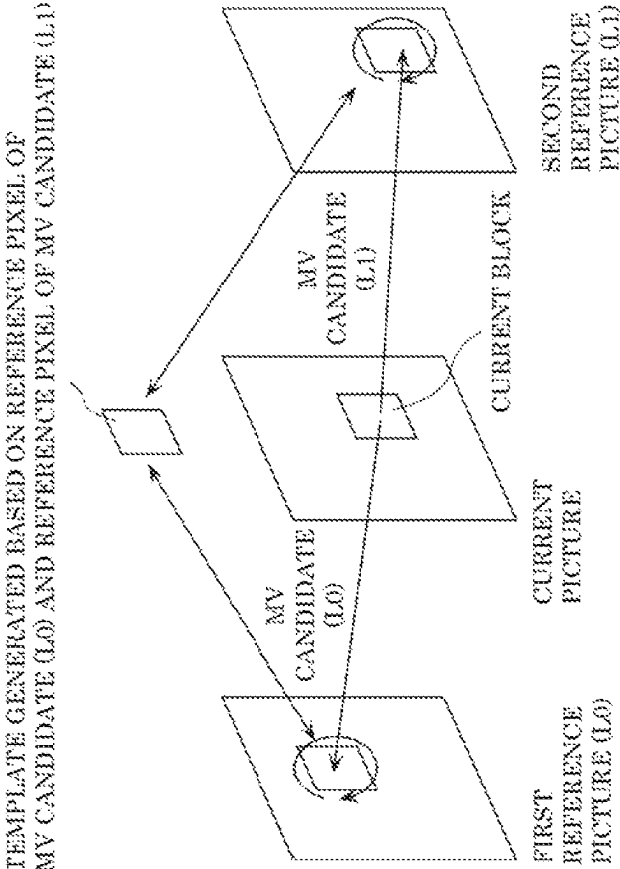


FIG. 57

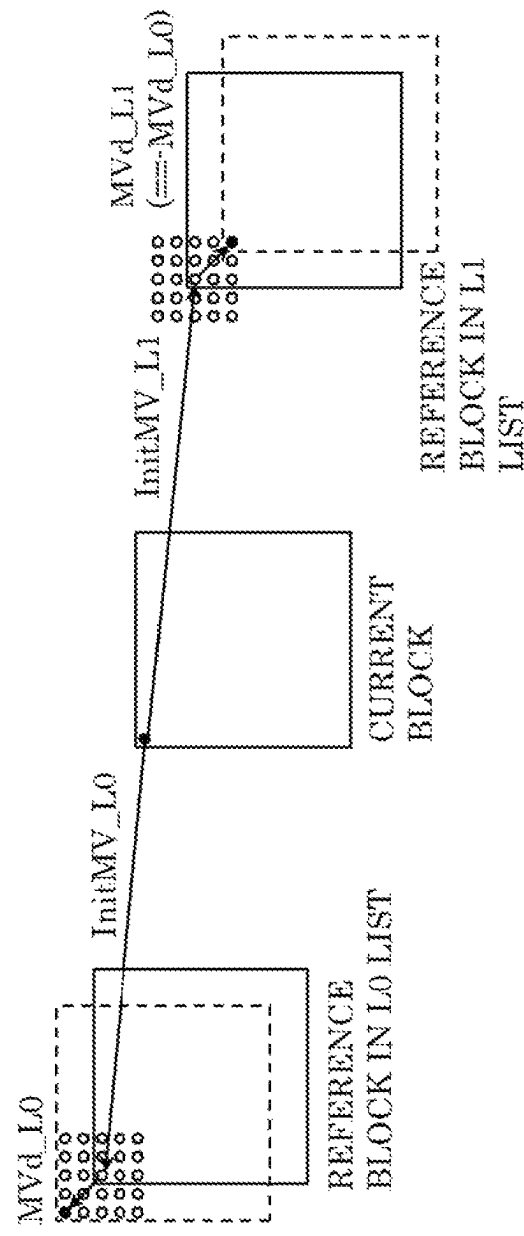


FIG. 58A

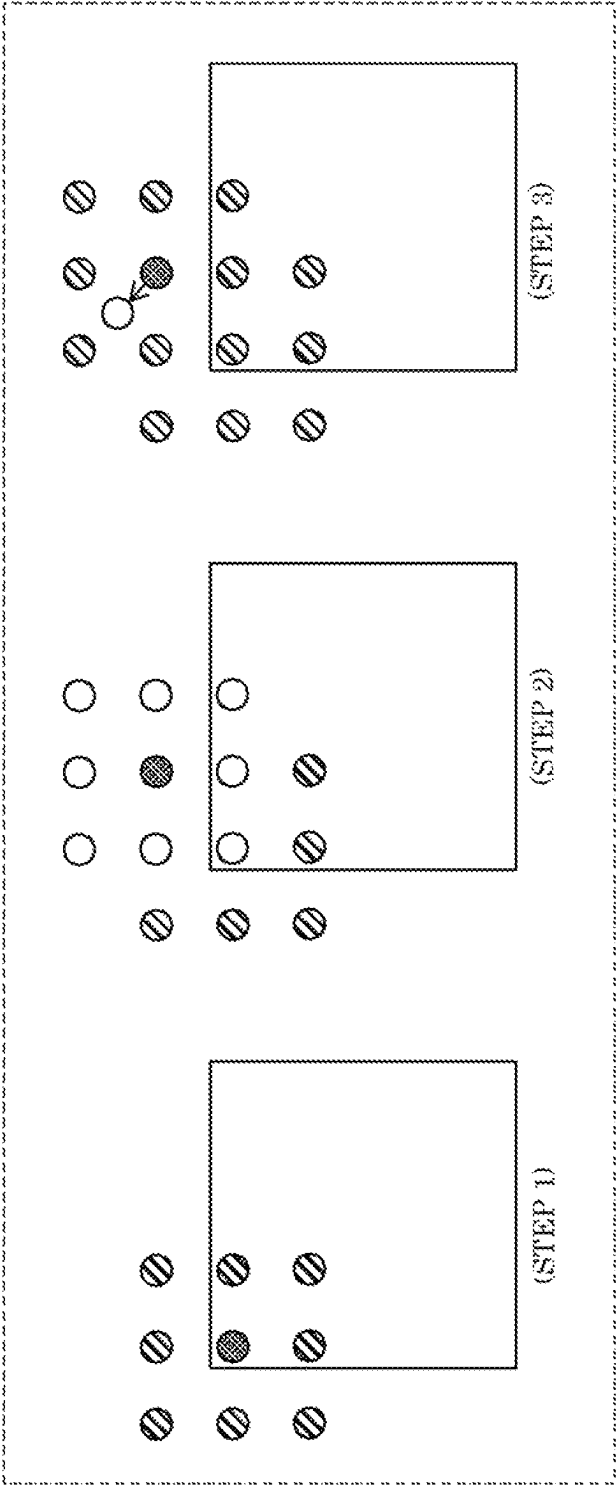


FIG. 58B

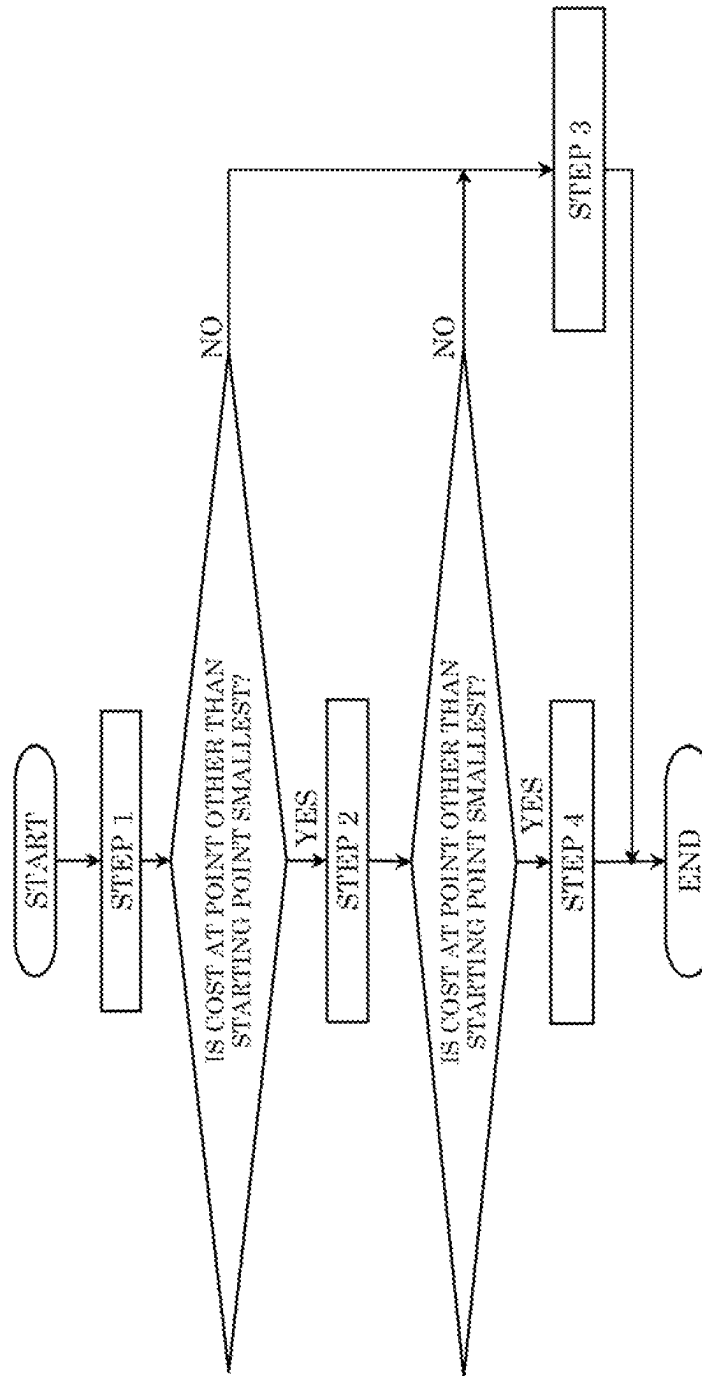


FIG. 59

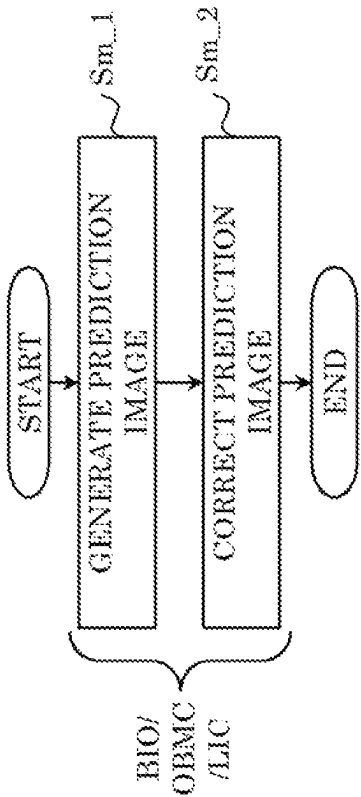


FIG. 60

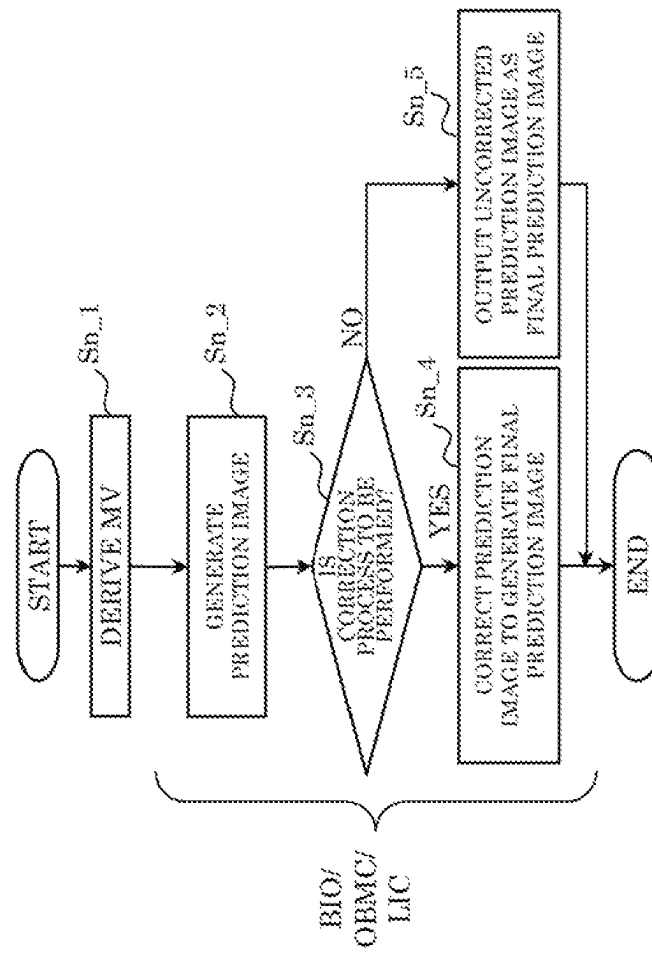


FIG. 61

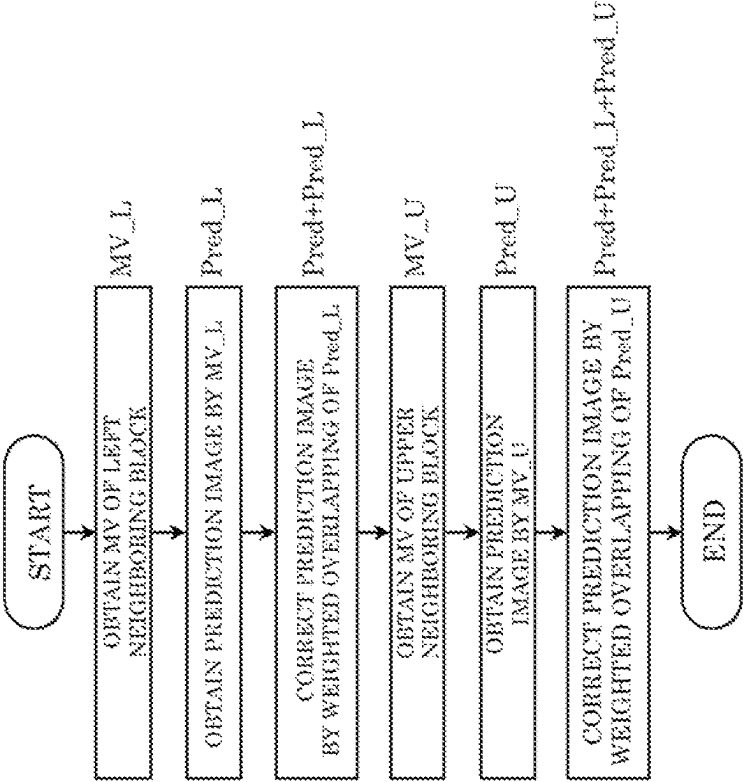


FIG. 62

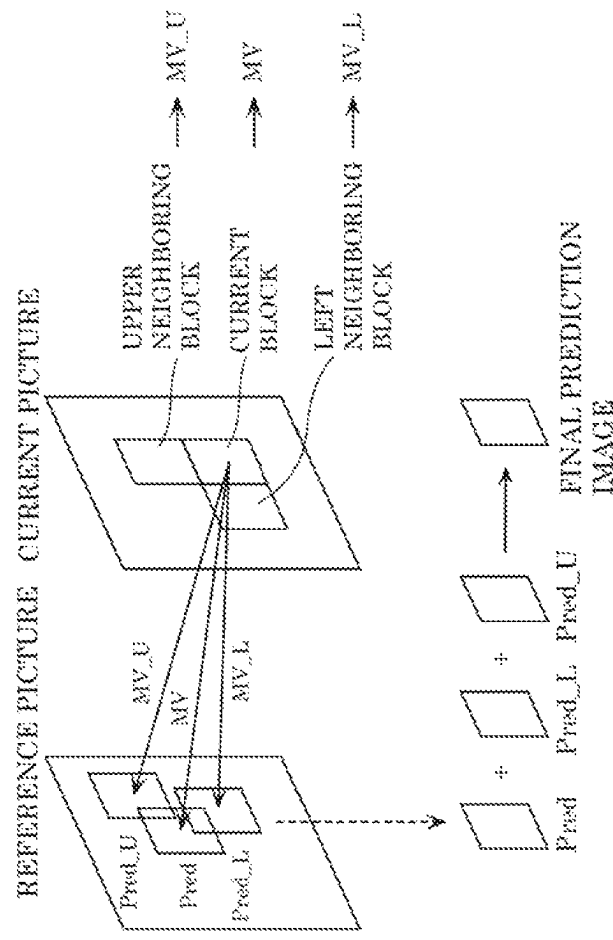


FIG. 63

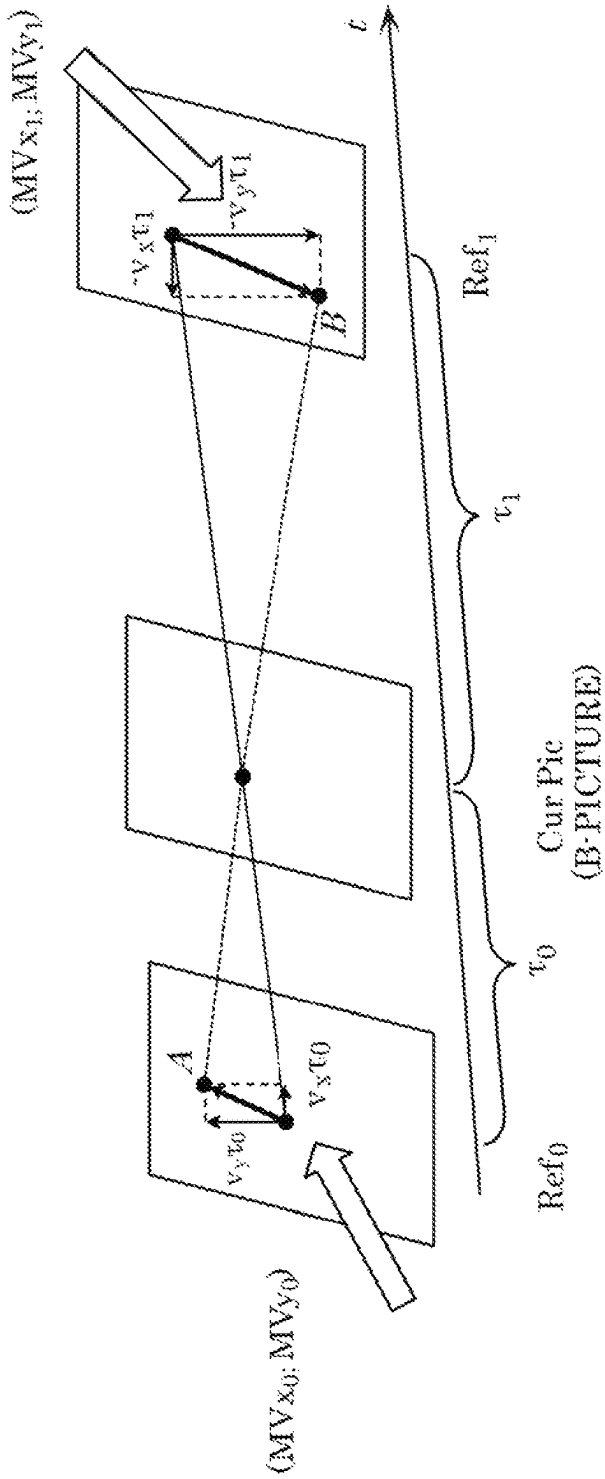


FIG. 64

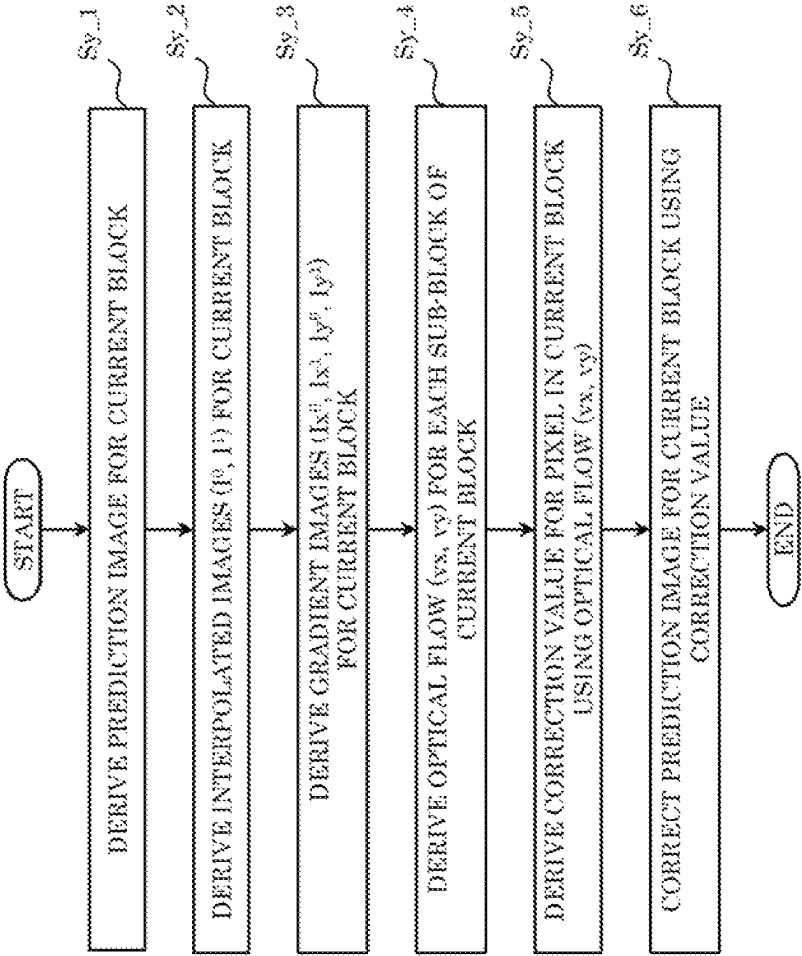
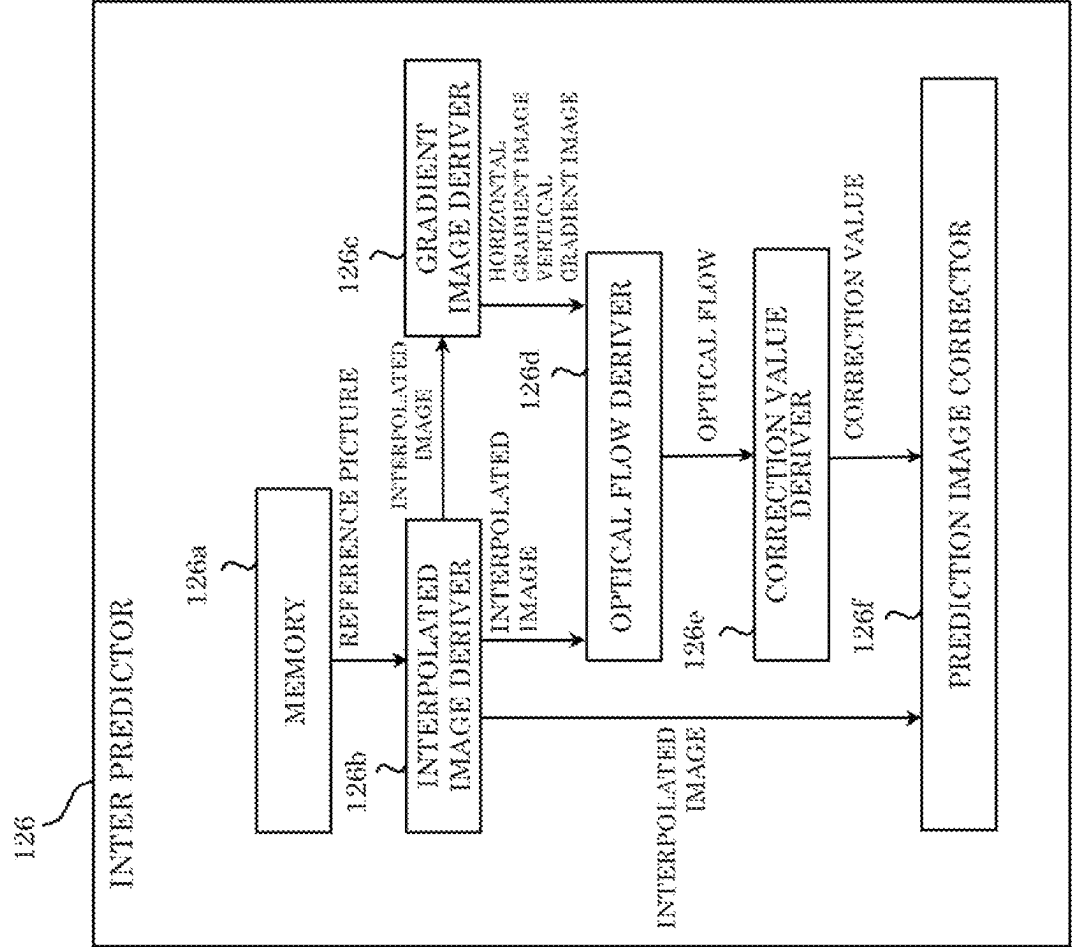


FIG. 65



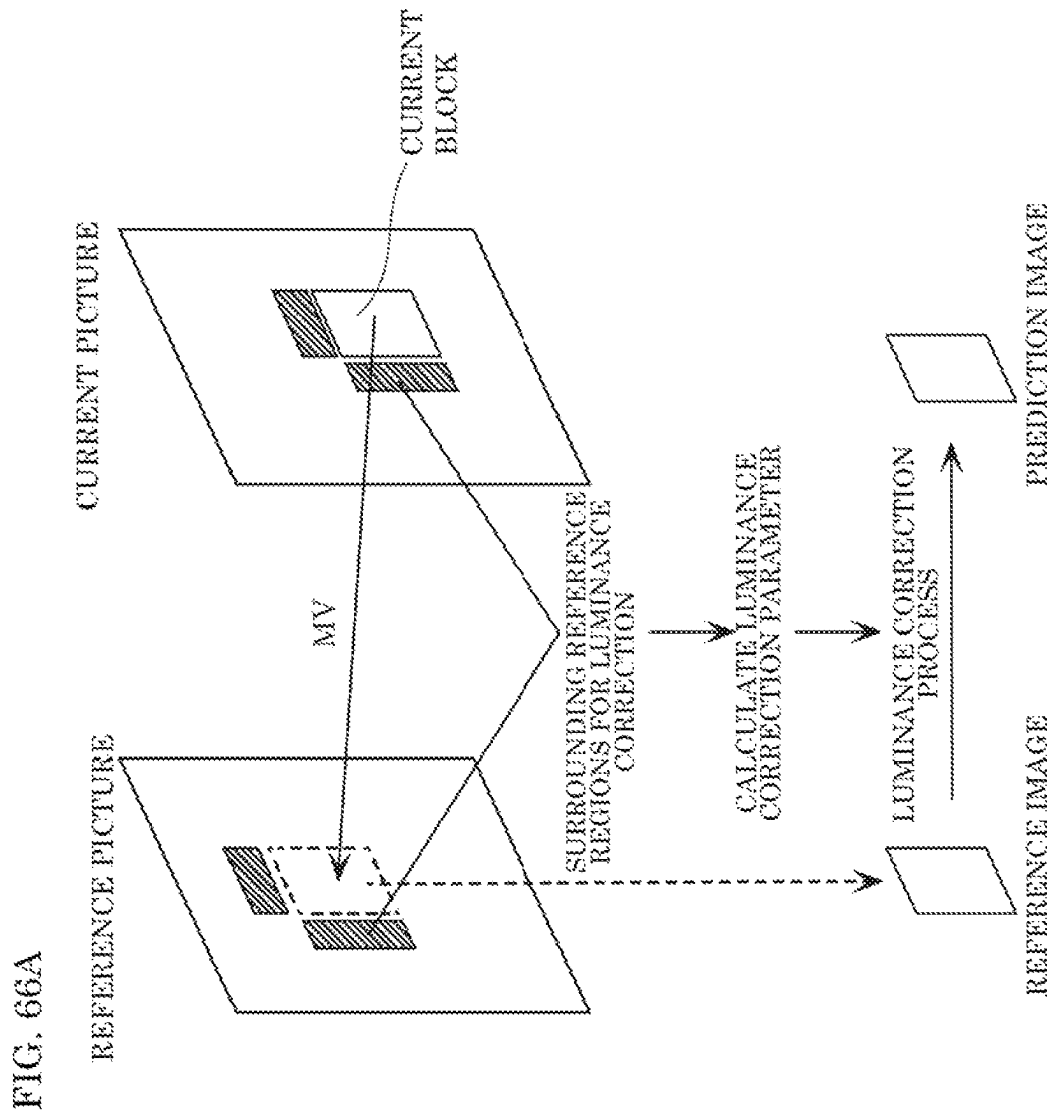


FIG. 66B

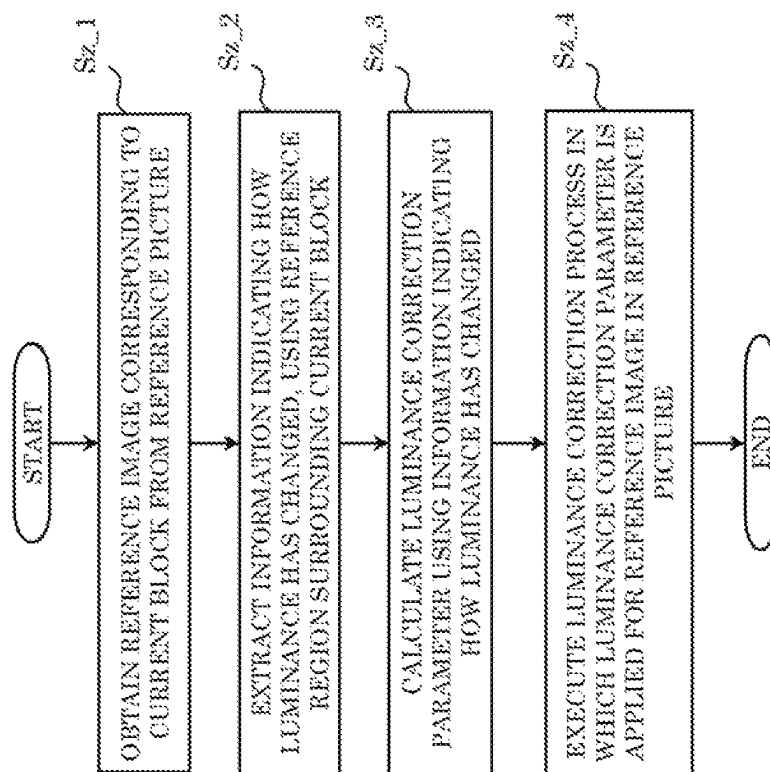


FIG. 67

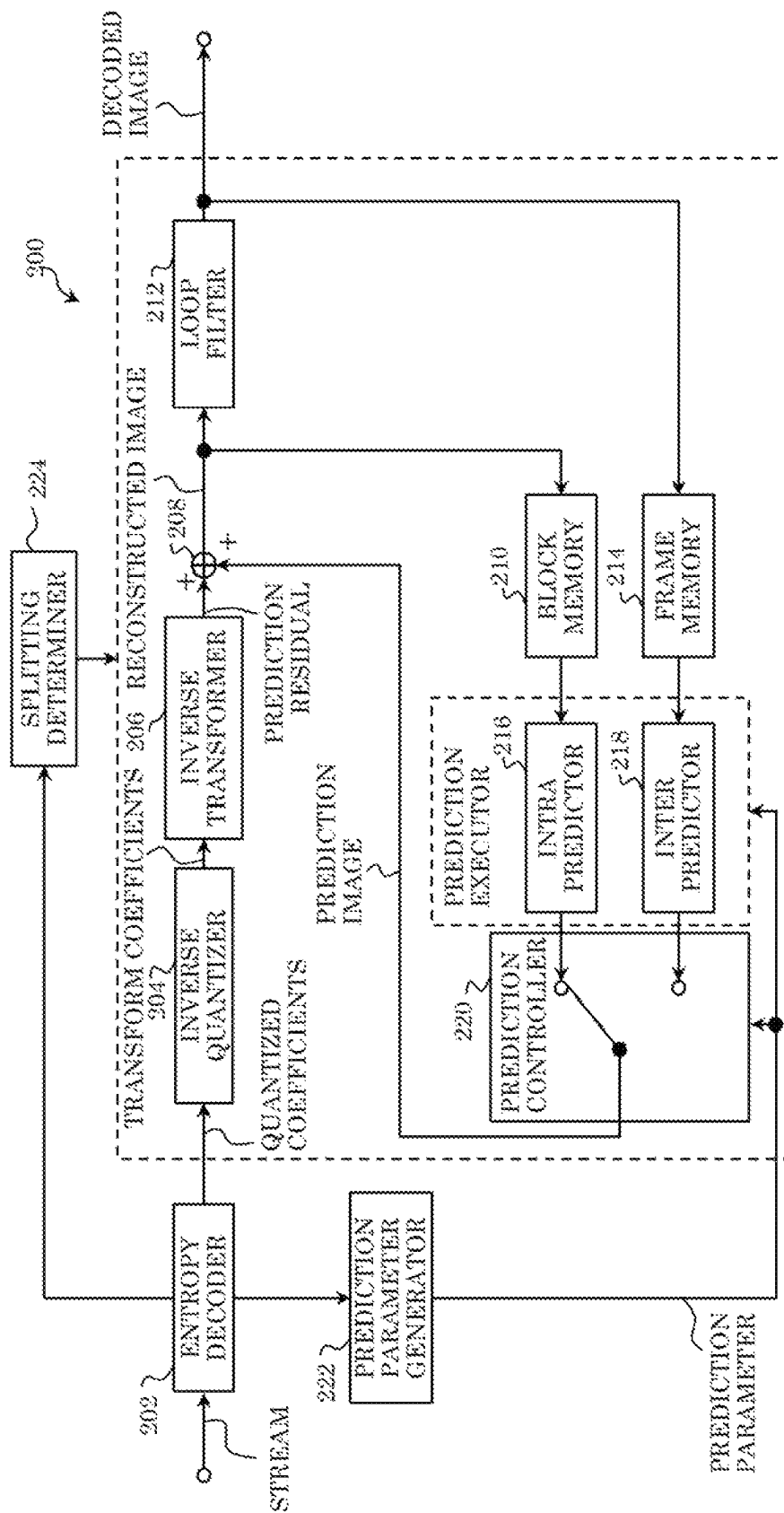


FIG. 68

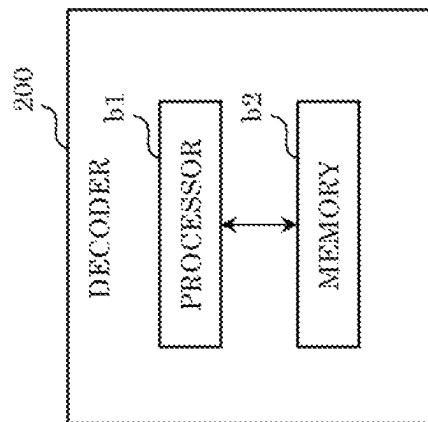


FIG. 69

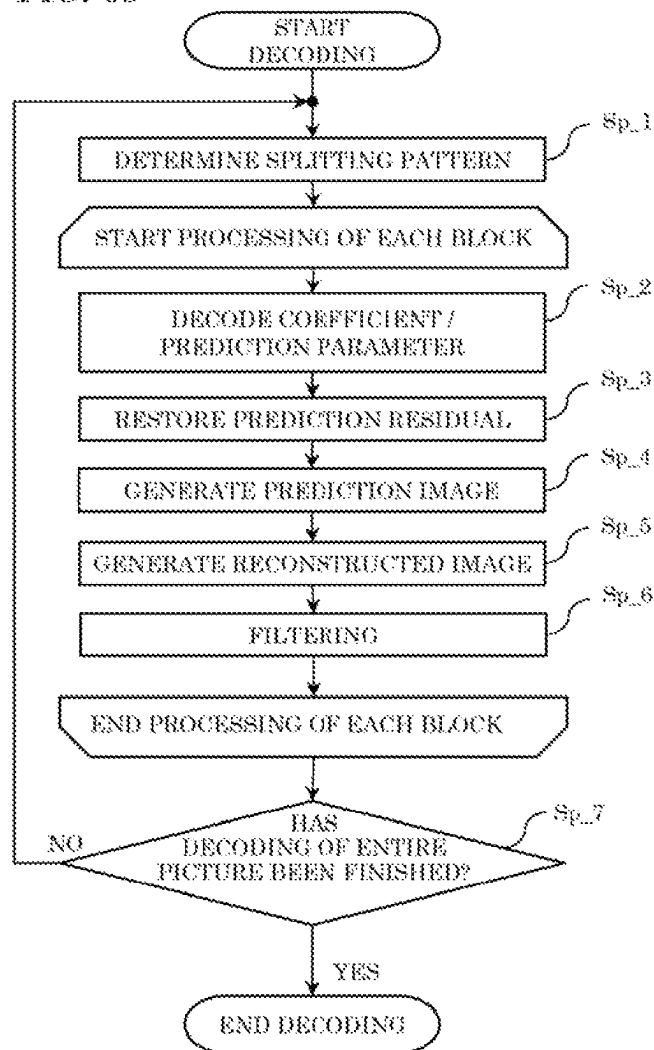


FIG. 70

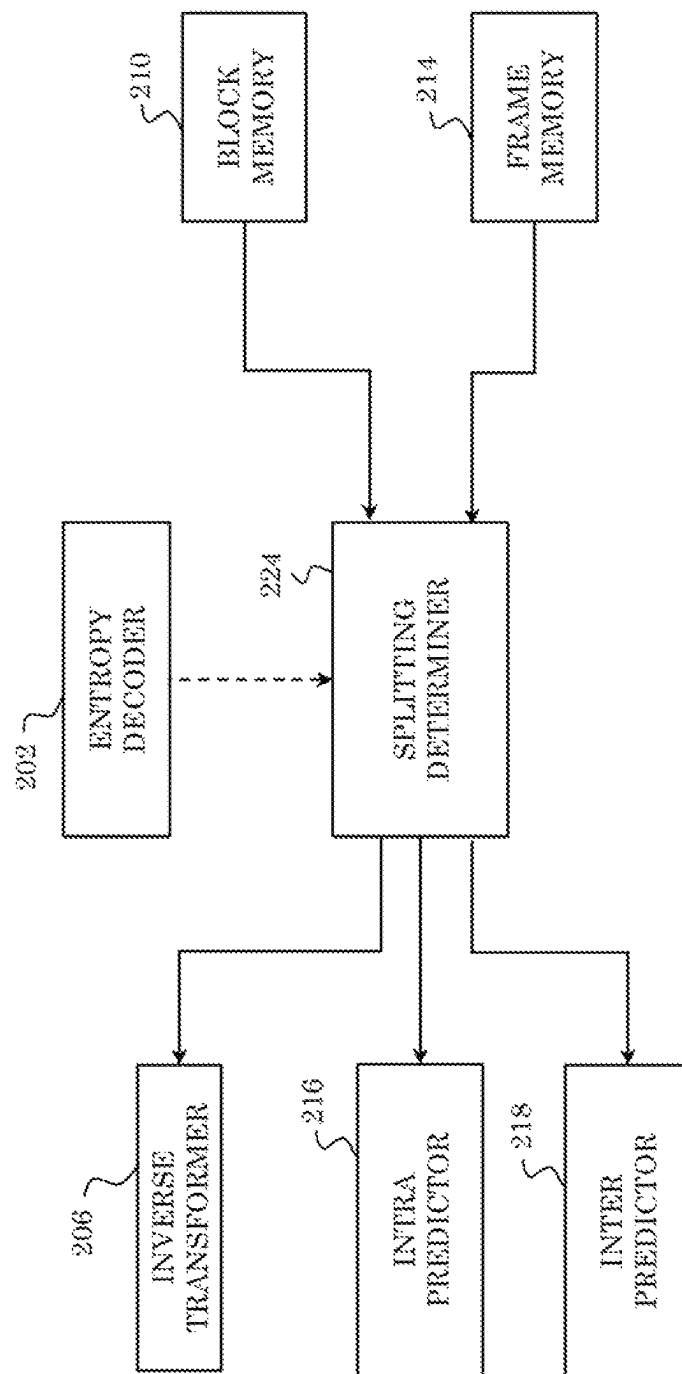


FIG. 71

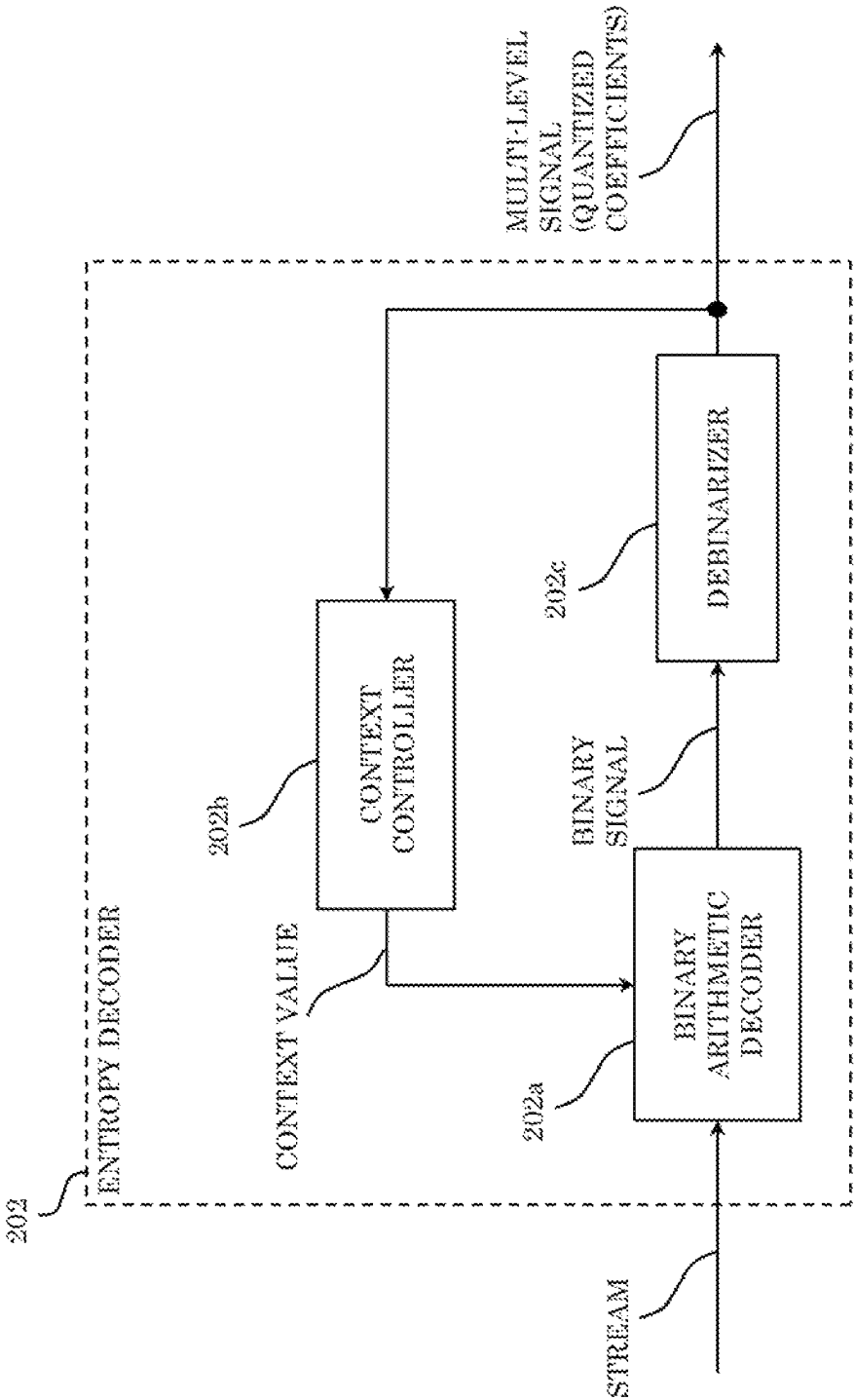


FIG. 72

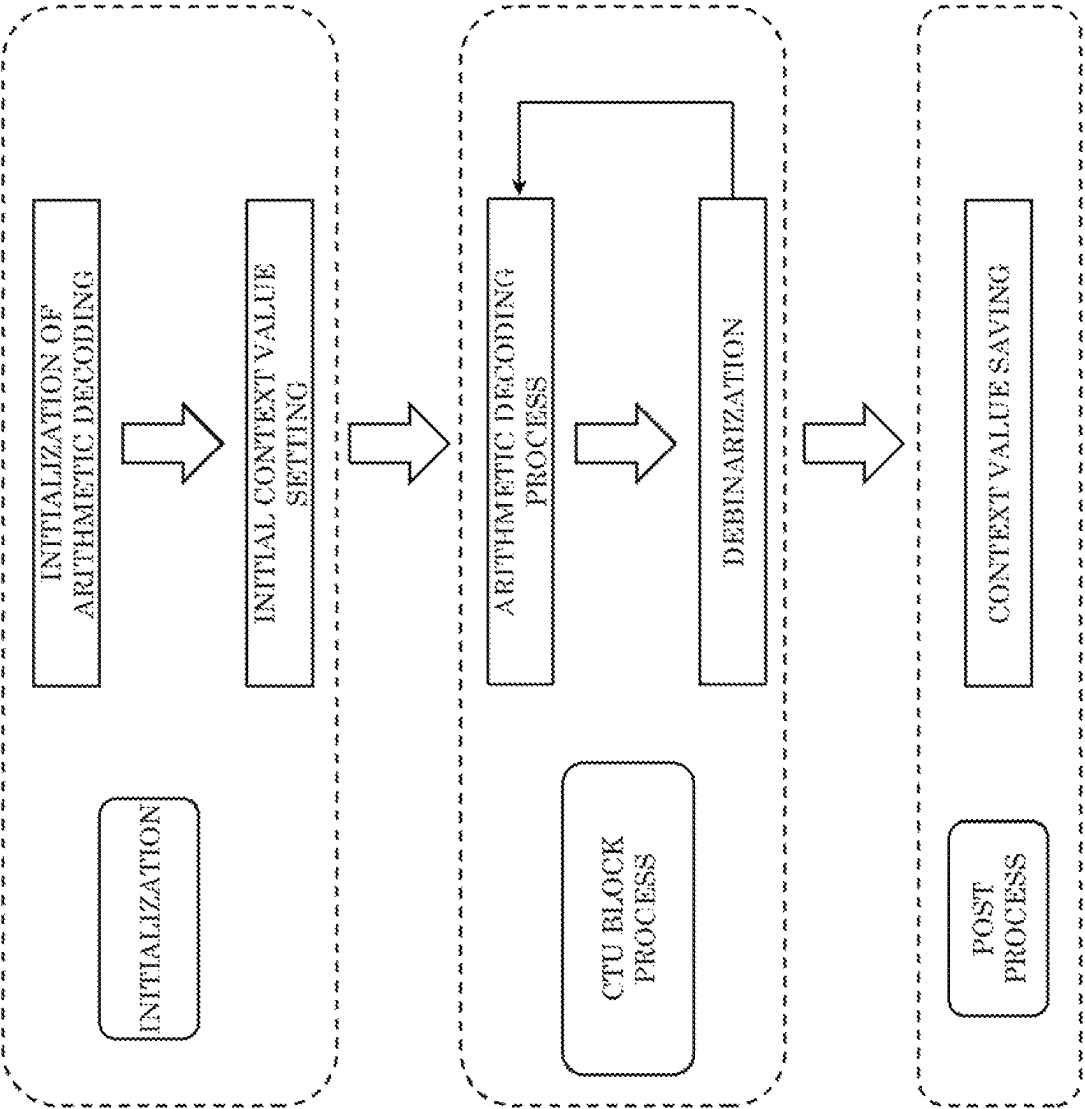


FIG. 73

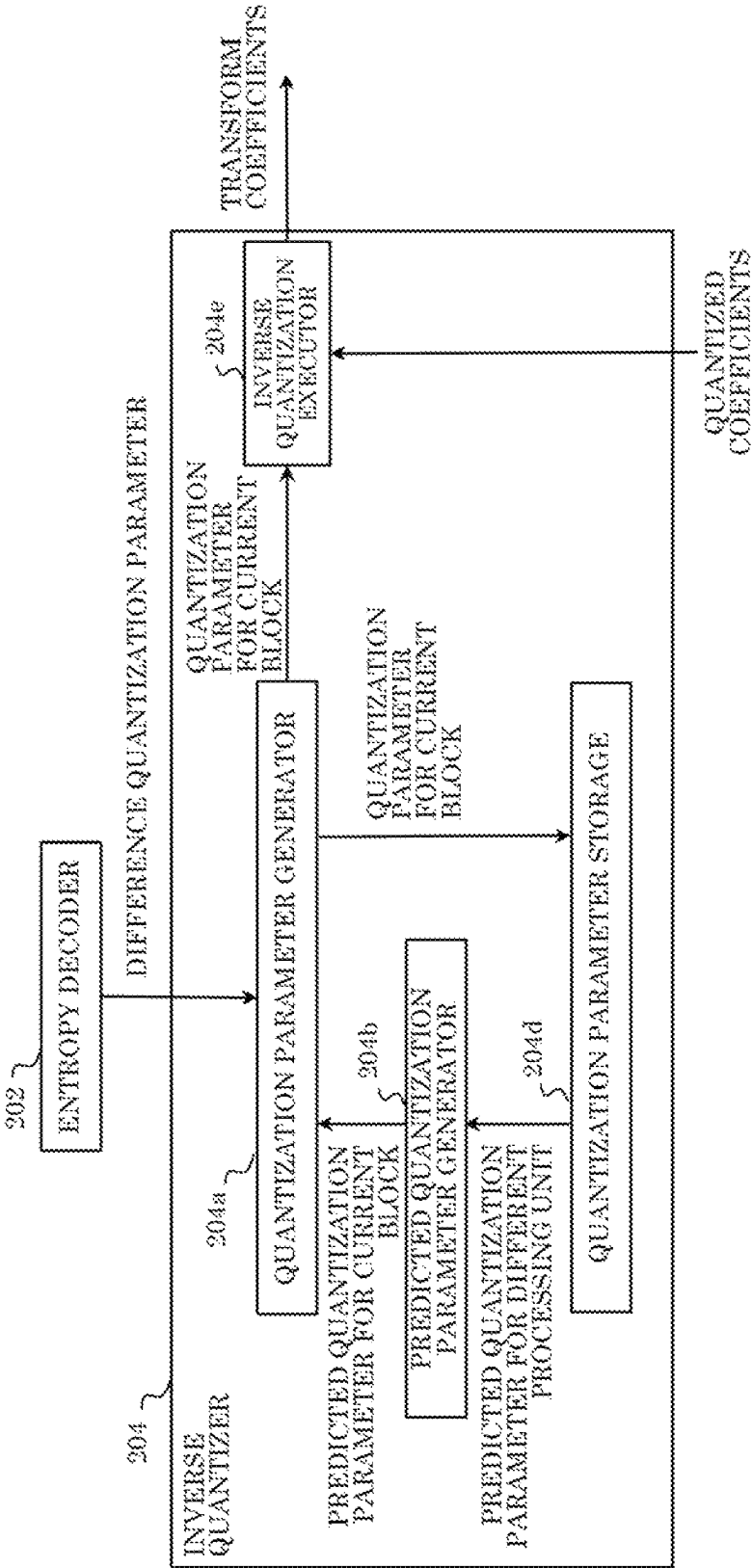


FIG. 74

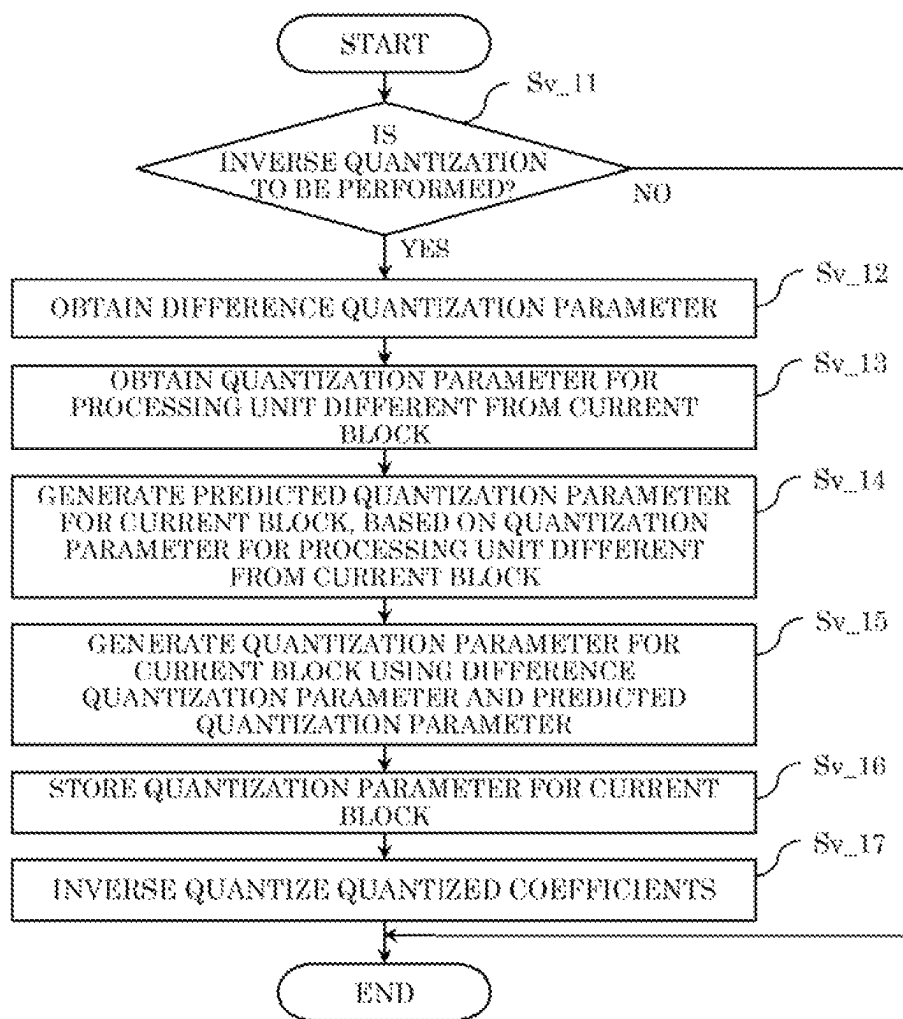


FIG. 75

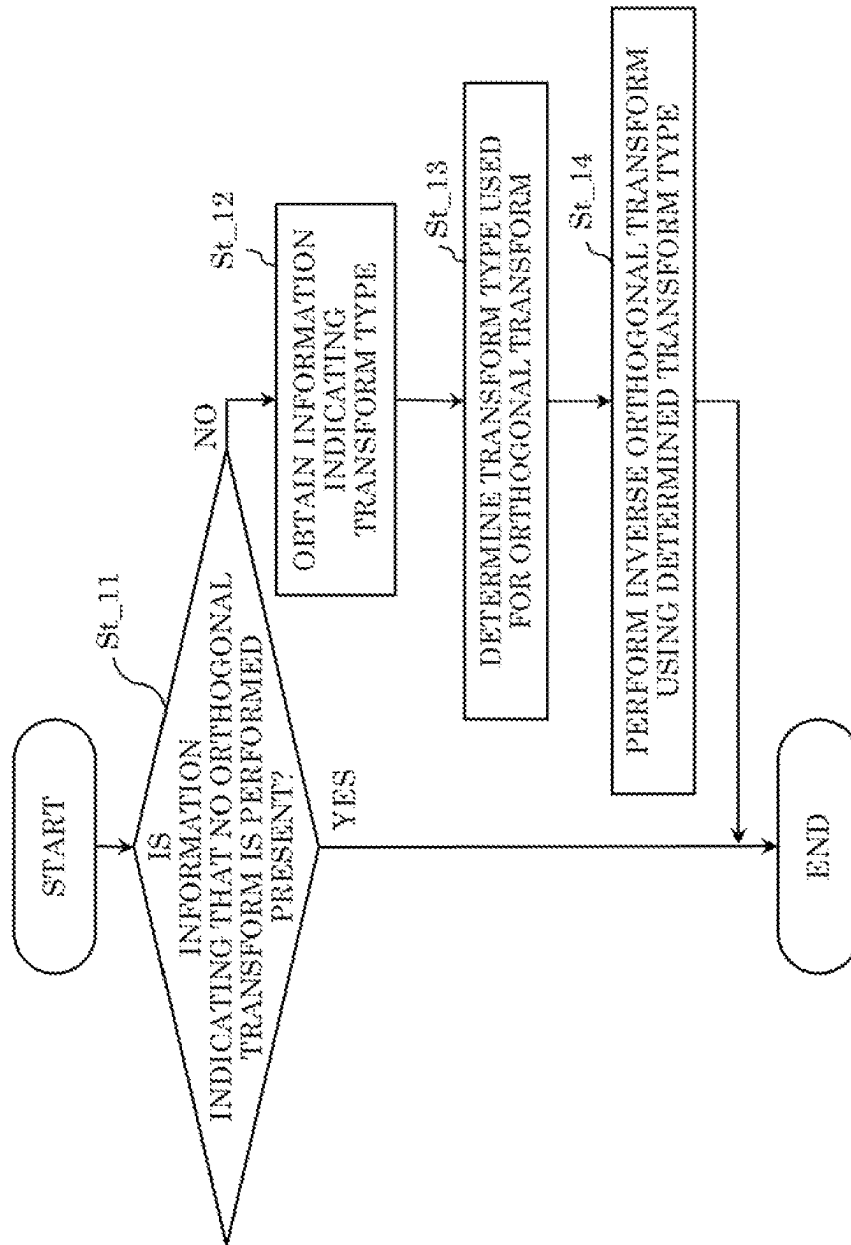
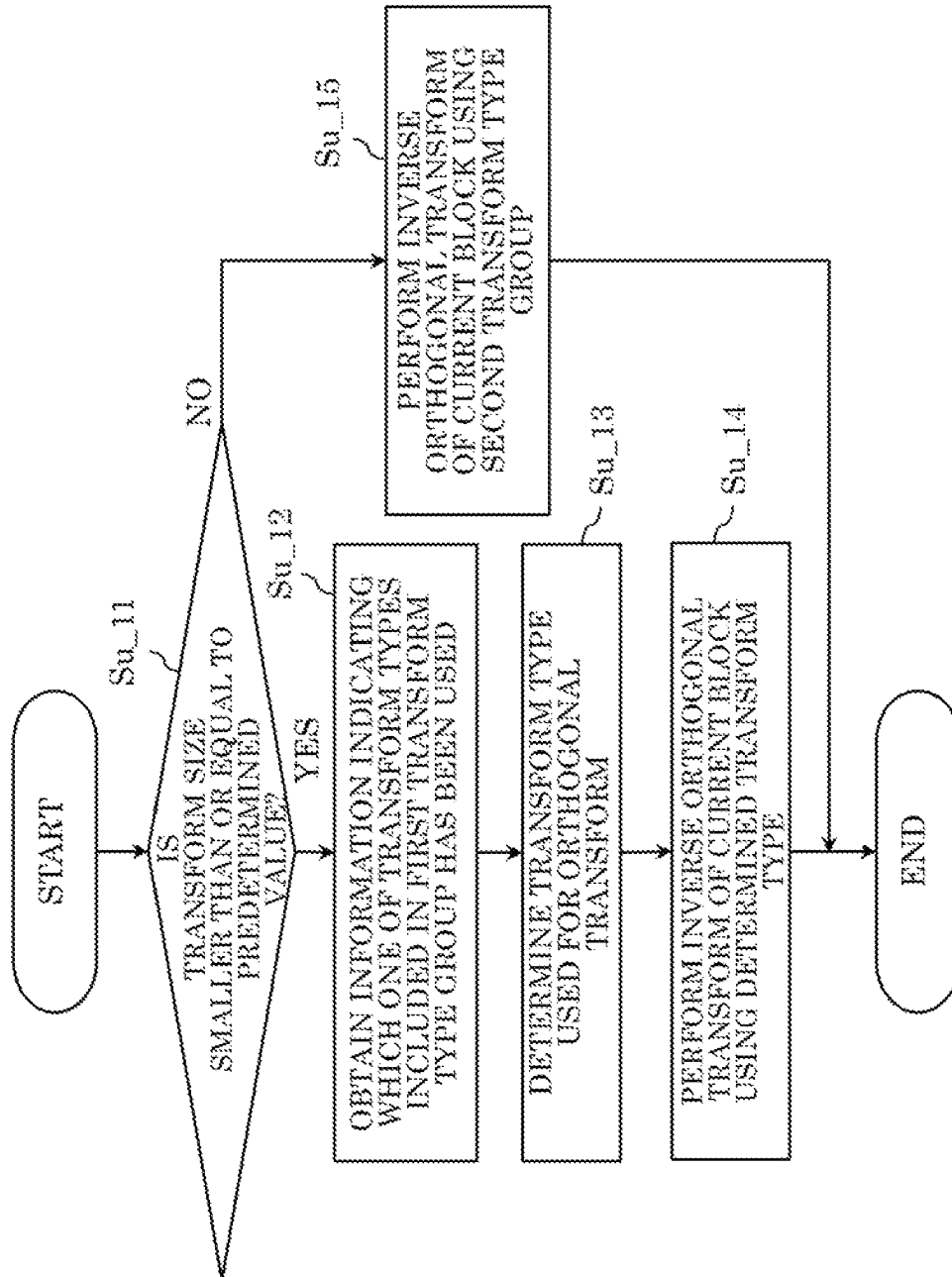


FIG. 76



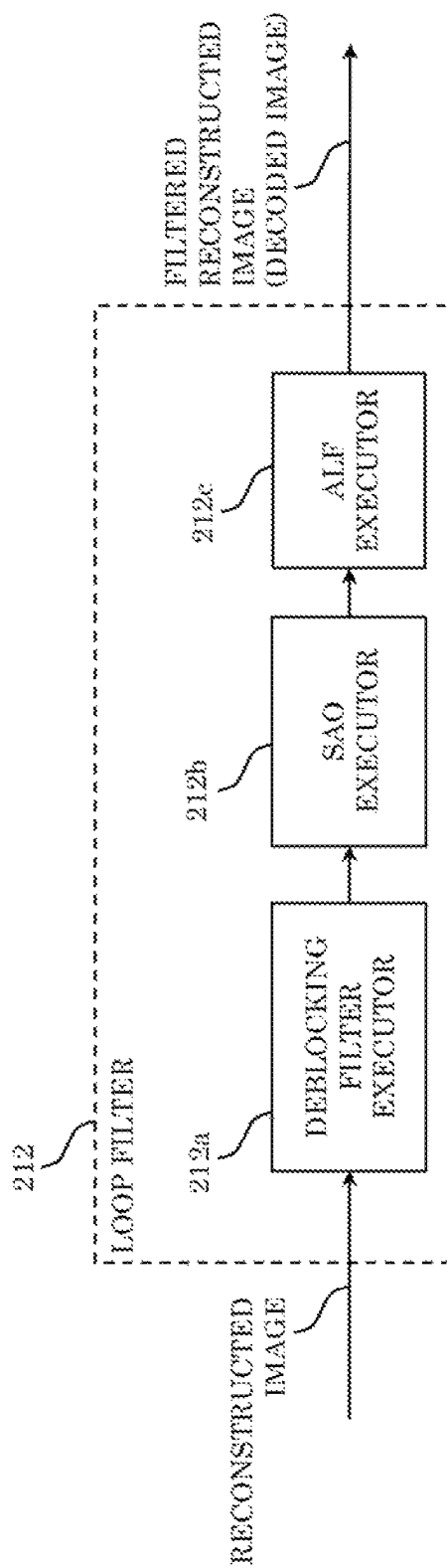


FIG. 77

FIG. 78

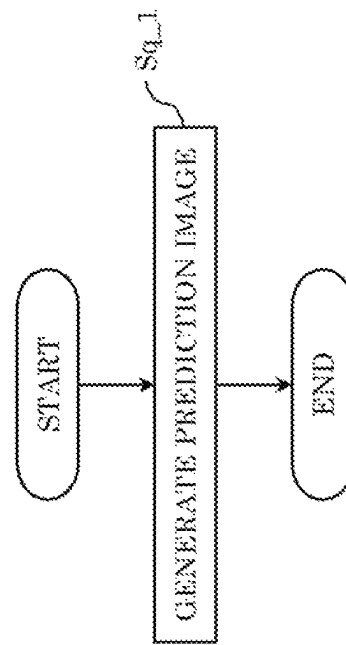


FIG. 79

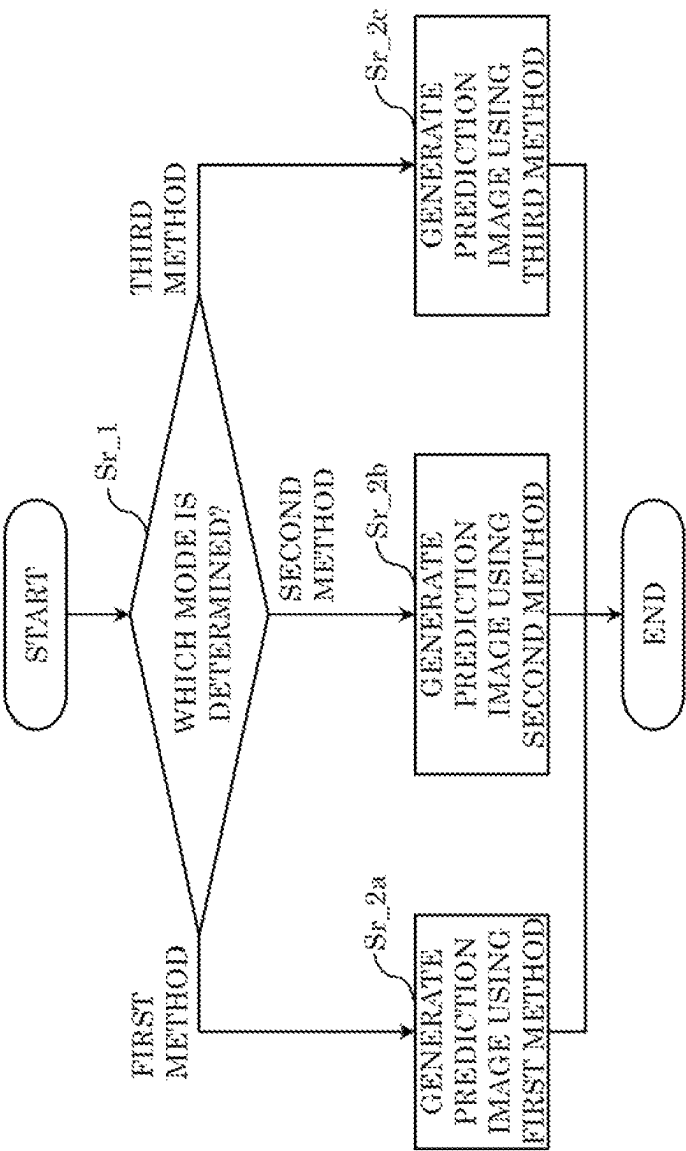


FIG. 80A

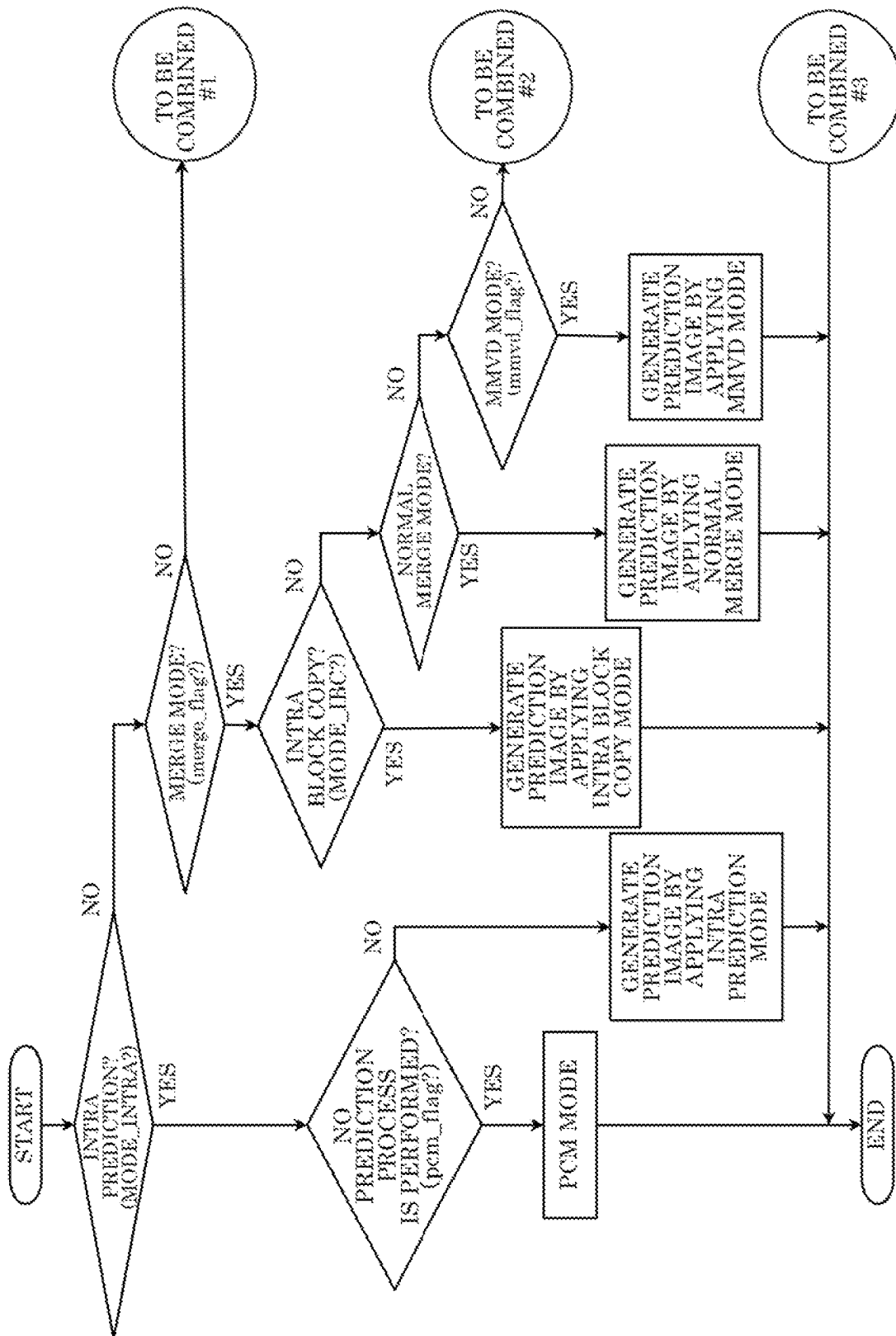


FIG. 80B

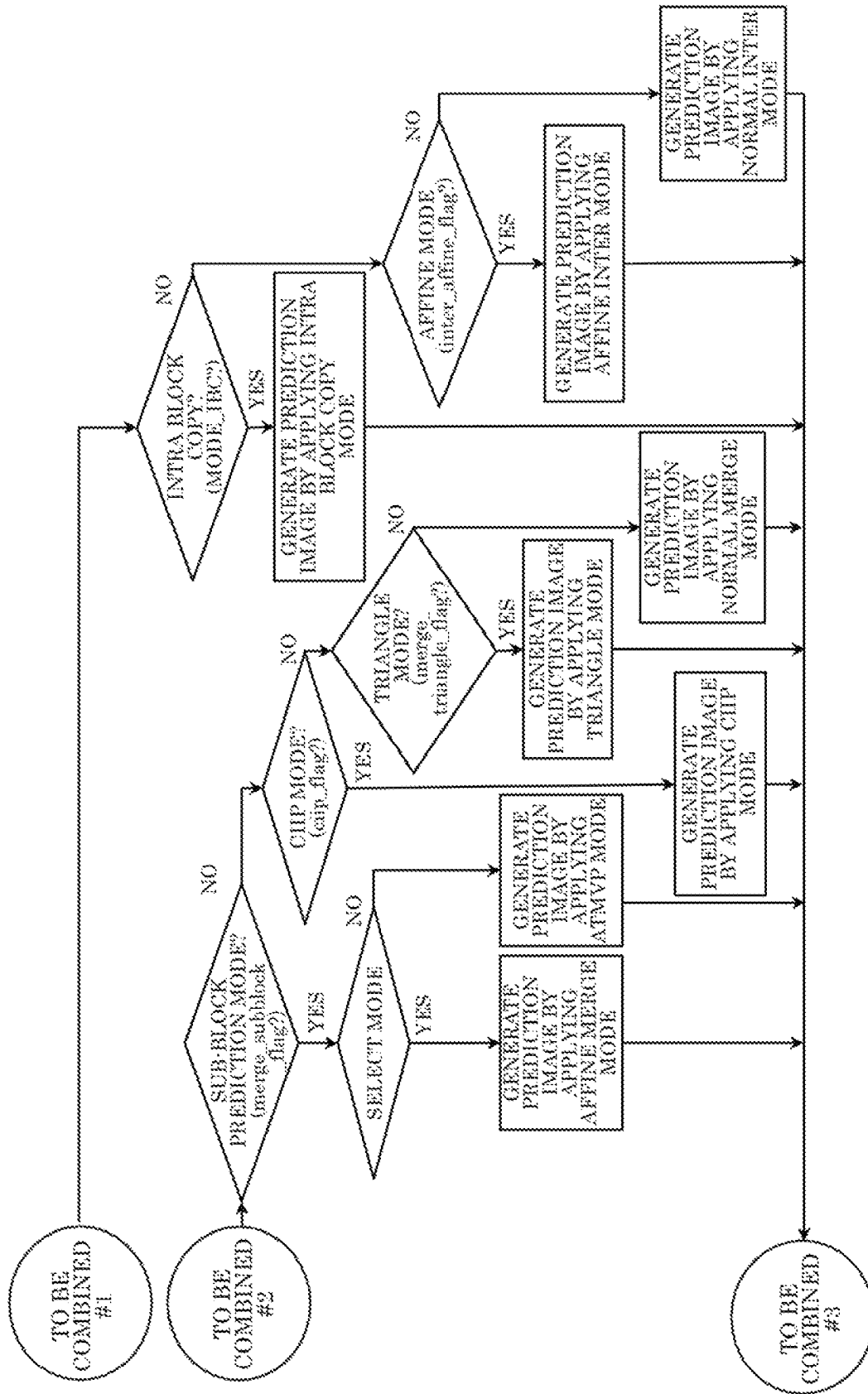


FIG. 81

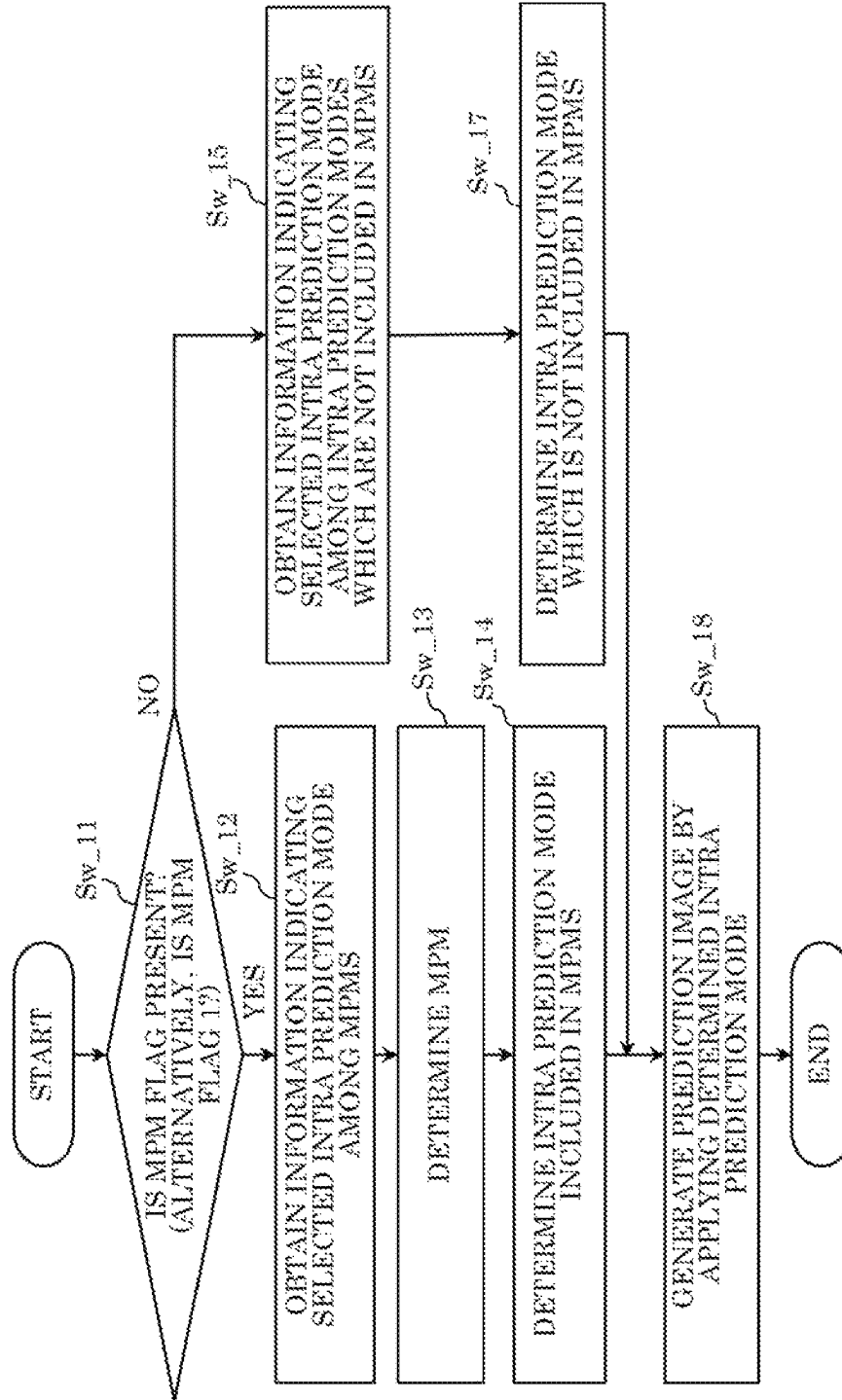


FIG. 82

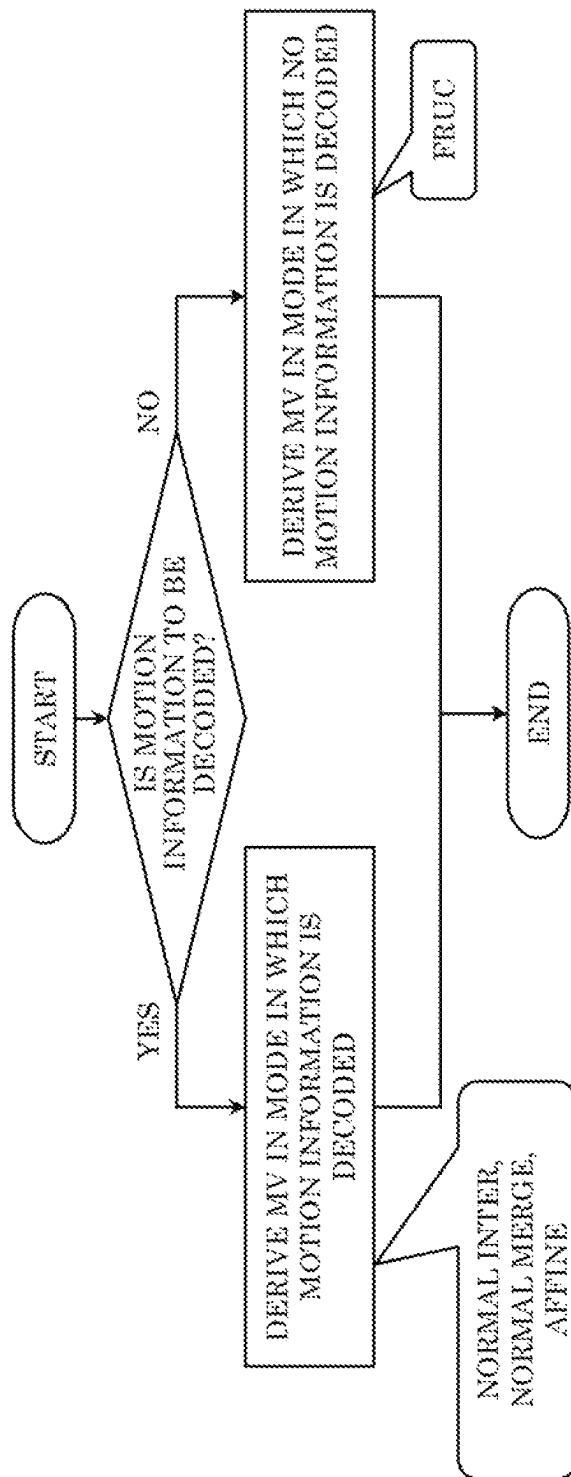


FIG. 83

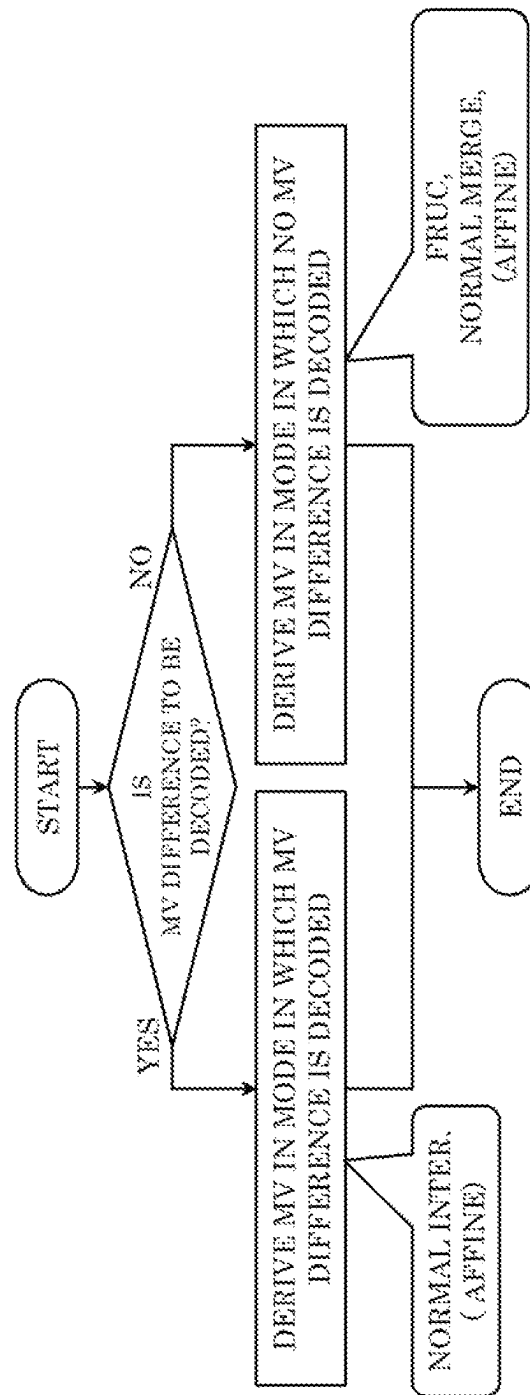


FIG. 84

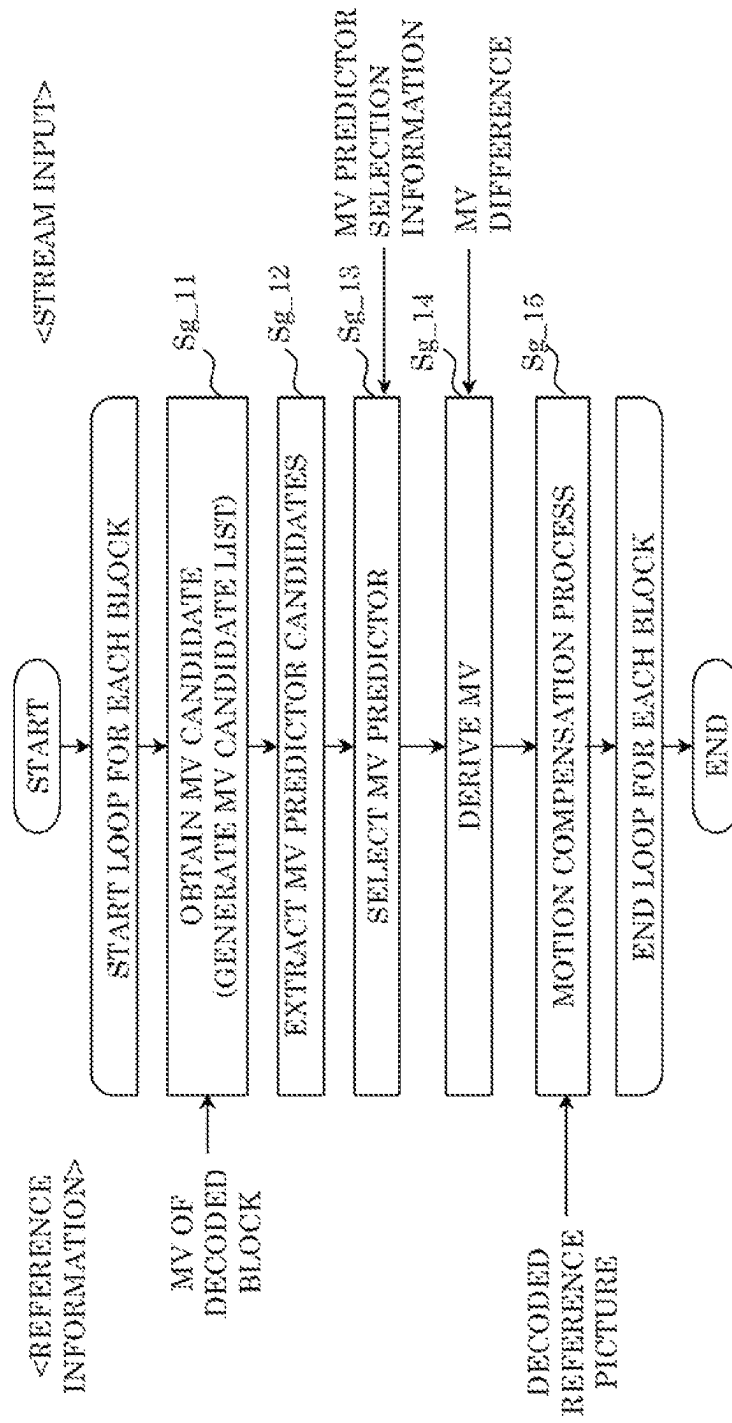


FIG. 85

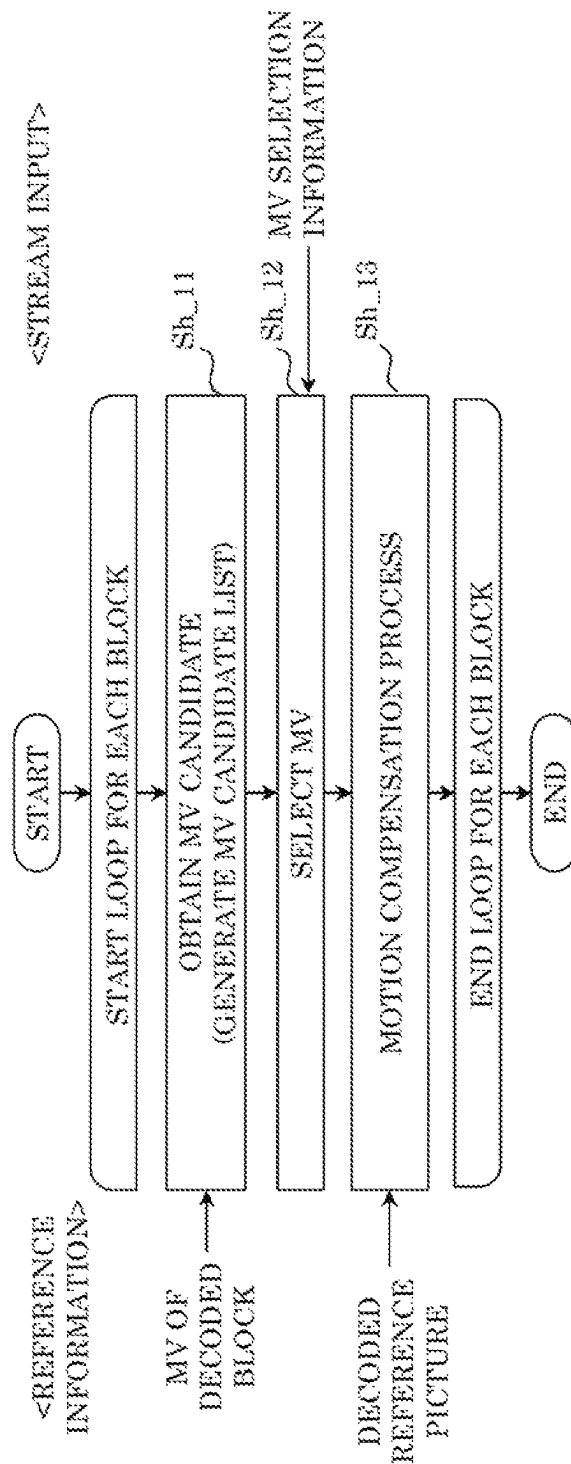


FIG. 86

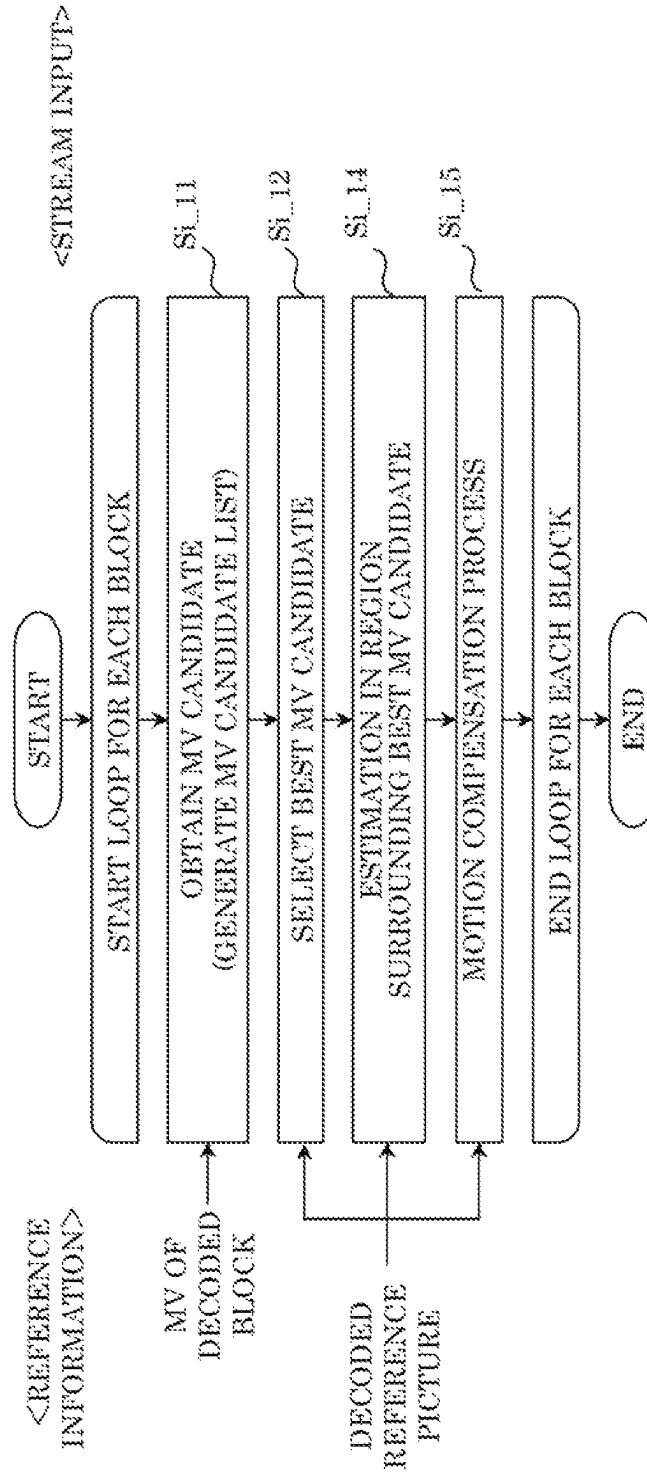


FIG. 87

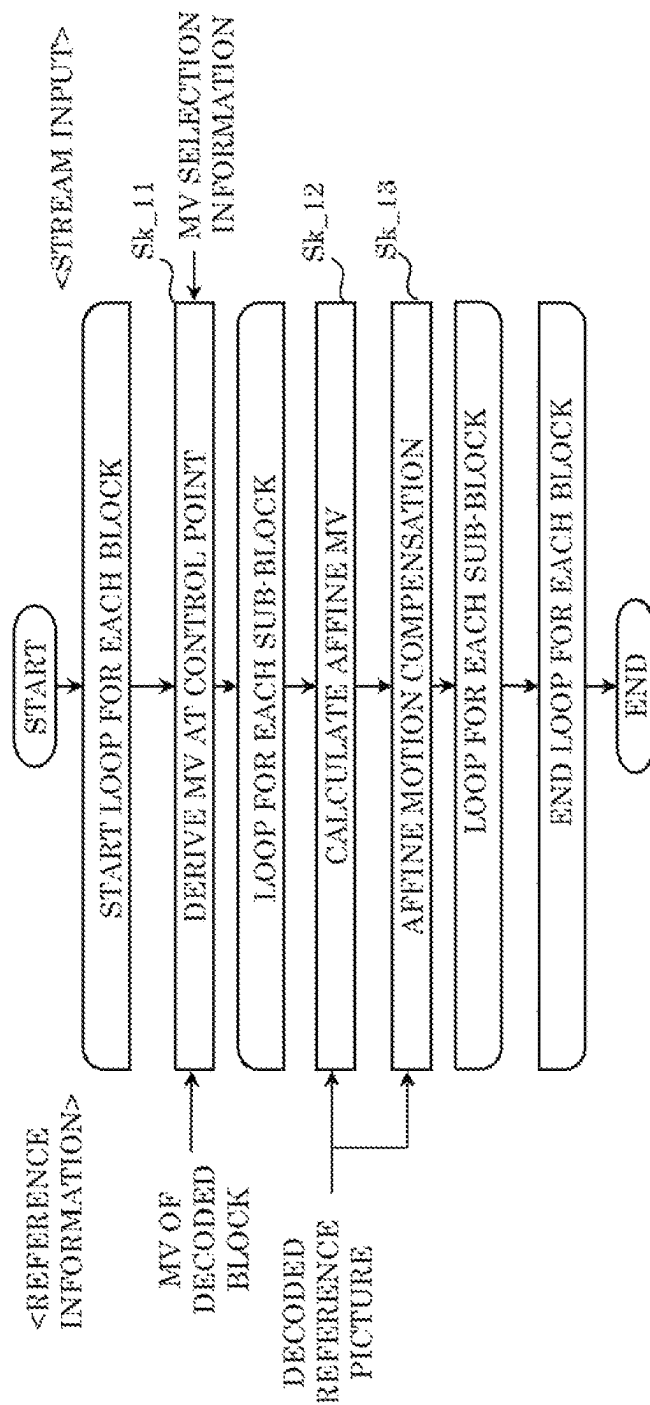


FIG. 88

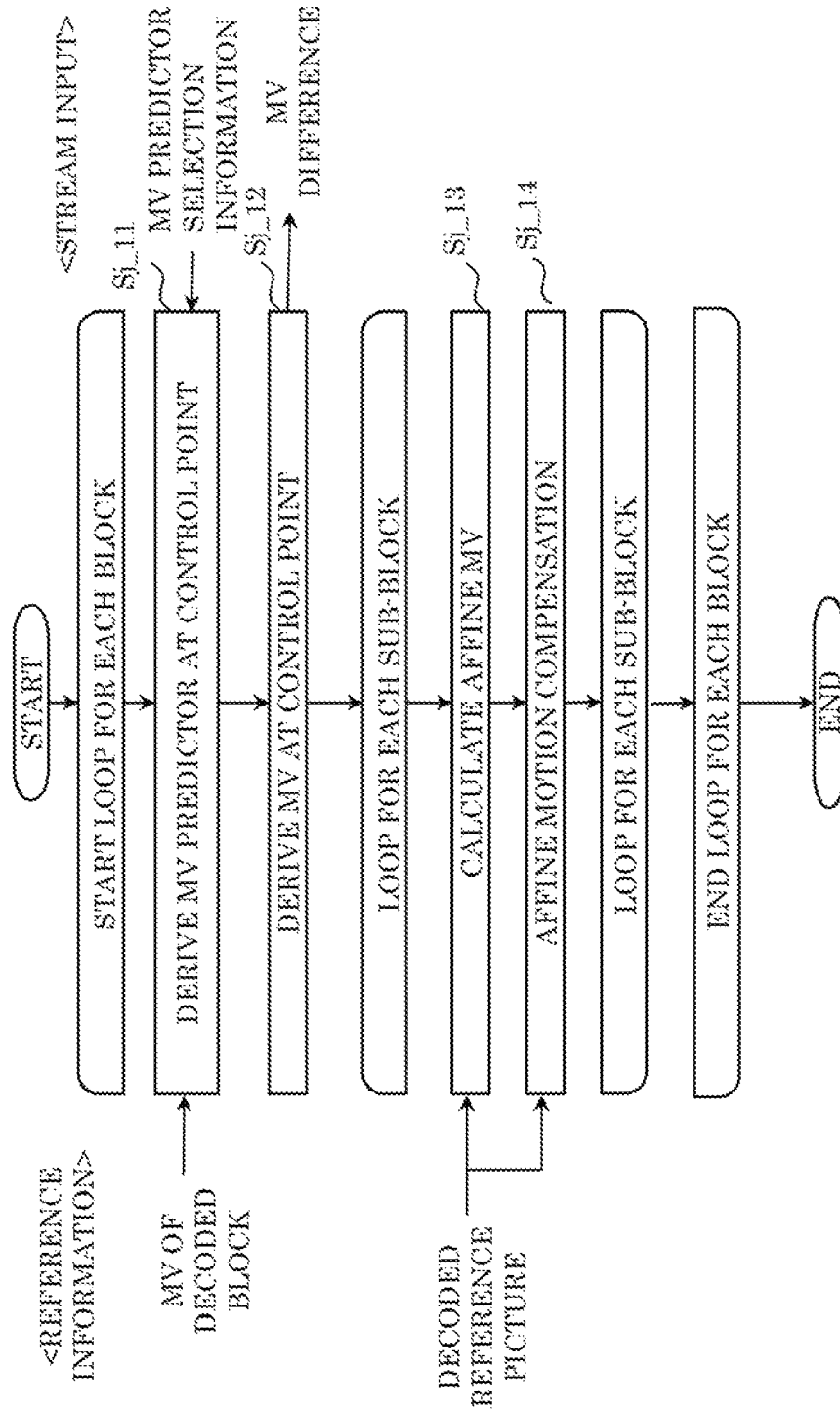


FIG. 89

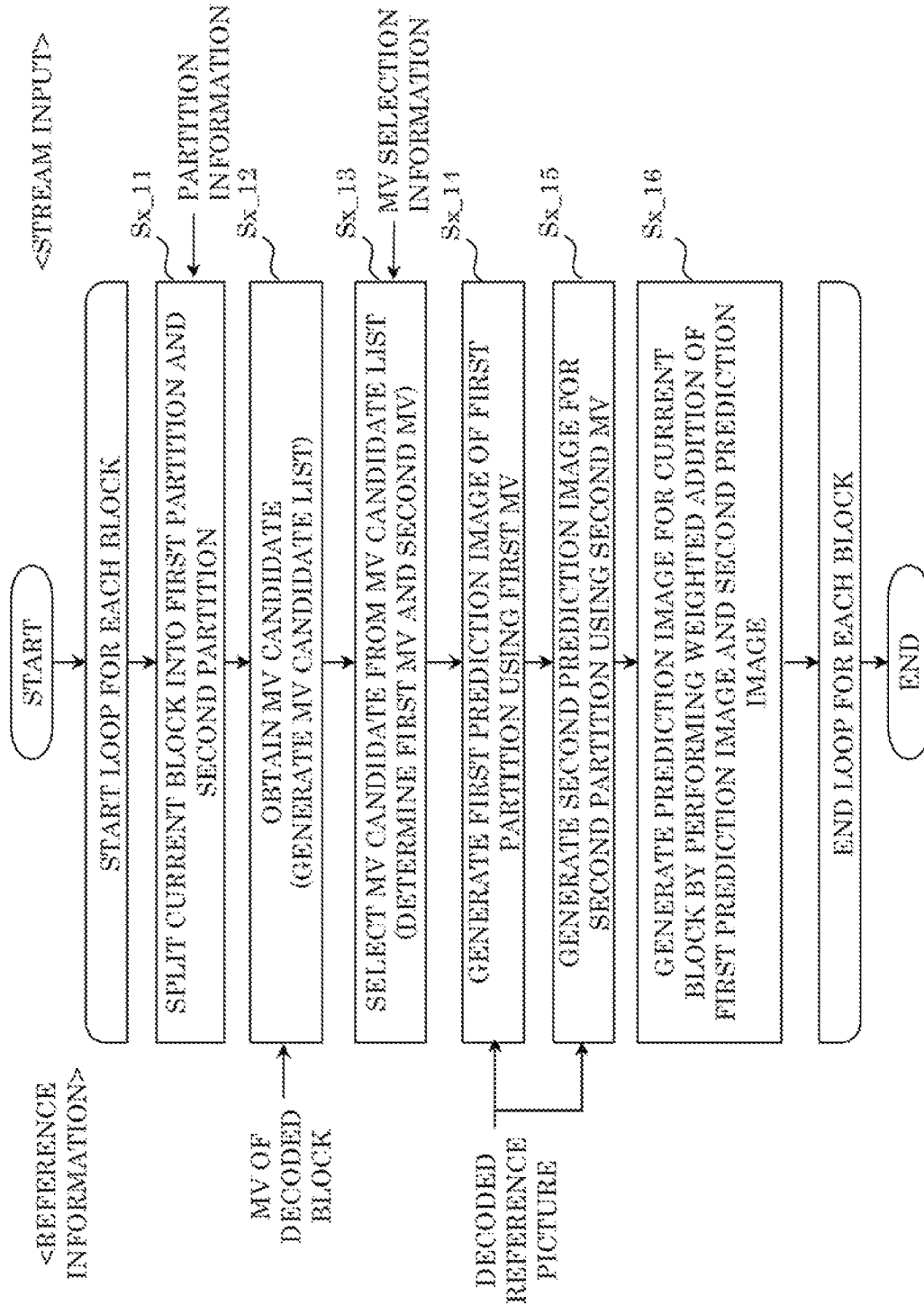


FIG. 90

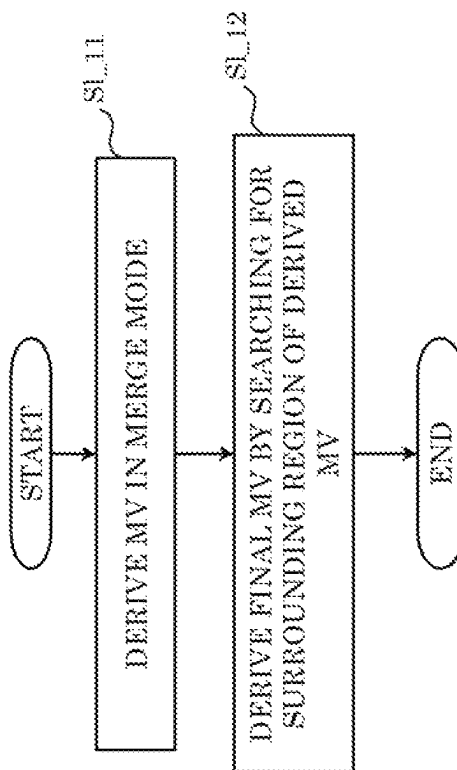


FIG. 91

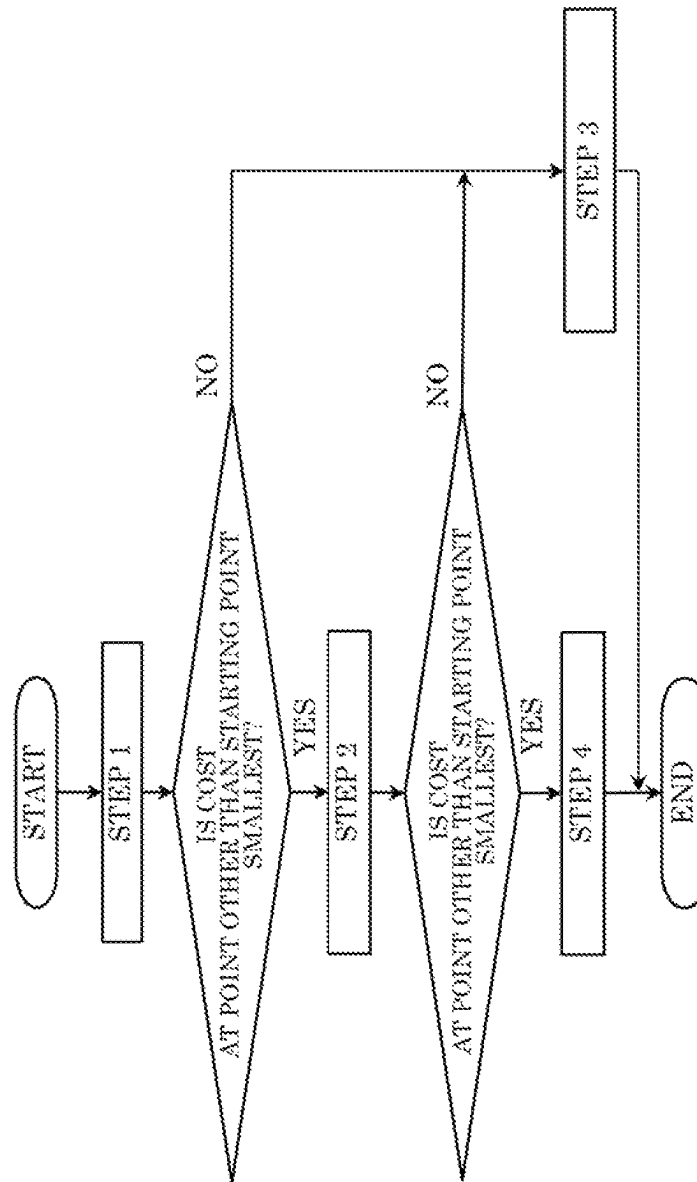


FIG. 92

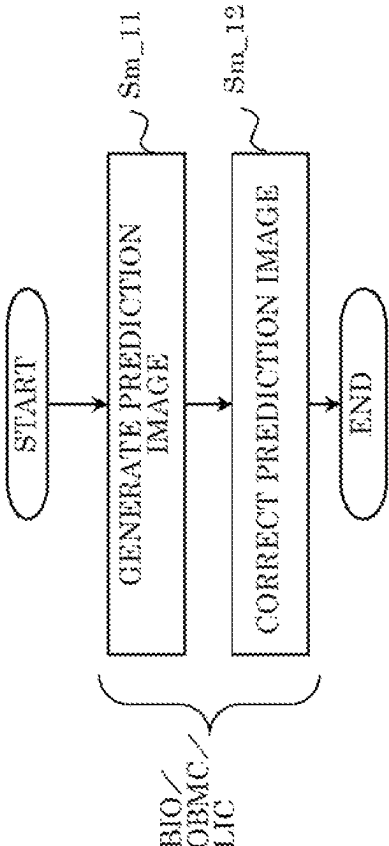


FIG. 93

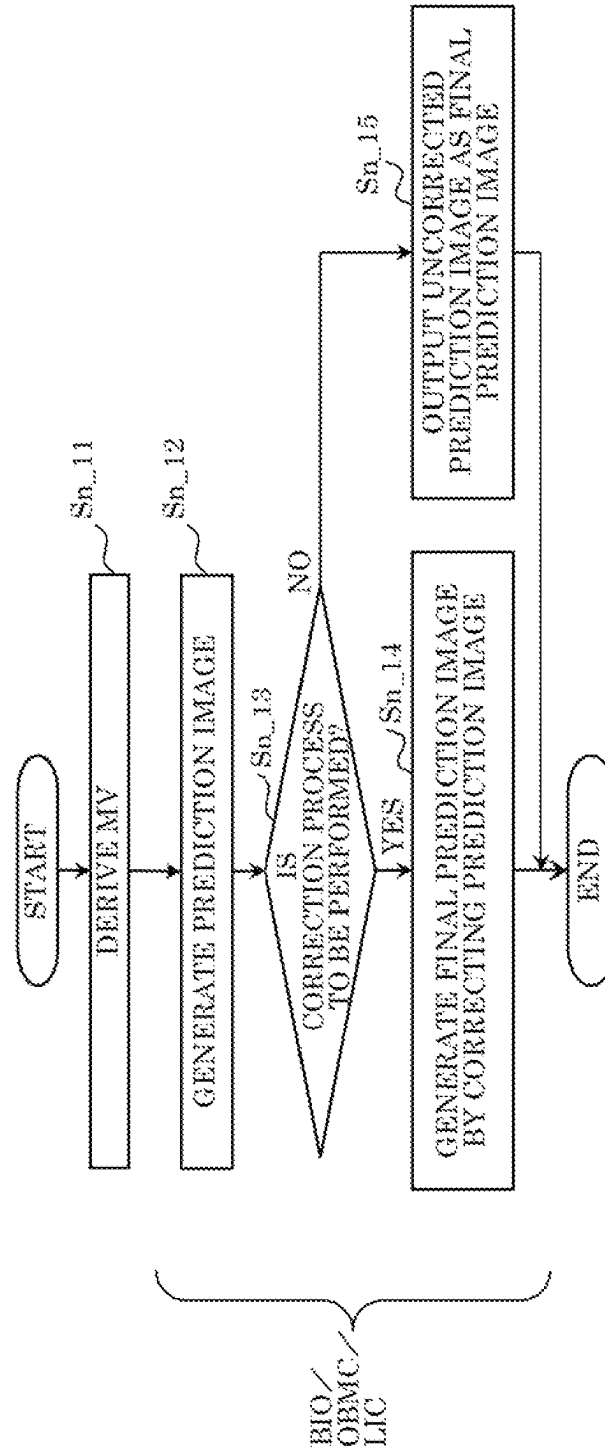


FIG. 94

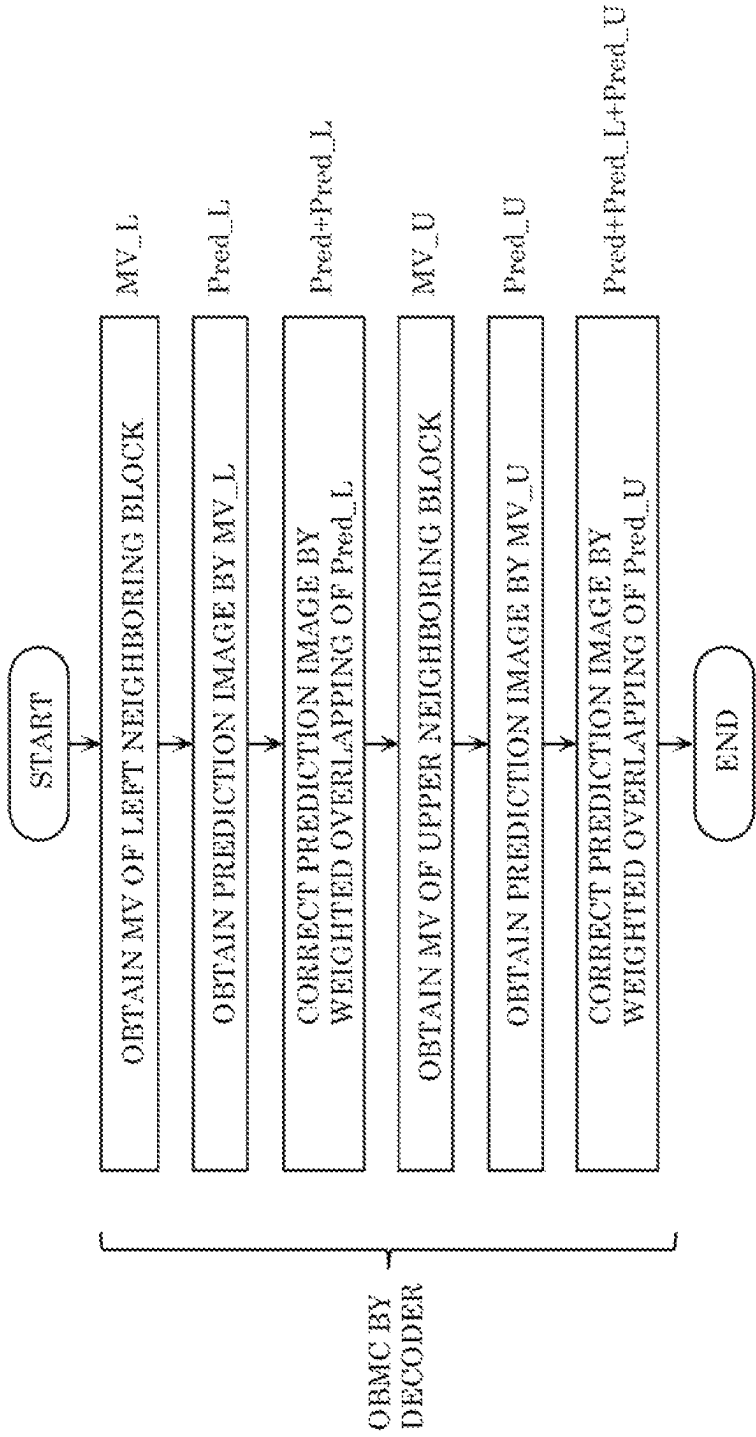


FIG. 95

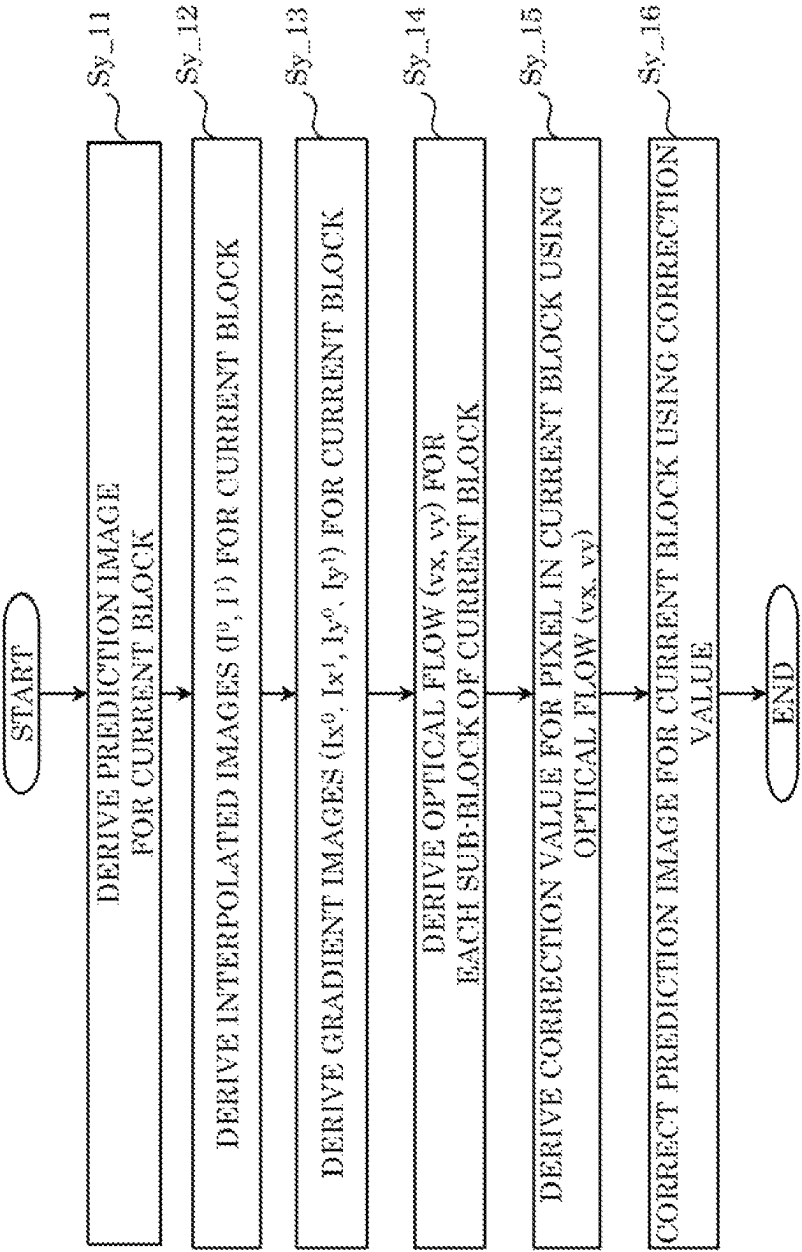


FIG. 96

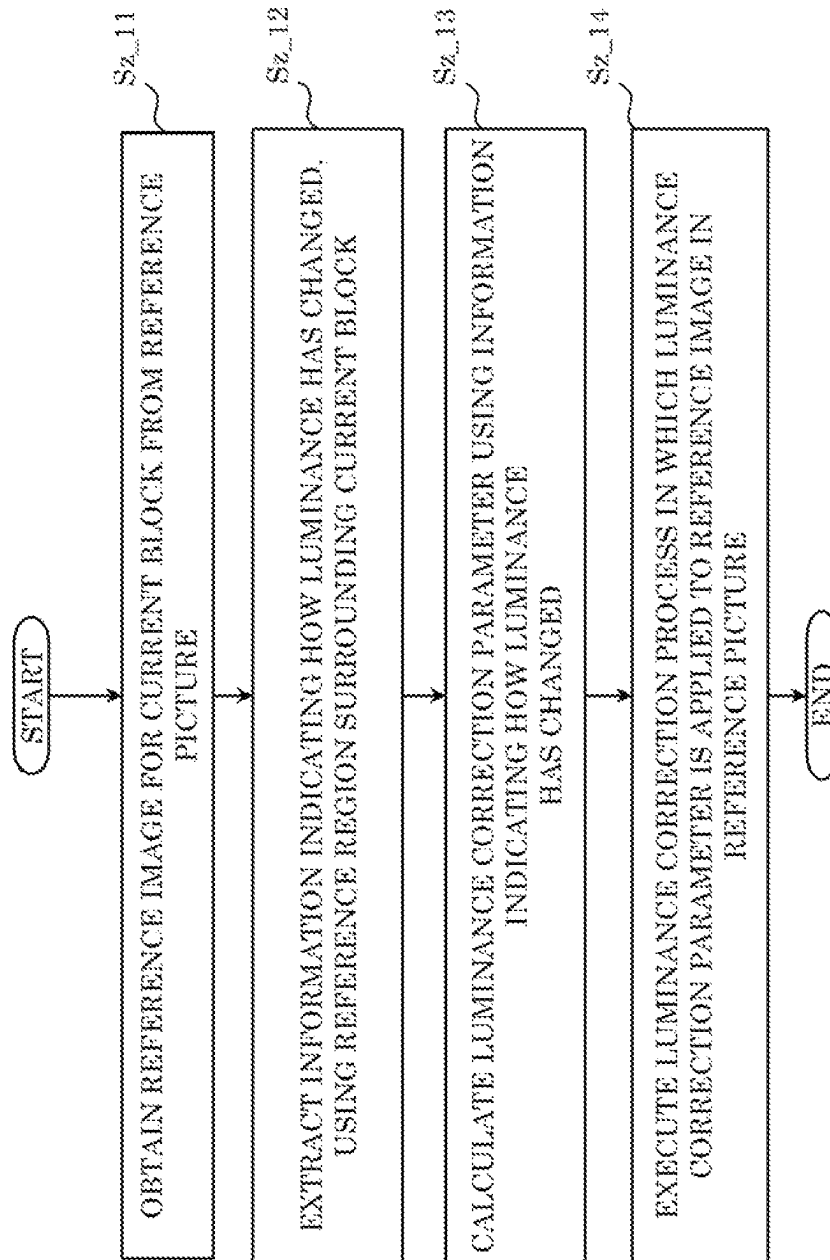


FIG. 97

buffering_period(payloadSize) {
...
for(tid = 0; tid <= sps_max_sublayers_minus1; tid++) {
bp_cpb_cnt_minus1[tid]
if(bp_nal_hrd_parameters_present_flag)
for(i = 0; i < bp_cpb_cnt_minus1[tid] + 1; i++) {
nal_initial_cpb_removal_delay[tid][i]
nal_initial_cpb_removal_offset[tid][i]
}
if(bp_vcl_hrd_parameters_present_flag)
for(i = 0; i < bp_cpb_cnt_minus1[tid] + 1; i++) {
vcl_initial_cpb_removal_delay[tid][i]
vcl_initial_cpb_removal_offset[tid][i]
}
}
}
...
}

FIG. 98

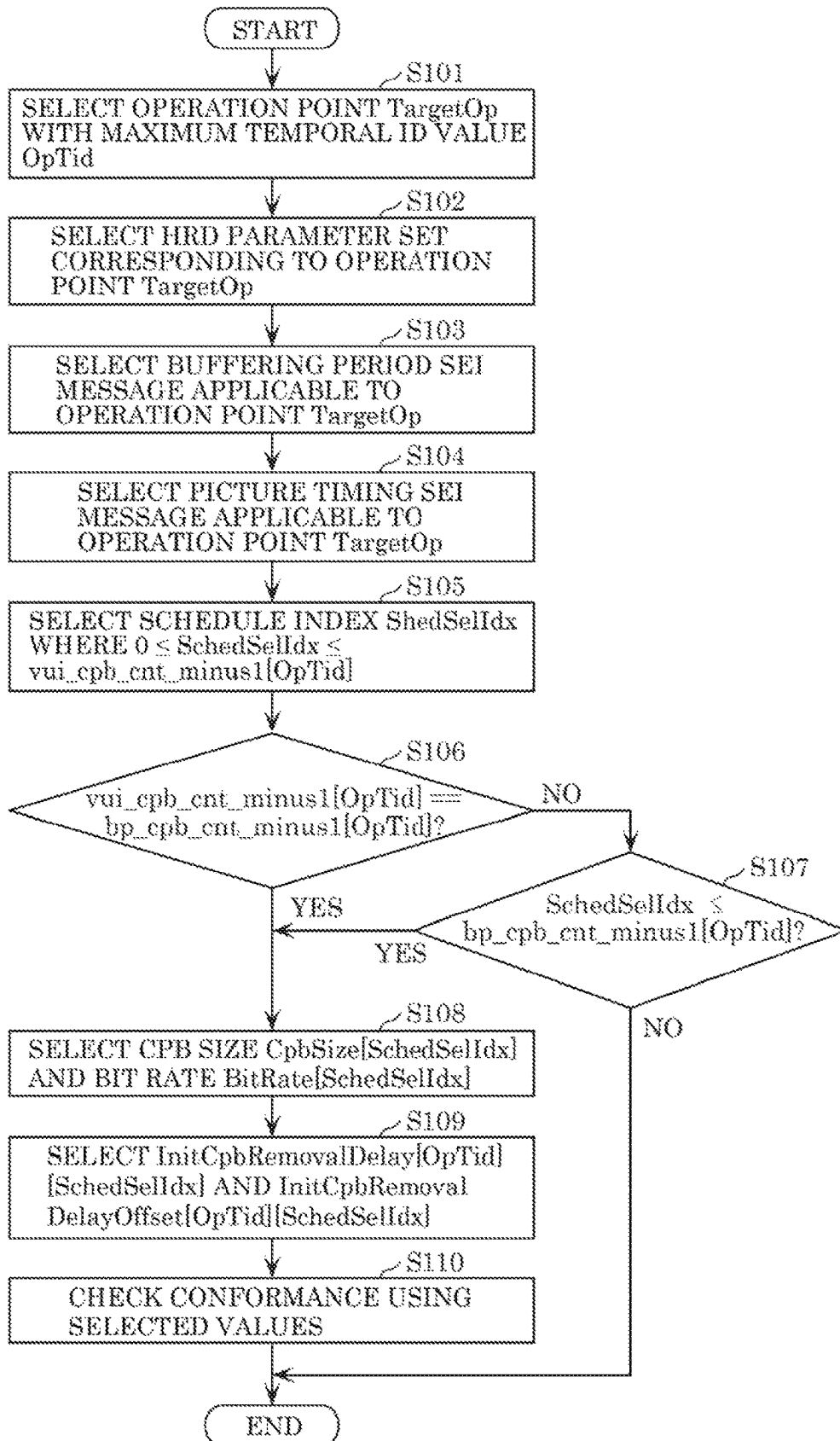


FIG. 99

buffering_period(payloadSize) {
...
bp_max_sublayers_minus1
for(tid = 0; tid <= bp_max_sublayers_minus1; tid++) {
bp_cpb_cnt_minus1[tid]
if(bp_nal_hrd_parameters_present_flag)
for(i = 0; i < bp_cpb_cnt_minus1[tid] + 1; i++) {
nal_initial_cpb_removal_delay[tid][i]
nal_initial_cpb_removal_offset[tid][i]
}
if(bp_vcl_hrd_parameters_present_flag)
for(i = 0; i < bp_cpb_cnt_minus1[tid] + 1; i++) {
vcl_initial_cpb_removal_delay[tid][i]
vcl_initial_cpb_removal_offset[tid][i]
}
}
}
...
}

FIG. 100

pic_timing(payloadSize) {
...
if(CpbDpbDelaysPresentFlag) {
pt_max_sublayers_minus1
cpb_removal_delay_minus1[pt_max_sublayers_minus1]
for(i = TemporalId; i < pt_max_sublayers_minus1; i++) {
sublayer_delays_present_flag[i]
if(sublayer_delays_present_flag[i]) {
cpb_removal_delay_delta_enabled_flag[i]
if(cpb_removal_delay_delta_enabled_flag[i])
cpb_removal_delay_delta_idx[i]
else
cpb_removal_delay_minus1[i]
}
}
dpb_output_delay
}
...
}

FIG. 101

video_parameter_set_rbsp() {
vps_video_parameter_set_id
vps_max_layers_minus1
vps_max_sublayers_minus1
...
vps_general_hrd_params_present_flag
if(vps_general_hrd_params_present_flag) {
general_hrd_parameters()
if(vps_max_sublayers_minus1 > 0)
vps_sublayer_cpb_params_present_flag
vps_num_ols_hrd_params_minus1
for(i= 0; i <= vps_num_ols_hrd_params_minus1; i++) {
if(!vps_all_layers_same_num_sublayers_flag)
vps_hrd_max_tid[i]
firstSubLayer = vps_sublayer_cpb_params_present_flag ? 0:
vps_hrd_max_tid[i]
ols_hrd_parameters(firstSubLayer, vps_hrd_max_tid[i])
}
...
}
...
}

FIG. 102

seq_parameter_set_rbsp() {
sps_seq_parameter_set_id
sps_video_parameter_set_id
sps_max_sublayers_minus1
...
sps_general_hrd_params_present_flag
if(sps_general_hrd_params_present_flag) {
general_hrd_parameters()
if(sps_max_sublayers_minus1 > 0)
sps_sublayer_cpb_params_present_flag
firstSubLayer = sps_sublayer_cpb_params_present_flag ? 0 :
sps_max_sublayers_minus1
ols_hrd_parameters(firstSubLayer, sps_max_sublayers_minus1)
}
...
}

FIG. 103

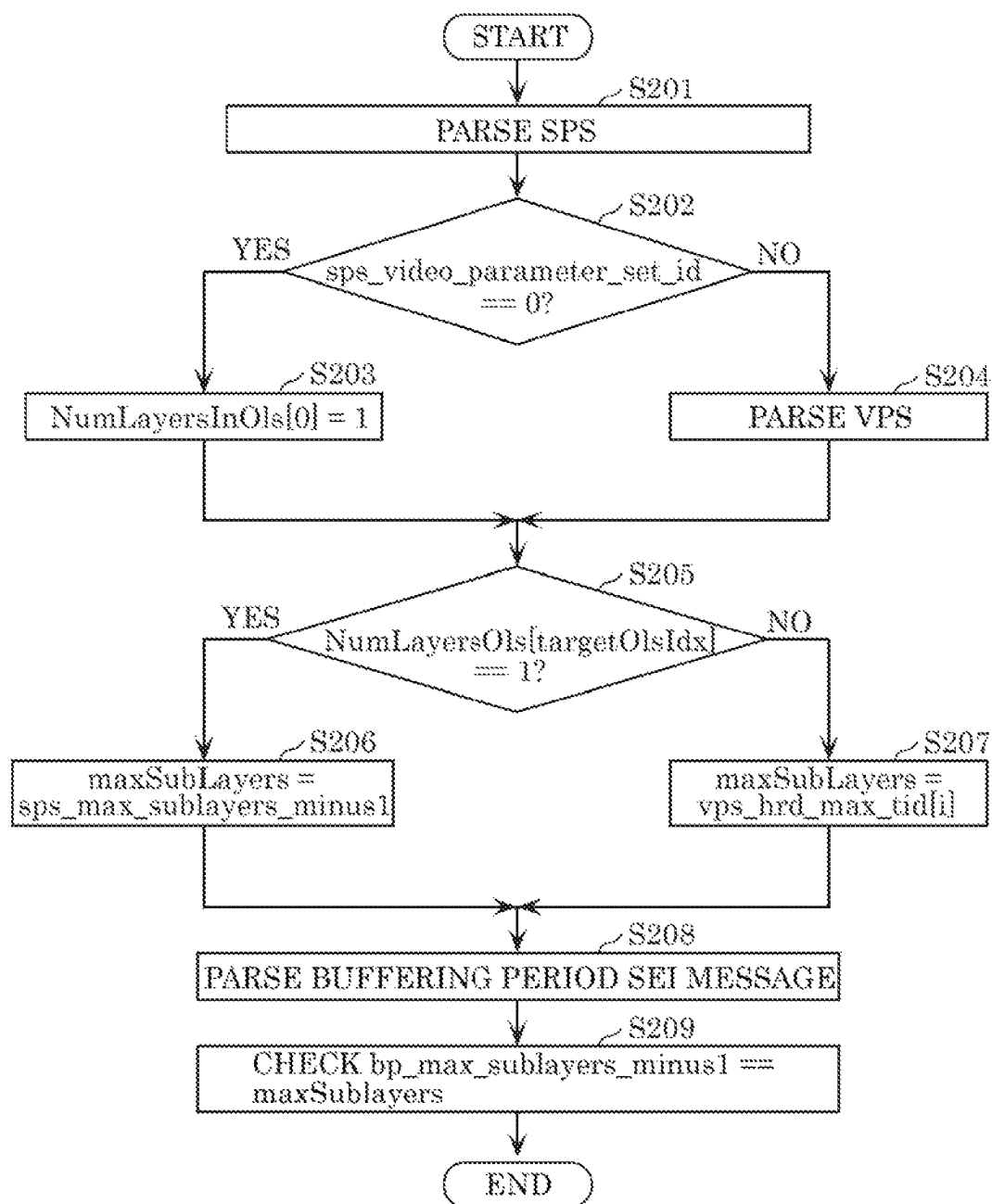


FIG. 104

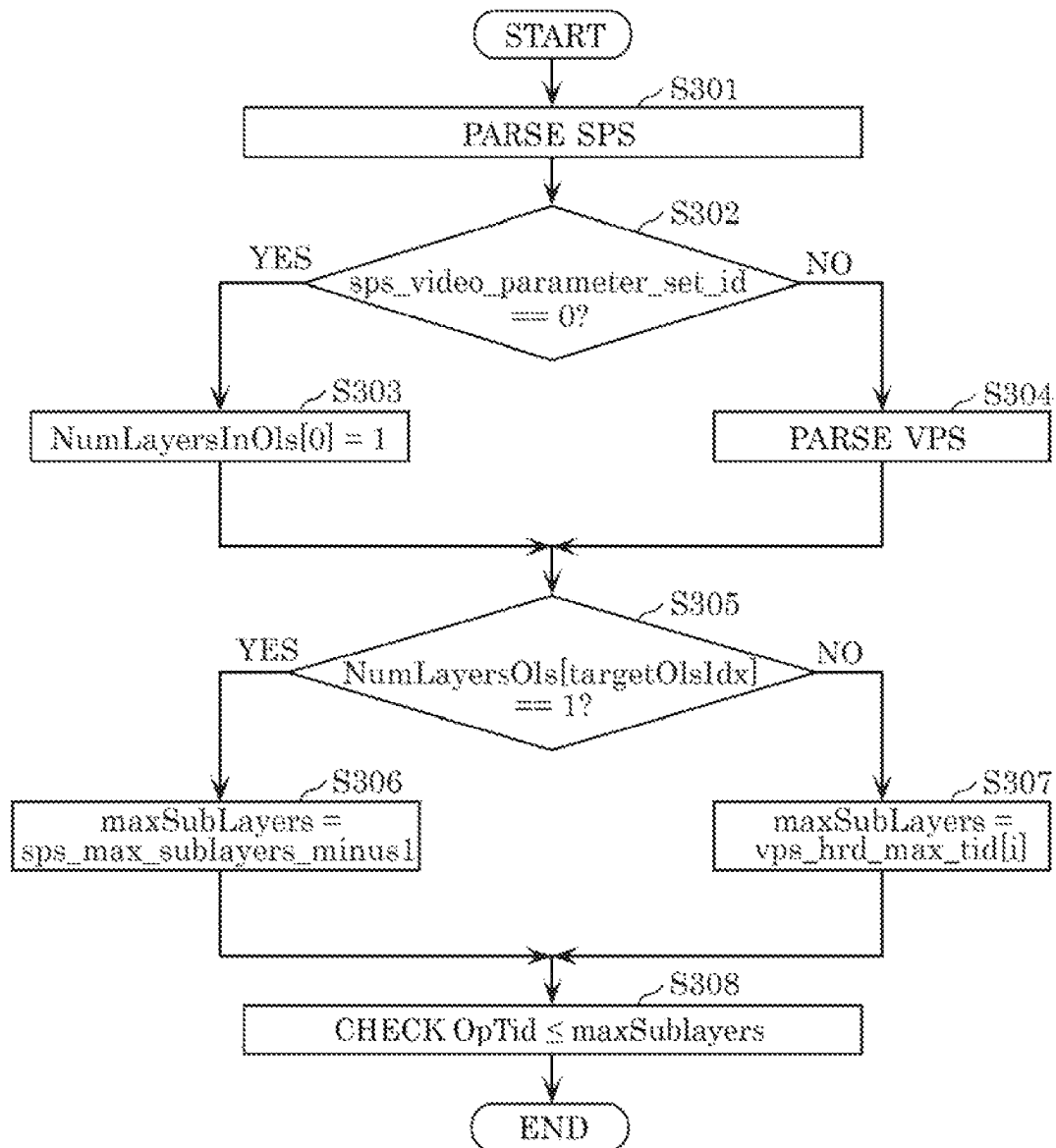


FIG. 105

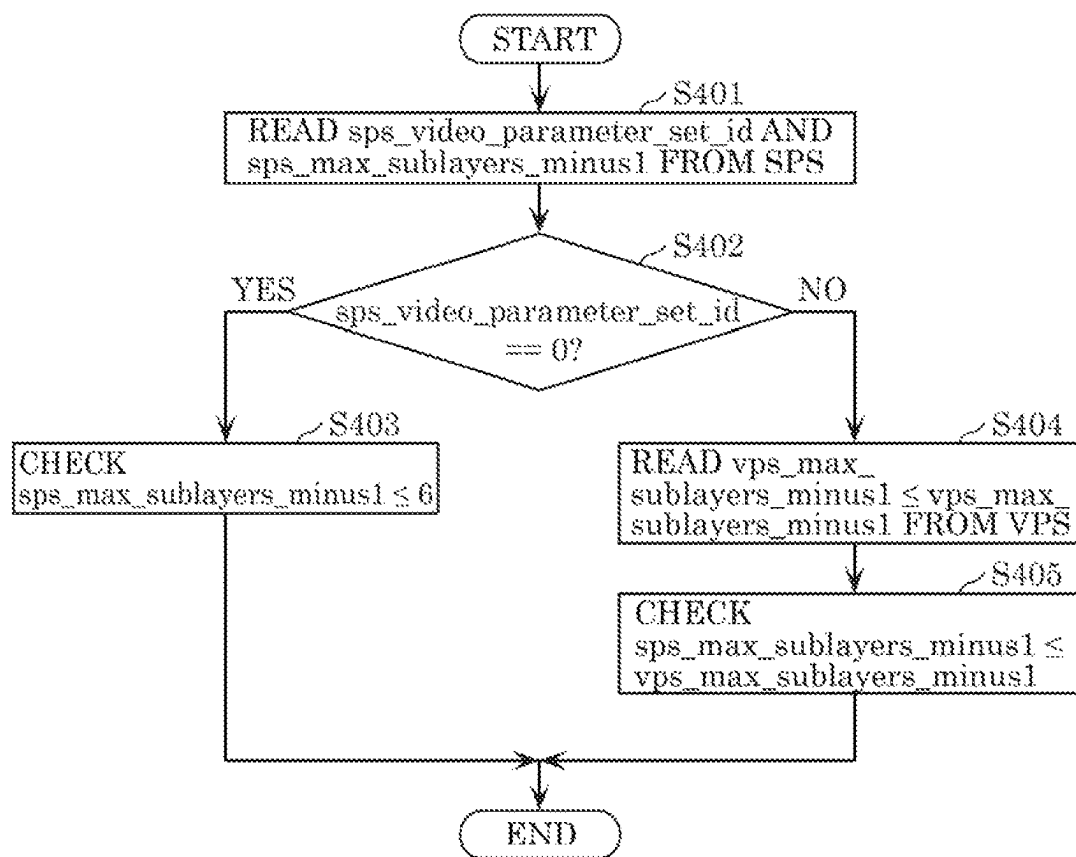


FIG. 106

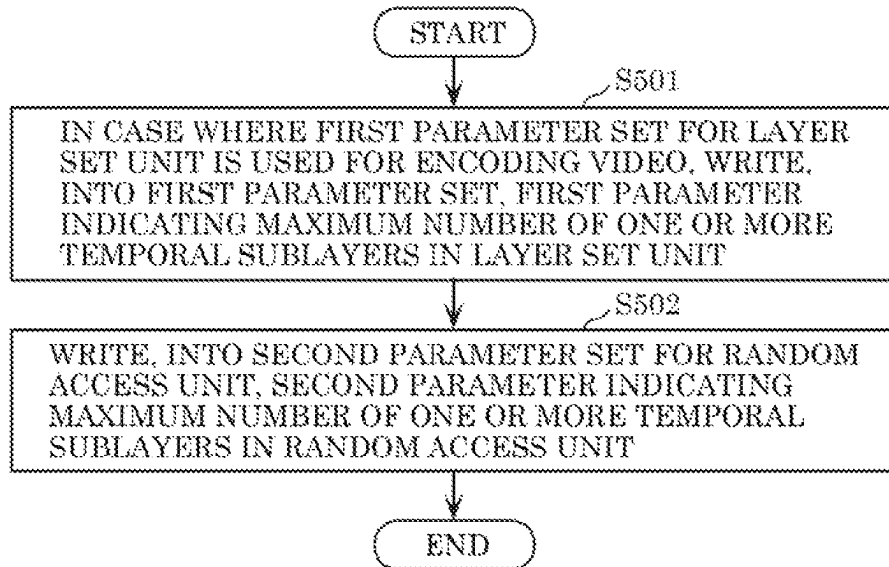


FIG. 107

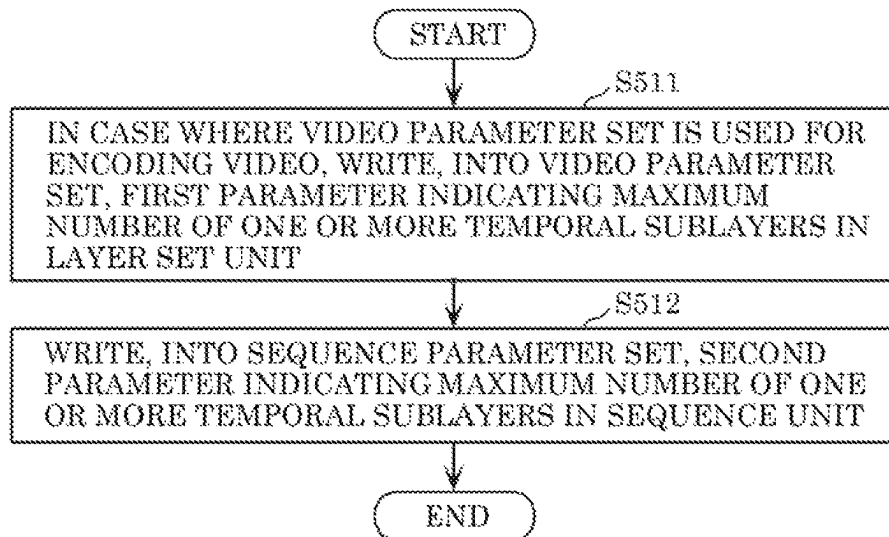


FIG. 108

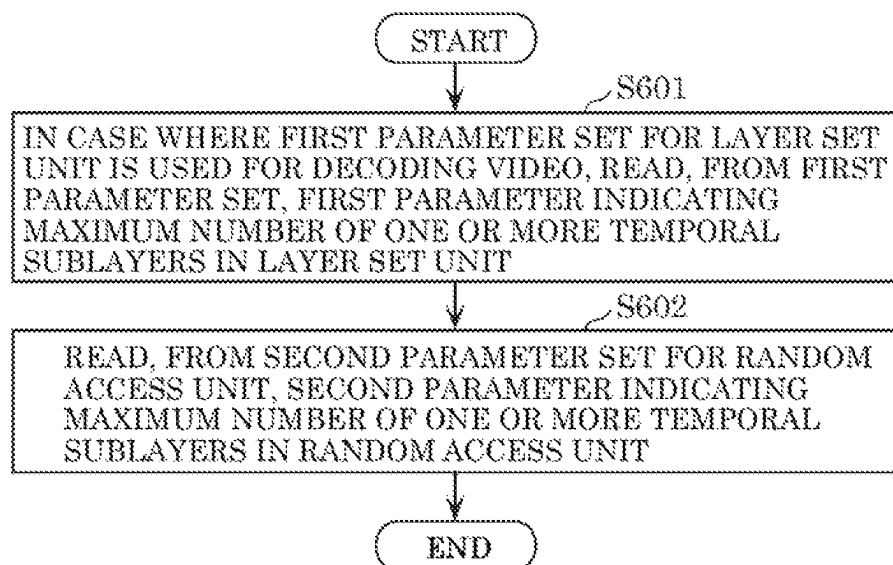


FIG. 109

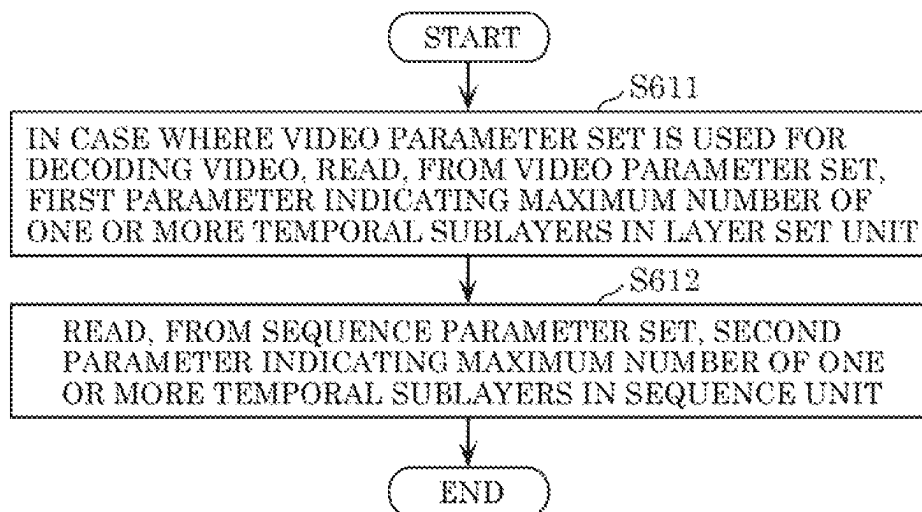


FIG. 110

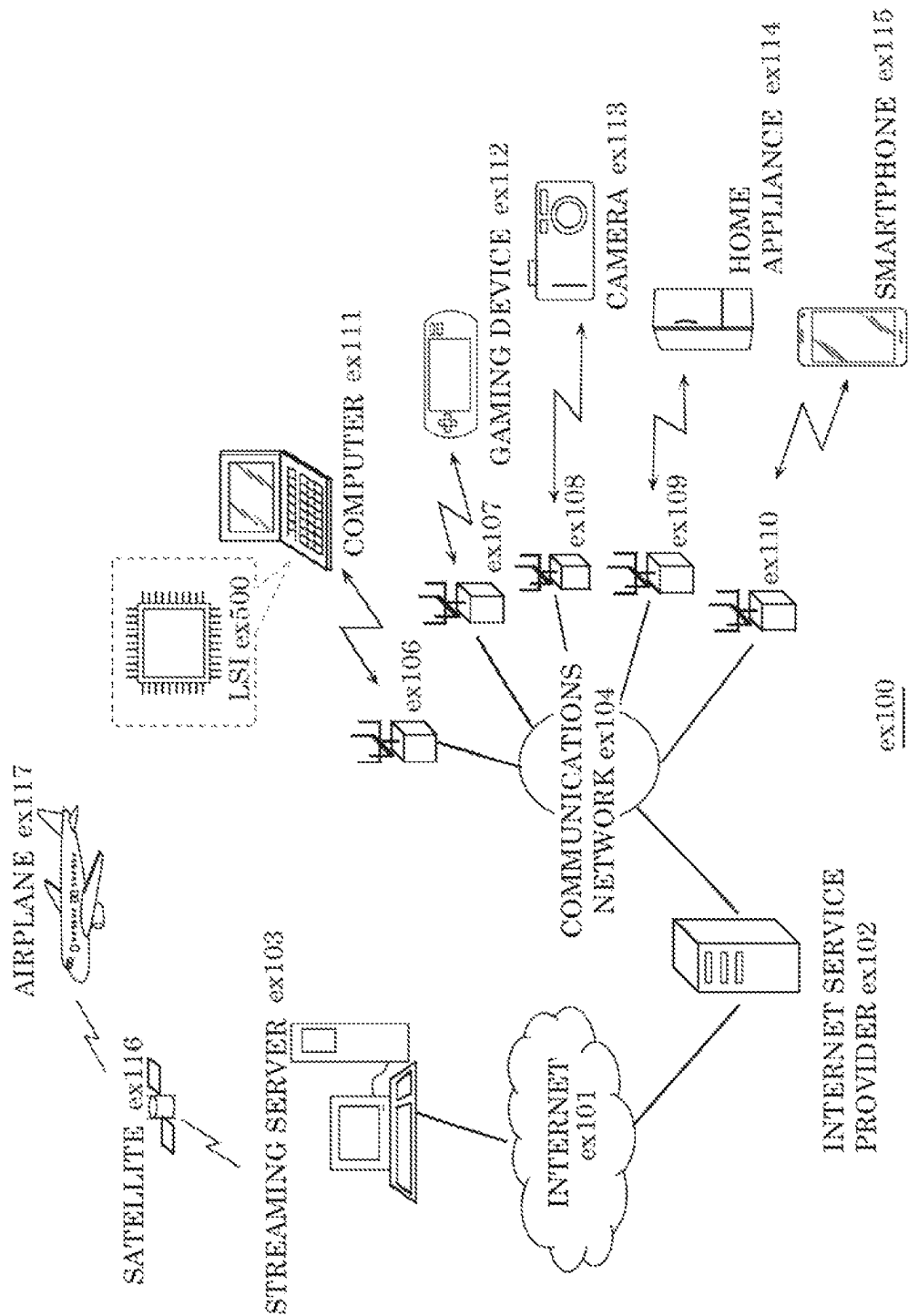


FIG. 111

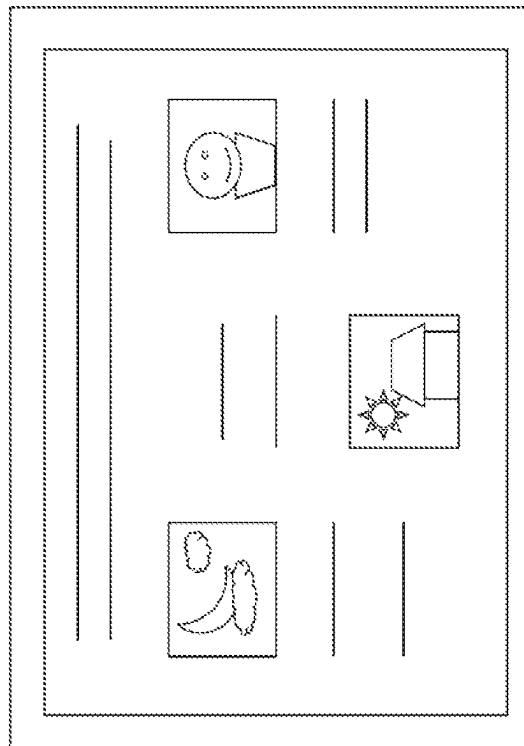


FIG. 112

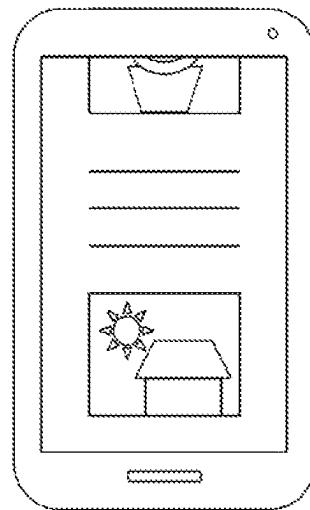


FIG. 113

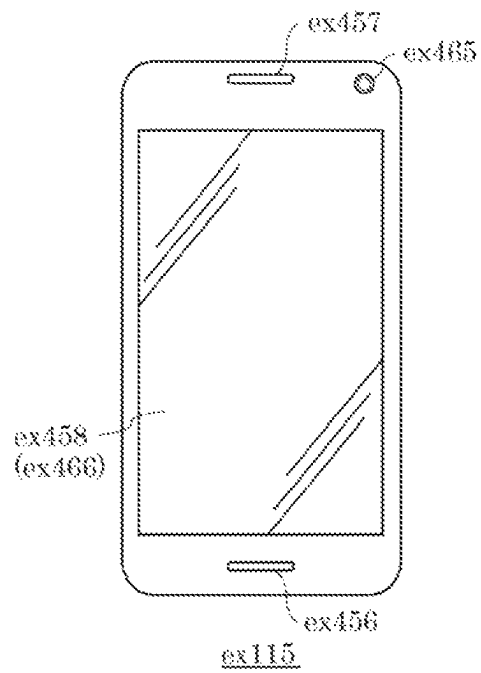
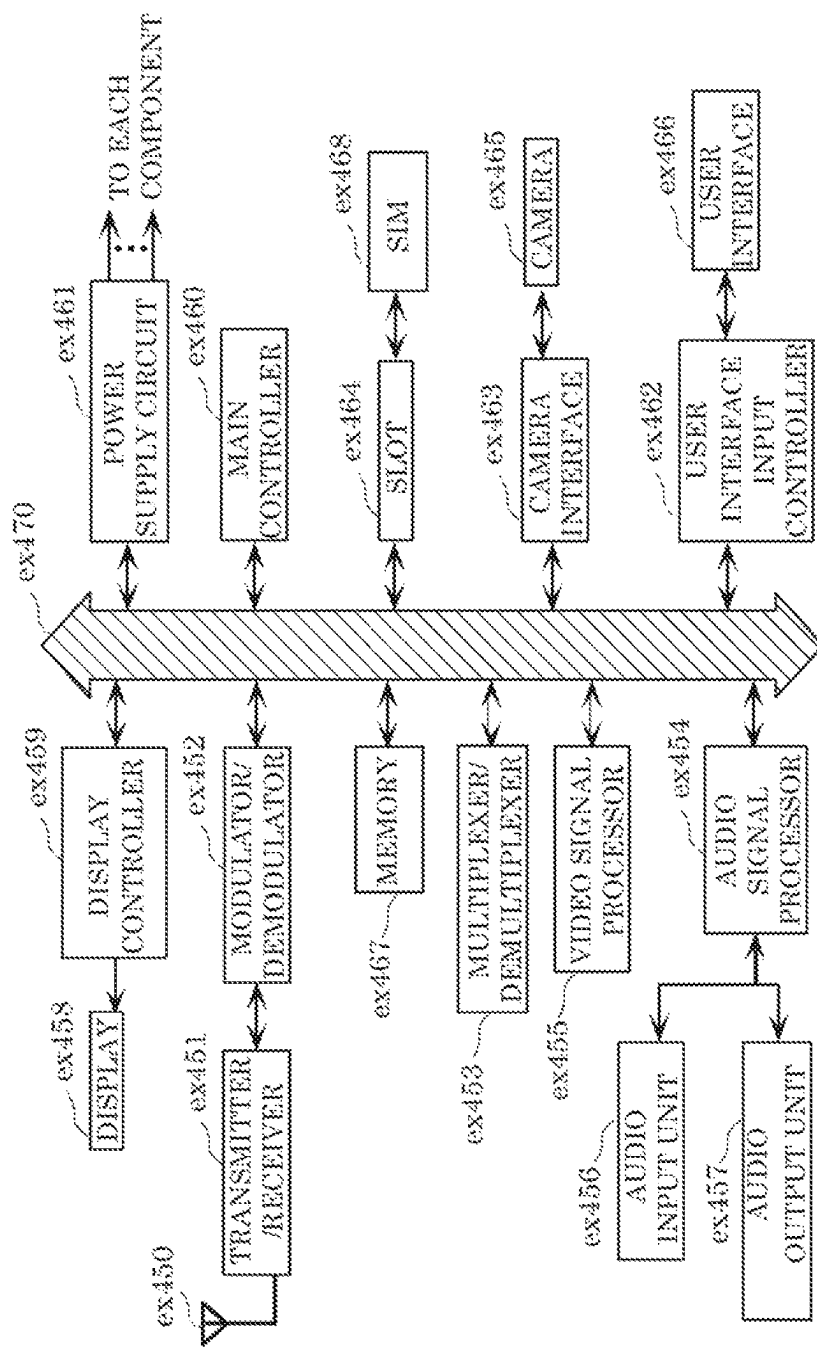


FIG. 114



ENCODER, DECODER, ENCODING METHOD, AND DECODING METHOD

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. continuation application of PCT International Patent Application Number PCT/JP2021/018301 filed on May 13, 2021, designating the United States of America, which is based on and claims priority of U.S. Provisional Patent Application No. 63/026,978 filed on May 19, 2020. The entire disclosures of the above-identified applications, including the specifications, drawings and claims are incorporated herein by reference in their entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to an encoder, a decoder, an encoding method, and a decoding method.

2. Description of the Related Art

With advancement in video coding technology, from H.261 and MPEG-1 to H.264/AVC (Advanced Video Coding), MPEG-LA, H.265/HEVC (High Efficiency Video Coding) and H.266/VVC (Versatile Video Codec), there remains a constant need to provide improvements and optimizations to the video coding technology to process an ever-increasing amount of digital video data in various applications. The present disclosure relates to further advancements, improvements, and optimizations in video coding.

Note that H.265 (ISO/IEC 23008-2 HEVC)/HEVC (High Efficiency Video Coding) relates to one example of a conventional standard regarding the above-described video coding technology.

SUMMARY

An encoder according to an aspect of the present disclosure includes circuitry and memory coupled to the circuitry. In operation, in the case where a video parameter set is used for encoding a video, the circuitry writes a first parameter into the video parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to a stream and includes one or more layers. The circuitry writes a second parameter into a sequence parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value without any reference to the video parameter set.

Each of embodiments or each of a configuration or a method that is part of the embodiment according to the present disclosure enables, for example, at least one of the following: an improvement in encoding efficiency, an improvement in image quality, a reduction in the processing amount of encoding/decoding, a reduction in the scale of circuitry, or an improvement in the processing speed of

encoding/decoding. Alternatively, each of embodiments or each of a configuration or a method that is part of the embodiment according to the present disclosure enables an appropriate selection of operations or constituent elements such as filters, block sizes, motion vectors, reference pictures, and reference blocks. It is to be noted that the present disclosure includes configurations and methods which may provide advantages other than above-described ones. Examples of such configurations and methods include a configuration or method for improving coding efficiency while reducing an increase in the processing amount.

Additional benefits and advantages according to an aspect of the present disclosure will become apparent from the specification and drawings. The benefits and/or advantages may be individually obtained by various embodiments and features of the specification and drawings, and not all of which need to be provided in order to obtain one or more of such benefits and/or advantages.

It should be noted that these general or specific aspects of the present disclosure may be implemented as a system, an integrated circuit, a computer program, a computer-readable medium (recording medium) such as a CD-ROM, or any combination of systems, methods, integrated circuits, computer programs, and media.

BRIEF DESCRIPTION OF DRAWINGS

These and other objects, advantages and features of the disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

FIG. 1 is a schematic diagram illustrating one example of a configuration of a transmission system according to an embodiment;

FIG. 2 is a diagram illustrating one example of a hierarchical structure of data in a stream;

FIG. 3 is a diagram illustrating one example of a slice configuration;

FIG. 4 is a diagram illustrating one example of a tile configuration;

FIG. 5 is a diagram illustrating one example of an encoding structure in scalable encoding;

FIG. 6 is a diagram illustrating one example of an encoding structure in scalable encoding;

FIG. 7 is a block diagram illustrating one example of a configuration of an encoder according to an embodiment;

FIG. 8 is a block diagram illustrating a mounting example of the encoder;

FIG. 9 is a flow chart illustrating one example of an overall encoding process performed by the encoder;

FIG. 10 is a diagram illustrating one example of block splitting;

FIG. 11 is a diagram illustrating one example of a configuration of a splitter;

FIG. 12 is a diagram illustrating examples of splitting patterns;

FIG. 13A is a diagram illustrating one example of a syntax tree of a splitting pattern;

FIG. 13B is a diagram illustrating another example of a syntax tree of a splitting pattern;

FIG. 14 is a chart illustrating transform basis functions for each transform type;

FIG. 15 is a diagram illustrating examples of SVT;

FIG. 16 is a flow chart illustrating one example of a process performed by a transformer;

3

FIG. 17 is a flow chart illustrating another example of a process performed by the transformer;

FIG. 18 is a block diagram illustrating one example of a configuration of a quantizer;

FIG. 19 is a flow chart illustrating one example of quantization performed by the quantizer;

FIG. 20 is a block diagram illustrating one example of a configuration of an entropy encoder;

FIG. 21 is a diagram illustrating a flow of CABAC in the entropy encoder;

FIG. 22 is a block diagram illustrating one example of a configuration of a loop filter;

FIG. 23A is a diagram illustrating one example of a filter shape used in an adaptive loop filter (ALF);

FIG. 23B is a diagram illustrating another example of a filter shape used in an ALF;

FIG. 23C is a diagram illustrating another example of a filter shape used in an ALF;

FIG. 23D is a diagram illustrating an example where Y samples (first component) are used for a cross component ALF (CCALF) for Cb and a CCALF for Cr (components different from the first component);

FIG. 23E is a diagram illustrating a diamond shaped filter;

FIG. 23F is a diagram illustrating an example for a joint chroma CCALF (JC-CCALF);

FIG. 23G is a diagram illustrating an example for JC-CCALF weight index candidates;

FIG. 24 is a block diagram illustrating one example of a specific configuration of a loop filter which functions as a DBF;

FIG. 25 is a diagram illustrating an example of a deblocking filter having a symmetrical filtering characteristic with respect to a block boundary;

FIG. 26 is a diagram for illustrating a block boundary on which a deblocking filter process is performed;

FIG. 27 is a diagram illustrating examples of Bs values;

FIG. 28 is a flow chart illustrating one example of a process performed by a predictor of the encoder;

FIG. 29 is a flow chart illustrating another example of a process performed by the predictor of the encoder;

FIG. 30 is a flow chart illustrating another example of a process performed by the predictor of the encoder;

FIG. 31 is a diagram illustrating one example of sixty-seven intra prediction modes used in intra prediction;

FIG. 32 is a flow chart illustrating one example of a process performed by an intra predictor;

FIG. 33 is a diagram illustrating examples of reference pictures;

FIG. 34 is a diagram illustrating examples of reference picture lists;

FIG. 35 is a flow chart illustrating a basic processing flow of inter prediction;

FIG. 36 is a flow chart illustrating one example of MV derivation;

FIG. 37 is a flow chart illustrating another example of MV derivation;

FIG. 38A is a diagram illustrating one example of categorization of modes for MV derivation;

FIG. 38B is a diagram illustrating one example of categorization of modes for MV derivation;

FIG. 39 is a flow chart illustrating an example of inter prediction by normal inter mode;

FIG. 40 is a flow chart illustrating an example of inter prediction by normal merge mode;

FIG. 41 is a diagram for illustrating one example of an MV derivation process by normal merge mode;

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FIG. 42 is a diagram for illustrating one example of an MV derivation process by a history-based motion vector prediction/predictor (HMVP) mode;

FIG. 43 is a flow chart illustrating one example of frame rate up conversion (FRUC);

FIG. 44 is a diagram for illustrating one example of pattern matching (bilateral matching) between two blocks located along a motion trajectory;

FIG. 45 is a diagram for illustrating one example of pattern matching (template matching) between a template in a current picture and a block in a reference picture;

FIG. 46A is a diagram for illustrating one example of MV derivation in units of a sub-block in affine mode in which two control points are used;

FIG. 46B is a diagram for illustrating one example of MV derivation in units of a sub-block in affine mode in which three control points are used;

FIG. 47A is a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode;

FIG. 47B is a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode;

FIG. 47C is a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode;

FIG. 48A is a diagram for illustrating an affine mode in which two control points are used;

FIG. 48B is a diagram for illustrating an affine mode in which three control points are used;

FIG. 49A is a conceptual diagram for illustrating one example of a method for MV derivation at control points when the number of control points for an encoded block and the number of control points for a current block are different from each other;

FIG. 49B is a conceptual diagram for illustrating another example of a method for MV derivation at control points when the number of control points for an encoded block and the number of control points for a current block are different from each other;

FIG. 50 is a flow chart illustrating one example of a process in affine merge mode;

FIG. 51 is a flow chart illustrating one example of a process in affine inter mode;

FIG. 52A is a diagram for illustrating generation of two triangular prediction images;

FIG. 52B is a conceptual diagram illustrating examples of a first portion of a first partition and first and second sets of samples;

FIG. 52C is a conceptual diagram illustrating a first portion of a first partition;

FIG. 53 is a flow chart illustrating one example of a triangle mode;

FIG. 54 is a diagram illustrating one example of an advanced temporal motion vector prediction/predictor (ATMVP) mode in which an MV is derived in units of a sub-block;

FIG. 55 is a diagram illustrating a relationship between a merge mode and dynamic motion vector refreshing (DMVR);

FIG. 56 is a conceptual diagram for illustrating one example of DMVR;

FIG. 57 is a conceptual diagram for illustrating another example of DMVR for determining an MV;

FIG. 58A is a diagram illustrating one example of motion estimation in DMVR;

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FIG. 58B is a flow chart illustrating one example of motion estimation in DMVR;

FIG. 59 is a flow chart illustrating one example of generation of a prediction image;

FIG. 60 is a flow chart illustrating another example of generation of a prediction image;

FIG. 61 is a flow chart illustrating one example of a correction process of a prediction image by overlapped block motion compensation (OBMC);

FIG. 62 is a conceptual diagram for illustrating one example of a prediction image correction process by OBMC;

FIG. 63 is a diagram for illustrating a model assuming uniform linear motion;

FIG. 64 is a flow chart illustrating one example of inter prediction according to BIO;

FIG. 65 is a diagram illustrating one example of a configuration of an inter predictor which performs inter prediction according to BIO;

FIG. 66A is a diagram for illustrating one example of a prediction image generation method using a luminance correction process by local illumination compensation (LIC);

FIG. 66B is a flow chart illustrating one example of a prediction image generation method using a luminance correction process by LIC;

FIG. 67 is a block diagram illustrating a configuration of a decoder according to an embodiment;

FIG. 68 is a block diagram illustrating a mounting example of a decoder;

FIG. 69 is a flow chart illustrating one example of an overall decoding process performed by the decoder;

FIG. 70 is a diagram illustrating a relationship between a splitting determiner and other constituent elements;

FIG. 71 is a block diagram illustrating one example of a configuration of an entropy decoder;

FIG. 72 is a diagram illustrating a flow of CABAC in the entropy decoder;

FIG. 73 is a block diagram illustrating one example of a configuration of an inverse quantizer;

FIG. 74 is a flow chart illustrating one example of inverse quantization performed by the inverse quantizer;

FIG. 75 is a flow chart illustrating one example of a process performed by an inverse transformer;

FIG. 76 is a flow chart illustrating another example of a process performed by the inverse transformer;

FIG. 77 is a block diagram illustrating one example of a configuration of a loop filter;

FIG. 78 is a flow chart illustrating one example of a process performed by a predictor of the decoder;

FIG. 79 is a flow chart illustrating another example of a process performed by the predictor of the decoder;

FIG. 80A is a flow chart illustrating a portion of other example of a process performed by the predictor of the decoder;

FIG. 80B is a flow chart illustrating the remaining portion of the other example of the process performed by the predictor of the decoder;

FIG. 81 is a diagram illustrating one example of a process performed by an intra predictor of the decoder;

FIG. 82 is a flow chart illustrating one example of MV derivation in the decoder;

FIG. 83 is a flow chart illustrating another example of MV derivation in the decoder;

FIG. 84 is a flow chart illustrating an example of inter prediction by normal inter mode in the decoder;

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FIG. 85 is a flow chart illustrating an example of inter prediction by normal merge mode in the decoder;

FIG. 86 is a flow chart illustrating an example of inter prediction by FRUC mode in the decoder;

FIG. 87 is a flow chart illustrating an example of inter prediction by affine merge mode in the decoder;

FIG. 88 is a flow chart illustrating an example of inter prediction by affine inter mode in the decoder;

FIG. 89 is a flow chart illustrating an example of inter prediction by triangle mode in the decoder;

FIG. 90 is a flow chart illustrating an example of motion estimation by DMVR in the decoder;

FIG. 91 is a flow chart illustrating one specific example of motion estimation by DMVR in the decoder;

FIG. 92 is a flow chart illustrating one example of generation of a prediction image in the decoder;

FIG. 93 is a flow chart illustrating another example of generation of a prediction image in the decoder;

FIG. 94 is a flow chart illustrating another example of correction of a prediction image by OBMC in the decoder;

FIG. 95 is a flow chart illustrating another example of correction of a prediction image by BIO in the decoder;

FIG. 96 is a flow chart illustrating another example of correction of a prediction image by LIC in the decoder;

FIG. 97 is a diagram illustrating syntax of HRD-related information in a buffering period SEI message;

FIG. 98 is a flow chart illustrating an example of a process of checking bitstream conformance using an HRD;

FIG. 99 is a diagram illustrating syntax of a buffering period SEI message according to a second aspect;

FIG. 100 is a diagram illustrating syntax of a picture timing SEI message according to the second aspect;

FIG. 101 is a diagram illustrating syntax of HRD-related information in VPS;

FIG. 102 is a diagram illustrating syntax of HRD-related information in SPS;

FIG. 103 is a flow chart illustrating an example of a process for checking conformance according to the second aspect;

FIG. 104 is a flow chart illustrating an example of a process for checking conformance according to a third aspect;

FIG. 105 is a flow chart illustrating an example of a process for checking conformance according to a fourth aspect;

FIG. 106 is a flow chart illustrating an operation performed by the encoder according to the embodiment;

FIG. 107 is a flow chart illustrating a more specific operation performed by the encoder according to the embodiment;

FIG. 108 is a flow chart illustrating an operation performed by the decoder according to the embodiment;

FIG. 109 is a flow chart illustrating a more specific operation performed by the decoder according to the embodiment;

FIG. 110 is a diagram illustrating an overall configuration of a content providing system for implementing a content distribution service;

FIG. 111 is a diagram illustrating an example of a display screen of a web page;

FIG. 112 is a diagram illustrating an example of a display screen of a web page;

FIG. 113 is a diagram illustrating one example of a smartphone; and

FIG. 114 is a block diagram illustrating an example of a configuration of a smartphone.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Introduction

In video coding and decoding, for example, a video parameter set (VPS) and a sequence parameter set (SPS) may be used. VPS is a parameter set including parameters for an entire stream. SPS is a parameter set including parameters for sequences included in a stream. SPS may be used on a random access unit basis. Moreover, VPS and SPS are included in a bitstream. Accordingly, the same parameters are used in encoding and decoding.

VPS may not be used in video coding and decoding. In this case, VPS is not included in a bitstream. Basically, in the case where layers defined by multi-layer coding such as spatially-scalable video coding or multi-view video coding are included in a bitstream, VPS is included in the bitstream. In the case where layers are not included in a bitstream, VPS is not included in the bitstream.

Temporal sublayers defined by temporally-scalable coding may be included in a bitstream. The temporal sublayers are different from layers defined by multi-layer coding such as spatially-scalable video coding or multi-view video coding.

A parameter indicating the maximum number of temporal sublayers in a stream may be included in VPS. A parameter indicating the maximum number of temporal sublayers in a sequence may be included in SPS.

When a parameter indicating the maximum number of temporal sublayers in SPS is not consistent with a parameter indicating the maximum number of temporal sublayers in VPS, however, malfunction may occur in encoding and decoding. Even in the case where VPS is not used in video coding and decoding, it may be difficult to efficiently perform video coding and decoding, depending on the parameter indicating the maximum number of temporal sublayers in SPS.

In view of this, an encoder according to an aspect of the present disclosure includes circuitry and memory. In operation, in the case where a video parameter set is used for encoding a video, the circuitry writes a first parameter into the video parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to a stream and includes one or more layers. The circuitry writes a second parameter into a sequence parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable the encoder to write, into SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, it may be possible for the encoder to inhibit inconsistency between a parameter in VPS and a parameter in SPS. In the case where VPS is not used, the encoder may be capable of writing, into the SPS, the second parameter that is limited to less than or

equal to a fixed value. Accordingly, it may be possible for the encoder to define the second parameter in a range where processing is performed efficiently.

Moreover, with the second parameter being limited by the first parameter or a fixed value, it may be possible for the encoder to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. The encoder thus may be capable of processing the one or more temporal sublayers according to the maximum number that is not an indefinite value.

For example, the fixed value is an upper limit of the first parameter.

This may enable the encoder to write, into SPS, the second parameter that is limited by the upper limit of the first parameter in VPS, even in the case where VPS is not used. Accordingly, even in the case where VPS is not used, it may be possible for the encoder to define the second parameter in a range where processing is performed efficiently, as is the case where VPS is used.

Stated differently, the encoder may be capable of adjusting the upper limit of the maximum number of one or more temporal sublayers in a sequence unit to the upper limit of the maximum number of one or more temporal sublayers in a layer set unit. This may enable the encoder to define the upper limit of the maximum number of one or more temporal sublayers in a sequence unit in a range where processing is performed efficiently in accordance with the upper limit of the maximum number of one or more temporal sublayers in a layer set unit.

For example, the first parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the layer set unit. The second parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the sequence unit. The fixed value is 6. In the case where the video parameter set is used for encoding the video, the second parameter is limited to a range from 0 to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to a range from 0 to 6.

This may enable the encoder to limit the maximum number of temporal sublayers in a sequence unit to a range from 1 to the maximum number of temporal sublayers in a layer set unit in the case where VPS is used. In the case where VPS is not used, the encoder may be capable of limiting the maximum number of temporal sublayers in a sequence unit to a range from 1 to 7. Accordingly, it may be possible for the encoder to define the maximum number of temporal sublayers in a sequence unit by a range where processing is performed efficiently.

For example, the circuitry writes a third parameter into the sequence parameter set. The third parameter (i) indicates that the video parameter set is used for encoding the video and indicates an identifier of the video parameter set, when the third parameter is greater than 0, and (ii) indicates that the video parameter set is not used for encoding the video, when the third parameter is 0. When the third parameter is greater than 0, the second parameter is limited to less than or equal to the first parameter. When the third parameter is 0, the second parameter is limited to less than or equal to the fixed value.

This may enable the encoder to switch between limiting the second parameter in SPS to less than or equal to the first parameter in VPS and limiting the second parameter in the SPS to less than or equal to a fixed value, in accordance with the third parameter indicating whether or not VPS is used. In

other words, the encoder may be capable of adaptively limiting the second parameter in accordance with the third parameter.

Moreover, since the encoder writes the third parameter into SPS, the encoder may be capable of switching, on a sequence basis, between limiting the second parameter to less than or equal to the first parameter in VPS and limiting the second parameter to less than or equal to a fixed value. In other words, the encoder may be capable of adaptively limiting the second parameter on a per sequence basis in accordance with the third parameter written into the SPS.

For example, a decoder according to an aspect of the present disclosure includes circuitry and memory coupled to the circuitry. In operation, in the case where a video parameter set is used for decoding a video, the circuitry reads a first parameter from the video parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to a stream and includes one or more layers. The circuitry reads a second parameter from a sequence parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for decoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable the decoder to read, from SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, it may be possible for the decoder to inhibit inconsistency between a parameter in VPS and a parameter in SPS. In the case where VPS is not used, the decoder may be capable of reading, from the SPS, the second parameter that is limited to less than or equal to a fixed value. Accordingly, it may be possible for the decoder to define the second parameter in a range where processing is performed efficiently.

Moreover, with the second parameter being limited by the first parameter or a fixed value, it may be possible for the decoder to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. The decoder thus may be capable of processing the one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, the fixed value is an upper limit of the first parameter.

This may enable the decoder to read, from SPS, the second parameter that is limited by the upper limit of the first parameter in VPS, even in the case where VPS is not used. Accordingly, even in the case where VPS is not used, it may be possible for the decoder to define the second parameter in a range where processing is performed efficiently, as is the case where VPS is used.

Stated differently, the decoder may be capable of adjusting the upper limit of the maximum number of one or more temporal sublayers in a sequence unit to the upper limit of the maximum number of one or more temporal sublayers in a layer set unit. This may enable the decoder to define the upper limit of the maximum number of one or more temporal sublayers in a sequence unit in a range where processing is performed efficiently in accordance with the upper limit of the maximum number of one or more temporal sublayers in a layer set unit.

For example, the first parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the layer set unit. The second parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the sequence unit. The fixed value is 6. In the case where the video parameter set is used for decoding the video, the second parameter is limited to a range from 0 to the first parameter. In the case where the video parameter set is not used for decoding the video, the second parameter is limited to a range from 0 to 6.

This may enable the decoder to limit the maximum number of temporal sublayers in a sequence unit to a range from 1 to the maximum number of temporal sublayers in a layer set unit in the case where VPS is used. In the case where VPS is not used, the decoder may be capable of limiting the maximum number of temporal sublayers in a sequence unit to a range from 1 to 7. Accordingly, it may be possible for the decoder to define the maximum number of temporal sublayers in a sequence unit by a range where the processing is performed efficiently.

For example, the circuitry reads a third parameter from the sequence parameter set. The third parameter (i) indicates that the video parameter set is used for decoding the video and indicates an identifier of the video parameter set, when the third parameter is greater than 0, and (ii) indicates that the video parameter set is not used for decoding the video, when the third parameter is 0. When the third parameter is greater than 0, the second parameter is limited to less than or equal to the first parameter. When the third parameter is 0, the second parameter is limited to less than or equal to the fixed value.

This may enable the decoder to switch between limiting the second parameter in SPS to less than or equal to the first parameter in VPS and limiting the second parameter in the SPS to less than or equal to a fixed value, in accordance with the third parameter indicating whether or not VPS is used. In other words, the decoder may be capable of adaptively limiting the second parameter in accordance with the third parameter.

Moreover, since the decoder reads the third parameter from SPS, the decoder may be capable of switching, on a sequence basis, between limiting the second parameter to less than or equal to the first parameter in VPS and limiting the second parameter to less than or equal to a fixed value. In other words, the decoder may be capable of adaptively limiting the second parameter on a per sequence basis in accordance with the third parameter read from the SPS.

For example, a decoder according to an aspect of the present disclosure includes circuitry and memory coupled to the circuitry. In operation, in the case where a first parameter set is used for decoding a video, the circuitry reads a first parameter from the first parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to a stream and includes one or more layers. The first parameter set is a parameter set for the layer set unit. The circuitry reads a second parameter from a second parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a random access unit. The random access unit is a unit which corresponds to at least a portion of the stream. The second parameter set is a parameter set for the random access unit. In the case where the first parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the first

parameter set is not used for decoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable the decoder to read, from the second parameter set for the random access unit, the second parameter consistent with the first parameter in the first parameter set for the layer set unit, in the case where the first parameter set for the layer set unit is used. Accordingly, it may be possible for the decoder to inhibit inconsistency between a parameter in the first parameter set for the layer set unit and a parameter in the second parameter set for the random access unit.

In the case where the first parameter set for the layer set unit is not used, the decoder may be capable of reading the second parameter that is limited to less than or equal to a fixed value, from the second parameter set for the random access unit. Accordingly, the decoder may be capable of defining the second parameter in a range where processing is performed efficiently.

Moreover, with the second parameter being limited by the first parameter or a fixed value, it may be possible for the decoder to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. The decoder thus may be capable of processing one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, an encoding method according to an aspect of the present disclosure includes: in the case where a video parameter set is used for encoding a video, writing a first parameter into the video parameter set, where the first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit, and the layer set unit is a unit which corresponds to a stream and includes one or more layers; and writing a second parameter into a sequence parameter set, where the second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit, and the sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable writing, into SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, it may be possible to inhibit inconsistency between a parameter in VPS and a parameter in SPS. In the case where VPS is not used, it may be possible to write the second parameter that is limited to less than or equal to a fixed value into the SPS. Accordingly, it may be possible to define the second parameter in a range where processing is performed efficiently.

Moreover, with the second parameter being limited by the first parameter or a fixed value, it may be possible to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. It may then be possible to process the one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, a decoding method according to an aspect of the present disclosure includes: in the case where the video parameter set is used for decoding a video, reading a first parameter from the video parameter set, where the first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit, and the layer set unit is a unit which corresponds to a stream and includes one or more layers; and reading a second parameter from a sequence parameter set, where the second parameter indi-

cates the maximum number of one or more temporal sublayers in a sequence unit, and the sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for decoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable reading, from SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, it may be possible to inhibit inconsistency between a parameter in VPS and a parameter in SPS. In the case where VPS is not used, it may be possible to read the second parameter that is limited to less than or equal to a fixed value from SPS. Accordingly, it may be possible to define the second parameter in a range where processing is performed efficiently.

Moreover, with the second parameter being limited by the first parameter or a fixed value, it may be possible to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. It may then be possible to process the one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, a non-transitory medium according to an aspect of the present disclosure is a non-transitory medium storing a bitstream. The bitstream includes a video parameter set and a sequence parameter set for causing a decoder that reads out the bitstream to perform a decoding process. The video parameter set includes a first parameter (i) read by the decoder in the case where the video parameter set is used in the decoding process and (ii) indicating the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to the bitstream and includes one or more layers. The sequence parameter set includes a second parameter indicating a maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit for which the sequence parameter set is referred to in the bitstream. In the case where the video parameter set is used in the decoding process, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used in the decoding process, the second parameter is limited to less than or equal to a fixed value.

This may enable reading, from SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, it may be possible to inhibit inconsistency between a parameter in VPS and a parameter in SPS. In the case where VPS is not used, it may be possible to read, from SPS, the second parameter that is limited to less than or equal to a fixed value. Accordingly, it may be possible to define the second parameter in a range where processing is performed efficiently.

Moreover, with the second parameter being limited by the first parameter or a fixed value, it may be possible to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. It may then be possible to process the one or more temporal sublayers according to the maximum number that is not an indefinite value.

For example, a bitstream transmitter according to an aspect of the present disclosure transmits a bitstream to a decoder. The bitstream includes a video parameter set and a sequence parameter set for causing the decoder to perform a decoding process. The video parameter set includes a first parameter (i) read by the decoder in the case where the video parameter set is used in the decoding process and (ii) indicating the maximum number of one or more temporal

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sublayers in a layer set unit. The layer set unit is a unit which corresponds to the bitstream and includes one or more layers. The sequence parameter set includes a second parameter indicating the maximum number of one or more temporal sublayers in a sequence unit for which the sequence parameter set is referred to in the bitstream. In the case where the video parameter set is used in the decoding process, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used in the decoding process, the second parameter is limited to less than or equal to a fixed value.

For example, a bitstream storage device according to an aspect of the present disclosure stores a bitstream. The bitstream includes a video parameter set and a sequence parameter set for causing the decoder to perform a decoding process. The video parameter set includes a first parameter (i) read by the decoder in the case where the video parameter set is used in the decoding process and (ii) indicating the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to the bitstream and includes one or more layers. The sequence parameter set includes a second parameter that indicates the maximum number of one or more temporal sublayers in a sequence unit for which the sequence parameter set is referred to in the bitstream. In the case where the video parameter set is used in the decoding process, the second parameter is limited to less than or equal to the first parameter, and in the case where the video parameter set is not used in the decoding process, the second parameter is limited to less than or equal to a fixed value.

An encoder according to an aspect of the present disclosure may include, for example, a splitter, an intra predictor, an inter predictor, a loop filter, a transformer, a quantizer, and an entropy encoder.

A current picture is input to the input unit. The splitter partitions the current picture into a plurality of blocks.

The intra predictor generates the prediction signal of a current block included in the current picture using a reference image included in the current picture. The inter predictor generates the prediction signal of a current block included in the current picture using a reference image included in a reference picture different from the current picture. The loop filter applies a filter to a reconstructed block of the current block included in the current picture.

The transformer transforms prediction errors between the original signal of the current block included in the current picture and the prediction signal generated by the intra predictor or the inter predictor, to generate transform coefficients. The quantizer quantizes the transform coefficients to generate quantized coefficients. The entropy encoder applies variable length encoding to the quantized coefficients to generate an encoded bitstream. The encoded bitstream including control information and the quantized coefficients to which the variable length encoding has been applied is then outputted from the output unit.

For example, in operation, in the case where a video parameter set is used for encoding a video, the entropy encoder writes a first parameter into the video parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to a stream and includes one or more layers. The entropy encoder writes a second parameter into a sequence parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where

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the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value.

A decoder according to an aspect of the present disclosure may include, for example, an input unit, an entropy decoder, an inverse quantizer, an inverse transformer, an intra predictor, an inter predictor, a loop filter, and an output unit.

An encoded bitstream is input to the input unit. The entropy decoder applies variable length decoding to the encoded bitstream to derive quantized coefficients. The inverse quantizer inverse quantizes the quantized coefficients to derive the transform coefficients. The inverse transformer generates the prediction signal of the current block included in the current picture, using a reference image included in a reference picture different from the current picture.

The intra predictor generates the prediction signal of a current block included in the current picture, using a reference image included in the current picture. The inter predictor generates the prediction signal of the current block included in the current picture, using a reference image included in a reference picture different from the current picture.

The loop filter applies a filter to a reconstructed block of the current block included in the current picture. The current picture is then outputted from the output unit.

For example, in operation, in the case where a video parameter set is used for decoding a video, the entropy decoder reads a first parameter from the video parameter set. The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit which corresponds to a stream and includes one or more layers. The entropy decoder reads a second parameter from a sequence parameter set. The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for decoding the video, the second parameter is limited to less than or equal to a fixed value.

It should be noted that these general or specific aspects may be implemented using a system, a device, a method, an integrated circuit, a computer program, a non-transitory computer-readable medium such as a CD-ROM, or any combination of systems, devices, methods, integrated circuits, computer programs, and media.

Definitions of Terms

The respective terms may be defined as indicated below as examples.

(1) Image

An image is a data unit configured with a set of pixels, is a picture or includes blocks smaller than a picture. Images include a still image in addition to a video.

(2) Picture

A picture is an image processing unit configured with a set of pixels, and is also referred to as a frame or a field.

(3) Block

A block is a processing unit which is a set of a particular number of pixels. The block is also referred to as indicated in the following examples. The shapes of blocks are not

limited. Examples include a rectangle shape of M×N pixels and a square shape of M×M pixels for the first place, and also include a triangular shape, a circular shape, and other shapes.

(Examples of Blocks)

slice/tile/brick

CTU/super block/basic splitting unit

VPDU/processing splitting unit for hardware

CU/processing block unit/prediction block unit (PU)/

orthogonal transform block unit (TU)/unit

sub-block

(4) Pixel/Sample

A pixel or sample is a smallest point of an image. Pixels or samples include not only a pixel at an integer position but also a pixel at a sub-pixel position generated based on a pixel at an integer position.

(5) Pixel Value/Sample Value

A pixel value or sample value is an eigen value of a pixel. Pixel or sample values naturally include a luma value, a chroma value, an RGB gradation level and also covers a depth value, or a binary value of 0 or 1.

(6) Flag

A flag indicates one or more bits, and may be, for example, a parameter or index represented by two or more bits. Alternatively, the flag may indicate not only a binary value represented by a binary number but also a multiple value represented by a number other than the binary number.

(7) Signal

A signal is the one symbolized or encoded to convey information. Signals include a discrete digital signal and an analog signal which takes a continuous value.

(8) Stream/Bitstream

A stream or bitstream is a digital data string or a digital data flow. A stream or bitstream may be one stream or may be configured with a plurality of streams having a plurality of hierarchical layers. A stream or bitstream may be transmitted in serial communication using a single transmission path, or may be transmitted in packet communication using a plurality of transmission paths.

(9) Difference

In the case of scalar quantity, it is only necessary that a simple difference ($x-y$) and a difference calculation be included. Differences include an absolute value of a difference ($|x-y|$), a squared difference (x^2-y^2), a square root of a difference ($\sqrt{x-y}$), a weighted difference ($ax-by$: a and b are constants), an offset difference ($x-y+a$: a is an offset).

(10) Sum

In the case of scalar quantity, it is only necessary that a simple sum ($x+y$) and a sum calculation be included. Sums include an absolute value of a sum ($|x+y|$), a squared sum (x^2+y^2), a square root of a sum ($\sqrt{x+y}$), a weighted difference ($ax+by$: a and b are constants), an offset sum ($x+y+a$: a is an offset).

(11) Based on

A phrase “based on something” means that a thing other than the something may be considered. In addition, “based on” may be used in a case in which a direct result is obtained or a case in which a result is obtained through an intermediate result.

(12) Used, Using

A phrase “something used” or “using something” means that a thing other than the something may be considered. In addition, “used” or “using” may be used in a case in which a direct result is obtained or a case in which a result is obtained through an intermediate result.

(13) Prohibit, Forbid

The term “prohibit” or “forbid” can be rephrased as “does not permit” or “does not allow”. In addition, “being not prohibited/forbidden” or “being permitted/allowed” does not always mean “obligation”.

(14) Limit, Restriction/Restrict/Restricted

The term “limit” or “restriction/restrict/restricted” can be rephrased as “does not permit/allow” or “being not permitted/allowed”. In addition, “being not prohibited/forbidden” or “being permitted/allowed” does not always mean “obligation”. Furthermore, it is only necessary that part of something be prohibited/forbidden quantitatively or qualitatively, and something may be fully prohibited/forbidden.

(15) Chroma

An adjective, represented by the symbols C_b and C_r , specifying that a sample array or single sample is representing one of the two color difference signals related to the primary colors. The term chroma may be used instead of the term chrominance.

(16) Luma

An adjective, represented by the symbol or subscript Y or L , specifying that a sample array or single sample is representing the monochrome signal related to the primary colors. The term luma may be used instead of the term luminance.

[Notes Related to the Descriptions]

In the drawings, same reference numbers indicate same or similar components. The sizes and relative locations of components are not necessarily drawn by the same scale.

Hereinafter, embodiments will be described with reference to the drawings. Note that the embodiments described below each show a general or specific example. The numerical values, shapes, materials, components, the arrangement and connection of the components, steps, the relation and order of the steps, etc., indicated in the following embodiments are mere examples, and are not intended to limit the scope of the claims.

Embodiments of an encoder and a decoder will be described below. The embodiments are examples of an encoder and a decoder to which the processes and/or configurations presented in the description of aspects of the present disclosure are applicable. The processes and/or configurations can also be implemented in an encoder and a decoder different from those according to the embodiments. For example, regarding the processes and/or configurations as applied to the embodiments, any of the following may be implemented:

(1) Any of the components of the encoder or the decoder according to the embodiments presented in the description of aspects of the present disclosure may be substituted or combined with another component presented anywhere in the description of aspects of the present disclosure.

(2) In the encoder or the decoder according to the embodiments, discretionary changes may be made to functions or processes performed by one or more components of the encoder or the decoder, such as addition, substitution, removal, etc., of the functions or processes. For example, any function or process may be substituted or combined with another function or process presented anywhere in the description of aspects of the present disclosure.

(3) In methods implemented by the encoder or the decoder according to the embodiments, discretionary changes may be made such as addition, substitution, and removal of one or more of the processes included in the method. For example, any process in the method may be substituted or combined with another process presented anywhere in the description of aspects of the present disclosure.

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(4) One or more components included in the encoder or the decoder according to embodiments may be combined with a component presented anywhere in the description of aspects of the present disclosure, may be combined with a component including one or more functions presented anywhere in the description of aspects of the present disclosure, and may be combined with a component that implements one or more processes implemented by a component presented in the description of aspects of the present disclosure.

(5) A component including one or more functions of the encoder or the decoder according to the embodiments, or a component that implements one or more processes of the encoder or the decoder according to the embodiments, may be combined or substituted with a component presented anywhere in the description of aspects of the present disclosure, with a component including one or more functions presented anywhere in the description of aspects of the present disclosure, or with a component that implements one or more processes presented anywhere in the description of aspects of the present disclosure.

(6) In methods implemented by the encoder or the decoder according to the embodiments, any of the processes included in the method may be substituted or combined with a process presented anywhere in the description of aspects of the present disclosure or with any corresponding or equivalent process.

(7) One or more processes included in methods implemented by the encoder or the decoder according to the embodiments may be combined with a process presented anywhere in the description of aspects of the present disclosure.

(8) The implementation of the processes and/or configurations presented in the description of aspects of the present disclosure is not limited to the encoder or the decoder according to the embodiments. For example, the processes and/or configurations may be implemented in a device used for a purpose different from the moving picture encoder or the moving picture decoder disclosed in the embodiments. [System Configuration]

FIG. 1 is a schematic diagram illustrating one example of a configuration of a transmission system according to an embodiment.

Transmission system Trs is a system which transmits a stream generated by encoding an image and decodes the transmitted stream.

Transmission system Trs like this includes, for example, encoder **100**, network Nw, and decoder **200** as illustrated in FIG. 1.

An image is input to encoder **100**. Encoder **100** generates a stream by encoding the input image, and outputs the stream to network Nw. The stream includes, for example, the encoded image and control information for decoding the encoded image. The image is compressed by the encoding.

It is to be noted that a previous image before being encoded and being input to encoder **100** is also referred to as the original image, the original signal, or the original sample. The image may be a video or a still image. The image is a generic concept of a sequence, a picture, and a block, and thus is not limited to a spatial region having a particular size and to a temporal region having a particular size unless otherwise specified. The image is an array of pixels or pixel values, and the signal representing the image or pixel values are also referred to as samples. The stream may be referred to as a bitstream, an encoded bitstream, a compressed bitstream, or an encoded signal. Furthermore, the encoder may be referred to as an image encoder or a video encoder. The encoding method performed by encoder

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100 may be referred to as an encoding method, an image encoding method, or a video encoding method.

Network Nw transmits the stream generated by encoder **100** to decoder **200**. Network Nw may be the Internet, the Wide Area Network (WAN), the Local Area Network (LAN), or any combination of these networks. Network Nw is not always limited to a bi-directional communication network, and may be a uni-directional communication network which transmits broadcast waves of digital terrestrial broadcasting, satellite broadcasting, or the like. Alternatively, network Nw may be replaced by a medium such as a Digital Versatile Disc (DVD) and a Blu-Ray Disc (BD) I, etc. on which a stream is recorded.

Decoder **200** generates, for example, a decoded image which is an uncompressed image by decoding a stream transmitted by network Nw. For example, the decoder decodes a stream according to a decoding method corresponding to an encoding method by encoder **100**.

It is to be noted that the decoder may also be referred to as an image decoder or a video decoder, and that the decoding method performed by decoder **200** may also be referred to as a decoding method, an image decoding method, or a video decoding method.

[Data Structure]

FIG. 2 is a diagram illustrating one example of a hierarchical structure of data in a stream. A stream includes, for example, a video sequence. As illustrated in (a) of FIG. 2, the video sequence includes a video parameter set (VPS), a sequence parameter set (SPS), a picture parameter set (PPS), supplemental enhancement information (SEI), and a plurality of pictures.

In a video having a plurality of layers, a VPS includes: a coding parameter which is common between some of the plurality of layers; and a coding parameter related to some of the plurality of layers included in the video or an individual layer.

An SPS includes a parameter which is used for a sequence, that is, a coding parameter which decoder **200** refers to in order to decode the sequence. For example, the coding parameter may indicate the width or height of a picture. It is to be noted that a plurality of SPSs may be present.

A PPS includes a parameter which is used for a picture, that is, a coding parameter which decoder **200** refers to in order to decode each of the pictures in the sequence. For example, the coding parameter may include a reference value for the quantization width which is used to decode a picture and a flag indicating application of weighted prediction. It is to be noted that a plurality of PPSs may be present. Each of the SPS and the PPS may be simply referred to as a parameter set.

As illustrated in (b) of FIG. 2, a picture may include a picture header and at least one slice. A picture header includes a coding parameter which decoder **200** refers to in order to decode the at least one slice.

As illustrated in I of FIG. 2, a slice includes a slice header and at least one brick. A slice header includes a coding parameter which decoder **200** refers to in order to decode the at least one brick.

As illustrated in (d) of FIG. 2, a brick includes at least one coding tree unit (CTU).

It is to be noted that a picture may not include any slice and may include a tile group instead of a slice. In this case, the tile group includes at least one tile. In addition, a brick may include a slice.

A CTU is also referred to as a super block or a basis splitting unit. As illustrated in I of FIG. 2, a CTU like this

includes a CTU header and at least one coding unit (CU). A CTU header includes a coding parameter which decoder 200 refers to in order to decode the at least one CU.

A CU may be split into a plurality of smaller CUs. As illustrated in (f) of FIG. 2, a CU includes a CU header, prediction information, and residual coefficient information. Prediction information is information for predicting the CU, and the residual coefficient information is information indicating a prediction residual to be described later. Although a CU is basically the same as a prediction unit (PU) and a transform unit (TU), it is to be noted that, for example, an SBT to be described later may include a plurality of TUs smaller than the CU. In addition, the CU may be processed for each virtual pipeline decoding unit (VPDU) included in the CU. The VPDU is, for example, a fixed unit which can be processed at one stage when pipeline processing is performed in hardware.

It is to be noted that a stream may not include part of the hierarchical layers illustrated in FIG. 2. The order of the hierarchical layers may be exchanged, or any of the hierarchical layers may be replaced by another hierarchical layer. Here, a picture which is a target for a process which is about to be performed by a device such as encoder 100 or decoder 200 is referred to as a current picture. A current picture means a current picture to be encoded when the process is an encoding process, and a current picture means a current picture to be decoded when the process is a decoding process. Likewise, for example, a CU or a block of CUs which is a target for a process which is about to be performed by a device such as encoder 100 or decoder 200 is referred to as a current block. A current block means a current block to be encoded when the process is an encoding process, and a current block means a current block to be decoded when the process is a decoding process.

[Picture Structure: Slice/Tile]

A picture may be configured with one or more slice units or tile units in order to decode the picture in parallel.

Slices are basic encoding units included in a picture. A picture may include, for example, one or more slices. In addition, a slice includes one or more successive coding tree units (CTUs).

FIG. 3 is a diagram illustrating one example of a slice configuration. For example, a picture includes 11×8 CTUs, and is split into four slices (slices 1 to 4). Slice 1 includes sixteen CTUs, slice 2 includes twenty-one CTUs, slice 3 includes twenty-nine CTUs, and slice 4 includes twenty-two CTUs. Here, each CTU in the picture belongs to one of the slices. The shape of each slice is a shape obtained by splitting the picture horizontally. A boundary of each slice does not need to coincide with an image end, and may coincide with any of the boundaries between CTUs in the image. The processing order of the CTUs in a slice (an encoding order or a decoding order) is, for example, a raster-scan order. A slice includes a slice header and encoded data. Features of the slice may be written in the slice header. The features include a CTU address of a top CTU in the slice, a slice type, etc.

A tile is a unit of a rectangular region included in a picture. Each of tiles may be assigned with a number referred to as TileId in raster-scan order.

FIG. 4 is a diagram illustrating one example of a tile configuration. For example, a picture includes 11×8 CTUs, and is split into four tiles of rectangular regions (tiles 1 to 4). When tiles are used, the processing order of CTUs is changed from the processing order in the case where no tile is used. When no tile is used, a plurality of CTUs in a picture are processed in raster-scan order. When a plurality of tiles

are used, at least one CTU in each of the plurality of tiles is processed in raster-scan order. For example, as illustrated in FIG. 4, the processing order of the CTUs included in tile 1 is the order which starts from the left-end of the first column of tile 1 toward the right-end of the first column of tile 1 and then starts from the left-end of the second column of tile 1 toward the right-end of the second column of tile 1.

It is to be noted that one tile may include one or more slices, and one slice may include one or more tiles.

It is to be noted that a picture may be configured with one or more tile sets. A tile set may include one or more tile groups, or one or more tiles. A picture may be configured with only one of a tile set, a tile group, and a tile. For example, an order for scanning a plurality of tiles for each tile set in raster scan order is assumed to be a basic encoding order of tiles. A set of one or more tiles which are continuous in the basic encoding order in each tile set is assumed to be a tile group. Such a picture may be configured by splitter 102 (see FIG. 7) to be described later.

[Scalable Encoding]

FIGS. 5 and 6 are diagrams illustrating examples of scalable stream structures.

As illustrated in FIG. 5, encoder 100 may generate a temporally/spatially scalable stream by dividing each of a plurality of pictures into any of a plurality of layers and encoding the picture in the layer. For example, encoder 100 encodes the picture for each layer, thereby achieving scalability where an enhancement layer is present above a base layer. Such encoding of each picture is also referred to as scalable encoding. In this way, decoder 200 is capable of switching image quality of an image which is displayed by decoding the stream. In other words, decoder 200 determines up to which layer to decode based on internal factors such as the processing ability of decoder 200 and external factors such as a state of a communication bandwidth. As a result, decoder 200 is capable of decoding a content while freely switching between low resolution and high resolution. For example, the user of the stream watches a video of the stream halfway using a smartphone on the way to home, and continues watching the video at home on a device such as a TV connected to the Internet. It is to be noted that each of the smartphone and the device described above includes decoder 200 having the same or different performances. In this case, when the device decodes layers up to the higher layer in the stream, the user can watch the video at high quality at home. In this way, encoder 100 does not need to generate a plurality of streams having different image qualities of the same content, and thus the processing load can be reduced.

Furthermore, the enhancement layer may include meta information based on statistical information on the image. Decoder 200 may generate a video whose image quality has been enhanced by performing super-resolution imaging on a picture in the base layer based on the metadata. Super-resolution imaging may be any of improvement in the Signal-to-Noise (SN) ratio in the same resolution and increase in resolution. Metadata may include information for identifying a linear or a non-linear filter coefficient, as used in a super-resolution process, or information identifying a parameter value in a filter process, machine learning, or a least squares method used in super-resolution processing.

Alternatively, a configuration may be provided in which a picture is divided into, for example, tiles in accordance with, for example, the meaning of an object in the picture. In this case, decoder 200 may decode only a partial region in a picture by selecting a tile to be decoded. In addition, an attribute of the object (person, car, ball, etc.) and a position

of the object in the picture (coordinates in identical images) may be stored as metadata. In this case, decoder 200 is capable of identifying the position of a desired object based on the metadata, and determining the tile including the object. For example, as illustrated in FIG. 6, the metadata may be stored using a data storage structure different from image data, such as SEI in HEVC. This metadata indicates, for example, the position, size, or color of a main object.

Metadata may be stored in units of a plurality of pictures, such as a stream, a sequence, or a random access unit. In this way, decoder 200 is capable of obtaining, for example, the time at which a specific person appears in the video, and by fitting the time information with picture unit information, is capable of identifying a picture in which the object is present and determining the position of the object in the picture. [Encoder]

Next, encoder 100 according to this embodiment is described. FIG. 7 is a block diagram illustrating one example of a configuration of encoder 100 according to this embodiment. Encoder 100 encodes an image in units of a block.

As illustrated in FIG. 7, encoder 100 is an apparatus which encodes an image in units of a block, and includes splitter 102, subtractor 104, transformer 106, quantizer 108, entropy encoder 110, inverse quantizer 112, inverse transformer 114, adder 116, block memory 118, loop filter 120, frame memory 122, intra predictor 124, inter predictor 126, prediction controller 128, and prediction parameter generator 130. It is to be noted that intra predictor 124 and inter predictor 126 are configured as part of a prediction executor. [Mounting Example of Encoder]

FIG. 8 is a block diagram illustrating a mounting example of encoder 100. Encoder 100 includes processor a1 and memory a2. For example, the plurality of constituent elements of encoder 100 illustrated in FIG. 7 are mounted on processor a1 and memory a2 illustrated in FIG. 8.

Processor a1 is circuitry which performs information processing and is accessible to memory a2. For example, processor a1 is dedicated or general electronic circuitry which encodes an image. Processor a1 may be a processor such as a CPU. In addition, processor a1 may be an aggregate of a plurality of electronic circuits. In addition, for example, processor a1 may take the roles of two or more constituent elements other than a constituent element for storing information out of the plurality of constituent elements of encoder 100 illustrated in FIG. 7, etc.

Memory a2 is dedicated or general memory for storing information that is used by processor a1 to encode the image. Memory a2 may be electronic circuitry, and may be connected to processor a1. In addition, memory a2 may be included in processor a1. In addition, memory a2 may be an aggregate of a plurality of electronic circuits. In addition, memory a2 may be a magnetic disc, an optical disc, or the like, or may be represented as storage, a medium, or the like. In addition, memory a2 may be non-volatile memory, or volatile memory.

For example, memory a2 may store an image to be encoded or a stream corresponding to an encoded image. In addition, memory a2 may store a program for causing processor a1 to encode an image.

In addition, for example, memory a2 may take the roles of two or more constituent elements for storing information out of the plurality of constituent elements of encoder 100 illustrated in FIG. 7. More specifically, memory a2 may take the roles of block memory 118 and frame memory 122 illustrated in FIG. 7. More specifically, memory a2 may

store a reconstructed image (specifically, a reconstructed block, a reconstructed picture, or the like).

It is to be noted that, in encoder 100, not all of the plurality of constituent elements indicated in FIG. 7, etc. may be implemented, and not all the processes described above may be performed. Part of the constituent elements indicated in FIG. 7 may be included in another device, or part of the processes described above may be performed by another device.

Hereinafter, an overall flow of processes performed by encoder 100 is described, and then each of constituent elements included in encoder 100 is described.

[Overall Flow of Encoding Process]

FIG. 9 is a flow chart illustrating one example of an overall encoding process performed by encoder 100.

First, splitter 102 of encoder 100 splits each of pictures included in an original image into a plurality of blocks having a fixed size (128×128 pixels) (Step Sa_1). Splitter 102 then selects a splitting pattern for the fixed-size block (Step Sa_2). In other words, splitter 102 further splits the fixed-size block into a plurality of blocks which form the selected splitting pattern. Encoder 100 performs, for each of the plurality of blocks, Steps Sa_3 to Sa_9 for the block.

Prediction controller 128 and a prediction executor which is configured with intra predictor 124 and inter predictor 126 generate a prediction image of a current block (Step Sa_3). It is to be noted that the prediction image is also referred to as a prediction signal, a prediction block, or prediction samples.

Next, subtractor 104 generates the difference between a current block and a prediction image as a prediction residual (Step Sa_4). It is to be noted that the prediction residual is also referred to as a prediction error.

Next, transformer 106 transforms the prediction image and quantizer 108 quantizes the result, to generate a plurality of quantized coefficients (Step Sa_5).

Next, entropy encoder 110 encodes (specifically, entropy encodes) the plurality of quantized coefficients and a prediction parameter related to generation of a prediction image to generate a stream (Step Sa_6).

Next, inverse quantizer 112 performs inverse quantization of the plurality of quantized coefficients and inverse transformer 114 performs inverse transform of the result, to restore a prediction residual (Step Sa_7).

Next, adder 116 adds the prediction image to the restored prediction residual to reconstruct the current block (Step Sa_8). In this way, the reconstructed image is generated. It is to be noted that the reconstructed image is also referred to as a reconstructed block, and, in particular, that a reconstructed image generated by encoder 100 is also referred to as a local decoded block or a local decoded image.

When the reconstructed image is generated, loop filter 120 performs filtering of the reconstructed image as necessary (Step Sa_9).

Encoder 100 then determines whether encoding of the entire picture has been finished (Step Sa_10). When determining that the encoding has not yet been finished (No in Step Sa_10), processes from Step Sa_2 are executed repeatedly.

Although encoder 100 selects one splitting pattern for a fixed-size block, and encodes each block according to the splitting pattern in the above-described example, it is to be noted that each block may be encoded according to a corresponding one of a plurality of splitting patterns. In this case, encoder 100 may evaluate a cost for each of the plurality of splitting patterns, and, for example, may select

the stream obtained by encoding according to the splitting pattern which yields the smallest cost as a stream which is output finally.

Alternatively, the processes in Steps Sa_1 to Sa_10 may be performed sequentially by encoder 100, or two or more of the processes may be performed in parallel or may be reordered.

The encoding process by encoder 100 is hybrid encoding using prediction encoding and transform encoding. In addition, prediction encoding is performed by an encoding loop configured with subtractor 104, transformer 106, quantizer 108, inverse quantizer 112, inverse transformer 114, adder 116, loop filter 120, block memory 118, frame memory 122, intra predictor 124, inter predictor 126, and prediction controller 128. In other words, the prediction executor configured with intra predictor 124 and inter predictor 126 is part of the encoding loop.

[Splitter]

Splitter 102 splits each of pictures included in the original image into a plurality of blocks, and outputs each block to subtractor 104. For example, splitter 102 first splits a picture into blocks of a fixed size (for example, 128×128 pixels). The fixed-size block is also referred to as a coding tree unit (CTU). Splitter 102 then splits each fixed-size block into blocks of variable sizes (for example, 64×64 pixels or smaller), based on recursive quadtree and/or binary tree block splitting. In other words, splitter 102 selects a splitting pattern. The variable-size block is also referred to as a coding unit (CU), a prediction unit (PU), or a transform unit (TU). It is to be noted that, in various kinds of mounting examples, there is no need to differentiate between CU, PU, and TU; all or some of the blocks in a picture may be processed in units of a CU, a PU, or a TU.

FIG. 10 is a diagram illustrating one example of block splitting according to this embodiment. In FIG. 10, the solid lines represent block boundaries of blocks split by quadtree block splitting, and the dashed lines represent block boundaries of blocks split by binary tree block splitting.

Here, block 10 is a square block having 128×128 pixels. This block 10 is first split into four square 64×64 pixel blocks (quadtree block splitting).

The upper-left 64×64 pixel block is further vertically split into two rectangle 32×64 pixel blocks, and the left 32×64 pixel block is further vertically split into two rectangle 16×64 pixel blocks (binary tree block splitting). As a result, the upper-left square 64×64 pixel block is split into two 16×64 pixel blocks 11 and 12 and one 32×64 pixel block 13.

The upper-right square 64×64 pixel block is horizontally split into two rectangle 64×32 pixel blocks 14 and 15 (binary tree block splitting).

The lower-left square 64×64 pixel block is first split into four square 32×32 pixel blocks (quadtree block splitting). The upper-left block and the lower-right block among the four square 32×32 pixel blocks are further split. The upper-left square 32×32 pixel block is vertically split into two rectangle 16×32 pixel blocks, and the right 16×32 pixel block is further horizontally split into two 16×16 pixel blocks (binary tree block splitting). The lower-right 32×32 pixel block is horizontally split into two 32×16 pixel blocks (binary tree block splitting). The upper-right square 32×32 pixel block is horizontally split into two rectangle 32×16 pixel blocks (binary tree block splitting). As a result, the lower-left square 64×64 pixel block is split into rectangle 16×32 pixel block 16, two square 16×16 pixel blocks 17 and 18, two square 32×32 pixel blocks 19 and 20, and two rectangle 32×16 pixel blocks 21 and 22.

The lower-right 64×64 pixel block 23 is not split.

As described above, in FIG. 10, block 10 is split into thirteen variable-size blocks 11 through 23 based on recursive quadtree and binary tree block splitting. Such splitting is also referred to as quad-tree plus binary tree splitting (QTBT).

It is to be noted that, in FIG. 10, one block is split into four or two blocks (quadtree or binary tree block splitting), but splitting is not limited to these examples. For example, one block may be split into three blocks (ternary block splitting). Splitting including such ternary block splitting is also referred to as multi type tree (MBT) splitting.

FIG. 11 is a diagram illustrating one example of a configuration of splitter 102. As illustrated in FIG. 11, splitter 102 may include block splitting determiner 102a. Block splitting determiner 102a may perform the following processes as examples.

For example, block splitting determiner 102a collects block information from either block memory 118 or frame memory 122, and determines the above-described splitting pattern based on the block information. Splitter 102 splits the original image according to the splitting pattern, and outputs at least one block obtained by the splitting to subtractor 104.

In addition, for example, block splitting determiner 102a outputs a parameter indicating the above-described splitting pattern to transformer 106, inverse transformer 114, intra predictor 124, inter predictor 126, and entropy encoder 110. Transformer 106 may transform a prediction residual based on the parameter. Intra predictor 124 and inter predictor 126 may generate a prediction image based on the parameter. In addition, entropy encoder 110 may entropy encode the parameter.

The parameter related to the splitting pattern may be written in a stream as indicated below as one example.

FIG. 12 is a diagram illustrating examples of splitting patterns. Examples of splitting patterns include: splitting into four regions (QT) in which a block is split into two regions both horizontally and vertically; splitting into three regions (HT or VT) in which a block is split in the same direction in a ratio of 1:2:1; splitting into two regions (HB or VB) in which a block is split in the same direction in a ratio of 1:1; and no splitting (NS).

It is to be noted that the splitting pattern does not have any block splitting direction in the case of splitting into four regions and no splitting, and that the splitting pattern has splitting direction information in the case of splitting into two regions or three regions.

FIGS. 13A and 13B are each a diagram illustrating one example of a syntax tree of a splitting pattern. In the example of FIG. 13A, first, information indicating whether to perform splitting (S: Split flag) is present, and information indicating whether to perform splitting into four regions (QT: QT flag) is present next. Information indicating which one of splitting into three regions and two regions is to be performed (TT: TT flag or BT: BT flag) is present next, and lastly, information indicating a division direction (Ver: Vertical flag or Hor: Horizontal flag) is present. It is to be noted that each of at least one block obtained by splitting according to such a splitting pattern may be further split repeatedly in a similar process. In other words, as one example, whether splitting is performed, whether splitting into four regions is performed, which one of the horizontal direction and the vertical direction is the direction in which a splitting method is to be performed, which one of splitting into three regions and splitting into two regions is to be performed may be recursively determined, and the determination results may

be encoded in a stream according to the encoding order disclosed by the syntax tree illustrated in FIG. 13A.

In addition, although information items respectively indicating S, QT, TT, and Ver are arranged in the listed order in the syntax tree illustrated in FIG. 13A, information items respectively indicating S, QT, Ver, and BT may be arranged in the listed order. In other words, in the example of FIG. 13B, first, information indicating whether to perform splitting (S: Split flag) is present, and information indicating whether to perform splitting into four regions (QT: QT flag) is present next. Information indicating the splitting direction (Ver: Vertical flag or Hor: Horizontal flag) is present next, and lastly, information indicating which one of splitting into two regions and splitting into three regions is to be performed (BT: BT flag or TT: TT flag) is present.

It is to be noted that the splitting patterns described above are examples, and splitting patterns other than the described splitting patterns may be used, or part of the described splitting patterns may be used.

[Subtractor]

Subtractor **104** subtracts a prediction image (prediction image that is input from prediction controller **128**) from the original image in units of a block input from splitter **102** and split by splitter **102**. In other words, subtractor **104** calculates prediction residuals of a current block. Subtractor **104** then outputs the calculated prediction residuals to transformer **106**.

The original signal is an input signal which has been input to encoder **100** and represents an image of each picture included in a video (for example, a luma signal and two chroma signals).

[Transformer]

Transformer **106** transforms prediction residuals in spatial domain into transform coefficients in frequency domain, and outputs the transform coefficients to quantizer **108**. More specifically, transformer **106** applies, for example, a pre-defined discrete cosine transform (DCT) or discrete sine transform (DST) to prediction residuals in spatial domain.

It is to be noted that transformer **106** may adaptively select a transform type from among a plurality of transform types, and transform prediction residuals into transform coefficients by using a transform basis function corresponding to the selected transform type. This sort of transform is also referred to as explicit multiple core transform (EMT) or adaptive multiple transform (AMT). In addition, a transform basis function is also simply referred to as a basis.

The transform types include, for example, DCT-II, DCT-V, DCT-VIII, DST-I, and DST-VII. It is to be noted that these transform types may be represented as DCT2, DCT5, DCT8, DST1, and DST7. FIG. 14 is a chart illustrating transform basis functions for each transform type. In FIG. 14, N indicates the number of input pixels. For example, selection of a transform type from among the plurality of transform types may depend on a prediction type (one of intra prediction and inter prediction), and may depend on an intra prediction mode.

Information indicating whether to apply such EMT or AMT (referred to as, for example, an EMT flag or an AMT flag) and information indicating the selected transform type is normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, or CTU level).

In addition, transformer **106** may re-transform the transform coefficients (which are transform results). Such re-transform is also referred to as adaptive secondary transform

(AST) or non-separable secondary transform (NSST). For example, transformer **106** performs re-transform in units of a sub-block (for example, 4×4 pixel sub-block) included in a transform coefficient block corresponding to an intra prediction residual. Information indicating whether to apply NSST and information related to a transform matrix for use in NSST are normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, or CTU level).

Transformer **106** may employ a separable transform and a non-separable transform. A separable transform is a method in which a transform is performed a plurality of times by separately performing a transform for each of directions according to the number of dimensions of inputs. A non-separable transform is a method of performing a collective transform in which two or more dimensions in multidimensional inputs are collectively regarded as a single dimension.

In one example of the non-separable transform, when an input is a 4×4 pixel block, the 4×4 pixel block is regarded as a single array including sixteen elements, and the transform applies a 16×16 transform matrix to the array.

In another example of the non-separable transform, an input block of 4×4 pixels is regarded as a single array including sixteen elements, and then a transform (hypercube gives revolution) in which gives revolution is performed on the array a plurality of times may be performed.

In the transform in transformer **106**, the transform types of transform basis functions to be transformed into the frequency domain according to regions in a CU can be switched. Examples include a spatially varying transform (SVT).

FIG. 15 is a diagram illustrating one example of SVT.

In SVT, as illustrated in FIG. 15, CUs are split into two equal regions horizontally or vertically, and only one of the regions is transformed into the frequency domain. A transform type can be set for each region. For example, DST7 and DST8 are used. For example, among the two regions obtained by splitting a CU vertically into two equal regions, DST7 and DCT8 may be used for the region at position 0. Alternatively, among the two regions, DST7 is used for the region at position 1. Likewise, among the two regions obtained by splitting a CU horizontally into two equal regions, DST7 and DCT8 are used for the region at position 0. Alternatively, among the two regions, DST7 is used for the region at position 1. Although only one of the two regions in a CU is transformed and the other is not transformed in the example illustrated in FIG. 15, each of the two regions may be transformed. In addition, splitting method may include not only splitting into two regions but also splitting into four regions. In addition, the splitting method can be more flexible. For example, information indicating the splitting method may be encoded and may be signaled in the same manner as the CU splitting. It is to be noted that SVT is also referred to as sub-block transform (SBT).

The AMT and EMT described above may be referred to as MTS (multiple transform selection). When MTS is applied, a transform type that is DST7, DCT8, or the like can be selected, and the information indicating the selected transform type may be encoded as index information for each CU. There is another process referred to as IMTS (implicit MTS) as a process for selecting, based on the shape of a CU, a transform type to be used for orthogonal transform performed without encoding index information. When IMTS is applied, for example, when a CU has a

rectangle shape, orthogonal transform of the rectangle shape is performed using DST7 for the short side and DST2 for the long side. In addition, for example, when a CU has a square shape, orthogonal transform of the rectangle shape is performed using DCT2 when MTS is effective in a sequence and using DST7 when MTS is ineffective in the sequence. DCT2 and DST7 are mere examples. Other transform types may be used, and it is also possible to change the combination of transform types for use to a different combination of transform types. IMTS may be used only for intra prediction blocks, or may be used for both intra prediction blocks and inter prediction block.

The three processes of MTS, SBT, and IMTS have been described above as selection processes for selectively switching transform types for use in orthogonal transform. However, all of the three selection processes may be made effective, or only part of the selection processes may be selectively made effective. Whether each of the selection processes is made effective can be identified based on flag information or the like in a header such as an SPS. For example, when all of the three selection processes are effective, one of the three selection processes is selected for each CU and orthogonal transform of the CU is performed. It is to be noted that the selection processes for selectively switching the transform types may be selection processes different from the above three selection processes, or each of the three selection processes may be replaced by another process as long as at least one of the following four functions [1] to [4] can be achieved. Function [1] is a function for performing orthogonal transform of the entire CU and encoding information indicating the transform type used in the transform. Function [2] is a function for performing orthogonal transform of the entire CU and determining the transform type based on a predetermined rule without encoding information indicating the transform type. Function [3] is a function for performing orthogonal transform of a partial region of a CU and encoding information indicating the transform type used in the transform. Function [4] is a function for performing orthogonal transform of a partial region of a CU and determining the transform type based on a predetermined rule without encoding information indicating the transform type used in the transform.

It is to be noted that whether each of MTS, IMTS, and SBT is applied may be determined for each processing unit. For example, whether each of MTS, IMTS, and SBT is applied may be determined for each sequence, picture, brick, slice, CTU, or CU.

It is to be noted that a tool for selectively switching transform types in the present disclosure may be rephrased by a method for selectively selecting a basis for use in a transform process, a selection process, or a process for selecting a basis. In addition, the tool for selectively switching transform types may be rephrased by a mode for adaptively selecting a transform type.

FIG. 16 is a flow chart illustrating one example of a process performed by transformer 106.

For example, transformer 106 determines whether to perform orthogonal transform (Step St_1). Here, when determining to perform orthogonal transform (Yes in Step St_1), transformer 106 selects a transform type for use in orthogonal transform from a plurality of transform types (Step St_2). Next, transformer 106 performs orthogonal transform by applying the selected transform type to the prediction residual of a current block (Step St_3). Transformer 106 then outputs information indicating the selected transform type to entropy encoder 110, so as to allow entropy encoder 110 to encode the information (Step St_4).

On the other hand, when determining not to perform orthogonal transform (No in Step St_1), transformer 106 outputs information indicating that no orthogonal transform is performed, so as to allow entropy encoder 110 to encode the information (Step St_5). It is to be noted that whether to perform orthogonal transform in Step St_1 may be determined based on, for example, the size of a transform block, a prediction mode applied to the CU, etc.

Alternatively, orthogonal transform may be performed using a predefined transform type without encoding information indicating the transform type for use in orthogonal transform.

FIG. 17 is a flow chart illustrating another example of a process performed by transformer 106. It is to be noted that the example illustrated in FIG. 17 is an example of orthogonal transform in the case where transform types for use in orthogonal transform are selectively switched as in the case of the example illustrated in FIG. 16.

As one example, a first transform type group may include DCT2, DST7, and DCT8. As another example, a second transform type group may include DCT2. The transform types included in the first transform type group and the transform types included in the second transform type group may partly overlap with each other, or may be totally different from each other.

More specifically, transformer 106 determines whether a transform size is smaller than or equal to a predetermined value (Step Su_1). Here, when determining that the transform size is smaller than or equal to the predetermined value (Yes in Step Su_1), transformer 106 performs orthogonal transform of the prediction residual of the current block using the transform type included in the first transform type group (Step Su_2). Next, transformer 106 outputs information indicating the transform type to be used among at least one transform type included in the first transform type group to entropy encoder 110, so as to allow entropy encoder 110 to encode the information (Step Su_3). On the other hand, when determining that the transform size is not smaller than or equal to the predetermined value (No in Step Su_1), transformer 106 performs orthogonal transform of the prediction residual of the current block using the second transform type group (Step Su_4).

In Step Su_3, the information indicating the transform type for use in orthogonal transform may be information indicating a combination of the transform type to be applied vertically in the current block and the transform type to be applied horizontally in the current block. The first type group may include only one transform type, and the information indicating the transform type for use in orthogonal transform may not be encoded. The second transform type group may include a plurality of transform types, and information indicating the transform type for use in orthogonal transform among the one or more transform types included in the second transform type group may be encoded.

Alternatively, a transform type may be determined based only on a transform size. It is to be noted that such determinations are not limited to the determination as to whether the transform size is smaller than or equal to the predetermined value, and other processes are also possible as long as the processes are for determining a transform type for use in orthogonal transform based on the transform size. [Quantizer]

Quantizer 108 quantizes the transform coefficients output from transformer 106. More specifically, quantizer 108 scans, in a determined scanning order, the transform coefficients of the current block, and quantizes the scanned

transform coefficients based on quantization parameters (QP) corresponding to the transform coefficients. Quantizer **108** then outputs the quantized transform coefficients (hereinafter also referred to as quantized coefficients) of the current block to entropy encoder **110** and inverse quantizer **112**.

A determined scanning order is an order for quantizing/inverse quantizing transform coefficients. For example, a determined scanning order is defined as ascending order of frequency (from low to high frequency) or descending order of frequency (from high to low frequency).

A quantization parameter (QP) is a parameter defining a quantization step (quantization width). For example, when the value of the quantization parameter increases, the quantization step also increases. In other words, when the value of the quantization parameter increases, an error in quantized coefficients (quantization error) increases.

In addition, a quantization matrix may be used for quantization. For example, several kinds of quantization matrices may be used correspondingly to frequency transform sizes such as 4x4 and 8x8, prediction modes such as intra prediction and inter prediction, and pixel components such as luma and chroma pixel components. It is to be noted that quantization means digitalizing values sampled at predetermined intervals correspondingly to predetermined levels. In this technical field, quantization may be represented as other expressions such as rounding and scaling.

Methods using quantization matrices include a method using a quantization matrix which has been set directly at the encoder **100** side and a method using a quantization matrix which has been set as a default (default matrix). At the encoder **100** side, a quantization matrix suitable for features of an image can be set by directly setting a quantization matrix. This case, however, has a disadvantage of increasing a coding amount for encoding the quantization matrix. It is to be noted that a quantization matrix to be used to quantize the current block may be generated based on a default quantization matrix or an encoded quantization matrix, instead of directly using the default quantization matrix or the encoded quantization matrix.

There is a method for quantizing a high-frequency coefficient and a low-frequency coefficient in the same manner without using a quantization matrix.

It is to be noted that this method is equivalent to a method using a quantization matrix (flat matrix) whose all coefficients have the same value.

The quantization matrix may be encoded, for example, at the sequence level, picture level, slice level, brick level, or CTU level.

When using a quantization matrix, quantizer **108** scales, for each transform coefficient, for example a quantization width which can be calculated based on a quantization parameter, etc., using the value of the quantization matrix. The quantization process performed without using any quantization matrix may be a process of quantizing transform coefficients based on a quantization width calculated based on a quantization parameter, etc. It is to be noted that, in the quantization process performed without using any quantization matrix, the quantization width may be multiplied by a predetermined value which is common for all the transform coefficients in a block.

FIG. **18** is a block diagram illustrating one example of a configuration of quantizer **108**.

For example, quantizer **108** includes difference quantization parameter generator **108a**, predicted quantization

parameter generator **108b**, quantization parameter generator **108c**, quantization parameter storage **108d**, and quantization executor **108e**.

FIG. **19** is a flow chart illustrating one example of quantization performed by quantizer **108**.

As one example, quantizer **108** may perform quantization for each CU based on the flow chart illustrated in FIG. **19**. More specifically quantization parameter generator **108c** determines whether to perform quantization (Step Sv_1). Here, when determining to perform quantization (Yes in Step Sv_1), quantization parameter generator **108c** generates a quantization parameter for a current block (Step Sv_2), and stores the quantization parameter into quantization parameter storage **108d** (Step Sv_3).

Next, quantization executor **108e** quantizes transform coefficients of the current block using the quantization parameter generated in Step Sv_2 (Step Sv_4). Predicted quantization parameter generator **108b** then obtains a quantization parameter for a processing unit different from the current block from quantization parameter storage **108d** (Step Sv_5). Predicted quantization parameter generator **108b** generates a predicted quantization parameter of the current block based on the obtained quantization parameter (Step Sv_6). Difference quantization parameter generator **108a** calculates the difference between the quantization parameter of the current block generated by quantization parameter generator **108c** and the predicted quantization parameter of the current block generated by predicted quantization parameter generator **108b** (Step Sv_7). The difference quantization parameter is generated by calculating the difference. Difference quantization parameter generator **108a** outputs the difference quantization parameter to entropy encoder **110**, so as to allow entropy encoder **110** to encode the difference quantization parameter (Step Sv_8).

It is to be noted that the difference quantization parameter may be encoded, for example, at the sequence level, picture level, slice level, brick level, or CTU level. In addition, the initial value of the quantization parameter may be encoded at the sequence level, picture level, slice level, brick level, or CTU level. At this time, the quantization parameter may be generated using the initial value of the quantization parameter and the difference quantization parameter.

It is to be noted that quantizer **108** may include a plurality of quantizers, and may apply dependent quantization in which transform coefficients are quantized using a quantization method selected from a plurality of quantization methods.

[Entropy Encoder]

FIG. **20** is a block diagram illustrating one example of a configuration of entropy encoder **110**.

Entropy encoder **110** generates a stream by entropy encoding the quantized coefficients input from quantizer **108** and a prediction parameter input from prediction parameter generator **130**. For example, context-based adaptive binary arithmetic coding (CABAC) is used as the entropy encoding. More specifically, entropy encoder **110** includes binarizer **110a**, context controller **110b**, and binary arithmetic encoder **110c**. Binarizer **110a** performs binarization in which multi-level signals such as quantized coefficients and a prediction parameter are transformed into binary signals. Examples of binarization methods include truncated Rice binarization, exponential Golomb codes, and fixed length binarization. Context controller **110b** derives a context value according to a feature or a surrounding state of a syntax element, that is, an occurrence probability of a binary signal. Examples of methods for deriving a context value include bypass, referring to a syntax element, referring to an upper

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and left adjacent blocks, referring to hierarchical information, and others. Binary arithmetic encoder **110c** arithmetically encodes the binary signal using the derived context value.

FIG. **21** is a diagram illustrating a flow of CABAC in entropy encoder **110**.

First, initialization is performed in CABAC in entropy encoder **110**. In the initialization, initialization in binary arithmetic encoder **110c** and setting of an initial context value are performed. For example, binarizer **110a** and binary arithmetic encoder **110c** execute binarization and arithmetic encoding of a plurality of quantization coefficients in a CTU sequentially. At this time, context controller **110b** updates the context value each time arithmetic encoding is performed. Context controller **110b** then saves the context value as a post process. The saved context value is used, for example, to initialize the context value for the next CTU. [Inverse Quantizer]

Inverse quantizer **112** inverse quantizes quantized coefficients which have been input from quantizer **108**. More specifically, inverse quantizer **112** inverse quantizes, in a determined scanning order, quantized coefficients of the current block. Inverse quantizer **112** then outputs the inverse quantized transform coefficients of the current block to inverse transformer **114**.

[Inverse Transformer]

Inverse transformer **114** restores prediction errors by inverse transforming the transform coefficients which have been input from inverse quantizer **112**. More specifically, inverse transformer **114** restores the prediction residuals of the current block by performing an inverse transform corresponding to the transform applied to the transform coefficients by transformer **106**. Inverse transformer **114** then outputs the restored prediction residuals to adder **116**.

It is to be noted that since information is normally lost in quantization, the restored prediction residuals do not match the prediction errors calculated by subtractor **104**. In other words, the restored prediction residuals normally include quantization errors.

[Adder]

Adder **116** reconstructs the current block by adding the prediction residuals which have been input from inverse transformer **114** and prediction images which have been input from prediction controller **128**. Consequently, a reconstructed image is generated. Adder **116** then outputs the reconstructed image to block memory **118** and loop filter **120**.

[Block Memory]

Block memory **118** is storage for storing a block which is included in a current picture and is referred to in intra prediction. More specifically, block memory **118** stores a reconstructed image output from adder **116**.

[Frame Memory]

Frame memory **122** is, for example, storage for storing reference pictures for use in inter prediction, and is also referred to as a frame buffer. More specifically, frame memory **122** stores a reconstructed image filtered by loop filter **120**.

[Loop Filter]

Loop filter **120** applies a loop filter to a reconstructed image output by adder **116**, and outputs the filtered reconstructed image to frame memory **122**. A loop filter is a filter used in an encoding loop (in-loop filter). Examples of loop filters include, for example, an adaptive loop filter (ALF), a deblocking filter (DF or DBF), a sample adaptive offset (SAO), etc.

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FIG. **22** is a block diagram illustrating one example of a configuration of loop filter **120**.

For example, as illustrated in FIG. **22**, loop filter **120** includes deblocking filter executor **120a**, SAO executor **120b**, and ALF executor **120c**. Deblocking filter executor **120a** performs a deblocking filter process of the reconstructed image. SAO executor **120b** performs a SAO process of the reconstructed image after being subjected to the deblocking filter process. ALF executor **120c** performs an ALF process of the reconstructed image after being subjected to the SAO process. The ALF and deblocking filter processes are described later in detail. The SAO process is a process for enhancing image quality by reducing ringing (a phenomenon in which pixel values are distorted like waves around an edge) and correcting deviation in pixel value. Examples of SAO processes include an edge offset process and a band offset process. It is to be noted that loop filter **120** does not always need to include all the constituent elements disclosed in FIG. **22**, and may include only part of the constituent elements. In addition, loop filter **120** may be configured to perform the above processes in a processing order different from the one disclosed in FIG. **22**. [Loop Filter > Adaptive Loop Filter]

In an ALF, a least square error filter for removing compression artifacts is applied. For example, one filter selected from among a plurality of filters based on the direction and activity of local gradients is applied for each of 2x2 pixel sub-blocks in the current block.

More specifically, first, each sub-block (for example, each 2x2 pixel sub-block) is categorized into one out of a plurality of classes (for example, fifteen or twenty-five classes). The categorization of the sub-block is based on, for example, gradient directionality and activity. In a specific example, category index C (for example, $C=5D+A$) is calculated based on gradient directionality D (for example, 0 to 2 or 0 to 4) and gradient activity A (for example, 0 to 4). Then, based on category index C, each sub-block is categorized into one out of a plurality of classes.

For example, gradient directionality D is calculated by comparing gradients of a plurality of directions (for example, the horizontal, vertical, and two diagonal directions). Moreover, for example, gradient activity A is calculated by adding gradients of a plurality of directions and quantizing the result of the addition.

The filter to be used for each sub-block is determined from among the plurality of filters based on the result of such categorization.

The filter shape to be used in an ALF is, for example, a circular symmetric filter shape. FIG. **23A** through FIG. **23C** illustrate examples of filter shapes used in ALFs. FIG. **23A** illustrates a 5x5 diamond shape filter, FIG. **23B** illustrates a 7x7 diamond shape filter, and FIG. **23C** illustrates a 9x9 diamond shape filter. Information indicating the filter shape is normally signaled at the picture level. It is to be noted that the signaling of such information indicating the filter shape does not necessarily need to be performed at the picture level, and may be performed at another level (for example, at the sequence level, slice level, brick level, CTU level, or CU level).

The ON or OFF of the ALF is determined, for example, at the picture level or CU level. For example, the decision of whether to apply the ALF to luma may be made at the CU level, and the decision of whether to apply ALF to chroma may be made at the picture level. Information indicating ON or OFF of the ALF is normally signaled at the picture level or CU level. It is to be noted that the signaling of information indicating ON or OFF of the ALF does not necessarily need

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to be performed at the picture level or CU level, and may be performed at another level (for example, at the sequence level, slice level, brick level, or CTU level).

In addition, as described above, one filter is selected from the plurality of filters, and an ALF process of a sub-block is performed. A coefficient set of coefficients to be used for each of the plurality of filters (for example, up to the fifteenth or twenty-fifth filter) is normally signaled at the picture level. It is to be noted that the coefficient set does not always need to be signaled at the picture level, and may be signaled at another level (for example, the sequence level, slice level, brick level, CTU level, CU level, or sub-block level).

[Loop Filter >Cross Component Adaptive Loop Filter]

FIG. 23D is a diagram illustrating an example where Y samples (first component) are used for a cross component ALF (CCALF) for Cb and a CCALF for Cr (components different from the first component). FIG. 23E is a diagram illustrating a diamond shaped filter.

One example of CC-ALF operates by applying a linear, diamond shaped filter (FIGS. 23D, 23E) to a luma channel for each chroma component. The filter coefficients, for example, may be transmitted in the APS, scaled by a factor of 2^{10} , and rounded for fixed point representation. The application of the filters is controlled on a variable block size and signaled by a context-coded flag received for each block of samples. The block size along with a CC-ALF enabling flag is received at the slice-level for each chroma component. Syntax and semantics for CC-ALF are provided in the Appendix. In the contribution, the following block sizes (in chroma samples) were supported: 16×16, 32×32, 64×64, and 128×128.

[Loop Filter >Joint Chroma Cross Component Adaptive Loop Filter]

FIG. 23F is a diagram illustrating an example for a joint chroma CCALF (JC-CCALF).

One example of JC-CCALF, where only one CCALF filter will be used to generate one CCALF filtered output as a chroma refinement signal for one color component only, while a properly weighted version of the same chroma refinement signal will be applied to the other color component. In this way, the complexity of existing CCALF is reduced roughly by half.

The weight value is coded into a sign flag and a weight index. The weight index (denoted as weight_index) is coded into 3 bits, and specifies the magnitude of the JC-CCALF weight JcCcWeight. It cannot be equal to 0. The magnitude of JcCcWeight is determined as follows.

If weight_index is less than or equal to 4, JcCcWeight is equal to weight_index >>2.

Otherwise, JcCcWeight is equal to 4/(weight_index-4).

The block-level on/off control of ALF filtering for Cb and Cr are separate. This is the same as in CCALF, and two separate sets of block-level on/off control flags will be coded. Different from CCALF, herein, the Cb, Cr on/off control block sizes are the same, and thus, only one block size variable is coded.

[Loop Filter >Deblocking Filter]

In a deblocking filter process, loop filter 120 performs a filter process on a block boundary in a reconstructed image so as to reduce distortion which occurs at the block boundary.

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FIG. 24 is a block diagram illustrating one example of a specific configuration of deblocking filter executor 120a.

For example, deblocking filter executor 120a includes: boundary determiner 1201; filter determiner 1203; filter executor 1205; process determiner 1208; filter characteristic determiner 1207; and switches 1202, 1204, and 1206.

Boundary determiner 1201 determines whether a pixel to be deblock filtered (that is, a current pixel) is present around a block boundary. Boundary determiner 1201 then outputs the determination result to switch 1202 and process determiner 1208.

In the case where boundary determiner 1201 has determined that a current pixel is present around a block boundary, switch 1202 outputs an unfiltered image to switch 1204. In the opposite case where boundary determiner 1201 has determined that no current pixel is present around a block boundary, switch 1202 outputs an unfiltered image to switch 1206. It is to be noted that the unfiltered image is an image configured with a current pixel and at least one surrounding pixel located around the current pixel.

Filter determiner 1203 determines whether to perform deblocking filtering of the current pixel, based on the pixel value of at least one surrounding pixel located around the current pixel. Filter determiner 1203 then outputs the determination result to switch 1204 and process determiner 1208.

In the case where filter determiner 1203 has determined to perform deblocking filtering of the current pixel, switch 1204 outputs the unfiltered image obtained through switch 1202 to filter executor 1205. In the opposite case where filter determiner 1203 has determined not to perform deblocking filtering of the current pixel, switch 1204 outputs the unfiltered image obtained through switch 1202 to switch 1206.

When obtaining the unfiltered image through switches 1202 and 1204, filter executor 1205 executes, for the current pixel, deblocking filtering having the filter characteristic determined by filter characteristic determiner 1207. Filter executor 1205 then outputs the filtered pixel to switch 1206.

Under control by process determiner 1208, switch 1206 selectively outputs a pixel which has not been deblock filtered and a pixel which has been deblock filtered by filter executor 1205.

Process determiner 1208 controls switch 1206 based on the results of determinations made by boundary determiner 1201 and filter determiner 1203. In other words, process determiner 1208 causes switch 1206 to output the pixel which has been deblock filtered when boundary determiner 1201 has determined that the current pixel is present around the block boundary and filter determiner 1203 has determined to perform deblocking filtering of the current pixel. In addition, in a case other than the above case, process determiner 1208 causes switch 1206 to output the pixel which has not been deblock filtered. A filtered image is output from switch 1206 by repeating output of a pixel in this way. It is to be noted that the configuration illustrated in FIG. 24 is one example of a configuration in deblocking filter executor 120a. Deblocking filter executor 120a may have another configuration.

FIG. 25 is a diagram illustrating an example of a deblocking filter having a symmetrical filtering characteristic with respect to a block boundary.

In a deblocking filter process, one of two deblocking filters having different characteristics, that is, a strong filter and a weak filter is selected using pixel values and quantization parameters, for example. In the case of the strong filter, pixels p0 to p2 and pixels q0 to q2 are present across a block boundary as illustrated in FIG. 25, the pixel values

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of the respective pixels q_0 to q_2 are changed to pixel values q'_0 to q'_2 by performing computations according to the expressions below.

$$q'_0 = (p_1 + 2 \times p_0 + 2 \times q_0 + 2 \times q_1 + q_2 + 4) / 8$$

$$q'_1 = (p_0 + q_0 + q_1 + q_2 + 2) / 4$$

$$q'_2 = (p_0 + q_0 + q_1 + 3 \times q_2 + 2 \times q_3 + 4) / 8$$

It is to be noted that, in the above expressions, p_0 to p_2 and q_0 to q_2 are the pixel values of respective pixels p_0 to p_2 and pixels q_0 to q_2 . In addition, q_3 is the pixel value of neighboring pixel q_3 located at the opposite side of pixel q_2 with respect to the block boundary. In addition, in the right side of each of the expressions, coefficients which are multiplied with the respective pixel values of the pixels to be used for deblocking filtering are filter coefficients.

Furthermore, in the deblocking filtering, clipping may be performed so that the calculated pixel values do not change over a threshold value. In the clipping process, the pixel values calculated according to the above expressions are clipped to a value obtained according to "a pre-computation pixel value $\pm 2 \times$ a threshold value" using the threshold value determined based on a quantization parameter. In this way, it is possible to prevent excessive smoothing.

FIG. 26 is a diagram for illustrating one example of a block boundary on which a deblocking filter process is performed. FIG. 27 is a diagram illustrating examples of Bs values.

The block boundary on which the deblocking filter process is performed is, for example, a boundary between CUs, PUs, or TUs having 8×8 pixel blocks as illustrated in FIG. 26. The deblocking filter process is performed, for example, in units of four rows or four columns. First, boundary strength (Bs) values are determined as indicated in FIG. 27 for block P and block Q illustrated in FIG. 26.

According to the Bs values in FIG. 27, whether to perform deblocking filter processes of block boundaries belonging to the same image using different strengths may be determined. The deblocking filter process for a chroma signal is performed when a Bs value is 2. The deblocking filter process for a luma signal is performed when a Bs value is 1 or more and a determined condition is satisfied. It is to be noted that conditions for determining Bs values are not limited to those indicated in FIG. 27, and a Bs value may be determined based on another parameter.

[Predictor (Intra Predictor, Inter Predictor, Prediction Controller)]

FIG. 28 is a flow chart illustrating one example of a process performed by a predictor of encoder 100. It is to be noted that the predictor, as one example, includes all or part of the following constituent elements: intra predictor 124; inter predictor 126; and prediction controller 128. The prediction executor includes, for example, intra predictor 124 and inter predictor 126.

The predictor generates a prediction image of a current block (Step Sb_1). It is to be noted that the prediction image is, for example, an intra prediction image (intra prediction signal) or an inter prediction image (inter prediction signal). More specifically the predictor generates the prediction image of the current block using a reconstructed image which has been already obtained for another block through generation of a prediction image, generation of a prediction residual, generation of quantized coefficients, restoring of a prediction residual, and addition of a prediction image.

The reconstructed image may be, for example, an image in a reference picture or an image of an encoded block (that

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is, the other block described above) in a current picture which is the picture including the current block. The encoded block in the current picture is, for example, a neighboring block of the current block.

FIG. 29 is a flow chart illustrating another example of a process performed by the predictor of encoder 100.

The predictor generates a prediction image using a first method (Step Sc_1a), generates a prediction image using a second method (Step Sc_1b), and generates a prediction image using a third method (Step Sc_1c). The first method, the second method, and the third method may be mutually different methods for generating a prediction image. Each of the first to third methods may be an inter prediction method, an intra prediction method, or another prediction method. The above-described reconstructed image may be used in these prediction methods.

Next, the predictor evaluates the prediction images generated in Steps Sc_1a, Sc_1b, and Sc_1c (Step Sc_2). For example, the predictor calculates costs C for the prediction images generated in Step Sc_1a, Sc_1b, and Sc_1c, and evaluates the prediction images by comparing the costs C of the prediction images. It is to be noted that cost C is calculated according to an expression of an R-D optimization model, for example, $C = D + \lambda \times R$. In this expression, D indicates compression artifacts of a prediction image, and is represented as, for example, a sum of absolute differences between the pixel value of a current block and the pixel value of a prediction image. In addition, R indicates a bit rate of a stream. In addition, X indicates, for example, a multiplier according to the method of Lagrange multiplier.

The predictor then selects one of the prediction images generated in Steps Sc_1a, Sc_1b, and Sc_1c (Step Sc_3). In other words, the predictor selects a method or a mode for obtaining a final prediction image. For example, the predictor selects the prediction image having the smallest cost C, based on costs C calculated for the prediction images. Alternatively, the evaluation in Step Sc_2 and the selection of the prediction image in Step Sc_3 may be made based on a parameter which is used in an encoding process. Encoder 100 may transform information for identifying the selected prediction image, the method, or the mode into a stream. The information may be, for example, a flag or the like. In this way, decoder 200 is capable of generating a prediction image according to the method or the mode selected by encoder 100, based on the information. It is to be noted that, in the example illustrated in FIG. 29, the predictor selects any of the prediction images after the prediction images are generated using the respective methods. However, the predictor may select a method or a mode based on a parameter for use in the above-described encoding process before generating prediction images, and may generate a prediction image according to the method or mode selected.

For example, the first method and the second method may be intra prediction and inter prediction, respectively, and the predictor may select a final prediction image for a current block from prediction images generated according to the prediction methods.

FIG. 30 is a flow chart illustrating another example of a process performed by the predictor of encoder 100.

First, the predictor generates a prediction image using intra prediction (Step Sd_1a), and generates a prediction image using inter prediction (Step Sd_1b). It is to be noted that the prediction image generated by intra prediction is also referred to as an intra prediction image, and the prediction image generated by inter prediction is also referred to as an inter prediction image.

Next, the predictor evaluates each of the intra prediction image and the inter prediction image (Step Sd_2). Cost C described above may be used in the evaluation. The predictor may then select the prediction image for which the smallest cost C has been calculated among the intra prediction image and the inter prediction image, as the final prediction image for the current block (Step Sd_3). In other words, the prediction method or the mode for generating the prediction image for the current block is selected.

[Intra Predictor]

Intra predictor **124** generates a prediction image (that is, intra prediction image) of a current block by performing intra prediction (also referred to as intra frame prediction) of the current block by referring to a block or blocks in the current picture which is or are stored in block memory **118**. More specifically, intra predictor **124** generates an intra prediction image by performing intra prediction by referring to pixel values (for example, luma and/or chroma values) of a block or blocks neighboring the current block, and then outputs the intra prediction image to prediction controller **128**.

For example, intra predictor **124** performs intra prediction by using one mode from among a plurality of intra prediction modes which have been predefined. The intra prediction modes normally include one or more non-directional prediction modes and a plurality of directional prediction modes.

The one or more non-directional prediction modes include, for example, planar prediction mode and DC prediction mode defined in the H.265/HEVC standard.

The plurality of directional prediction modes include, for example, the thirty-three directional prediction modes defined in the H.265/HEVC standard. It is to be noted that the plurality of directional prediction modes may further include thirty-two directional prediction modes in addition to the thirty-three directional prediction modes (for a total of sixty-five directional prediction modes). FIG. **31** is a diagram illustrating sixty-seven intra prediction modes in total used in intra prediction (two non-directional prediction modes and sixty-five directional prediction modes). The solid arrows represent the thirty-three directions defined in the H.265/HEVC standard, and the dashed arrows represent the additional thirty-two directions (the two non-directional prediction modes are not illustrated in FIG. **31**).

In various kinds of mounting examples, a luma block may be referred to in intra prediction of a chroma block. In other words, a chroma component of the current block may be predicted based on a luma component of the current block. Such intra prediction is also referred to as cross-component linear model (CCLM). The intra prediction mode for a chroma block in which such a luma block is referred to (also referred to as, for example, a CCLM mode) may be added as one of the intra prediction modes for chroma blocks.

Intra predictor **124** may correct intra-predicted pixel values based on horizontal/vertical reference pixel gradients. The intra prediction which accompanies this sort of correcting is also referred to as position dependent intra prediction combination (PDPC). Information indicating whether to apply PDPC (referred to as, for example, a PDPC flag) is normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, or CTU level).

FIG. **32** is a flow chart illustrating one example of a process performed by intra predictor **124**.

Intra predictor **124** selects one intra prediction mode from a plurality of intra prediction modes (Step Sw_1). Intra predictor **124** then generates a prediction image according to the selected intra prediction mode (Step Sw_2). Next, intra predictor **124** determines most probable modes (MPMs) (Step Sw_3). MPMs include, for example, six intra prediction modes. Two modes among the six intra prediction modes may be planar mode and DC prediction mode, and the other four modes may be directional prediction modes. Intra predictor **124** determines whether the intra prediction mode selected in Step Sw_1 is included in the MPMs (Step Sw_4).

Here, when determining that the intra prediction mode selected in Step Sw_1 is included in the MPMs (Yes in Step Sw_4), intra predictor **124** sets an MPM flag to 1 (Step Sw_5), and generates information indicating the selected intra prediction mode among the MPMs (Step Sw_6). It is to be noted that the MPM flag set to 1 and the information indicating the intra prediction mode are encoded as prediction parameters by entropy encoder **110**.

When determining that the selected intra prediction mode is not included in the MPMs (No in Step Sw_4), intra predictor **124** sets the MPM flag to 0 (Step Sw_7). Alternatively, intra predictor **124** does not set any MPM flag. Intra predictor **124** then generates information indicating the selected intra prediction mode among at least one intra prediction mode which is not included in the MPMs (Step Sw_8). It is to be noted that the MPM flag set to 0 and the information indicating the intra prediction mode are encoded as prediction parameters by entropy encoder **110**. The information indicating the intra prediction mode indicates, for example, any one of 0 to 60.

[Inter Predictor]

Inter predictor **126** generates a prediction image (inter prediction image) by performing inter prediction (also referred to as inter frame prediction) of the current block by referring to a block or blocks in a reference picture which is different from the current picture and is stored in frame memory **122**. Inter prediction is performed in units of a current block or a current sub-block in the current block. The sub-block is included in the block and is a unit smaller than the block. The size of the sub-block may be 4×4 pixels, 8×8 pixels, or another size. The size of the sub-block may be switched for a unit such as slice, brick, picture, etc.

For example, inter predictor **126** performs motion estimation in a reference picture for a current block or a current sub-block, and finds out a reference block or a reference sub-block which best matches the current block or current sub-block. Inter predictor **126** then obtains motion information (for example, a motion vector) which compensates a motion or a change from the reference block or the reference sub-block to the current block or the current sub-block. Inter predictor **126** generates an inter prediction image of the current block or the current sub-block by performing motion compensation (or motion prediction) based on the motion information. Inter predictor **126** outputs the generated inter prediction image to prediction controller **128**.

The motion information used in motion compensation may be signaled as inter prediction images in various forms. For example, a motion vector may be signaled. As another example, the difference between a motion vector and a motion vector predictor may be signaled.

[Reference Picture List]

FIG. **33** is a diagram illustrating examples of reference pictures. FIG. **34** is a conceptual diagram illustrating examples of reference picture lists. Each reference picture list is a list indicating at least one reference picture stored in

frame memory **122**. It is to be noted that, in FIG. **33**, each of rectangles indicates a picture, each of arrows indicates a picture reference relationship, the horizontal axis indicates time, I, P, and B in the rectangles indicate an intra prediction picture, a uni-prediction picture, and a bi-prediction picture, respectively, and numerals in the rectangles indicate a decoding order. As illustrated in FIG. **33**, the decoding order of the pictures is an order of I0, P1, B2, B3, and B4, and the display order of the pictures is an order of I0, B3, B2, B4, and P1. As illustrated in FIG. **34**, the reference picture list is a list representing reference picture candidates. For example, one picture (or a slice) may include at least one reference picture list. For example, one reference picture list is used when a current picture is a uni-prediction picture, and two reference picture lists are used when a current picture is a bi-prediction picture. In the examples of FIGS. **33** and **34**, picture B3 which is current picture currPic has two reference picture lists which are the L0 list and the L1 list. When current picture currPic is picture B3, reference picture candidates for current picture currPic are I0, P1, and B2, and the reference picture lists (which are the L0 list and the L1 list) indicate these pictures. Inter predictor **126** or prediction controller **128** specifies which picture in each reference picture list is to be actually referred to in form of a reference picture index refIdxLx. In FIG. **34**, reference pictures P1 and B2 are specified by reference picture indices refIdxL0 and refIdxL1.

Such a reference picture list may be generated for each unit such as a sequence, picture, slice, brick, CTU, or CU. In addition, among reference pictures indicated in reference picture lists, a reference picture index indicating a reference picture to be referred to in inter prediction may be signaled at the sequence level, picture level, slice level, brick level, CTU level, or CU level. In addition, a common reference picture list may be used in a plurality of inter prediction modes.

[Basic Flow of Inter Prediction]

FIG. **35** is a flow chart illustrating a basic processing flow of inter prediction.

First, inter predictor **126** generates a prediction signal (Steps Se_1 to Se_3). Next, subtractor **104** generates the difference between a current block and a prediction image as a prediction residual (Step Se_4).

Here, in the generation of the prediction image, inter predictor **126** generates the prediction image through, for example, determination of a motion vector (MV) of the current block (Steps Se_1 and Se_2) and motion compensation (Step Se_3). Furthermore, in determination of an MV inter predictor **126** determines the MV through, for example, selection of a motion vector candidate (MV candidate) (Step Se_1) and derivation of an MV (Step Se_2). The selection of the MV candidate is made by means of, for example, inter predictor **126** generating an MV candidate list and selecting at least one MV candidate from the MV candidate list. It is to be noted that MVs derived in the past may be added to the MV candidate list. Alternatively, in derivation of an MV, inter predictor **126** may further select at least one MV candidate from the at least one MV candidate, and determine the selected at least one MV candidate as the MV for the current block. Alternatively, inter predictor **126** may determine the MV for the current block by performing estimation in a reference picture region specified by each of the selected at least one MV candidate. It is to be noted that the estimation in the reference picture region may be referred to as motion estimation.

In addition, although Steps Se_1 to Se_3 are performed by inter predictor **126** in the above-described example, a pro-

cess that is, for example, Step Se_1, Step Se_2, or the like may be performed by another constituent element included in encoder **100**.

It is to be noted that an MV candidate list may be generated for each process in inter prediction mode, or a common MV candidate list may be used in a plurality of inter prediction modes. The processes in Steps Se_3 and Se_4 correspond to Steps Sa_3 and Sa_4 illustrated in FIG. **9**, respectively. The process in Step Se_3 corresponds to the process in Step Sd_1b in FIG. **30**.

[MV Derivation Flow]

FIG. **36** is a flow chart illustrating one example of MV derivation.

Inter predictor **126** may derive an MV for a current block in a mode for encoding motion information (for example, an MV). In this case, for example, the motion information may be encoded as a prediction parameter, and may be signaled. In other words, the encoded motion information is included in a stream.

Alternatively, inter predictor **126** may derive an MV in a mode in which motion information is not encoded. In this case, no motion information is included in the stream.

Here, MV derivation modes include a normal inter mode, a normal merge mode, a FRUC mode, an affine mode, etc. which are described later. Modes in which motion information is encoded among the modes include the normal inter mode, the normal merge mode, the affine mode (specifically, an affine inter mode and an affine merge mode), etc. It is to be noted that motion information may include not only an MV but also MV predictor selection information which is described later. Modes in which no motion information is encoded include the FRUC mode, etc. Inter predictor **126** selects a mode for deriving an MV of the current block from the plurality of modes, and derives the MV of the current block using the selected mode.

FIG. **37** is a flow chart illustrating another example of MV derivation.

Inter predictor **126** may derive an MV for a current block in a mode in which an MV difference is encoded. In this case, for example, the MV difference is encoded as a prediction parameter, and is signaled. In other words, the encoded MV difference is included in a stream. The MV difference is the difference between the MV of the current block and the MV predictor. It is to be noted that the MV predictor is a motion vector predictor.

Alternatively, inter predictor **126** may derive an MV in a mode in which no MV difference is encoded. In this case, no encoded MV difference is included in the stream.

Here, as described above, the MV derivation modes include the normal inter mode, the normal merge mode, the FRUC mode, the affine mode, etc. which are described later. Modes in which an MV difference is encoded among the modes include the normal inter mode, the affine mode (specifically, the affine inter mode), etc. Modes in which no MV difference is encoded include the FRUC mode, the normal merge mode, the affine mode (specifically, the affine merge mode), etc. Inter predictor **126** selects a mode for deriving an MV of the current block from the plurality of modes, and derives the MV for the current block using the selected mode.

[MV Derivation Modes]

FIGS. **38A** and **38B** are each a diagram illustrating one example of categorization of modes for MV derivation. For example, as illustrated in FIG. **38A**, MV derivation modes are roughly categorized into three modes according to whether to encode motion information and whether to encode MV differences. The three modes are inter mode,

merge mode, and frame rate up-conversion (FRUC) mode. The inter mode is a mode in which motion estimation is performed, and in which motion information and an MV difference are encoded. For example, as illustrated in FIG. 38B, the inter mode includes affine inter mode and normal inter mode. The merge mode is a mode in which no motion estimation is performed, and in which an MV is selected from an encoded surrounding block and an MV for the current block is derived using the MV. The merge mode is a mode in which, basically, motion information is encoded and no MV difference is encoded. For example, as illustrated in FIG. 38B, the merge modes include normal merge mode (also referred to as normal merge mode or regular merge mode), merge with motion vector difference (MMVD) mode, combined inter merge/intra prediction (CIIP) mode, triangle mode, ATMVP mode, and affine merge mode. Here, an MV difference is encoded exceptionally in the MMVD mode among the modes included in the merge modes. It is to be noted that the affine merge mode and the affine inter mode are modes included in the affine modes. The affine mode is a mode for deriving, as an MV of a current block, an MV of each of a plurality of sub-blocks included in the current block, assuming affine transform. The FRUC mode is a mode which is for deriving an MV of the current block by performing estimation between encoded regions, and in which neither motion information nor any MV difference is encoded. It is to be noted that the respective modes will be described later in detail.

It is to be noted that the categorization of the modes illustrated in FIGS. 38A and 38B are examples, and categorization is not limited thereto. For example, when an MV difference is encoded in CIIP mode, the CIIP mode is categorized into inter modes.

[MV Derivation >Normal Inter Mode]

The normal inter mode is an inter prediction mode for deriving an MV of a current block by finding out a block similar to the image of the current block from a reference picture region specified by an MV candidate. In this normal inter mode, an MV difference is encoded.

FIG. 39 is a flow chart illustrating an example of inter prediction by normal inter mode.

First, inter predictor 126 obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sg_1). In other words, inter predictor 126 generates an MV candidate list.

Next, inter predictor 126 extracts N (an integer of 2 or larger) MV candidates from the plurality of MV candidates obtained in Step Sg_1, as motion vector predictor candidates according to a predetermined priority order (Step Sg_2). It is to be noted that the priority order is determined in advance for each of the N MV candidates.

Next, inter predictor 126 selects one MV predictor candidate from the N MV predictor candidates as the MV predictor for the current block (Step Sg_3). At this time, inter predictor 126 encodes, in a stream, MV predictor selection information for identifying the selected MV predictor. In other words, inter predictor 126 outputs the MV predictor selection information as a prediction parameter to entropy encoder 110 through prediction parameter generator 130.

Next, inter predictor 126 derives an MV of a current block by referring to an encoded reference picture (Step Sg_4). At this time, inter predictor 126 further encodes, in the stream, the difference value between the derived MV and the MV predictor as an MV difference. In other words, inter predictor 126 outputs the MV difference as a prediction parameter

to entropy encoder 110 through prediction parameter generator 130. It is to be noted that the encoded reference picture is a picture including a plurality of blocks which have been reconstructed after being encoded.

Lastly, inter predictor 126 generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Sg_5). The processes in Steps Sg_1 to Sg_5 are executed on each block. For example, when the processes in Steps Sg_1 to Sg_5 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal inter mode finishes. For example, when the processes in Steps Sg_1 to Sg_5 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal inter mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Sg_1 to Sg_5, and inter prediction of the slice using the normal inter mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal inter mode may finish when the processes in Steps Sg_1 to Sg_5 are executed on part of the blocks in the picture.

It is to be noted that the prediction image is an inter prediction signal as described above. In addition, information indicating the inter prediction mode (normal inter mode in the above example) used to generate the prediction image is, for example, encoded as a prediction parameter in an encoded signal.

It is to be noted that the MV candidate list may be also used as a list for use in another mode. In addition, the processes related to the MV candidate list may be applied to processes related to the list for use in another mode. The processes related to the MV candidate list include, for example, extraction or selection of an MV candidate from the MV candidate list, reordering of MV candidates, or deletion of an MV candidate.

[MV Derivation >Normal Merge Mode]

The normal merge mode is an inter prediction mode for selecting an MV candidate from an MV candidate list as an MV for a current block, thereby deriving the MV. It is to be noted that the normal merge mode is a merge mode in a narrow meaning and is also simply referred to as a merge mode. In this embodiment, the normal merge mode and the merge mode are distinguished, and the merge mode is used in a broad meaning.

FIG. 40 is a flow chart illustrating an example of inter prediction by normal merge mode.

First, inter predictor 126 obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sh_1). In other words, inter predictor 126 generates an MV candidate list.

Next, inter predictor 126 selects one MV candidate from the plurality of MV candidates obtained in Step Sh_1, thereby deriving an MV for the current block (Step Sh_2). At this time, inter predictor 126 encodes, in a stream, MV selection information for identifying the selected MV candidate. In other words, inter predictor 126 outputs the MV selection information as a prediction parameter to entropy encoder 110 through prediction parameter generator 130.

Lastly, inter predictor 126 generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Sh_3). The processes in Steps Sh_1 to Sh_3 are executed, for example, on each block. For example, when the processes in Steps Sh_1 to Sh_3 are executed on each of all the blocks in the slice, inter predic-

tion of the slice using the normal merge mode finishes. In addition, when the processes in Steps Sh_1 to Sh_3 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal merge mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Sh_1 to Sh_3, and inter prediction of the slice using the normal merge mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal merge mode may finish when the processes in Steps Sh_1 to Sh_3 are executed on part of the blocks in the picture.

In addition, information indicating the inter prediction mode (normal merge mode in the above example) used to generate the prediction image is, for example, encoded as a prediction parameter in a stream.

FIG. 41 is a diagram for illustrating one example of an MV derivation process for a current picture by normal merge mode.

First, inter predictor 126 generates an MV candidate list in which MV candidates are registered. Examples of MV candidates include: spatially neighboring MV candidates which are MVs of a plurality of encoded blocks located spatially surrounding a current block; temporally neighboring MV candidates which are MVs of surrounding blocks on which the position of a current block in an encoded reference picture is projected; combined MV candidates which are MVs generated by combining the MV value of a spatially neighboring MV predictor and the MV value of a temporally neighboring MV predictor; and a zero MV candidate which is an MV having a zero value.

Next, inter predictor 126 selects one MV candidate from a plurality of MV candidates registered in an MV candidate list, and determines the MV candidate as the MV of the current block.

Furthermore, entropy encoder 110 writes and encodes, in a stream, merge_idx which is a signal indicating which MV candidate has been selected.

It is to be noted that the MV candidates registered in the MV candidate list described in FIG. 41 are examples. The number of MV candidates may be different from the number of MV candidates in the diagram, the MV candidate list may be configured in such a manner that some of the kinds of the MV candidates in the diagram may not be included, or that one or more MV candidates other than the kinds of MV candidates in the diagram are included.

A final MV may be determined by performing a dynamic motion vector refreshing (DMVR) to be described later using the MV of the current block derived by normal merge mode. It is to be noted that, in normal merge mode, no MV difference is encoded, but an MV difference is encoded. In MMVD mode, one MV candidate is selected from an MV candidate list as in the case of normal merge mode, an MV difference is encoded. As illustrated in FIG. 38B, MMVD may be categorized into merge modes together with normal merge mode. It is to be noted that the MV difference in MMVD mode does not always need to be the same as the MV difference for use in inter mode. For example, MV difference derivation in MMVD mode may be a process that requires a smaller amount of processing than the amount of processing required for MV difference derivation in inter mode.

In addition, a combined inter merge/intra prediction (CIIP) mode may be performed. The mode is for overlapping a prediction image generated in inter prediction and a prediction image generated in intra prediction to generate a prediction image for a current block.

It is to be noted that the MV candidate list may be referred to as a candidate list. In addition, merge_idx is MV selection information.

[MV Derivation >HMVP Mode]

FIG. 42 is a diagram for illustrating one example of an MV derivation process for a current picture by HMVP merge mode.

In normal merge mode, an MV for, for example, a CU which is a current block is determined by selecting one MV candidate from an MV candidate list generated by referring to an encoded block (for example, a CU). Here, another MV candidate may be registered in the MV candidate list. The mode in which such another MV candidate is registered is referred to as HMVP mode.

In HMVP mode, MV candidates are managed using a first-in first-out (FIFO) buffer for HMVP, separately from the MV candidate list for normal merge mode.

In FIFO buffer, motion information such as MVs of blocks processed in the past are stored newest first. In the management of the FIFO buffer, each time when one block is processed, the MV for the newest block (that is the CU processed immediately before) is stored in the FIFO buffer, and the MV of the oldest CU (that is, the CU processed earliest) is deleted from the FIFO buffer. In the example illustrated in FIG. 42, HMVP1 is the MV for the newest block, and HMVP5 is the MV for the oldest MV.

Inter predictor 126 then, for example, checks whether each MV managed in the FIFO buffer is an MV different from all the MV candidates which have been already registered in the MV candidate list for normal merge mode starting from HMVP1. When determining that the MV is different from all the MV candidates, inter predictor 126 may add the MV managed in the FIFO buffer in the MV candidate list for normal merge mode as an MV candidate. At this time, the MV candidate registered from the FIFO buffer may be one or more.

By using the HMVP mode in this way, it is possible to add not only the MV of a block which neighbors the current block spatially or temporally but also an MV for a block processed in the past. As a result, the variation of MV candidates for normal merge mode is expanded, which increases the probability that coding efficiency can be increased.

It is to be noted that the MV may be motion information. In other words, information stored in the MV candidate list and the FIFO buffer may include not only MV values but also reference picture information, reference directions, the numbers of pictures, etc. In addition, the block is, for example, a CU.

It is to be noted that the MV candidate list and the FIFO buffer illustrated in FIG. 42 are examples. The MV candidate list and FIFO buffer may be different in size from those in FIG. 42, or may be configured to register MV candidates in an order different from the one in FIG. 42. In addition, the process described here is common between encoder 100 and decoder 200.

It is to be noted that the HMVP mode can be applied for modes other than the normal merge mode. For example, it is also excellent that motion information such as MVs of blocks processed in affine mode in the past may be stored newest first, and may be used as MV candidates. The mode obtained by applying HMVP mode to affine mode may be referred to as history affine mode.

[MV Derivation >FRUC Mode]

Motion information may be derived at the decoder 200 side without being signaled from the encoder 100 side. For example, motion information may be derived by performing

motion estimation at the decoder 200 side. At this time, at the decoder 200 side, motion estimation is performed without using any pixel value in a current block. Modes in which motion estimation is performed at the decoder 200 side in this way include a frame rate up-conversion (FRUC) mode, a pattern matched motion vector derivation (PMMVD) mode, etc.

One example of a FRUC process is illustrated in FIG. 43. First, a list which indicates, as MV candidates, MVs for encoded blocks each of which neighbors the current block spatially or temporally is generated by referring to the MVs (the list may be an MV candidate list, and be also used as the MV candidate list for normal merge mode) (Step Si_1). Next, a best MV candidate is selected from the plurality of MV candidates registered in the MV candidate list (Step Si_2). For example, the evaluation values of the respective MV candidates included in the MV candidate list are calculated, and one MV candidate is selected as the best MV candidate based on the evaluation values. Based on the selected best MV candidate, a motion vector for the current block is then derived (Step Si_4). More specifically for example, the selected best MV candidate is directly derived as the MV for the current block. In addition, for example, the MV for the current block may be derived using pattern matching in a surrounding region of a position which is included in a reference picture and corresponds to the selected best MV candidate. In other words, estimation using the pattern matching in a reference picture and the evaluation values may be performed in the surrounding region of the best MV candidate, and when there is an MV that yields a better evaluation value, the best MV candidate may be updated to the MV that yields the better evaluation value, and the updated MV may be determined as the final MV for the current block. Update to the MV that yields the better evaluation value may not be performed.

Lastly inter predictor 126 generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Si_5). The processes in Steps Si_1 to Si_5 are executed, for example, on each block. For example, when the processes in Steps Si_1 to Si_5 are executed on each of all the blocks in the slice, inter prediction of the slice using the FRUC mode finishes. For example, when the processes in Steps Si_1 to Si_5 are executed on each of all the blocks in the picture, inter prediction of the picture using the FRUC mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Si_1 to Si_5, and inter prediction of the slice using the FRUC mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the FRUC mode may finish when the processes in Steps Si_1 to Si_5 are executed on part of the blocks included in the picture.

Each sub-block may be processed similarly to the above-described case of processing each block.

Evaluation values may be calculated according to various kinds of methods. For example, a comparison is made between a reconstructed image in a region in a reference picture corresponding to an MV and a reconstructed image in a determined region (the region may be, for example, a region in another reference picture or a region in a neighboring block of a current picture, as indicated below). The difference between the pixel values of the two reconstructed images may be used for an evaluation value of the MV. It is to be noted that an evaluation value may be calculated using information other than the value of the difference.

Next, pattern matching is described in detail. First, one MV candidate included in an MV candidate list (also referred to as a merge list) is selected as a starting point for estimation by pattern matching. As the pattern matching, either a first pattern matching or a second pattern matching may be used. The first pattern matching and the second pattern matching may be referred to as bilateral matching and template matching, respectively.

[MV Derivation >FRUC >Bilateral Matching]

In the first pattern matching, the pattern matching is performed between two blocks which are located along a motion trajectory of a current block and included in two different reference pictures. Accordingly, in the first pattern matching, a region in another reference picture located along the motion trajectory of the current block is used as a determined region for calculating the evaluation value of the above-described MV candidate.

FIG. 44 is a diagram for illustrating one example of the first pattern matching (bilateral matching) between the two blocks in the two reference pictures located along the motion trajectory. As illustrated in FIG. 44, in the first pattern matching, two motion vectors (MV0, MV1) are derived by estimating a pair which best matches among pairs of two blocks which are included in the two different reference pictures (Ref0, Ref1) and located along the motion trajectory of the current block (Cur block). More specifically, a difference between the reconstructed image at a specified position in the first encoded reference picture (Ref0) specified by an MV candidate and the reconstructed image at a specified position in the second encoded reference picture (Ref1) specified by a symmetrical MV obtained by scaling the MV candidate at a display time interval is derived for the current block, and an evaluation value is calculated using the value of the obtained difference. It is excellent to select, as the best MV, the MV candidate which yields the best evaluation value among the plurality of MV candidates.

In the assumption of a continuous motion trajectory, the motion vectors (MV0, MV1) specifying the two reference blocks are proportional to temporal distances (TD0, TD1) between the current picture (Cur Pic) and the two reference pictures (Ref0, Ref1). For example, when the current picture is temporally located between the two reference pictures and the temporal distances from the current picture to the respective two reference pictures are equal to each other, mirror-symmetrical bi-directional MVs are derived in the first pattern matching.

[MV Derivation >FRUC >Template Matching]

In the second pattern matching (template matching), pattern matching is performed between a block in a reference picture and a template in the current picture (the template is a block neighboring the current block in the current picture (the neighboring block is, for example, an upper and/or left neighboring block(s))). Accordingly, in the second pattern matching, the block neighboring the current block in the current picture is used as the determined region for calculating the evaluation value of the above-described MV candidate.

FIG. 45 is a diagram for illustrating one example of pattern matching (template matching) between a template in a current picture and a block in a reference picture. As illustrated in FIG. 45, in the second pattern matching, the MV for the current block (Cur block) is derived by estimating, in the reference picture (Ref0), the block which best matches the block neighboring the current block in the current picture (Cur Pic). More specifically, the difference between a reconstructed image in an encoded region which neighbors both left and above or either left or above and a

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reconstructed image which is in a corresponding region in the encoded reference picture (Ref0) and is specified by an MV candidate is derived, and an evaluation value is calculated using the value of the obtained difference. It is excellent to select, as the best MV candidate, the MV candidate which yields the best evaluation value among the plurality of MV candidates.

Such information indicating whether to apply the FRUC mode (referred to as, for example, a FRUC flag) may be signaled at the CU level. In addition, when the FRUC mode is applied (for example, when a FRUC flag is true), information indicating an applicable pattern matching method (either the first pattern matching or the second pattern matching) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, CTU level, or sub-block level). [MV Derivation >Affine Mode]

The affine mode is a mode for generating an MV using affine transform. For example, an MV may be derived in units of a sub-block based on motion vectors of a plurality of neighboring blocks. This mode is also referred to as an affine motion compensation prediction mode.

FIG. 46A is a diagram for illustrating one example of MV derivation in units of a sub-block based on MVs of a plurality of neighboring blocks. In FIG. 46A, the current block includes sixteen 4×4 pixel sub-blocks. Here, motion vector v_0 at an upper-left corner control point in the current block is derived based on an MV of a neighboring block, and likewise, motion vector v_1 at an upper-right corner control point in the current block is derived based on an MV of a neighboring sub-block. Two motion vectors v_0 and v_1 are projected according to an expression (1A) indicated below, and motion vectors (v_x, v_y) for the respective sub-blocks in the current block are derived.

[MATH. 1]

$$\begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x - \frac{(v_{1y} - v_{0y})}{w}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x + \frac{(v_{1x} - v_{0x})}{w}y + v_{0y} \end{cases} \quad (1A)$$

Here, x and y indicate the horizontal position and the vertical position of the sub-block, respectively, and w indicates a predetermined weighting coefficient.

Such information indicating the affine mode (for example, referred to as an affine flag) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, brick level, CTU level, or sub-block level).

In addition, the affine mode may include several modes for different methods for deriving MVs at the upper-left and upper-right corner control points. For example, the affine modes include two modes which are the affine inter mode (also referred to as an affine normal inter mode) and the affine merge mode.

FIG. 46B is a diagram for illustrating one example of MV derivation in units of a sub-block in affine mode in which three control points are used. In FIG. 46B, the current block includes, for example, sixteen 4×4 pixel sub-blocks. Here, motion vector v_0 at an upper-left corner control point in the current block is derived based on an MV of a neighboring

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block. Here, motion vector v_1 at an upper-right corner control point in the current block is derived based on an MV of a neighboring block, and likewise, motion vector v_2 at a lower-left corner control point for the current block is derived based on an MV of a neighboring block. Three motion vectors v_0 , v_1 , and v_2 are projected according to an expression (1B) indicated below, and motion vectors (v_x, v_y) for the respective sub-blocks in the current block are derived.

[MATH. 2]

$$\begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x + \frac{(v_{2x} - v_{0x})}{h}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x + \frac{(v_{2y} - v_{0y})}{h}y + v_{0y} \end{cases} \quad (1B)$$

Here, x and y indicate the horizontal position and the vertical position of the sub-block, respectively, and each of w and h indicates a predetermined weighting coefficient. Here, w may indicate the width of a current block, and h may indicate the height of the current block.

Affine modes in which different numbers of control points (for example, two and three control points) are used may be switched and signaled at the CU level. It is to be noted that information indicating the number of control points in affine mode used at the CU level may be signaled at another level (for example, the sequence level, picture level, slice level, brick level, CTU level, or sub-block level).

In addition, such an affine mode in which three control points are used may include different methods for deriving MVs at the upper-left, upper-right, and lower-left corner control points. For example, the affine modes in which three control points are used include two modes which are affine inter mode and affine merge mode, as in the case of affine modes in which two control points are used.

It is to be noted that, in the affine modes, the size of each sub-block included in the current block may not be limited to 4×4 pixels, and may be another size. For example, the size of each sub-block may be 8×8 pixels.

[MV Derivation >Affine Mode >Control Point]

FIGS. 47A, 47B, and 47C are each a conceptual diagram for illustrating one example of MV derivation at control points in an affine mode.

As illustrated in FIG. 47A, in the affine mode, for example, MV predictors at respective control points for a current block are calculated based on a plurality of MVs corresponding to blocks encoded according to the affine mode among encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) which neighbor the current block. More specifically, encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) are checked in the listed order, and the first effective block encoded according to the affine mode is identified. The MV at each control point for the current block is calculated based on the plurality of MVs corresponding to the identified block.

For example, as illustrated in FIG. 47B, when block A which neighbors to the left of the current block has been encoded according to an affine mode in which two control points are used, motion vectors v_3 and v_4 projected at the upper-left corner position and the upper-right corner position of the encoded block including block A are derived. Motion vector v_0 at the upper-left control point and motion

vector v_1 at the upper-right control point for the current block are then calculated from derived motion vectors v_3 and v_4 .

For example, as illustrated in FIG. 47C, when block A which neighbors to the left of the current block has been encoded according to an affine mode in which three control points are used, motion vectors v_3 , v_4 , and v_5 projected at the upper-left corner position, the upper-right corner position, and the lower-left corner position of the encoded block including block A are derived. Motion vector v_0 at the upper-left control point for the current block, motion vector v_1 at the upper-right control point for the current block, and motion vector v_2 at the lower-left control point for the current block are then calculated from derived motion vectors v_3 , v_4 , and v_5 .

The MV derivation methods illustrated in FIGS. 47A to 47C may be used in the MV derivation at each control point for the current block in Step Sk_1 illustrated in FIG. 50 described later, or may be used for MV predictor derivation at each control point for the current block in Step Sj_1 illustrated in FIG. 51 described later.

FIGS. 48A and 48B are each a conceptual diagram for illustrating another example of MV derivation at control points in affine mode.

FIG. 48A is a diagram for illustrating an affine mode in which two control points are used.

In the affine mode, as illustrated in FIG. 48A, an MV selected from MVs at encoded block A, block B, and block C which neighbor the current block is used as motion vector v_0 at the upper-left corner control point for the current block. Likewise, an MV selected from MVs of encoded block D and block E which neighbor the current block is used as motion vector v_1 at the upper-right corner control point for the current block.

FIG. 48B is a diagram for illustrating an affine mode in which three control points are used.

In the affine mode, as illustrated in FIG. 48B, an MV selected from MVs at encoded block A, block B, and block C which neighbor the current block is used as motion vector v_0 at the upper-left corner control point for the current block. Likewise, an MV selected from MVs of encoded block D and block E which neighbor the current block is used as motion vector v_1 at the upper-right corner control point for the current block. Furthermore, an MV selected from MVs of encoded block F and block G which neighbor the current block is used as motion vector v_2 at the lower-left corner control point for the current block.

It is to be noted that the MV derivation methods illustrated in FIGS. 48A and 48B may be used in the MV derivation at each control point for the current block in Step Sk_1 illustrated in FIG. 50 described later, or may be used for MV predictor derivation at each control point for the current block in Step Sj_1 illustrated in FIG. 51 described later.

Here, when affine modes in which different numbers of control points (for example, two and three control points) are used may be switched and signaled at the CU level, the number of control points for an encoded block and the number of control points for a current block may be different from each other.

FIGS. 49A and 49B are each a conceptual diagram for illustrating one example of a method for MV derivation at control points when the number of control points for an encoded block and the number of control points for a current block are different from each other.

For example, as illustrated in FIG. 49A, a current block has three control points at the upper-left corner, the upper-right corner, and the lower-left corner, and block A which

neighbors to the left of the current block has been encoded according to an affine mode in which two control points are used. In this case, motion vectors v_3 and v_4 projected at the upper-left corner position and the upper-right corner position in the encoded block including block A are derived. Motion vector v_0 at the upper-left corner control point and motion vector v_1 at the upper-right corner control point for the current block are then calculated from derived motion vectors v_3 and v_4 . Furthermore, motion vector v_2 at the lower-left corner control point is calculated from derived motion vectors v_0 and v_1 .

For example, as illustrated in FIG. 49B, a current block has two control points at the upper-left corner and the upper-right corner, and block A which neighbors to the left of the current block has been encoded according to an affine mode in which three control points are used. In this case, motion vectors v_3 , v_4 , and v_5 projected at the upper-left corner position in the encoded block including block A, the upper-right corner position in the encoded block, and the lower-left corner position in the encoded block are derived. Motion vector v_0 at the upper-left corner control point for the current block and motion vector v_1 at the upper-right corner control point for the current block are then calculated from derived motion vectors v_3 , v_4 , and v_5 .

It is to be noted that the MV derivation methods illustrated in FIGS. 49A and 49B may be used in the MV derivation at each control point for the current block in Step Sk_1 illustrated in FIG. 50 described later, or may be used for MV predictor derivation at each control point for the current block in Step Sj_1 illustrated in FIG. 51 described later.

[MV Derivation >Affine Mode >Affine Merge Mode]

FIG. 50 is a flow chart illustrating one example of the affine merge mode.

In the affine merge mode, first, inter predictor 126 derives MVs at respective control points for a current block (Step Sk_1). The control points are an upper-left corner point of the current block and an upper-right corner point of the current block as illustrated in FIG. 46A, or an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. 46B. At this time, inter predictor 126 may encode MV selection information for identifying two or three derived MVs in a stream.

For example, when MV derivation methods illustrated in FIGS. 47A to 47C are used, as illustrated in FIG. 47A, inter predictor 126 checks encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) in the listed order, and identifies the first effective block encoded according to the affine mode.

Inter predictor 126 derives the MV at the control point using the identified first effective block encoded according to the identified affine mode. For example, when block A is identified and block A has two control points, as illustrated in FIG. 47B, inter predictor 126 calculates motion vector v_0 at the upper-left corner control point of the current block and motion vector v_1 at the upper-right corner control point of the current block from motion vectors v_3 and v_4 at the upper-left corner of the encoded block including block A and the upper-right corner of the encoded block. For example, inter predictor 126 calculates motion vector v_0 at the upper-left corner control point of the current block and motion vector v_1 at the upper-right corner control point of the current block by projecting motion vectors v_3 and v_4 at the upper-left corner and the upper-right corner of the encoded block onto the current block.

Alternatively, when block A is identified and block A has three control points, as illustrated in FIG. 47C, inter predic-

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tor 126 calculates motion vector v_0 at the upper-left corner control point of the current block, motion vector v_1 at the upper-right corner control point of the current block, and motion vector v_2 at the lower-left corner control point of the current block from motion vectors v_3 , v_4 , and v_5 at the upper-left corner of the encoded block including block A, the upper-right corner of the encoded block, and the lower-left corner of the encoded block. For example, inter predictor 126 calculates motion vector v_0 at the upper-left corner control point of the current block, motion vector v_1 at the upper-right corner control point of the current block, and motion vector v_2 at the lower-left corner control point of the current block by projecting motion vectors v_a , v_b , and v_c at the upper-left corner, the upper-right corner, and the lower-left corner of the encoded block onto the current block.

It is to be noted that, as illustrated in FIG. 49A described above, MVs at three control points may be calculated when block A is identified and block A has two control points, and that, as illustrated in FIG. 49B described above, MVs at two control points may be calculated when block A is identified and block A has three control points.

Next, inter predictor 126 performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor 126 calculates an MV for each of the plurality of sub-blocks as an affine MV using either two motion vectors v_0 and v_1 and the above expression (1A) or three motion vectors v_0 , v_1 , and v_2 and the above expression (1B) (Step Sk_2). Inter predictor 126 then performs motion compensation of the sub-blocks using these affine MVs and encoded reference pictures (Step Sk_3). When the processes in Steps Sk_2 and Sk_3 are executed for each of all the sub-blocks included in the current block, the process for generating a prediction image using the affine merge mode for the current block finishes. In other words, motion compensation of the current block is performed to generate a prediction image of the current block.

It is to be noted that the above-described MV candidate list may be generated in Step Sk_1. The MV candidate list may be, for example, a list including MV candidates derived using a plurality of MV derivation methods for each control point. The plurality of MV derivation methods may be any combination of the MV derivation methods illustrated in FIGS. 47A to 47C, the MV derivation methods illustrated in FIGS. 48A and 48B, the MV derivation methods illustrated in FIGS. 49A and 49B, and other MV derivation methods.

It is to be noted that MV candidate lists may include MV candidates in a mode in which prediction is performed in units of a sub-block, other than the affine mode.

It is to be noted that, for example, an MV candidate list including MV candidates in an affine merge mode in which two control points are used and an affine merge mode in which three control points are used may be generated as an MV candidate list. Alternatively, an MV candidate list including MV candidates in the affine merge mode in which two control points are used and an MV candidate list including MV candidates in the affine merge mode in which three control points are used may be generated separately. Alternatively, an MV candidate list including MV candidates in one of the affine merge mode in which two control points are used and the affine merge mode in which three control points are used may be generated. The MV candidate(s) may be, for example, MVs for encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left), or an MV for an effective block among the blocks.

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It is to be noted that index indicating one of the MVs in an MV candidate list may be transmitted as MV selection information.

[MV Derivation >Affine Mode >Affine Inter Mode]

FIG. 51 is a flow chart illustrating one example of an affine inter mode.

In the affine inter mode, first, inter predictor 126 derives MV predictors (v_0 , v_1) or (v_0 , v_1 , v_2) of respective two or three control points for a current block (Step Sj_1). The control points are an upper-left corner point for the current block, an upper-right corner point of the current block, and a lower-left corner point for the current block as illustrated in FIG. 46A or FIG. 46B.

For example, when the MV derivation methods illustrated in FIGS. 48A and 48B are used, inter predictor 126 derives the MV predictors (v_0 , v_1) or (v_0 , v_1 , v_2) at respective two or three control points for the current block by selecting MVs of any of the blocks among encoded blocks in the vicinity of the respective control points for the current block illustrated in either FIG. 48A or FIG. 48B. At this time, inter predictor 126 encodes, in a stream, MV predictor selection information for identifying the selected two or three MV predictors.

For example, inter predictor 126 may determine, using a cost evaluation or the like, the block from which an MV as an MV predictor at a control point is selected from among encoded blocks neighboring the current block, and may write, in a bitstream, a flag indicating which MV predictor has been selected. In other words, inter predictor 126 outputs, as a prediction parameter, the MV predictor selection information such as a flag to entropy encoder 110 through prediction parameter generator 130.

Next, inter predictor 126 performs motion estimation (Steps Sj_3 and Sj_4) while updating the MV predictor selected or derived in Step Sj_1 (Step Sj_2). In other words, inter predictor 126 calculates, as an affine MV, an MV of each of sub-blocks which corresponds to an updated MV predictor, using either the expression (1A) or expression (1B) described above (Step Sj_3). Inter predictor 126 then performs motion compensation of the sub-blocks using these affine MVs and encoded reference pictures (Step Sj_4). The processes in Steps Sj_3 and Sj_4 are executed on all the blocks in the current block each time an MV predictor is updated in Step Sj_2. As a result, for example, inter predictor 126 determines the MV predictor which yields the smallest cost as the MV at a control point in a motion estimation loop (Step Sj_5). At this time, inter predictor 126 further encodes, in the stream, the difference value between the determined MV and the MV predictor as an MV difference. In other words, inter predictor 126 outputs the MV difference as a prediction parameter to entropy encoder 110 through prediction parameter generator 130.

Lastly inter predictor 126 generates a prediction image for the current block by performing motion compensation of the current block using the determined MV and the encoded reference picture (Step Sj_6).

It is to be noted that the above-described MV candidate list may be generated in Step Sj_1. The MV candidate list may be, for example, a list including MV candidates derived using a plurality of MV derivation methods for each control point. The plurality of MV derivation methods may be any combination of the MV derivation methods illustrated in FIGS. 47A to 47C, the MV derivation methods illustrated in FIGS. 48A and 48B, the MV derivation methods illustrated in FIGS. 49A and 49B, and other MV derivation methods.

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It is to be noted that the MV candidate list may include MV candidates in a mode in which prediction is performed in units of a sub-block, other than the affine mode.

It is to be noted that, for example, an MV candidate list including MV candidates in an affine inter mode in which two control points are used and an affine inter mode in which three control points are used may be generated as an MV candidate list. Alternatively, an MV candidate list including MV candidates in the affine inter mode in which two control points are used and an MV candidate list including MV candidates in the affine inter mode in which three control points are used may be generated separately. Alternatively, an MV candidate list including MV candidates in one of the affine inter mode in which two control points are used and the affine inter mode in which three control points are used may be generated. The MV candidate(s) may be, for example, MVs for encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left), or an MV for an effective block among the blocks.

It is to be noted that index indicating one of the MV candidates in an MV candidate list may be transmitted as MV predictor selection information.

[MV Derivation >Triangle Mode]

Inter predictor **126** generates one rectangular prediction image for a rectangular current block in the above example. However, inter predictor **126** may generate a plurality of prediction images each having a shape different from a rectangle for the rectangular current block, and may combine the plurality of prediction images to generate the final rectangular prediction image. The shape different from a rectangle may be, for example, a triangle.

FIG. **52A** is a diagram for illustrating generation of two triangular prediction images.

Inter predictor **126** generates a triangular prediction image by performing motion compensation of a first partition having a triangular shape in a current block by using a first MV of the first partition, to generate a triangular prediction image. Likewise, inter predictor **126** generates a triangular prediction image by performing motion compensation of a second partition having a triangular shape in a current block by using a second MV of the second partition, to generate a triangular prediction image. Inter predictor **126** then generates a prediction image having the same rectangular shape as the rectangular shape of the current block by combining these prediction images.

It is to be noted that a first prediction image having a rectangular shape corresponding to a current block may be generated as a prediction image for a first partition, using a first MV. In addition, a second prediction image having a rectangular shape corresponding to a current block may be generated as a prediction image for a second partition, using a second MV. A prediction image for the current block may be generated by performing a weighted addition of the first prediction image and the second prediction image. It is to be noted that the part which is subjected to the weighted addition may be a partial region across the boundary between the first partition and the second partition.

FIG. **52B** is a conceptual diagram for illustrating examples of a first portion of a first partition which overlaps with a second partition, and first and second sets of samples which may be weighted as part of a correction process. The first portion may be, for example, one fourth of the width or height of the first partition. In another example, the first portion may have a width corresponding to N samples adjacent to an edge of the first partition, where N is an integer greater than zero, and N may be, for example, the

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integer 2. As illustrated, the left example of FIG. **52B** shows a rectangular partition having a rectangular portion with a width which is one fourth of the width of the first partition, with the first set of samples including samples outside of the first portion and samples inside of the first portion, and the second set of samples including samples within the first portion. The center example of FIG. **52B** shows a rectangular partition having a rectangular portion with a height which is one fourth of the height of the first partition, with the first set of samples including samples outside of the first portion and samples inside of the first portion, and the second set of samples including samples within the first portion. The right example of FIG. **52B** shows a triangular partition having a polygonal portion with a height which corresponds to two samples, with the first set of samples including samples outside of the first portion and samples inside of the first portion, and the second set of samples including samples within the first portion.

The first portion may be a portion of the first partition which overlaps with an adjacent partition. FIG. **52C** is a conceptual diagram for illustrating a first portion of a first partition, which is a portion of the first partition that overlaps with a portion of an adjacent partition. For ease of illustration, a rectangular partition having an overlapping portion with a spatially adjacent rectangular partition is shown. Partitions having other shapes, such as triangular partitions, may be employed, and the overlapping portions may overlap with a spatially or temporally adjacent partition.

In addition, although an example is given in which a prediction image is generated for each of two partitions using inter prediction, a prediction image may be generated for at least one partition using intra prediction.

FIG. **53** is a flow chart illustrating one example of a triangle mode.

In the triangle mode, first, inter predictor **126** splits the current block into the first partition and the second partition (Step Sx_1). At this time, inter predictor **126** may encode, in a stream, partition information which is information related to the splitting into the partitions as a prediction parameter. In other words, inter predictor **126** may output the partition information as the prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

First, inter predictor **126** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sx_2). In other words, inter predictor **126** generates an MV candidate list.

Inter predictor **126** then selects the MV candidate for the first partition and the MV candidate for the second partition as a first MV and a second MV respectively, from the plurality of MV candidates obtained in Step Sx_2 (Step Sx_3). At this time, inter predictor **126** encodes, in a stream, MV selection information for identifying the selected MV candidate, as a prediction parameter. In other words, inter predictor **126** outputs the MV selection information as a prediction parameter to entropy encoder **110** through prediction parameter generator **130**.

Next, inter predictor **126** generates a first prediction image by performing motion compensation using the selected first MV and an encoded reference picture (Step Sx_4). Likewise, inter predictor **126** generates a second prediction image by performing motion compensation using the selected second MV and an encoded reference picture (Step Sx_5).

Lastly, inter predictor **126** generates a prediction image for the current block by performing a weighted addition of the first prediction image and the second prediction image (Step Sx_6).

It is to be noted that, although the first partition and the second partition are triangles in the example illustrated in FIG. 52A, the first partition and the second partition may be trapezoids, or other shapes different from each other. Furthermore, although the current block includes two partitions in the example illustrated in FIG. 52A, the current block may include three or more partitions.

In addition, the first partition and the second partition may overlap with each other. In other words, the first partition and the second partition may include the same pixel region. In this case, a prediction image for a current block may be generated using a prediction image in the first partition and a prediction image in the second partition.

In addition, although the example in which the prediction image is generated for each of the two partitions using inter prediction has been illustrated, a prediction image may be generated for at least one partition using intra prediction.

It is to be noted that the MV candidate list for selecting the first MV and the MV candidate list for selecting the second MV may be different from each other, or the MV candidate list for selecting the first MV may be also used as the MV candidate list for selecting the second MV.

It is to be noted that partition information may include an index indicating the splitting direction in which at least a current block is split into a plurality of partitions. The MV selection information may include an index indicating the selected first MV and an index indicating the selected second MV. One index may indicate a plurality of pieces of information. For example, one index collectively indicating a part or the entirety of partition information and a part or the entirety of MV selection information may be encoded.

[MV Derivation >ATMVP Mode]

FIG. 54 is a diagram illustrating one example of an ATMVP mode in which an MV is derived in units of a sub-block.

The ATMVP mode is a mode categorized into the merge mode. For example, in the ATMVP mode, an MV candidate for each sub-block is registered in an MV candidate list for use in normal merge mode.

More specifically, in the ATMVP mode, first, as illustrated in FIG. 54, a temporal MV reference block associated with a current block is identified in an encoded reference picture specified by an MV (MV0) of a neighboring block located at the lower-left position with respect to the current block. Next, in each sub-block in the current block, the MV used to encode the region corresponding to the sub-block in the temporal MV reference block is identified. The MV identified in this way is included in an MV candidate list as an MV candidate for the sub-block in the current block. When the MV candidate for each sub-block is selected from the MV candidate list, the sub-block is subjected to motion compensation in which the MV candidate is used as the MV for the sub-block. In this way, a prediction image for each sub-block is generated.

Although the block located at the lower-left position with respect to the current block is used as a surrounding MV reference block in the example illustrated in FIG. 54, it is to be noted that another block may be used. In addition, the size of the sub-block may be 4×4 pixels, 8×8 pixels, or another size. The size of the sub-block may be switched for a unit such as a slice, brick, picture, etc.

[Motion Estimation >DMVR]

FIG. 55 is a diagram illustrating a relationship between a merge mode and DMVR.

Inter predictor **126** derives an MV for a current block according to the merge mode (Step SI_1). Next, inter predictor **126** determines whether to perform estimation of an MV that is motion estimation (Step SI_2). Here, when determining not to perform motion estimation (No in Step SI_2), inter predictor **126** determines the MV derived in Step SI_1 as the final MV for the current block (Step SI_4). In other words, in this case, the MV for the current block is determined according to the merge mode.

When determining to perform motion estimation in Step SI_1 (Yes in Step SI_2), inter predictor **126** derives the final MV for the current block by estimating a surrounding region of the reference picture specified by the MV derived in Step SI_1 (Step SI_3). In other words, in this case, the MV for the current block is determined according to the DMVR.

FIG. 56 is a conceptual diagram for illustrating another example of DMVR for determining an MV.

First, in the merge mode for example, MV candidates (L0 and L1) are selected for the current block. A reference pixel is identified from a first reference picture (L0) which is an encoded picture in the L0 list according to the MV candidate (L0). Likewise, a reference pixel is identified from a second reference picture (L1) which is an encoded picture in the L1 list according to the MV candidate (L1). A template is generated by calculating an average of these reference pixels.

Next, each of the surrounding regions of MV candidates of the first reference picture (L0) and the second reference picture (L1) are estimated using the template, and the MV which yields the smallest cost is determined to be the final MV. It is to be noted that the cost may be calculated, for example, using a difference value between each of the pixel values in the template and a corresponding one of the pixel values in the estimation region, the values of MV candidates, etc.

Exactly the same processes described here do not always need to be performed. Any process for enabling derivation of the final MV by estimation in surrounding regions of MV candidates may be used.

FIG. 57 is a conceptual diagram for illustrating another example of DMVR for determining an MV. Unlike the example of DMVR illustrated in FIG. 56, in the example illustrated in FIG. 57, costs are calculated without generating any template.

First, inter predictor **126** estimates a surrounding region of a reference block included in each of reference pictures in the L0 list and L1 list, based on an initial MV which is an MV candidate obtained from each MV candidate list. For example, as illustrated in FIG. 57, the initial MV corresponding to the reference block in the L0 list is InitMV_L0, and the initial MV corresponding to the reference block in the L1 list is InitMV_L1. In motion estimation, inter predictor **126** firstly sets a search position for the reference picture in the L0 list. Based on the position indicated by the vector difference indicating the search position to be set, specifically, the initial MV (that is, InitMV_L0), the vector difference to the search position is MVd_L0. Inter predictor **126** then determines the estimation position in the reference picture in the L1 list. This search position is indicated by the vector difference to the search position from the position indicated by the initial MV (that is, InitMV_L1). More specifically, inter predictor **126** determines the vector difference as MVd_L1 by mirroring of MVd_L0. In other words, inter predictor **126** determines the position which is symmetrical with respect to the position indicated by the

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initial MV to be the search position in each reference picture in the L0 list and the L1 list. Inter predictor 126 calculates, for each search position, the total sum of the absolute differences (SADs) between values of pixels at search positions in blocks as a cost, and finds out the search position that yields the smallest cost.

FIG. 58A is a diagram illustrating one example of motion estimation in DMVR, and FIG. 58B is a flow chart illustrating one example of the motion estimation.

First, in Step 1, inter predictor 126 calculates the cost between the search position (also referred to as a starting point) indicated by the initial MV and eight surrounding search positions. Inter predictor 126 then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at the search position other than the starting point is the smallest, inter predictor 126 changes a target to the search position at which the smallest cost is obtained, and performs the process in Step 2. When the cost at the starting point is the smallest, inter predictor 126 skips the process in Step 2 and performs the process in Step 3.

In Step 2, inter predictor 126 performs the search similar to the process in Step 1, regarding, as a new starting point, the search position after the target change according to the result of the process in Step 1. Inter predictor 126 then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at the search position other than the starting point is the smallest, inter predictor 126 performs the process in Step 4. When the cost at the starting point is the smallest, inter predictor 126 performs the process in Step 3.

In Step 4, inter predictor 126 regards the search position at the starting point as the final search position, and determines the difference between the position indicated by the initial MV and the final search position to be a vector difference.

In Step 3, inter predictor 126 determines the pixel position at sub-pixel accuracy at which the smallest cost is obtained, based on the costs at the four points located at upper, lower, left, and right positions with respect to the starting point in Step 1 or Step 2, and regards the pixel position as the final search position. The pixel position at the sub-pixel accuracy is determined by performing weighted addition of each of the four upper, lower, left, and right vectors ((0, 1), (0, -1), (-1, 0), and (1, 0)), using, as a weight, the cost at a corresponding one of the four search positions. Inter predictor 126 then determines the difference between the position indicated by the initial MV and the final search position to be the vector difference.

[Motion Compensation >BIO/OBMC/LIC]

Motion compensation involves a mode for generating a prediction image, and correcting the prediction image. The mode is, for example, BIO, OBMC, and LIC to be described later.

FIG. 59 is a flow chart illustrating one example of generation of a prediction image.

Inter predictor 126 generates a prediction image (Step Sm_1), and corrects the prediction image according to any of the modes described above (Step Sm_2).

FIG. 60 is a flow chart illustrating another example of generation of a prediction image.

Inter predictor 126 derives an MV of a current block (Step Sn_1). Next, inter predictor 126 generates a prediction image using the MV (Step Sn_2), and determines whether to perform a correction process (Step Sn_3). Here, when determining to perform a correction process (Yes in Step

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Sn_3), inter predictor 126 generates the final prediction image by correcting the prediction image (Step Sn_4). It is to be noted that, in LIC described later, luminance and chrominance may be corrected in Step Sn_4. When determining not to perform a correction process (No in Step Sn_3), inter predictor 126 outputs the prediction image as the final prediction image without correcting the prediction image (Step Sn_5).

[Motion Compensation >OBMC]

It is to be noted that an inter prediction image may be generated using motion information for a neighboring block in addition to motion information for the current block obtained by motion estimation. More specifically, an inter prediction image may be generated for each sub-block in a current block by performing weighted addition of a prediction image based on the motion information obtained by motion estimation (in a reference picture) and a prediction image based on the motion information of the neighboring block (in the current picture). Such inter prediction (motion compensation) is also referred to as overlapped block motion compensation (OBMC) or an OBMC mode.

In OBMC mode, information indicating a sub-block size for OBMC (referred to as, for example, an OBMC block size) may be signaled at the sequence level. Moreover, information indicating whether to apply the OBMC mode (referred to as, for example, an OBMC flag) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the sequence level and CU level, and may be performed at another level (for example, at the picture level, slice level, brick level, CTU level, or sub-block level).

The OBMC mode will be described in further detail. FIGS. 61 and 62 are a flow chart and a conceptual diagram for illustrating an outline of a prediction image correction process performed by OBMC.

First, as illustrated in FIG. 62, a prediction image (Pred) by normal motion compensation is obtained using an MV assigned to a current block. In FIG. 62, the arrow "MV" points a reference picture, and indicates what the current block of the current picture refers to in order to obtain the prediction image.

Next, a prediction image (Pred_L) is obtained by applying a motion vector (MV_L) which has been already derived for the encoded block neighboring to the left of the current block to the current block (re-using the motion vector for the current block). The motion vector (MV_L) is indicated by an arrow "MV_L" indicating a reference picture from a current block. A first correction of a prediction image is performed by overlapping two prediction images Pred and Pred_L. This provides an effect of blending the boundary between neighboring blocks.

Likewise, a prediction image (Pred_U) is obtained by applying an MV (MV_U) which has been already derived for the encoded block neighboring above the current block to the current block (re-using the MV for the current block). The MV (MV_U) is indicated by an arrow "MV_U" indicating a reference picture from a current block. A second correction of a prediction image is performed by overlapping the prediction image Pred_U to the prediction images (for example, Pred and Pred_L) on which the first correction has been performed. This provides an effect of blending the boundary between neighboring blocks. The prediction image obtained by the second correction is the one in which the boundary between the neighboring blocks has been blended (smoothed), and thus is the final prediction image of the current block.

Although the above example is a two-path correction method using left and upper neighboring blocks, it is to be noted that the correction method may be three- or more-path correction method using also the right neighboring block and/or the lower neighboring block.

It is to be noted that the region in which such overlapping is performed may be only part of a region near a block boundary instead of the pixel region of the entire block.

It is to be noted that the prediction image correction process according to OBMC for obtaining one prediction image Pred from one reference picture by overlapping additional prediction images Pred_L and Pred_U has been described above. However, when a prediction image is corrected based on a plurality of reference images, a similar process may be applied to each of the plurality of reference pictures. In such a case, after corrected prediction images are obtained from the respective reference pictures by performing OBMC image correction based on the plurality of reference pictures, the obtained corrected prediction images are further overlapped to obtain the final prediction image.

It is to be noted that, in OBMC, a current block unit may be a PU or a sub-block unit obtained by further splitting the PU.

One example of a method for determining whether to apply OBMC is a method for using an obmc_flag which is a signal indicating whether to apply OBMC. As one specific example, encoder 100 may determine whether the current block belongs to a region having complicated motion. Encoder 100 sets the obmc_flag to a value of "1" when the block belongs to a region having complicated motion and applies OBMC when encoding, and sets the obmc_flag to a value of "0" when the block does not belong to a region having complicated motion and encodes the block without applying OBMC. Decoder 200 switches between application and non-application of OBMC by decoding the obmc_flag written in a stream.

[Motion Compensation >BIO]

Next, an MV derivation method is described. First, a mode for deriving an MV based on a model assuming uniform linear motion is described. This mode is also referred to as a bi-directional optical flow (BIO) mode. In addition, this bi-directional optical flow may be written as BDOF instead of BIO.

FIG. 63 is a diagram for illustrating a model assuming uniform linear motion. In FIG. 63, (vx, vy) indicates a velocity vector, and τ_0 and τ_1 indicate temporal distances between a current picture (Cur Pic) and two reference pictures (Ref₀, Ref₁). (MVx₀, Mvy₀) indicates an MV corresponding to reference picture Ref₀, and (MVx₁, Mvy₁) indicates an MV corresponding to reference picture Ref₁.

Here, under the assumption of uniform linear motion exhibited by a velocity vector (vx, vy), (MVx₀, Mvy₀) and (MVx₁, Mvy₁) are represented as (vx τ_0 , vy τ_0) and (-vx τ_1 , -vy τ_1), respectively, and the following optical flow equation (2) is given.

[MATH. 3]

$$\partial I^{(k)} / \partial t + v_x \partial I^{(k)} / \partial x + v_y \partial I^{(k)} / \partial y = 0 \quad (2)$$

Here, I(k) denotes a luma value from reference image k (k=0, 1) after motion compensation. This optical flow equation shows that the sum of (i) the time derivative of the luma value, (ii) the product of the horizontal velocity and the horizontal component of the spatial gradient of a reference image, and (iii) the product of the vertical velocity and the vertical component of the spatial gradient of a reference image is equal to zero. A motion vector of each block

obtained from, for example, an MV candidate list may be corrected in units of a pixel, based on a combination of the optical flow equation and Hermite interpolation.

It is to be noted that a motion vector may be derived on the decoder 200 side using a method other than deriving a motion vector based on a model assuming uniform linear motion. For example, a motion vector may be derived in units of a sub-block based on MVs of a plurality of neighboring blocks.

FIG. 64 is a flow chart illustrating one example of inter prediction according to BIO. FIG. 65 is a diagram illustrating one example of a configuration of inter predictor 126 which performs inter prediction according to BIO.

As illustrated in FIG. 65, inter predictor 126 includes, for example, memory 126a, interpolated image deriver 126b, gradient image deriver 126c, optical flow deriver 126d, correction value deriver 126e, and prediction image corrector 126f. It is to be noted that memory 126a may be frame memory 122.

Inter predictor 126 derives two motion vectors (M0, M1), using two reference pictures (Ref0, Ref1) different from the picture (Cur Pic) including a current block. Inter predictor 126 then derives a prediction image for the current block using the two motion vectors (M0, M1) (Step Sy_1). It is to be noted that motion vector M0 is motion vector (MVx₀, Mvy₀) corresponding to reference picture Ref₀, and motion vector M1 is motion vector (MVx₁, Mvy₁) corresponding to reference picture Ref₁.

Next, interpolated image deriver 126b derives interpolated image I⁰ for the current block, using motion vector M0 and reference picture L0 by referring to memory 126a. Next, interpolated image deriver 126b derives interpolated image I¹ for the current block, using motion vector M1 and reference picture L1 by referring to memory 126a (Step Sy_2). Here, interpolated image I⁰ is an image included in reference picture Ref₀ and to be derived for the current block, and interpolated image I¹ is an image included in reference picture Ref₁ and to be derived for the current block. Each of interpolated image I⁰ and interpolated image I¹ may be the same in size as the current block. Alternatively, each of interpolated image I⁰ and interpolated image I¹ may be an image larger than the current block. Furthermore, interpolated image I⁰ and interpolated image I¹ may include a prediction image obtained by using motion vectors (M0, M1) and reference pictures (L0, L1) and applying a motion compensation filter.

In addition, gradient image deriver 126c derives gradient images (Ix⁰, Ix¹, Iy⁰, Iy¹) of the current block, from interpolated image I⁰ and interpolated image I¹. It is to be noted that the gradient images in the horizontal direction are (Ix⁰, Ix¹), and the gradient images in the vertical direction are (Iy⁰, Iy¹). Gradient image deriver 126c may derive each gradient image by, for example, applying a gradient filter to the interpolated images. It is only necessary that a gradient image indicate the amount of spatial change in pixel value along the horizontal direction or the vertical direction.

Next, optical flow deriver 126d derives, for each sub-block of the current block, an optical flow (vx, vy) which is a velocity vector, using the interpolated images (I⁰, I¹) and the gradient images (Ix⁰, Ix¹, Iy⁰, Iy¹). The optical flow indicates coefficients for correcting the amount of spatial pixel movement, and may be referred to as a local motion estimation value, a corrected motion vector, or a corrected weighting vector. As one example, a sub-block may be 4×4 pixel sub-CU. It is to be noted that the optical flow derivation may be performed for each pixel unit, or the like, instead of being performed for each sub-block.

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Next, inter predictor **126** corrects a prediction image for the current block using the optical flow (vx, vy). For example, correction value derivator **126e** derives a correction value for the value of a pixel included in a current block, using the optical flow (vx, vy) (Step Sy_5). Prediction image corrector **126f** may then correct the prediction image for the current block using the correction value (Step Sy_6). It is to be noted that the correction value may be derived in units of a pixel, or may be derived in units of a plurality of pixels or in units of a sub-block.

It is to be noted that the BIO process flow is not limited to the process disclosed in FIG. 64. Only part of the processes disclosed in FIG. 64 may be performed, or a different process may be added or used as a replacement, or the processes may be executed in a different processing order.

[Motion Compensation >LIC]

Next, one example of a mode for generating a prediction image (prediction) using a local illumination compensation (LIC) is described.

FIG. 66A is a diagram for illustrating one example of a prediction image generation method using a luminance correction process performed by LIC. FIG. 66B is a flow chart illustrating one example of a prediction image generation method using the LIC.

First, inter predictor **126** derives an MV from an encoded reference picture, and obtains a reference image corresponding to the current block (Step Sz_1).

Next, inter predictor **126** extracts, for the current block, information indicating how the luma value has changed between the current block and the reference picture (Step Sz_2). This extraction is performed based on the luma pixel values of the encoded left neighboring reference region (surrounding reference region) and the encoded upper neighboring reference region (surrounding reference region) in the current picture, and the luma pixel values at the corresponding positions in the reference picture specified by the derived MVs. Inter predictor **126** calculates a luminance correction parameter, using the information indicating how the luma value has changed (Step Sz_3).

Inter predictor **126** generates a prediction image for the current block by performing a luminance correction process in which the luminance correction parameter is applied to the reference image in the reference picture specified by the MV (Step Sz_4). In other words, the prediction image which is the reference image in the reference picture specified by the MV is subjected to the correction based on the luminance correction parameter. In this correction, luminance may be corrected, or chrominance may be corrected. In other words, a chrominance correction parameter may be calculated using information indicating how chrominance has changed, and a chrominance correction process may be performed.

It is to be noted that the shape of the surrounding reference region illustrated in FIG. 66A is one example; another shape may be used.

Moreover, although the process in which a prediction image is generated from a single reference picture has been described here, cases in which a prediction image is generated from a plurality of reference pictures can be described in the same manner. The prediction image may be generated after performing a luminance correction process of the reference images obtained from the reference pictures in the same manner as described above.

One example of a method for determining whether to apply LIC is a method for using a lic_flag which is a signal indicating whether to apply the LIC. As one specific example, encoder **100** determines whether the current block

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belongs to a region having a luminance change. Encoder **100** sets the lic_flag to a value of "1" when the block belongs to a region having a luminance change and applies LIC when encoding, and sets the lic_flag to a value of "0" when the block does not belong to a region having a luminance change and performs encoding without applying LIC. Decoder **200** may decode the lic_flag written in the stream and decode the current block by switching between application and non-application of LIC in accordance with the flag value.

One example of a different method of determining whether to apply a LIC process is a determining method in accordance with whether a LIC process has been applied to a surrounding block. As one specific example, when a current block has been processed in merge mode, inter predictor **126** determines whether an encoded surrounding block selected in MV derivation in merge mode has been encoded using LIC. Inter predictor **126** performs encoding by switching between application and non-application of LIC according to the result. It is to be noted that, also in this example, the same processes are applied to processes at the decoder **200** side.

The luminance correction (LIC) process has been described with reference to FIGS. 66A and 66B, and is further described below.

First, inter predictor **126** derives an MV for obtaining a reference image corresponding to a current block from a reference picture which is an encoded picture.

Next, inter predictor **126** extracts information indicating how the luma value of the reference picture has been changed to the luma value of the current picture, using the luma pixel values of encoded surrounding reference regions which neighbor to the left of and above the current block and the luma pixel values in the corresponding positions in the reference pictures specified by MVs, and calculates a luminance correction parameter. For example, it is assumed that the luma pixel value of a given pixel in the surrounding reference region in the current picture is p0, and that the luma pixel value of the pixel corresponding to the given pixel in the surrounding reference region in the reference picture is p1. Inter predictor **126** calculates coefficients A and B for optimizing $A \times p1 + B = p0$ as the luminance correction parameter for a plurality of pixels in the surrounding reference region.

Next, inter predictor **126** performs a luminance correction process using the luminance correction parameter for the reference image in the reference picture specified by the MV, to generate a prediction image for the current block. For example, it is assumed that the luma pixel value in the reference image is p2, and that the luminance-corrected luma pixel value of the prediction image is p3. Inter predictor **126** generates the prediction image after being subjected to the luminance correction process by calculating $A \times p2 + B = p3$ for each of the pixels in the reference image.

It is to be noted that part of the surrounding reference regions illustrated in FIG. 66A may be used. For example, a region having a determined number of pixels extracted from each of upper neighboring pixels and left neighboring pixels may be used as a surrounding reference region. In addition, the surrounding reference region is not limited to a region which neighbors the current block, and may be a region which does not neighbor the current block. In the example illustrated in FIG. 66A, the surrounding reference region in the reference picture may be a region specified by another MV in a current picture, from a surrounding reference region in the current picture. For example, the other MV may be an MV in a surrounding reference region in the current picture.

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Although operations performed by encoder **100** have been described here, it is to be noted that decoder **200** performs similar operations.

It is to be noted that LIC may be applied not only to luma but also to chroma. At this time, a correction parameter may be derived individually for each of Y, Cb, and Cr, or a common correction parameter may be used for any of Y, Cb, and Cr.

In addition, the LIC process may be applied in units of a sub-block. For example, a correction parameter may be derived using a surrounding reference region in a current sub-block and a surrounding reference region in a reference sub-block in a reference picture specified by an MV of the current sub-block.

[Prediction Controller]

Prediction controller **128** selects one of an intra prediction image (an image or a signal output from intra predictor **124**) and an inter prediction image (an image or a signal output from inter predictor **126**), and outputs the selected prediction image to subtractor **104** and adder **116**.

[Prediction Parameter Generator]

Prediction parameter generator **130** may output information related to intra prediction, inter prediction, selection of a prediction image in prediction controller **128**, etc. as a prediction parameter to entropy encoder **110**. Entropy encoder **110** may generate a stream, based on the prediction parameter which is input from prediction parameter generator **130** and quantized coefficients which are input from quantizer **108**. The prediction parameter may be used in decoder **200**. Decoder **200** may receive and decode the stream, and perform the same processes as the prediction processes performed by intra predictor **124**, inter predictor **126**, and prediction controller **128**. The prediction parameter may include (i) a selection prediction signal (for example, an MV, a prediction type, or a prediction mode used by intra predictor **124** or inter predictor **126**), or (ii) an optional index, a flag, or a value which is based on a prediction process performed in each of intra predictor **124**, inter predictor **126**, and prediction controller **128**, or which indicates the prediction process.

[Decoder]

Next, decoder **200** capable of decoding a stream output from encoder **100** described above is described. FIG. **67** is a block diagram illustrating a configuration of decoder **200** according to this embodiment. Decoder **200** is an apparatus which decodes a stream that is an encoded image in units of a block.

As illustrated in FIG. **67**, decoder **200** includes entropy decoder **202**, inverse quantizer **204**, inverse transformer **206**, adder **208**, block memory **210**, loop filter **212**, frame memory **214**, intra predictor **216**, inter predictor **218**, prediction controller **220**, prediction parameter generator **222**, and splitting determiner **224**. It is to be noted that intra predictor **216** and inter predictor **218** are configured as part of a prediction executor.

[Mounting Example of Decoder]

FIG. **68** is a block diagram illustrating a mounting example of decoder **200**. Decoder **200** includes processor **b1** and memory **b2**. For example, the plurality of constituent elements of decoder **200** illustrated in FIG. **67** are mounted on processor **b1** and memory **b2** illustrated in FIG. **68**.

Processor **b1** is circuitry which performs information processing and is accessible to memory **b2**. For example, processor **b1** is a dedicated or general electronic circuit which decodes a stream. Processor **b1** may be a processor such as a CPU. In addition, processor **b1** may be an aggregate of a plurality of electronic circuits. In addition, for

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example, processor **b1** may take the roles of two or more constituent elements other than a constituent element for storing information out of the plurality of constituent elements of decoder **200** illustrated in FIG. **67**, etc.

Memory **b2** is dedicated or general memory for storing information that is used by processor **b1** to decode a stream. Memory **b2** may be electronic circuitry, and may be connected to processor **b1**. In addition, memory **b2** may be included in processor **b1**. In addition, memory **b2** may be an aggregate of a plurality of electronic circuits. In addition, memory **b2** may be a magnetic disc, an optical disc, or the like, or may be represented as a storage, a medium, or the like. In addition, memory **b2** may be non-volatile memory, or volatile memory.

For example, memory **b2** may store an image or a stream. In addition, memory **b2** may store a program for causing processor **b1** to decode a stream.

In addition, for example, memory **b2** may take the roles of two or more constituent elements for storing information out of the plurality of constituent elements of decoder **200** illustrated in FIG. **67**, etc. More specifically, memory **b2** may take the roles of block memory **210** and frame memory **214** illustrated in FIG. **67**. More specifically, memory **b2** may store a reconstructed image (specifically, a reconstructed block, a reconstructed picture, or the like).

It is to be noted that, in decoder **200**, not all of the plurality of constituent elements illustrated in FIG. **67**, etc. may be implemented, and not all the processes described above may be performed. Part of the constituent elements indicated in FIG. **67**, etc. may be included in another device, or part of the processes described above may be performed by another device.

Hereinafter, an overall flow of the processes performed by decoder **200** is described, and then each of the constituent elements included in decoder **200** is described. It is to be noted that, some of the constituent elements included in decoder **200** perform the same processes as performed by some of the constituent elements included in encoder **100**, and thus the same processes are not repeatedly described in detail. For example, inverse quantizer **204**, inverse transformer **206**, adder **208**, block memory **210**, frame memory **214**, intra predictor **216**, inter predictor **218**, prediction controller **220**, and loop filter **212** included in decoder **200** perform similar processes as performed by inverse quantizer **112**, inverse transformer **114**, adder **116**, block memory **118**, frame memory **122**, intra predictor **124**, inter predictor **126**, prediction controller **128**, and loop filter **120** included in encoder **100**, respectively.

[Overall Flow of Decoding Process]

FIG. **69** is a flow chart illustrating one example of an overall decoding process performed by decoder **200**.

First, splitting determiner **224** in decoder **200** determines a splitting pattern of each of a plurality of fixed-size blocks (128×128 pixels) included in a picture, based on a parameter which is input from entropy decoder **202** (Step Sp_1). This splitting pattern is a splitting pattern selected by encoder **100**. Decoder **200** then performs processes of Steps Sp_2 to Sp_6 for each of a plurality of blocks of the splitting pattern.

Entropy decoder **202** decodes (specifically, entropy decodes) encoded quantized coefficients and a prediction parameter of a current block (Step Sp_2).

Next, inverse quantizer **204** performs inverse quantization of the plurality of quantized coefficients and inverse transformer **206** performs inverse transform of the result, to restore prediction residuals of the current block (Step Sp_3).

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Next, the prediction executor including all or part of intra predictor **216**, inter predictor **218**, and prediction controller **220** generates a prediction image of the current block (Step Sp_4).

Next, adder **208** adds the prediction image to a prediction residual to generate a reconstructed image (also referred to as a decoded image block) of the current block (Step Sp_5).

When the reconstructed image is generated, loop filter **212** performs filtering of the reconstructed image (Step Sp_6).

Decoder **200** then determines whether decoding of the entire picture has been finished (Step Sp_7). When determining that the decoding has not yet been finished (No in Step Sp_7), decoder **200** repeatedly executes the processes starting with Step Sp_1.

It is to be noted that the processes of these Steps Sp_1 to Sp_7 may be performed sequentially by decoder **200**, or two or more of the processes may be performed in parallel. The processing order of the two or more of the processes may be modified.

[Splitting Determiner]

FIG. 70 is a diagram illustrating a relationship between splitting determiner **224** and other constituent elements. Splitting determiner **224** may perform the following processes as examples.

For example, splitting determiner **224** collects block information from block memory **210** or frame memory **214**, and furthermore obtains a parameter from entropy decoder **202**. Splitting determiner **224** may then determine the splitting pattern of a fixed-size block, based on the block information and the parameter. Splitting determiner **224** may then output information indicating the determined splitting pattern to inverse transformer **206**, intra predictor **216**, and inter predictor **218**. Inverse transformer **206** may perform inverse transform of transform coefficients, based on the splitting pattern indicated by the information from splitting determiner **224**. Intra predictor **216** and inter predictor **218** may generate a prediction image, based on the splitting pattern indicated by the information from splitting determiner **224**.

[Entropy Decoder]

FIG. 71 is a block diagram illustrating one example of a configuration of entropy decoder **202**.

Entropy decoder **202** generates quantized coefficients, a prediction parameter, and a parameter related to a splitting pattern, by entropy decoding the stream. For example, CABAC is used in the entropy decoding. More specifically, entropy decoder **202** includes, for example, binary arithmetic decoder **202a**, context controller **202b**, and debinarizer **202c**. Binary arithmetic decoder **202a** arithmetically decodes the stream using a context value derived by context controller **202b** to a binary signal. Context controller **202b** derives a context value according to a feature or a surrounding state of a syntax element, that is, an occurrence probability of a binary signal, in the same manner as performed by context controller **110b** of encoder **100**. Debinarizer **202c** performs debinarization for transforming the binary signal output from binary arithmetic decoder **202a** to a multi-level signal indicating quantized coefficients as described above. This binarization is performed according to the binarization method described above.

With this, entropy decoder **202** outputs quantized coefficients of each block to inverse quantizer **204**. Entropy decoder **202** may output a prediction parameter included in a stream (see FIG. 1) to intra predictor **216**, inter predictor **218**, and prediction controller **220**. Intra predictor **216**, inter predictor **218**, and prediction controller **220** are capable of

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executing the same prediction processes as those performed by intra predictor **124**, inter predictor **126**, and prediction controller **128** at the encoder **100** side.

[Entropy Decoder] FIG. 72 is a diagram illustrating a flow of CABAC in entropy decoder **202**.

First, initialization is performed in CABAC in entropy decoder **202**. In the initialization, initialization in binary arithmetic decoder **202a** and setting of an initial context value are performed. Binary arithmetic decoder **202a** and debinarizer **202c** then execute arithmetic decoding and debinarization of, for example, encoded data of a CTU. At this time, context controller **202b** updates the context value each time arithmetic decoding is performed. Context controller **202b** then saves the context value as a post process. The saved context value is used, for example, to initialize the context value for the next CTU.

[Inverse Quantizer]

Inverse quantizer **204** inverse quantizes quantized coefficients of a current block which are inputs from entropy decoder **202**. More specifically, inverse quantizer **204** inverse quantizes the quantized coefficients of the current block, based on quantization parameters corresponding to the quantized coefficients. Inverse quantizer **204** then outputs the inverse quantized transform coefficients (that are transform coefficients) of the current block to inverse transformer **206**.

FIG. 73 is a block diagram illustrating one example of a configuration of inverse quantizer **204**.

Inverse quantizer **204** includes, for example, quantization parameter generator **204a**, predicted quantization parameter generator **204b**, quantization parameter storage **204d**, and inverse quantization executor **204e**.

FIG. 74 is a flow chart illustrating one example of inverse quantization performed by inverse quantizer **204**.

Inverse quantizer **204** may perform an inverse quantization process as one example for each CU based on the flow illustrated in FIG. 74. More specifically, quantization parameter generator **204a** determines whether to perform inverse quantization (Step Sv_11). Here, when determining to perform inverse quantization (Yes in Step Sv_11), quantization parameter generator **204a** obtains a difference quantization parameter for the current block from entropy decoder **202** (Step Sv_12).

Next, predicted quantization parameter generator **204b** then obtains a quantization parameter for a processing unit different from the current block from quantization parameter storage **204d** (Step Sv_13). Predicted quantization parameter generator **204b** generates a predicted quantization parameter of the current block based on the obtained quantization parameter (Step Sv_14).

Quantization parameter generator **204a** then adds the difference quantization parameter for the current block obtained from entropy decoder **202** and the predicted quantization parameter for the current block generated by predicted quantization parameter generator **204b** (Step Sv_15). This addition generates a quantization parameter for the current block. In addition, quantization parameter generator **204a** stores the quantization parameter for the current block in quantization parameter storage **204d** (Step Sv_16).

Next, inverse quantization executor **204e** inverse quantizes the quantized coefficients of the current block into transform coefficients, using the quantization parameter generated in Step Sv_15 (Step Sv_17).

It is to be noted that the difference quantization parameter may be decoded at the bit sequence level, picture level, slice level, brick level, or CTU level. In addition, the initial value of the quantization parameter may be decoded at the

sequence level, picture level, slice level, brick level, or CTU level. At this time, the quantization parameter may be generated using the initial value of the quantization parameter and the difference quantization parameter.

It is to be noted that inverse quantizer **204** may include a plurality of inverse quantizers, and may inverse quantize the quantized coefficients using an inverse quantization method selected from a plurality of inverse quantization methods. [Inverse Transformer]

Inverse transformer **206** restores prediction residuals by inverse transforming the transform coefficients which are inputs from inverse quantizer **204**.

For example, when information parsed from a stream indicates that EMT or AMT is to be applied (for example, when an AMT flag is true), inverse transformer **206** inverse transforms the transform coefficients of the current block based on information indicating the parsed transform type.

Moreover, for example, when information parsed from a stream indicates that NSST is to be applied, inverse transformer **206** applies a secondary inverse transform to the transform coefficients.

FIG. **75** is a flow chart illustrating one example of a process performed by inverse transformer **206**.

For example, inverse transformer **206** determines whether information indicating that no orthogonal transform is performed is present in a stream (Step St_11). Here, when determining that no such information is present (No in Step St_11), inverse transformer **206** obtains information indicating the transform type decoded by entropy decoder **202** (Step St_12). Next, based on the information, inverse transformer **206** determines the transform type used for the orthogonal transform in encoder **100** (Step St_13). Inverse transformer **206** then performs inverse orthogonal transform using the determined transform type (Step St_14).

FIG. **76** is a flow chart illustrating another example of a process performed by inverse transformer **206**.

For example, inverse transformer **206** determines whether a transform size is smaller than or equal to a predetermined value (Step Su_11). Here, when determining that the transform size is smaller than or equal to a predetermined value (Yes in Step Su_11), inverse transformer **206** obtains, from entropy decoder **202**, information indicating which transform type has been used by encoder **100** among at least one transform type included in the first transform type group (Step Su_12). It is to be noted that such information is decoded by entropy decoder **202** and output to inverse transformer **206**.

Based on the information, inverse transformer **206** determines the transform type used for the orthogonal transform in encoder **100** (Step Su_13). Inverse transformer **206** then inverse orthogonal transforms the transform coefficients of the current block using the determined transform type (Step Su_14). When determining that a transform size is not smaller than or equal to the predetermined value (No in Step Su_11), inverse transformer **206** inverse transforms the transform coefficients of the current block using the second transform type group (Step Su_15).

It is to be noted that the inverse orthogonal transform by inverse transformer **206** may be performed according to the flow illustrated in FIG. **75** or FIG. **76** for each TU as one example. In addition, inverse orthogonal transform may be performed by using a predefined transform type without decoding information indicating a transform type used for orthogonal transform. In addition, the transform type is specifically DST7, DCT8, or the like. In inverse orthogonal transform, an inverse transform basis function corresponding to the transform type is used.

[Adder]

Adder **208** reconstructs the current block by adding a prediction residual which is an input from inverse transformer **206** and a prediction image which is an input from prediction controller **220**. In other words, a reconstructed image of the current block is generated. Adder **208** then outputs the reconstructed image of the current block to block memory **210** and loop filter **212**.

[Block Memory]

Block memory **210** is storage for storing a block which is included in a current picture and is referred to in intra prediction. More specifically, block memory **210** stores a reconstructed image output from adder **208**.

[Loop Filter]

Loop filter **212** applies a loop filter to the reconstructed image generated by adder **208**, and outputs the filtered reconstructed image to frame memory **214** and a display device, etc.

When information indicating ON or OFF of an ALF parsed from a stream indicates that an ALF is ON, one filter from among a plurality of filters is selected based on the direction and activity of local gradients, and the selected filter is applied to the reconstructed image.

FIG. **77** is a block diagram illustrating one example of a configuration of loop filter **212**. It is to be noted that loop filter **212** has a configuration similar to the configuration of loop filter **120** of encoder **100**.

For example, as illustrated in FIG. **77**, loop filter **212** includes deblocking filter executor **212a**, SAO executor **212b**, and ALF executor **212c**. Deblocking filter executor **212a** performs a deblocking filter process of the reconstructed image. SAO executor **212b** performs a SAO process of the reconstructed image after being subjected to the deblocking filter process. ALF executor **212c** performs an ALF process of the reconstructed image after being subjected to the SAO process. It is to be noted that loop filter **212** does not always need to include all the constituent elements disclosed in FIG. **77**, and may include only part of the constituent elements. In addition, loop filter **212** may be configured to perform the above processes in a processing order different from the one disclosed in FIG. **77**.

[Frame Memory]

Frame memory **214** is, for example, storage for storing reference pictures for use in inter prediction, and is also referred to as a frame buffer. More specifically, frame memory **214** stores a reconstructed image filtered by loop filter **212**.

[Predictor (Intra Predictor, Inter Predictor, Prediction Controller)]

FIG. **78** is a flow chart illustrating one example of a process performed by a predictor of decoder **200**. It is to be noted that the prediction executor includes all or part of the following constituent elements: intra predictor **216**; inter predictor **218**; and prediction controller **220**. The prediction executor includes, for example, intra predictor **216** and inter predictor **218**.

The predictor generates a prediction image of a current block (Step Sq_1). This prediction image is also referred to as a prediction signal or a prediction block. It is to be noted that the prediction signal is, for example, an intra prediction signal or an inter prediction signal. More specifically, the predictor generates the prediction image of the current block using a reconstructed image which has been already obtained for another block through generation of a prediction image, restoration of a prediction residual, and addition of a prediction image. The predictor of decoder **200** generates the same prediction image as the prediction image

generated by the predictor of encoder **100**. In other words, the prediction images are generated according to a method common between the predictors or mutually corresponding methods.

The reconstructed image may be, for example, an image in a reference picture, or an image of a decoded block (that is, the other block described above) in a current picture which is the picture including the current block. The decoded block in the current picture is, for example, a neighboring block of the current block.

FIG. **79** is a flow chart illustrating another example of a process performed by the predictor of decoder **200**.

The predictor determines either a method or a mode for generating a prediction image (Step **Sr_1**). For example, the method or mode may be determined based on, for example, a prediction parameter, etc.

When determining a first method as a mode for generating a prediction image, the predictor generates a prediction image according to the first method (Step **Sr_2a**). When determining a second method as a mode for generating a prediction image, the predictor generates a prediction image according to the second method (Step **Sr_2b**). When determining a third method as a mode for generating a prediction image, the predictor generates a prediction image according to the third method (Step **Sr_2c**).

The first method, the second method, and the third method may be mutually different methods for generating a prediction image. Each of the first to third methods may be an inter prediction method, an intra prediction method, or another prediction method. The above-described reconstructed image may be used in these prediction methods.

FIG. **80A** and FIG. **80B** illustrate a flow chart illustrating another example of a process performed by a predictor of decoder **200**.

The predictor may perform a prediction process according to the flow illustrated in FIG. **80A** and FIG. **80B** as one example. It is to be noted that intra block copy illustrated in FIG. **80A** and FIG. **80B** is one mode which belongs to inter prediction, and in which a block included in a current picture is referred to as a reference image or a reference block. In other words, no picture different from the current picture is referred to in intra block copy. In addition, the PCM mode illustrated in FIG. **80A** is one mode which belongs to intra prediction, and in which no transform and quantization is performed.

[Intra Predictor]

Intra predictor **216** performs intra prediction by referring to a block in a current picture stored in block memory **210**, based on the intra prediction mode parsed from the stream, to generate a prediction image of a current block (that is, an intra prediction image). More specifically, intra predictor **216** performs intra prediction by referring to pixel values (for example, luma and/or chroma values) of a block or blocks neighboring the current block to generate an intra prediction image, and then outputs the intra prediction image to prediction controller **220**.

It is to be noted that when an intra prediction mode in which a luma block is referred to in intra prediction of a chroma block is selected, intra predictor **216** may predict the chroma component of the current block based on the luma component of the current block.

Moreover, when information parsed from a stream indicates that PDPC is to be applied, intra predictor **216** corrects intra predicted pixel values based on horizontal/vertical reference pixel gradients.

FIG. **81** is a diagram illustrating one example of a process performed by intra predictor **216** of decoder **200**.

Intra predictor **216** firstly determines whether an MPM flag indicating **1** is present in the stream (Step **Sw_11**). Here, when determining that the MPM flag indicating **1** is present (Yes in Step **Sw_11**), intra predictor **216** obtains, from entropy decoder **202**, information indicating the intra prediction mode selected in encoder **100** among MPMs (Step **Sw_12**). It is to be noted that such information is decoded by entropy decoder **202** and output to intra predictor **216**. Next, intra predictor **216** determines an MPM (Step **Sw_13**). MPMs include, for example, six intra prediction modes. Intra predictor **216** then determines the intra prediction mode which is included in a plurality of intra prediction modes included in the MPMs and is indicated by the information obtained in Step **Sw_12** (Step **Sw_14**).

When determining that no MPM flag indicating **1** is present (No in Step **Sw_11**), intra predictor **216** obtains information indicating the intra prediction mode selected in encoder **100** (Step **Sw_15**). In other words, intra predictor **216** obtains, from entropy decoder **202**, information indicating the intra prediction mode selected in encoder **100** from among at least one intra prediction mode which is not included in the MPMs. It is to be noted that such information is decoded by entropy decoder **202** and output to intra predictor **216**. Intra predictor **216** then determines the intra prediction mode which is not included in a plurality of intra prediction modes included in the MPMs and is indicated by the information obtained in Step **Sw_15** (Step **Sw_17**).

Intra predictor **216** generates a prediction image according to the intra prediction mode determined in Step **Sw_14** or Step **Sw_17** (Step **Sw_18**).

[Inter Predictor]

Inter predictor **218** predicts the current block by referring to a reference picture stored in frame memory **214**. Prediction is performed in units of a current block or a current sub-block in the current block. It is to be noted that the sub-block is included in the block and is a unit smaller than the block. The size of the sub-block may be 4×4 pixels, 8×8 pixels, or another size. The size of the sub-block may be switched for a unit such as a slice, brick, picture, etc.

For example, inter predictor **218** generates an inter prediction image of a current block or a current sub-block by performing motion compensation using motion information (for example, an MV) parsed from a stream (for example, a prediction parameter output from entropy decoder **202**), and outputs the inter prediction image to prediction controller **220**.

When the information parsed from the stream indicates that the OBMC mode is to be applied, inter predictor **218** generates the inter prediction image using motion information of a neighboring block in addition to motion information of the current block obtained through motion estimation.

Moreover, when the information parsed from the stream indicates that the FRUC mode is to be applied, inter predictor **218** derives motion information by performing motion estimation in accordance with the pattern matching method (bilateral matching or template matching) parsed from the stream. Inter predictor **218** then performs motion compensation (prediction) using the derived motion information.

Moreover, when the BIO mode is to be applied, inter predictor **218** derives an MV based on a model assuming uniform linear motion. In addition, when the information parsed from the stream indicates that the affine mode is to be applied, inter predictor **218** derives an MV for each sub-block, based on the MVs of a plurality of neighboring blocks.

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[MV Derivation Flow]

FIG. 82 is a flow chart illustrating one example of MV derivation in decoder 200.

Inter predictor 218 determines, for example, whether to decode motion information (for example, an MV). For example, inter predictor 218 may make the determination according to the prediction mode included in the stream, or may make the determination based on other information included in the stream. Here, when determining to decode motion information, inter predictor 218 derives an MV for a current block in a mode in which the motion information is decoded. When determining not to decode motion information, inter predictor 218 derives an MV in a mode in which no motion information is decoded.

Here, MV derivation modes include a normal inter mode, a normal merge mode, a FRUC mode, an affine mode, etc. which are described later. Modes in which motion information is decoded among the modes include the normal inter mode, the normal merge mode, the affine mode (specifically, an affine inter mode and an affine merge mode), etc. It is to be noted that motion information may include not only an MV but also MV predictor selection information which is described later. Modes in which no motion information is decoded include the FRUC mode, etc. Inter predictor 218 selects a mode for deriving an MV for the current block from the plurality of modes, and derives the MV for the current block using the selected mode.

FIG. 83 is a flow chart illustrating another example of MV derivation in decoder 200.

For example, inter predictor 218 may determine whether to decode an MV difference, that is for example, may make the determination according to the prediction mode included in the stream, or may make the determination based on other information included in the stream. Here, when determining to decode an MV difference, inter predictor 218 may derive an MV for a current block in a mode in which the MV difference is decoded. In this case, for example, the MV difference included in the stream is decoded as a prediction parameter.

When determining not to decode any MV difference, inter predictor 218 derives an MV in a mode in which no MV difference is decoded. In this case, no encoded MV difference is included in the stream.

Here, as described above, the MV derivation modes include the normal inter mode, the normal merge mode, the FRUC mode, the affine mode, etc. which are described later. Modes in which an MV difference is encoded among the modes include the normal inter mode and the affine mode (specifically, the affine inter mode), etc. Modes in which no MV difference is encoded include the FRUC mode, the normal merge mode, the affine mode (specifically, the affine merge mode), etc. Inter predictor 218 selects a mode for deriving an MV for the current block from the plurality of modes, and derives the MV for the current block using the selected mode.

[MV Derivation >Normal Inter Mode]

For example, when information parsed from a stream indicates that the normal inter mode is to be applied, inter predictor 218 derives an MV based on the information parsed from the stream and performs motion compensation (prediction) using the MV.

FIG. 84 is a flow chart illustrating an example of inter prediction by normal inter mode in decoder 200.

Inter predictor 218 of decoder 200 performs motion compensation for each block. At this time, first, inter predictor 218 obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of

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decoded blocks temporally or spatially surrounding the current block (Step Sg_11). In other words, inter predictor 218 generates an MV candidate list.

Next, inter predictor 218 extracts N (an integer of 2 or larger) MV candidates from the plurality of MV candidates obtained in Step Sg_11, as motion vector predictor candidates (also referred to as MV predictor candidates) according to the predetermined ranks in priority order (Step Sg_12). It is to be noted that the ranks in priority order are determined in advance for the respective N MV predictor candidates.

Next, inter predictor 218 decodes the MV predictor selection information from the input stream, and selects one MV predictor candidate from the N MV predictor candidates as the MV predictor for the current block using the decoded MV predictor selection information (Step Sg_13).

Next, inter predictor 218 decodes an MV difference from the input stream, and derives an MV for the current block by adding a difference value which is the decoded MV difference and the selected MV predictor (Step Sg_14).

Lastly, inter predictor 218 generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Sg_15). The processes in Steps Sg_11 to Sg_15 are executed on each block. For example, when the processes in Steps Sg_11 to Sg_15 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal inter mode finishes. For example, when the processes in Steps Sg_11 to Sg_15 are executed on each of all the blocks in the picture, inter prediction of the picture using the normal inter mode finishes. It is to be noted that not all the blocks included in the slice may be subjected to the processes in Steps Sg_11 to Sg_15, and inter prediction of the slice using the normal inter mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal inter mode may finish when the processes in Steps Sg_11 to Sg_15 are executed on part of the blocks in the picture.

[MV Derivation >Normal Merge Mode]

For example, when information parsed from a stream indicates that the normal merge mode is to be applied, inter predictor 218 derives an MV and performs motion compensation (prediction) using the MV.

FIG. 85 is a flow chart illustrating an example of inter prediction by normal merge mode in decoder 200.

At this time, first, inter predictor 218 obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of decoded blocks temporally or spatially surrounding the current block (Step Sh_11). In other words, inter predictor 218 generates an MV candidate list.

Next, inter predictor 218 selects one MV candidate from the plurality of MV candidates obtained in Step Sh_11, thereby deriving an MV for the current block (Step Sh_12). More specifically inter predictor 218 obtains MV selection information included as a prediction parameter in a stream, and selects the MV candidate identified by the MV selection information as the MV for the current block.

Lastly, inter predictor 218 generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Sh_13). The processes in Steps Sh_11 to Sh_13 are executed, for example, on each block. For example, when the processes in Steps Sh_11 to Sh_13 are executed on each of all the blocks in the slice, inter prediction of the slice using the normal merge mode finishes. In addition, when the processes in Steps Sh_11 to Sh_13 are

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executed on each of all the blocks in the picture, inter prediction of the picture using the normal merge mode finishes. It is to be noted that not all the blocks included in the slice are subjected to the processes in Steps Sh_11 to Sh_13, and inter prediction of the slice using the normal merge mode may finish when part of the blocks are subjected to the processes. Likewise, inter prediction of the picture using the normal merge mode may finish when the processes in Steps Sh_11 to Sh_13 are executed on part of the blocks in the picture.

[MV Derivation >FRUC Mode]

For example, when information parsed from a stream indicates that the FRUC mode is to be applied, inter predictor 218 derives an MV in the FRUC mode and performs motion compensation (prediction) using the MV. In this case, the motion information is derived at the decoder 200 side without being signaled from the encoder 100 side. For example, decoder 200 may derive the motion information by performing motion estimation. In this case, decoder 200 performs motion estimation without using any pixel value in a current block.

FIG. 86 is a flow chart illustrating an example of inter prediction by FRUC mode in decoder 200.

First, inter predictor 218 generates a list indicating MVs of decoded blocks spatially or temporally neighboring the current block by referring to the MVs as MV candidates (the list is an MV candidate list, and may be used also as an MV candidate list for normal merge mode (Step Si_11)). Next, a best MV candidate is selected from the plurality of MV candidates registered in the MV candidate list (Step Si_12). For example, inter predictor 218 calculates the evaluation value of each MV candidate included in the MV candidate list, and selects one of the MV candidates as the best MV candidate based on the evaluation values. Based on the selected best MV candidate, inter predictor 218 then derives an MV for the current block (Step Si_14). More specifically, for example, the selected best MV candidate is directly derived as the MV for the current block. In addition, for example, the MV for the current block may be derived using pattern matching in a surrounding region of a position which is included in a reference picture and corresponds to the selected best MV candidate. In other words, estimation using the pattern matching in a reference picture and the evaluation values may be performed in the surrounding region of the best MV candidate, and when there is an MV that yields a better evaluation value, the best MV candidate may be updated to the MV that yields the better evaluation value, and the updated MV may be determined as the final MV for the current block. Update to the MV that yields the better evaluation value may not be performed.

Lastly, inter predictor 218 generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Si_15). The processes in Steps Si_11 to Si_15 are executed, for example, on each block. For example, when the processes in Steps Si_11 to Si_15 are executed on each of all the blocks in the slice, inter prediction of the slice using the FRUC mode finishes. For example, when the processes in Steps Si_11 to Si_15 are executed on each of all the blocks in the picture, inter prediction of the picture using the FRUC mode finishes. Each sub-block may be processed similarly to the above-described case of processing each block.

[MV Derivation >Affine Merge Mode]

For example, when information parsed from a stream indicates that the affine merge mode is to be applied, inter

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predictor 218 derives an MV in the affine merge mode and performs motion compensation (prediction) using the MV.

FIG. 87 is a flow chart illustrating an example of inter prediction by the affine merge mode in decoder 200.

In the affine merge mode, first, inter predictor 218 derives MVs at respective control points for a current block (Step Sk_11). The control points are an upper-left corner point of the current block and an upper-right corner point of the current block as illustrated in FIG. 46A, or an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. 46B.

For example, when the MV derivation methods illustrated in FIGS. 47A to 47C are used, as illustrated in FIG. 47A, inter predictor 218 checks decoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) in this order, and identifies the first effective block decoded according to the affine mode.

Inter predictor 218 derives the MV at the control point using the identified first effective block decoded according to the affine mode. For example, when block A is identified and block A has two control points, as illustrated in FIG. 47B, inter predictor 218 calculates motion vector v_0 at the upper-left corner control point of the current block and motion vector v_1 at the upper-right corner control point of the current block by projecting motion vectors v_3 and v_4 at the upper-left corner and the upper-right corner of the decoded block including block A onto the current block. In this way, the MV at each control point is derived.

It is to be noted that, as illustrated in FIG. 49A, MVs at three control points may be calculated when block A is identified and block A has two control points, and that, as illustrated in FIG. 49B, MVs at two control points may be calculated when block A is identified and when block A has three control points.

In addition, when MV selection information is included as a prediction parameter in a stream, inter predictor 218 may derive the MV at each control point for the current block using the MV selection information.

Next, inter predictor 218 performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor 218 calculates an MV for each of the plurality of sub-blocks as an affine MV, using either two motion vectors v_0 and v_1 and the above expression (1A) or three motion vectors v_0 , v_1 , and v_2 and the above expression (1B) (Step Sk_12). Inter predictor 218 then performs motion compensation of the sub-blocks using these affine MVs and decoded reference pictures (Step Sk_13). When the processes in Steps Sk_12 and Sk_13 are executed for each of all the sub-blocks included in the current block, the inter prediction using the affine merge mode for the current block finishes. In other words, motion compensation of the current block is performed to generate a prediction image of the current block.

It is to be noted that the above-described MV candidate list may be generated in Step Sk_11. The MV candidate list may be, for example, a list including MV candidates derived using a plurality of MV derivation methods for each control point. The plurality of MV derivation methods may be any combination of the MV derivation methods illustrated in FIGS. 47A to 47C, the MV derivation methods illustrated in FIGS. 48A and 48B, the MV derivation methods illustrated in FIGS. 49A and 49B, and other MV derivation methods.

It is to be noted that an MV candidate list may include MV candidates in a mode in which prediction is performed in units of a sub-block, other than the affine mode.

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It is to be noted that, for example, an MV candidate list including MV candidates in an affine merge mode in which two control points are used and an affine merge mode in which three control points are used may be generated as an MV candidate list. Alternatively, an MV candidate list including MV candidates in the affine merge mode in which two control points are used and an MV candidate list including MV candidates in the affine merge mode in which three control points are used may be generated separately. Alternatively, an MV candidate list including MV candidates in one of the affine merge mode in which two control points are used and the affine merge mode in which three control points are used may be generated.

[MV Derivation >Affine Inter Mode]

For example, when information parsed from a stream indicates that the affine inter mode is to be applied, inter predictor **218** derives an MV in the affine inter mode and performs motion compensation (prediction) using the MV.

FIG. **88** is a flow chart illustrating an example of inter prediction by the affine inter mode in decoder **200**.

In the affine inter mode, first, inter predictor **218** derives MV predictors (v_0 , v_1) or (v_0 , v_1 , v_2) of respective two or three control points for a current block (Step S_j_11). The control points are an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. **46A** or FIG. **46B**.

Inter predictor **218** obtains MV predictor selection information included as a prediction parameter in the stream, and derives the MV predictor at each control point for the current block using the MV identified by the MV predictor selection information. For example, when the MV derivation methods illustrated in FIGS. **48A** and **48B** are used, inter predictor **218** derives the motion vector predictors (v_0 , v_1) or (v_0 , v_1 , v_2) at control points for the current block by selecting the MV of the block identified by the MV predictor selection information among decoded blocks in the vicinity of the respective control points for the current block illustrated in either FIG. **48A** or FIG. **48B**.

Next, inter predictor **218** obtains each MV difference included as a prediction parameter in the stream, and adds the MV predictor at each control point for the current block and the MV difference corresponding to the MV predictor (Step S_j_12). In this way, the MV at each control point for the current block is derived.

Next, inter predictor **218** performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor **218** calculates an MV for each of the plurality of sub-blocks as an affine MV, using either two motion vectors v_0 and v_1 and the above expression (1A) or three motion vectors v_0 , v_1 , and v_2 and the above expression (1B) (Step S_j_13). Inter predictor **218** then performs motion compensation of the sub-blocks using these affine MVs and decoded reference pictures (Step S_j_14). When the processes in Steps S_j_13 and S_j_14 are executed for each of all the sub-blocks included in the current block, the inter prediction using the affine merge mode for the current block finishes. In other words, motion compensation of the current block is performed to generate a prediction image of the current block.

It is to be noted that the above-described MV candidate list may be generated in Step S_j_11 as in Step S_k_11.

[MV Derivation >Triangle Mode]

For example, when information parsed from a stream indicates that the triangle mode is to be applied, inter predictor **218** derives an MV in the triangle mode and performs motion compensation (prediction) using the MV.

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FIG. **89** is a flow chart illustrating an example of inter prediction by the triangle mode in decoder **200**.

In the triangle mode, first, inter predictor **218** splits the current block into a first partition and a second partition (Step S_x_11). At this time, inter predictor **218** may obtain, from the stream, partition information which is information related to the splitting as a prediction parameter. Inter predictor **218** may then split a current block into a first partition and a second partition according to the partition information.

Next, first, inter predictor **218** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of decoded blocks temporally or spatially surrounding the current block (Step S_x_12). In other words, inter predictor **218** generates an MV candidate list.

Inter predictor **218** then selects the MV candidate for the first partition and the MV candidate for the second partition as a first MV and a second MV respectively, from the plurality of MV candidates obtained in Step S_x_11 (Step S_x_13). At this time, inter predictor **218** may obtain, from the stream, MV selection information for identifying each selected MV candidate, as a prediction parameter. Inter predictor **218** may then select the first MV and the second MV according to the MV selection information.

Next, inter predictor **218** generates a first prediction image by performing motion compensation using the selected first MV and a decoded reference picture (Step S_x_14). Likewise, inter predictor **218** generates a second prediction image by performing motion compensation using the selected second MV and a decoded reference picture (Step S_x_15).

Lastly, inter predictor **218** generates a prediction image for the current block by performing a weighted addition of the first prediction image and the second prediction image (Step S_x_16).

[Motion Estimation >DMVR]

For example, information parsed from a stream indicates that DMVR is to be applied, inter predictor **218** performs motion estimation using DMVR.

FIG. **90** is a flow chart illustrating an example of motion estimation by DMVR in decoder **200**.

Inter predictor **218** derives an MV for a current block according to the merge mode (Step S_L_11). Next, inter predictor **218** derives the final MV for the current block by searching the region surrounding the reference picture indicated by the MV derived in S_L_11 (Step S_L_12). In other words, the MV of the current block is determined according to the DMVR.

FIG. **91** is a flow chart illustrating a specific example of motion estimation by DMVR in decoder **200**.

First, in Step **1** illustrated in FIG. **58A**, inter predictor **218** calculates the cost between the search position (also referred to as a starting point) indicated by the initial MV and eight surrounding search positions. Inter predictor **218** then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at one of the search positions other than the starting point is the smallest, inter predictor **218** changes a target to the search position at which the smallest cost is obtained, and performs the process in Step **2** illustrated in FIG. **58A**. When the cost at the starting point is the smallest, inter predictor **218** skips the process in Step **2** illustrated in FIG. **58A** and performs the process in Step **3**.

In Step **2** illustrated in FIG. **58A**, inter predictor **218** performs search similar to the process in Step **1**, regarding the search position after the target change as a new starting point according to the result of the process in Step **1**. Inter

predictor **218** then determines whether the cost at each of the search positions other than the starting point is the smallest. Here, when determining that the cost at one of the search positions other than the starting point is the smallest, inter predictor **218** performs the process in Step **4**. When the cost at the starting point is the smallest, inter predictor **218** performs the process in Step **3**.

In Step **4**, inter predictor **218** regards the search position at the starting point as the final search position, and determines the difference between the position indicated by the initial MV and the final search position to be a vector difference.

In Step **3** illustrated in FIG. **58A**, inter predictor **218** determines the pixel position at sub-pixel accuracy at which the smallest cost is obtained, based on the costs at the four points located at upper, lower, left, and right positions with respect to the starting point in Step **1** or Step **2**, and regards the pixel position as the final search position. The pixel position at the sub-pixel accuracy is determined by performing weighted addition of each of the four upper, lower, left, and right vectors $((0, 1), (0, -1), (-1, 0), \text{ and } (1, 0))$, using, as a weight, the cost at a corresponding one of the four search positions. Inter predictor **218** then determines the difference between the position indicated by the initial MV and the final search position to be the vector difference. [Motion Compensation >BIO/OBMC/LIC]

For example, when information parsed from a stream indicates that correction of a prediction image is to be performed, upon generating a prediction image, inter predictor **218** corrects the prediction image based on the mode for the correction. The mode is, for example, one of BIO, OBMC, and LIC described above.

FIG. **92** is a flow chart illustrating one example of generation of a prediction image in decoder **200**.

Inter predictor **218** generates a prediction image (Step Sm_11), and corrects the prediction image according to any of the modes described above (Step Sm_12).

FIG. **93** is a flow chart illustrating another example of generation of a prediction image in decoder **200**.

Inter predictor **218** derives an MV for a current block (Step Sn_11). Next, inter predictor **218** generates a prediction image using the MV (Step Sn_12), and determines whether to perform a correction process (Step Sn_13). For example, inter predictor **218** obtains a prediction parameter included in the stream, and determines whether to perform a correction process based on the prediction parameter. This prediction parameter is, for example, a flag indicating whether each of the above-described modes is to be applied. Here, when determining to perform a correction process (Yes in Step Sn_13), inter predictor **218** generates the final prediction image by correcting the prediction image (Step Sn_14). It is to be noted that, in LIC, the luminance and chrominance of the prediction image may be corrected in Step Sn_14. When determining not to perform a correction process (No in Step Sn_13), inter predictor **218** outputs the final prediction image without correcting the prediction image (Step Sn_15).

[Motion Compensation >OBMC]

For example, when information parsed from a stream indicates that OBMC is to be performed, upon generating a prediction image, inter predictor **218** corrects the prediction image according to the OBMC.

FIG. **94** is a flow chart illustrating an example of correction of a prediction image by OBMC in decoder **200**. It is to be noted that the flow chart in FIG. **94** indicates the correction flow of a prediction image using the current picture and the reference picture illustrated in FIG. **62**.

First, as illustrated in FIG. **62**, inter predictor **218** obtains a prediction image (Pred) by normal motion compensation using an MV assigned to the current block.

Next, inter predictor **218** obtains a prediction image (Pred_L) by applying a motion vector (MV_L) which has been already derived for the decoded block neighboring to the left of the current block to the current block (re-using the motion vector for the current block). Inter predictor **218** then performs a first correction of a prediction image by overlapping two prediction images Pred and Pred_L. This provides an effect of blending the boundary between neighboring blocks.

Likewise, inter predictor **218** obtains a prediction image (Pred_U) by applying an MV (MV_U) which has been already derived for the decoded block neighboring above the current block to the current block (re-using the motion vector for the current block). Inter predictor **218** then performs a second correction of the prediction image by overlapping the prediction image Pred_U to the prediction images (for example, Pred and Pred_L) on which the first correction has been performed. This provides an effect of blending the boundary between neighboring blocks. The prediction image obtained by the second correction is the one in which the boundary between the neighboring blocks has been blended (smoothed), and thus is the final prediction image of the current block.

[Motion Compensation >BIO]

For example, when information parsed from a stream indicates that BIO is to be performed, upon generating a prediction image, inter predictor **218** corrects the prediction image according to the BIO.

FIG. **95** is a flow chart illustrating an example of correction of a prediction image by the BIO in decoder **200**.

As illustrated in FIG. **63**, inter predictor **218** derives two motion vectors (M0, M1), using two reference pictures (Ref₀, Ref₁) different from the picture (Cur Pic) including a current block. Inter predictor **218** then derives a prediction image for the current block using the two motion vectors (M0, M1) (Step Sy_11). It is to be noted that motion vector M0 is a motion vector (MV_{x0}, MV_{y0}) corresponding to reference picture Ref₀, and motion vector M1 is a motion vector (MV_{x1}, MV_{y1}) corresponding to reference picture Ref₁.

Next, inter predictor **218** derives interpolated image I⁰ for the current block using motion vector M0 and reference picture L0. In addition, inter predictor **218** derives interpolated image I¹ for the current block using motion vector M1 and reference picture L1 (Step Sy_12). Here, interpolated image I⁰ is an image included in reference picture Ref₀ and to be derived for the current block, and interpolated image I¹ is an image included in reference picture Ref₁ and to be derived for the current block. Each of interpolated image I⁰ and interpolated image I¹ may be the same in size as the current block. Alternatively, each of interpolated image I⁰ and interpolated image I¹ may be an image larger than the current block. Furthermore, interpolated image I⁰ and interpolated image I¹ may include a prediction image obtained by using motion vectors (M0, M1) and reference pictures (L0, L1) and applying a motion compensation filter.

In addition, inter predictor **218** derives gradient images (Ix⁰, Ix¹, Iy⁰, Iy¹) of the current block, from interpolated image I⁰ and interpolated image I¹ (Step Sy_13). It is to be noted that the gradient images in the horizontal direction are (Ix⁰, Ix¹), and the gradient images in the vertical direction are (Iy⁰, Iy¹). Inter predictor **218** may derive the gradient images by, for example, applying a gradient filter to the interpolated images. The gradient images may be the ones

each of which indicates the amount of spatial change in pixel value along the horizontal direction or the amount of spatial change in pixel value along the vertical direction.

Next, inter predictor **218** derives, for each sub-block of the current block, an optical flow (vx, vy) which is a velocity vector, using the interpolated images (I^0 , I^1) and the gradient images (I_x^0 , I_x^1 , I_y^0 , I_y^1). As one example, a sub-block may be 4×4 pixel sub-CU.

Next, inter predictor **218** corrects a prediction image for the current block using the optical flow (vx, vy). For example, inter predictor **218** derives a correction value for the value of a pixel included in a current block, using the optical flow (vx, vy) (Step Sy_15). Inter predictor **218** may then correct the prediction image for the current block using the correction value (Step Sy_16). It is to be noted that the correction value may be derived in units of a pixel, or may be derived in units of a plurality of pixels or in units of a sub-block.

It is to be noted that the BIO process flow is not limited to the process disclosed in FIG. 95. Only part of the processes disclosed in FIG. 95 may be performed, or a different process may be added or used as a replacement, or the processes may be executed in a different processing order.

[Motion Compensation >LIC]

For example, when information parsed from a stream indicates that LIC is to be performed, upon generating a prediction image, inter predictor **218** corrects the prediction image according to the LIC.

FIG. 96 is a flow chart illustrating an example of correction of a prediction image by the LIC in decoder **200**.

First, inter predictor **218** obtains a reference image corresponding to a current block from a decoded reference picture using an MV (Step Sz_11).

Next, inter predictor **218** extracts, for the current block, information indicating how the luma value has changed between the current picture and the reference picture (Step Sz_12). This extraction is performed based on the luma pixel values for the decoded left neighboring reference region (surrounding reference region) and the decoded upper neighboring reference region (surrounding reference region), and the luma pixel values at the corresponding positions in the reference picture specified by the derived MVs. Inter predictor **218** calculates a luminance correction parameter, using the information indicating how the luma value changed (Step Sz_13).

Inter predictor **218** generates a prediction image for the current block by performing a luminance correction process in which the luminance correction parameter is applied to the reference image in the reference picture specified by the MV (Step Sz_14). In other words, the prediction image which is the reference image in the reference picture specified by the MV is subjected to the correction based on the luminance correction parameter. In this correction, luminance may be corrected, or chrominance may be corrected. [Prediction Controller]

Prediction controller **220** selects either an intra prediction image or an inter prediction image, and outputs the selected image to adder **208**. As a whole, the configurations, functions, and processes of prediction controller **220**, intra predictor **216**, and inter predictor **218** at the decoder **200** side may correspond to the configurations, functions, and processes of prediction controller **128**, intra predictor **124**, and inter predictor **126** at the encoder **100** side.

[Hypothetical Reference Decoder]

A hypothetical reference decoder (HRD) is a hypothetical model pertaining to the behavior of decoder **200** and par-

ticularly to the state of a buffer. Encoder **100** generates a bitstream so that the HRD does not fail. Decoder **200** may operate using the HRD adapted for HRD parameters defined by encoder **100**.

A bitstream may include multiple versions. For example, multi-layer coding such as spatially-scalable video coding or multi-view video coding may be applied to video coding. A layer set including at least one layer selected from layers defined in multi-layer coding, and then decoded may correspond to one of the versions. The layer set herein is also expressed as an output layer set (OLS). Layers included in an OLS are expressed also as output layers.

An HRD in video coding enables checking the conformance of decoder **200** regarding the output timings or output order of decodable versions included in a bitstream.

For example, the HRD calculates, for each picture, a coded picture buffer (CPB) removable time which corresponds to a time when a picture is removed from the CPB and then decoded. The CPB here is a buffer that stores pictures that have not been decoded. The CPB removable time may correspond to what is called a decoding timestamp (DTS) in some systems.

For example, the HRD calculates, for each picture, a decoded picture buffer (DPB) output time which corresponds to a time when a picture is output from the DPB, typically to be displayed. The DPB output time may correspond to what is called a presentation timestamp (PTS) in some systems.

In order to calculate the CPB removable time and the DPB output time for each of pictures included in a bitstream, the HRD uses timing information signaled in a buffering period SEI message and a picture timing SEI message.

The bitstream may be a single-layer bitstream including only one layer or a multi-layer bitstream including a plurality of layers. A multi-layer bitstream includes one or more OLSs. In other words, a multi-layer bitstream includes one or more versions. An OLS including a plurality of layers may be expressed as a multi-layer OLS, and an OLS including only one layer may be expressed as a single-layer OLS. A unit that corresponds to a bitstream and includes OLSs may be expressed as a layer set unit.

Information on each of the OLSs included in a bitstream is signaled in the video parameter set (VPS) of the bitstream together with a set of information related to the OLS. In other words, OLS may be specified by VPS. The set of information includes, for example, a maximum temporal ID value (vps_max_tid[i] indicated in FIG. 101 which is to be described later) for the OLS. A maximum temporal ID value for the OLS here means a maximum temporal ID value allowed for the OLS.

The maximum temporal ID value is used by the HRD to check conformance for the OLS, and may be expressed also as the highest temporal ID value. The maximum temporal ID value is equal to a value obtained by subtracting 1 from the maximum number of temporal layers.

A temporal ID is an ID for identifying a temporal sublayer. Temporal sublayers here correspond to the temporal layers illustrated in FIG. 5 or FIG. 6, and are different from layers in multi-layer coding such as spatially-scalable video coding or multi-view video coding. A maximum temporal ID value may be defined for each OLS.

A single-layer bitstream includes only a single layer that composes a single OLS. In the case of a single-layer bitstream, HRD-related information is signaled in a sequence parameter set (SPS) as HRD parameters. In this case, information related to the maximum temporal ID value

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(the syntax element `sps_max_sublayers_minus1` indicated in FIG. 102 which is to be described later) of the bitstream is signaled in the SPS.

In the case of a single-layer bitstream, the bitstream may not include VPS. For example, when the syntax element `sps_video_parameter_set_id` indicated in FIG. 102 which is to be described later is equal to 0, VPS is not used.

A maximum temporal ID value that is available for use in the HRD may be repeatedly signaled in the buffering period SEI message by the syntax element `bp_max_sublayers_minus1`, independently from SPS or VPS. In other words, the maximum temporal ID value signaled in SPS or VPS may be signaled also in the buffering period SEI message.

This enables parsing of HRD-related SEI messages, such as buffering period SEI messages and picture timing SEI messages, independently without referring to SPS or VPS. Accordingly, it is easier to parse the HRD-related SEI messages.

Note that syntax elements may be expressed as coding parameters or simply as parameters. The expression that “the value of a parameter is x” may be expressed as “a parameter is x”.

[First Aspect Regarding Parameters]

In video coding, for example, HRD is used for checking conformance between a bitstream and decoder 200. HRD-related information is stored in the bitstream for the conformance checking.

Specifically, the HRD-related information is information related to timings in the relationship between buffers and pictures, and more specifically, is information related to the CPB removal time and the DPB output time described above. Schedules related to such timings are signaled and a schedule selected from among the signaled schedules is applied to the processing. Each of the schedules also depends on, for instance, a maximum temporal ID value, a bit rate, and a CPB size.

Most of the HRD-related information in a single-layer bitstream or a single-layer OLS is signaled in an HRD parameter syntax structure, a buffering period SEI message, and a picture timing SEI message. The HRD parameter syntax structure here is included in, for example, SPS and includes HRD parameters as syntax elements.

More specifically information on the entire sequence, such as the number of schedules signaled per temporal sublayer as well as the bit rate of each schedule and a CPB size, is included in the HRD parameter syntax structure in the SPS.

Information on the buffering period of the sequence, such as the initial CPB removal delay and the initial CPB removal offset of each schedule, is included in the buffering period SEI message. The initial CPB removal delay and the initial CPB removal offset correspond to a buffering period required for decoder 200 and are parameters for determining the initial CPB removal time.

Moreover, information on a picture associated with the picture timing SEI message, such as the CPB removal delay and the DPB output delay, is included in the picture timing SEI message.

In a first aspect of the present disclosure, a loop for temporal sublayers is introduced on the top of a loop for schedules within a buffering period SEI message.

Specifically, one or more schedules are signaled per temporal sublayer in an HRD parameter syntax structure in SPS. In response to this, two loops for signaling, per temporal sublayer and per schedule, the initial CPB removal delay and the initial CPB removal offset are introduced into the buffering period SEI message.

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FIG. 97 is a diagram illustrating syntax of HRD-related information in a buffering period SEI message. “for(tid=0; tid<=sps_max_sublayers_minus1;tid++)” in FIG. 97 is a loop for temporal sublayers. `bp_cpb_cnt_minus1[tid]` is the number of schedules signaled about temporal sublayers whose temporal IDs are tid.

Moreover, “for(i=0;i<bp_cpb_cnt_minus1[tid]+1;i++)” is a loop for schedules. The initial CPB removal delay and the initial CPB removal offset are signaled per temporal sublayer and per schedule in accordance with the loop for temporal sublayers and the loop for schedules.

In the first aspect, HRD parameters for each temporal sublayer are signaled in the HRD parameter syntax structure in the SPS, the buffering period SEI message, and the picture timing SEI message. This makes the design of an HRD consistent regarding such signaling.

Furthermore, the buffering period SEI message does not need to be held for each temporal sublayer. In other words, it is possible to utilize the same buffering period SEI message for temporal sublayers. Accordingly, parsing of the HRD-related information in the buffering period SEI message is simplified.

FIG. 98 is a flow chart illustrating an example of a process of checking conformance of a bitstream using an HRD. For example, decoder 200 performs the process shown in FIG. 98 before a decoding process. Alternatively, encoder 100 may perform the process shown in FIG. 98 for checking the conformance between a bitstream and decoder 200.

First, an operation point TargetOp corresponding to a maximum temporal ID value OpTid in a range from 0 to `sps_max_sublayers_minus1` is selected (S101).

Subsequently, an HRD parameter set regarding the operation point TargetOp is selected (S102). The selected HRD parameter set includes an HRD parameter set sublayer_hrd_parameters(OpTid) regarding a temporal sublayer with the maximum temporal ID value OpTid.

Subsequently, a buffering period SEI message applicable to the operation point TargetOp is selected (S103), and an access unit associated with the buffering period SEI message is selected as an HRD initialization point. In addition, a picture timing SEI message applicable to the operation point TargetOp is selected for each access unit (S104).

Subsequently, a schedule index SchedSelIdx is selected in a range from 0 to `vui_cpb_cnt_minus1[OpTid]` (S105). The schedule index SchedSelIdx is an index indicating a schedule. `vui_cpb_cnt_minus1[OpTid]` is a syntax element signaled in the HRD parameter set that has already been selected.

Subsequently, two determination steps follow. In one determination step, whether the number of schedules signaled in the selected HRD parameter set is equal to the number of schedules signaled in the selected buffering period SEI message is checked. In the other determination step, if these numbers of schedules are not equal to each other, whether the schedule index SchedSelIdx is less than or equal to the number of schedules signaled in the selected buffering period SEI message is checked.

Specifically, whether `vui_cpb_cnt_minus1[OpTid]==bp_cpb_cnt_minus1[OpTid]` is satisfied is checked (S106). When `vui_cpb_cnt_minus1[OpTid]==bp_cpb_cnt_minus1[OpTid]` is not satisfied (No in S106), whether `SchedSelIdx<bp_cpb_cnt_minus1[OpTid]` is satisfied is determined (S107).

When either of the above two determination steps is satisfied (Yes in S106 or Yes in S107), a CPB size `CpbSize[SchedSelIdx]` and a bit rate `BitRate[SchedSelIdx]` are selected (S108). The CPB size `CpbSize[SchedSelIdx]` and

the bit rate $\text{BitRate}[\text{SchedSelIdx}]$ are determined in the HRD parameter set $\text{sublayer_hrd_parameters}(\text{OpTid})$.

The CPB size $\text{CpbSize}[\text{SchedSelIdx}]$ here is a CPB size for the schedule of the schedule index SchedSelIdx . The bit rate $\text{BitRate}[\text{SchedSelIdx}]$ is a bit rate for the schedule of the schedule index SchedSelIdx .

Subsequently, an initial CPB removal delay $\text{InitCpbRemovalDelay}[\text{OpTid}][\text{SchedSelIdx}]$ and an initial CPB removal offset $\text{InitCpbRemovalDelayOffset}[\text{OpTid}][\text{SchedSelIdx}]$ are selected based on the signaling in the buffering period SEI message (S109).

The initial CPB removal delay $\text{InitCpbRemovalDelay}[\text{OpTid}][\text{SchedSelIdx}]$ is an initial CPB removal delay for the maximum temporal ID value OpTid and the schedule index SchedSelIdx .

The initial CPB removal offset $\text{InitCpbRemovalDelayOffset}[\text{OpTid}][\text{SchedSelIdx}]$ is an initial CPB removal offset for the maximum temporal ID value OpTid and the schedule index SchedSelIdx .

Then, conformance is checked using values selected in the above processes (S108 and S109) (S110). Conformance may be checked in accordance with, for example, a process such as the one described in HEVC or VVC appendix C.

In such a process, the decoding timestamp of each picture is typically derived based on a CPB removal delay signaled for the maximum temporal ID value OpTid in the picture timing SEI message. Moreover, the presentation timestamp of each picture is derived based on a DPB output delay signaled for the maximum temporal ID value OpTid in the picture timing SEI message.

After that, the process ends.

When neither of the two determination steps described above is satisfied (No in S106 and No in S107), the process ends as an error since there is an issue related to the schedule of the schedule index SchedSelIdx .

An alternative process to avoid the two determination steps is to select a schedule index SchedSelIdx in a range from 0 to $\min(\text{vui_cpb_cnt_minus1}[\text{OpTid}], \text{bp_cpb_cnt_minus1}[\text{OpTid}])$. With such a selection being made, the subsequent selection processes (S108 and S109) and the conformance checking (S110) may be performed without the two determination steps.

As a variation of the first aspect, the number of schedules per temporal sublayer may be signaled only in the HRD parameter syntax structure in the SPS and not in the buffering period SEI message.

This generates parsing dependency of the buffering period SEI message on the SPS which includes the HRD parameter syntax structure. In this case, the syntax element $\text{bp_cpb_cnt_minus1}[\text{tid}]$ may not be signaled in the buffering period SEI message and may be simply derived based on the SPS signaling.

It is desirable, however, to avoid the parsing dependency between the buffering period SEI message and the SPS. For this purpose, it makes sense to signal the syntax element $\text{bp_cpb_cnt_minus1}[\text{tid}]$ in the buffering period SEI message.

A constraint that the value of the syntax element $\text{bp_cpb_cnt_minus1}[\text{tid}]$ should be equal to the value of the corresponding syntax element signaled in the HRD parameter syntax structure in the SPS may be introduced.

This ensures that the number of schedules signaled in the HRD parameter syntax structure in the SPS is equal to the number of schedules signaled in the buffering period SEI message. Note that the corresponding syntax element signaled in the HRD parameter syntax structure in the SPS is typically named $\text{vui_cpb_cnt_minus1}[\text{tid}]$.

[Second Aspect Regarding Parameters]

In a second aspect of the present disclosure, the maximum number of temporal sublayers in a bitstream is signaled in a buffering period SEI message and a picture timing SEI message in order to avoid parsing dependency on SPS.

Note that the maximum number of temporal sublayers corresponds to a maximum temporal ID value in the bitstream. More specifically, a value obtained by subtracting 1 from the maximum number of temporal sublayers in the bitstream is equal to the maximum temporal ID value in the bitstream.

FIG. 99 is a diagram illustrating syntax of a buffering period SEI message according to the second aspect.

A newly added syntax element $\text{bp_max_sublayers_minus1}$ indicates a value obtained by subtracting 1 from the maximum number of temporal sublayers in the bitstream. A loop for temporal sublayers therefore depends on $\text{bp_max_sublayers_minus1}$ instead of $\text{sps_max_sublayers_minus1}$ in FIG. 97.

Accordingly, parsing dependency on SPS is removed.

FIG. 100 is a diagram illustrating syntax of a picture timing SEI message according to the second aspect.

A newly added syntax element $\text{pt_max_sublayers_minus1}$ indicates a value obtained by subtracting 1 from the maximum number of temporal sublayers in the bitstream. A loop for temporal sublayers therefore depends on $\text{pt_max_sublayers_minus1}$ instead of $\text{sps_max_sublayers_minus1}$. Accordingly, parsing dependency on SPS is removed.

In the second aspect, it is possible to parse both the buffering period SEI message and the picture timing SEI message independently from SPS.

As a variation of the second aspect, a constraint that the syntax element $\text{bp_max_sublayers_minus1}$ should be equal to the corresponding syntax element $\text{sps_max_sublayers_minus1}$ signaled in the SPS may be introduced. Accordingly, the same maximum temporal ID value is signaled in the buffering period SEI message and the SPS.

Similarly, a constraint that the syntax element $\text{pt_max_sublayers_minus1}$ should be equal to the corresponding syntax element $\text{sps_max_sublayers_minus1}$ signaled in the SPS may be introduced. Accordingly, the same maximum temporal ID value is signaled in the picture timing SEI message and the SPS.

In the case of a multi-layer OLS, the HRD parameter syntax structure is included in VPS and the maximum temporal ID value is also signaled in the VPS. In this case, the same maximum temporal ID value may be signaled in the buffering period SEI message and the VPS.

In other words, the maximum temporal ID value signaled in the buffering period SEI message may be same as the maximum temporal ID value signaled in the VPS or the SPS.

Specifically, regarding a single-layer bitstream or a single-layer OLS, the syntax element $\text{bp_max_sublayers_minus1}$ is equal to the corresponding syntax element $\text{sps_max_tid}[i]$ in the SPS. Regarding a multi-layer bitstream in a multi-layer OLS, the syntax element $\text{bp_max_sublayers_minus1}$ is equal to the corresponding syntax element $\text{vps_hrd_max_tid}[i]$.

FIG. 101 is a diagram illustrating syntax of HRD-related information in VPS. For example, $\text{vps_hrd_max_tid}[i]$ indicates a maximum temporal ID value associated with the i -th HRD parameter set. $\text{ols_hrd_parameters}$ (firstSubLayers , $\text{vps_hrd_max_tid}[i]$) is syntax for determining an HRD parameter set that is related to each of temporal ID values from firstSubLayers to $\text{vps_hrd_max_tid}[i]$.

FIG. 102 is a diagram illustrating syntax of HRD-related information in SPS. For example, when sps_video_param

eter_set_id has a value greater than or equal to 0 and indicates a value greater than 0, sps_video_parameter_set_id indicates the ID of a current VPS to be referred to. When sps_video_parameter_set_id indicates 0, sps_video_parameter_set_id indicates that no VPS is referred to.

Moreover, sps_max_sublayers_minus1 indicates a maximum temporal ID value in a sequence. The maximum temporal ID value in the sequence is equal to a value obtained by subtracting 1 from the maximum number of temporal sublayers in the sequence.

Moreover, ols_hrd_parameters (firstSubLayers, sps_max_sublayers_minus1) syntaxes for determining an HRD parameter set related to each of the temporal ID values from firstSubLayers to sps_max_sublayers_minus1.

A constraint that bp_max_sublayers_minus1 in the buffering period SEI message should be equal to a value given to ols_hrd_parameters() syntax structure as a second input may be introduced. The second input of ols_hrd_parameters() may be represented as maxSubLayers.

When ols_hrd_parameters() syntax structure is included in SPS (in the case of a single-layer bitstream or a single-layer OLS), maxSubLayers is equal to sps_max_sublayers_minus1. When ols_hrd_parameters() syntax structure is included in VPS, maxSubLayers is equal to vps_hrd_max_tid[i].

Accordingly, the constraint that bp_max_sublayers_minus1 should be equal to maxSubLayers ensures that the signaling of the HRD-related information is consistent between the buffering period SEI message and the SPS or VPS.

FIG. 103 is a flow chart illustrating an example of a process for checking conformance according to the second aspect. For example, decoder 200 performs the process shown in FIG. 103 before decoding. Alternatively, encoder 100 may perform the process shown in FIG. 103 to check the conformance between a bitstream and decoder 200. The index targetOlsIdx of a target OLS on which the conformance checking is to be performed is input to the conformance checking process.

In the conformance checking, first, a current SPS to be referred to is parsed (S201). In the parsing of the SPS, the syntax element sps_video_parameter_set_id is read from the SPS. Subsequently, whether sps_video_parameter_set_id is equal to 0 is determined (S202).

When sps_video_parameter_set_id is 0 (Yes in S202), a bitstream is, in definition, a single-layer bitstream that includes only one OLS. In other words, the number of OLSs in the bitstream is 1 and the number of layers in that OLS is 1.

Accordingly, only one element having a value equal to 1 is defined for an array NumLayersInOls indicating the number of layers in each OLS. In other words, only NumLayersInOls[0]=1 is defined for the array NumLayersInOls (S203). Note that the value of the index targetOlsIdx that is input for the conformance checking of such a single-layer bitstream is expected to be equal to 0.

When sps_video_parameter_set_id is not 0 (No in S202), sps_video_parameter_set_id is greater than 0. In this case, the bitstream includes VPS. Although the bitstream is typically a multi-layer bitstream, the bitstream may be a single-layer bitstream. Then, VPS having vps_video_parameter_set_id equal to sps_video_parameter_set_id is referred to.

The referred VPS is then parsed (S204). In the parsing of the VPS, the array NumLayersInOls is derived from the VPS, for example.

Subsequently, whether NumLayersInOls[targetOlsIdx] is equal to 1 is checked (S205). When NumLayersInOls[targetOlsIdx] is equal to 1 (Yes in S205), the target OLS is a single-layer OLS. In this case, a variable maxSubLayers indicating a maximum temporal ID value to be used for conformance checking is set to a value equal to the value of sps_max_sublayers_minus1 from the SPS (S206).

When NumLayersInOls[targetOlsIdx] is not equal to 1 (No in S205), the variable maxSubLayers is set to a value equal to the value of vps_hrd_max_tid[i] from the VPS (S207). Here, i is an index to the list of one or more ols_hrd_parameters() syntax structures signaled in the VPS of a particular ols_hrd_parameters() syntax structure that applies to the target OLS.

The ols_hrd_parameters() syntax structure described above corresponds to an HRD parameter set. In other words, the i described above may be interpreted as an index to the list of one or more HRD parameter sets, from the VPS, of the HRD parameter set applied to the target OLS.

Subsequently, the buffering period SEI message is parsed (S208). In the parsing of the buffering period SEI message, bp_max_sublayers_minus1 is read from the buffering period SEI message.

Subsequently, whether bp_max_sublayers_minus1 is equal to the variable maxSubLayers is checked (S209). When bp_max_sublayers_minus1 is not equal to the variable maxSubLayers, the bitstream is determined to be non-conforming to decoder 200. When bp_max_sublayers_minus1 is equal to the variable maxSubLayers, the process ends with a positive result regarding the conformance checking.

In the second aspect, the HRD parameters are signaled so that HRD parameters in the HRD parameter syntax structure are consistent with HRD parameters in the buffering period SEI message for both a single-layer bitstream and a multi-layer bitstream. Accordingly, timings related to the HRD may be smoothly defined.

The present aspect may be implemented in combination with at least one or more of the other aspects according to the present disclosure. In addition, one or more of the processes in any one of the flowcharts, part of the configuration of any one of the devices, or part of any of the syntaxes described in the present aspect may be implemented in combination with the other aspects according to the present disclosure. The checking related to the consistency of the HRD parameters may be performed not only by decoder 200, but also by encoder 100 in a similar manner.

Not all of the constituent elements and the processes described in the present aspect are always necessary, and only one or more of the constituent elements and one or more of the processes of the present aspect may be implemented. Moreover, each of the names of the parameters described above may be changed.

[Third Aspect Regarding Parameters] A third aspect of the present disclosure relates to the allowed range of a maximum temporal ID value to be selected for conformance checking.

Specifically, the upper limit of the maximum temporal ID value is sps_max_sublayers_minus1 for single-layer bitstreams or single-layer OLSs. The upper limit of the maximum temporal ID value is vps_hrd_max_tid[i] for a multi-layer OLS in a multi-layer bitstream. As a consequence, the upper limits of the maximum temporal ID values depend on a target OLS index selected for conformance checking.

FIG. 104 is a flow chart illustrating an example of a process of checking conformance according to the third aspect. For example, decoder 200 performs the process

shown in FIG. 104 before decoding. Alternatively, encoder 100 may perform the process shown in FIG. 104 for checking the conformance between a bitstream and decoder 200. The index targetOlsIdx and the maximum temporal ID value OpTid of a target OLS on which the conformance checking is to be performed are input to the conformance checking process.

In the conformance checking, first, a current SPS to be referred to is parsed (S301). In the parsing of the SPS, the syntax element `sps_video_parameter_set_id` is read from the SPS. Subsequently, whether `sps_video_parameter_set_id` is equal to 0 is determined (S302).

When `sps_video_parameter_set_id` is 0 (Yes in S302), a bitstream is, in definition, a single-layer bitstream that includes only one OLS. In other words, the number of OLSs in the bitstream is 1 and the number of layers in that OLS is 1.

Accordingly, only one element having a value equal to 1 is defined for an array `NumLayersInOls` that indicates the number of layers in each OLS. In other words, only `NumLayersInOls[0]=1` is defined for the array `NumLayersInOls` (S303). Note that the value of the index `targetOlsIdx` that is input for the conformance checking of such a single-layer bitstream is expected to be equal to 0.

When `sps_video_parameter_set_id` is not 0 (No in S302), `sps_video_parameter_set_id` is greater than 0. In this case, the bitstream includes VPS. Although the bitstream is typically a multi-layer bitstream, the bitstream may be a single-layer bitstream. VPS having `vps_video_parameter_set_id` equal to `sps_video_parameter_set_id` is then referred to.

The referred VPS is then parsed (S304). In the parsing of the VPS, the array `NumLayersInOls` is derived from the VPS, for example.

Subsequently, whether `NumLayersInOls[targetOlsIdx]` is equal to 1 is checked (S305). When `NumLayersInOls[targetOlsIdx]` is equal to 1 (Yes in S305), the target OLS is a single-layer OLS. In this case, a variable `maxSubLayers` indicating a maximum temporal ID value to be used for conformance checking is set to a value equal to the value of `sps_max_sublayers_minus1` from the SPS (S306).

When `NumLayersInOls[targetOlsIdx]` is not equal to 1 (No in S305), the variable `maxSubLayers` is set to a value equal to the value of `vps_hrd_max_tid[i]` from the VPS (S307). Here, `i` is an index to the list of one or more `ols_hrd_parameters()` syntax structures signaled in the VPS of a particular `ols_hrd_parameters()` syntax structure that applies to the target OLS.

The `ols_hrd_parameters()` syntax structure described above corresponds to an HRD parameter set. In other words, the `i` described above may be interpreted as an index to the list of one or more HRD parameter sets, from the VPS, of the HRD parameter set applied to the target OLS.

Subsequently, whether the maximum temporal ID value OpTid that has been input to the conformance checking process is less than or equal to the variable `maxSubLayers` is checked (S308).

When the maximum temporal ID value OpTid is not less than or equal to the variable `maxSubLayers`, the combination of the index `targetOlsIdx` and the maximum temporal ID value OpTid is not appropriate. In this case, the process ends with a negative result regarding the conformance checking. When the maximum temporal ID value OpTid is less than or equal to the variable `maxSubLayers`, the combination of the index `targetOlsIdx` and the maximum temporal ID value OpTid is appropriate. In this case, the process ends with a positive result regarding the conformance checking.

The third aspect enables checking that a maximum temporal ID value selected for conformance checking is less than or equal to a maximum temporal ID value defined in SPS or VPS. Accordingly, it is possible to check that the maximum temporal ID value selected for conformance checking is available.

The present aspect may be implemented in combination with at least one or more of the other aspects according to the present disclosure. In addition, one or more of the processes in any one of the flow charts, part of the configuration of any of the devices, and part of any of the syntaxes described in the present aspect may be implemented in combination with the other aspects according to the present disclosure. The conformance checking described above may be performed not only by decoder 200, but also by encoder 100 in a similar manner.

Not all of the constituent elements and the processes described in the present aspect are always necessary, and only one or more of the constituent elements and one or more of the processes of the present aspect may be implemented. Moreover, each of the names of the parameters described above may be changed.

[Fourth Aspect Regarding Parameters] A fourth aspect of the present disclosure relates to the allowed range of the syntax element `sps_max_sublayers_minus1` from SPS shown in FIG. 102. `sps_max_sublayers_minus1` indicates a value obtained by subtracting 1 from a maximum temporal ID value in a sequence, that is, a maximum number of temporal sublayers in a bitstream.

Specifically, in the case where the bitstream includes VPS, `sps_max_sublayers_minus1` has a value in a range from 0 to the syntax element `vps_max_sublayers_minus1`. `vps_max_sublayers_minus1` indicates a value obtained by subtracting 1 from the maximum temporal ID value in the bitstream that includes one or more sequences, that is, the maximum number of temporal sublayers in the bitstream.

The aforementioned `vps_max_sublayers_minus1` is a syntax element included in VPS. In other words, in the case where VPS is used in video coding, `vps_max_sublayers_minus1` is written into the VPS. In this case, `vps_max_sublayers_minus1` is parsed from the VPS.

In the case where the bitstream does not include VPS, `sps_max_sublayers_minus1` has a value in a range from 0 to 6. In other words, in the case where VPS is not used in video coding and decoding, `sps_max_sublayers_minus1` is limited not by VPS but by a fixed value. Including VPS in a bitstream means using VPS for video coding and decoding and also means that the syntax element `sps_video_parameter_set_id` from SPS shown in FIG. 102 is greater than 0. Not including VPS in a bitstream, on the other hand, means not using VPS for video coding and decoding and also means that the syntax element `sps_video_parameter_set_id` from the SPS is equal to 0.

As a consequence, the allowed range of the syntax element `sps_max_sublayers_minus1` depends on the value of the syntax element `sps_video_parameter_set_id`.

Note that VPS is typically included in a multi-layer bitstream, but not so in a single-layer bitstream. VPS, however, may be included in a single-layer bitstream.

FIG. 105 is a flow chart illustrating a process for checking conformance according to the fourth aspect. For example, decoder 200 performs the process shown in FIG. 105 before decoding. Alternatively, encoder 100 may perform the process shown in FIG. 105 to check the conformance between a bitstream and decoder 200.

First, SPS is parsed and the syntax elements `sps_video_parameter_set_id` and `sps_max_sublayers_mi-`

nus1 are read from the SPS (S401). Subsequently, whether sps_video_parameter_set_id is equal to 0 is determined (S402).

When sps_video_parameter_set_id is equal to 0 (Yes in S402), the bitstream is a single-layer bitstream that does not include VPS. In this case, VPS is not referred to. Whether sps_max_sublayers_minus1 is less than or equal to 6 is then checked (S403).

When sps_max_sublayers_minus1 is not less than or equal to 6, the bitstream is determined to be non-conforming. When sps_max_sublayers_minus1 is less than or equal to 6, the process ends with a positive result of the conformance checking that the bitstream has conformance.

Since sps_video_parameter_set_id is greater than or equal to 0, when sps_video_parameter_set_id is not equal to 0 (No in S402), sps_video_parameter_set_id is greater than 0.

When sps_video_parameter_set_id is greater than 0, VPS with vps_video_parameter_set_id equal to sps_video_parameter_set_id is referred to. The referred VPS is then parsed and vps_max_sublayers_minus1 is read (S404).

Whether sps_max_sublayers_minus1 is less than or equal to vps_max_sublayers_minus1 is then checked (S405).

When sps_max_sublayers_minus1 is not less than or equal to vps_max_sublayers_minus1, the bitstream is determined to be non-conforming.

When sub_max_sublayers_minus1 is less than or equal to vps_max_sublayers_minus1, the process ends with a positive result of the conformance checking that the bitstream has conformance.

Moreover, in the checking of sps_max_sublayers_minus1 (S403 or S405), an abnormal value such as a negative value may be checked. Specifically, whether sps_max_sublayers_minus1 is greater than or equal to 0 may be checked. When sps_max_sublayers_minus1 is not greater than or equal to 0, the bitstream may be determined to be non-conforming.

In the fourth aspect, when VPS is referred to, the upper limit of sps_max_sublayers_minus1 is defined to be consistent with vps_max_sublayers_minus1. Accordingly, it is possible to inhibit inconsistency in parameters between sps_max_sublayers_minus1 and vps_max_sublayers_minus1.

When VPS is not referred to, the upper limit of sps_max_sublayers_minus1 is defined to be a fixed value. Accordingly, it is possible to define parameters in a range where processing is effectively performed.

When the upper limit of sps_max_sublayers_minus1 is not determined, for example, it is likely that sps_max_sublayers_minus1 is an indefinite value. This may generate an event that one or more specific temporal sublayers are not decoded. By defining the upper limit of sps_max_sublayers_minus1, sps_max_sublayers_minus1 is inhibited from being an indefinite value and the event that one or more specific temporal sublayers are not decoded is also inhibited.

The present aspect may be implemented in combination with at least one or more of the other aspects according to the present disclosure. In addition, one or more of the processes in any one of the flow charts, part of the configuration of any of the devices, and part of any of the syntaxes described in the present aspect may be implemented in combination with the other aspects according to the present disclosure.

When VPS is not referred to in the process described above, the upper limit for sps_max_sublayers_minus1 is defined to be 6. The number 6 here, however, is an example, and the upper limit may be defined to be a different fixed

value. The number 6 is also used for an example of a fixed upper limit for vps_max_sublayers_minus1. In other words, when VPS is not referred to, the upper limit for sps_max_sublayers_minus1 may also be defined as a fixed upper limit for vps_max_sublayers_minus1.

The fixed upper limit for sps_max_sublayers_minus1 may be a maximum value among values sps_max_sublayers_minus1 may take. The fixed upper limit for sps_max_sublayers_minus1 may be a maximum value among values vps_max_sublayers_minus1 may take.

In the process described above, a method for defining the upper limit for sps_max_sublayers_minus1 is switched according to whether sps_video_parameter_set_id is 0. Whether sps_video_parameter_set_id is 0 corresponds to whether VPS is referred to. Accordingly, the upper limit defining method may be switched according to whether VPS is referred to, without using sps_video_parameter_set_id.

Moreover, a lower limit for sps_max_sublayers_minus1 may be defined to be 0 irrespective of whether or not VPS is referred to. In the present aspect, it is possible to inhibit inconsistency between SPS and VPS. Accordingly, it is possible to adaptively refer to SPS and VPS to perform the processing effectively. It is therefore possible to utilize VPS even when the bitstream is a single-layer bitstream.

The conformance checking described above may be performed not only by decoder 200, but also by encoder 100 in a similar manner.

Not all of the constituent elements and the processes described in the present aspect are always necessary, and only one or more of the constituent elements and one or more of the processes of the present aspect may be implemented. Moreover, each of the names of the parameters described above may be changed.

[Representative Examples of Configuration and Processing]

The following describes representative examples of the configuration and processing of encoder 100 or decoder 200 that are described above.

FIG. 106 is a flow chart illustrating an example of an operation performed by encoder 100. For example, encoder 100 includes circuitry and memory coupled to the circuitry. The circuitry and the memory included in encoder 100 may correspond to processor a1 and memory a2 illustrated in FIG. 8. In operation, the circuitry in encoder 100 performs the following.

In the case where a first parameter set is used for encoding a video, for example, the circuitry in encoder 100 writes a first parameter into the first parameter set (S501). The first parameter set is a parameter set for a layer set unit. The layer set unit is a unit that corresponds to a stream and includes one or more layers. The first parameter indicates the maximum number of one or more temporal sublayers in the layer set unit.

The circuitry in encoder 100 writes a second parameter into a second parameter set (S502). The second parameter set is a parameter set for a random access unit, and indicates the maximum number of one or more temporal sublayers in the random access unit.

In the case where the first parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the first parameter set is not used for encoding the video, on the other hand, the second parameter is limited to less than or equal to a fixed value.

This may enable encoder 100 to write the second parameter consistent with the first parameter in the first parameter set for the layer set unit into the second parameter set for the random access unit, in the case where the first parameter set for the layer set unit is used. Accordingly, encoder 100 may be capable of inhibiting inconsistency between a parameter

in the first parameter set for the layer set unit and a parameter in the second parameter set for the random access unit.

In the case where the first parameter set for the layer set unit is not used, encoder **100** may be capable of writing the second parameter limited to less than or equal to a fixed value into the second parameter set for the random access unit. Accordingly, encoder **100** may be capable of defining the second parameter in a range in which processing is performed efficiently.

Encoder **100** may be capable of inhibiting the maximum number of one or more temporal sublayers from being an indefinite value by limiting the second parameter to the first parameter or a fixed value. Encoder **100** may be thus able to process one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, the second parameter may be limited to at least a lower limit. The lower limit may be a fixed value indicating **1** for the maximum number of one or more temporal sublayers in a random access unit.

In the above description, the first parameter set may be a video parameter set (VPS). The second parameter set may be a sequence parameter set (SPS).

The first parameter may be `vps_max_sublayers_minus1`. Alternatively, the first parameter may be a parameter expressed by `vps_max_sublayers`. In other words, the first parameter may be a parameter having the maximum number per se as a value instead of having a value obtained by subtracting 1 from the maximum number.

The second parameter may be `sps_max_sublayers_minus1`.

Alternatively, the second parameter may be a parameter expressed by `sps_max_sublayers`. In other words, the second parameter may be a parameter having the maximum number per se as a value instead of `sps_max_sublayers_minus1` having a value obtained by subtracting 1 from the maximum number.

A layer set unit may be expressed as a stream unit or a first data unit. A random access unit may be expressed as a second data unit. A stream may be expressed as a bitstream or a layer set including one or more layers. For example, a layer included in the layer set is defined by multi-layer coding such as spatially-scalable video coding or multi-view video coding.

FIG. **107** is a flow chart illustrating a more specific operation performed by encoder **100**. In operation, the circuitry in encoder **100** may perform the following.

In the case where a video parameter set is used for encoding a video, for example, the circuitry in encoder **100** writes a first parameter into the video parameter set (**S511**). The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit that corresponds to a stream and includes one or more layers.

The circuitry in encoder **100** writes a second parameter into a sequence parameter set (**S512**). The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to.

In the case where the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable encoder **100** to write the second parameter consistent with the first parameter in VPS into SPS in

the case where VPS is used. Accordingly, encoder **100** may be capable of inhibiting inconsistency between a parameter in the VPS and a parameter in the SPS.

Encoder **100** may be capable of writing the second parameter that is limited to less than or equal to a fixed value into the SPS in the case where VPS is not used. Accordingly, encoder **100** may be capable of defining the second parameter in a range where processing is performed efficiently.

By limiting the second parameter to the first parameter or a fixed value, encoder **100** may be capable of inhibiting the maximum number of one or more temporal sublayers from being an indefinite value. Encoder **100** thus may be capable of processing one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, a fixed value may be the upper limit of the first parameter. This may enable encoder **100** to write the second parameter that is limited to the upper limit of the first parameter in VPS into SPS even in the case where VPS is not used. Accordingly, even in the case where VPS is not used, encoder **100** may be capable of defining the second parameter in the range where processing is performed efficiently, as is the case where VPS is used.

Stated differently, encoder **100** may be capable of adjusting the upper limit of the maximum number of one or more temporal sublayers in a sequence unit to the upper limit of the maximum number of one or more temporal sublayers in a layer set unit. This may enable encoder **100** to define the upper limit of the maximum number of one or more temporal sublayers in a sequence unit in the range where processing is performed efficiently in accordance with the upper limit of the maximum number of one or more temporal sublayers in a layer set unit.

For example, the first parameter may have a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in a layer set unit. The second parameter may have a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in a sequence unit. The fixed value may be 6.

In the case where the video parameter set is used for encoding the video, the second parameter may be limited to a range from 0 to the first parameter. In the case where the video parameter set is not used for encoding the video, on the other hand, the second parameter may be limited to a range from 0 to 6.

This may enable encoder **100** to limit the maximum number of temporal sublayers in a sequence unit to a range from 1 to the maximum number of temporal sublayers in a layer set unit in the case where VPS is used. In the case where VPS is not used, encoder **100** may be capable of limiting the maximum number of temporal sublayers in a sequence unit to a range from 1 to 7. Accordingly, encoder **100** may be capable of defining the maximum number of temporal sublayers in a sequence unit in the range where processing is performed efficiently.

For example, the circuitry in encoder **100** may write a third parameter into a sequence parameter set. The third parameter here indicates that the video parameter set is used for encoding the video and also indicates the identifier of the video parameter set, when the third parameter is greater than 0. When the third parameter is 0, the third parameter indicates that no video parameter set is used for encoding the video.

When the third parameter is greater than 0, the second parameter may be limited to less than or equal to the first parameter. When the third parameter is 0, on the other hand, the second parameter may be limited to less than or equal to a fixed value.

This may enable encoder **100** to switch between limiting the second parameter in SPS to less than or equal to the first parameter in VPS or limiting the second parameter in the SPS to a fixed value, in accordance with the third parameter indicating whether VPS is used or not. In other words, encoder **100** may be capable of adaptively limiting the second parameter in accordance with the third parameter.

Moreover, since encoder **100** writes the third parameter into SPS, encoder **100** may be capable of switching, on a per sequence basis, between limiting the second parameter to less than or equal to the first parameter in VPS or limiting the second parameter to less than or equal to a fixed value. In other words, encoder **100** may be capable of adaptively limiting the second parameter in accordance with the third parameter written into the SPS.

For example, entropy encoder **110** in encoder **100** may perform the operation described above as the circuitry in encoder **100**. Entropy encoder **110** may perform the operation described above in cooperation with other components.

In the above description, not using any video parameter set means not using any of video parameter sets.

In the above description, the first parameter may be `vps_max_sublayers_minus1`. Alternatively, the first parameter may be a parameter expressed by `vps_max_sublayers`. In other words, the first parameter may be a parameter having the maximum value per se as a value instead of `sps_max_sublayers_minus1` having a value obtained by subtracting 1 from the maximum number.

The second parameter may be `sps_max_sublayers_minus1`.

Alternatively, the second parameter may be a parameter expressed by `vps_max_sublayers`. In other words, the second parameter may be a parameter having the maximum number as a value instead of `sps_max_sublayers_minus1` having a value obtained by subtracting 1 from the maximum number.

The third parameter may be `sps_video_parameter_set_id`. Alternatively, the third parameter may be a parameter indicating the order of video parameter sets as an identifier. Alternatively, when there is only one video parameter set at maximum, the third parameter may be a flag that may take two values. For example, when the flag indicates 0, no video parameter set is used, and when the flag indicates 1, 1 may be indicated by the flag as the identifier of the only video parameter set available.

A layer set unit may be expressed as a stream unit or a first data unit. A sequence unit may be expressed as a second data unit. A stream may be expressed as a bitstream or a layer set including one or more layers. For example, a layer included in the layer set is defined by multi-layer coding such as spatially-scalable video coding or multi-view video coding, separately from temporal sublayers.

FIG. **108** is a flow chart illustrating an example of an operation performed by decoder **200**. For example, decoder **200** includes circuitry and memory coupled to the circuitry. The circuitry and the memory included in decoder **200** may correspond to processor **b1** and memory **b2** illustrated in FIG. **68**. In operation, the circuitry in decoder **200** performs the following.

In the case where a first parameter set is used for decoding a video, for example, the circuitry in decoder **200** reads a first parameter from the first parameter set (**S601**). The first parameter set is a parameter set for a layer set unit. The layer set unit is a unit that corresponds to a stream and includes one or more layers. The first parameter indicates the maximum number of one or more temporal sublayers in the layer set unit.

The circuitry in decoder **200** reads a second parameter from a second parameter set (**S602**). The second parameter set is a parameter set for a random access unit, and indicates the maximum number of one or more temporal sublayers in the random access unit.

In the case where the first parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. When the first parameter set is not used for decoding the video, on the other hand, the second parameter is limited to less than or equal to a fixed value.

This may enable decoder **200** to read the second parameter consistent with the first parameter in the first parameter set for the layer set unit from the second parameter set for the random access unit, in the case where the first parameter set for the layer set unit is used. Accordingly, decoder **200** may be capable of inhibiting inconsistency between a parameter in the first parameter set for the layer set unit and a parameter in the second parameter set for the random access unit.

In the case where the first parameter set for the layer set unit is not used, decoder **200** may be capable of reading the second parameter limited to less than or equal to a fixed value from the second parameter set for the random access unit. Accordingly, decoder **200** may be capable of defining the second parameter in a range in which processing is performed efficiently.

Decoder **200** may be capable of inhibiting the maximum number of one or more temporal sublayers from being an indefinite value by limiting the second parameter to the first parameter or a fixed value. Decoder **200** may be thus able to process one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, the second parameter may be limited to at least a lower limit. The lower limit may be a fixed value indicating 1 for the maximum number of one or more temporal sublayers in a random access unit.

In the above description, the first parameter set may be a video parameter set (VPS). The second parameter set may be a sequence parameter set (SPS).

The first parameter may be `vps_max_sublayers_minus1`. Alternatively, the first parameter may be a parameter expressed by `vps_max_sublayers`. In other words, the first parameter may be a parameter having the maximum number per se as a value instead of having a value obtained by subtracting 1 from the maximum number.

The second parameter may be `sps_max_sublayers_minus1`.

Alternatively, the second parameter may be a parameter expressed by `sps_max_sublayers`. In other words, the second parameter may be a parameter having the maximum number per se as a value instead of `sps_max_sublayers_minus1` having a value obtained by subtracting 1 from the maximum number.

A layer set unit may be expressed as a stream unit or a first data unit. A random access unit may be expressed as a second data unit. A stream may be expressed as a bitstream or a layer set including one or more layers. For example, a layer included in the layer set is defined by multi-layer coding such as spatially-scalable video coding or multi-view video coding.

FIG. **109** is a flow chart illustrating a more specific operation performed by decoder **200**. In operation, the circuitry in decoder **200** may perform the following.

In the case where a video parameter set is used for decoding a video, for example, the circuitry in decoder **200** reads a first parameter from a video parameter set (**S611**).

The first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit is a unit that corresponds to a stream and includes one or more layers.

The circuitry in decoder **200** reads a second parameter from a sequence parameter set (**S612**). The second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit. The sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to.

In the case where the video parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for decoding the video, the second parameter is limited to less than or equal to a fixed value.

This may enable decoder **200** to read, from SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, decoder **200** may be capable of inhibiting inconsistency between a parameter in the VPS and a parameter in the SPS. Moreover, decoder **200** may be capable of reading, from the SPS, the second parameter that is limited to less than or equal to a fixed value in the case where VPS is not used. Accordingly, decoder **200** may be capable of defining the second parameter in a range where processing is performed efficiently.

By limiting the second parameter to the first parameter or a fixed value, decoder **200** may be capable of inhibiting the maximum number of one or more temporal sublayers from being an indefinite value. Decoder **200** thus may be capable of processing one or more temporal sublayers in accordance with the maximum number that is not an indefinite value.

For example, a fixed value may be the upper limit of the first parameter. This may enable decoder **200** to read, from SPS, the second parameter that is limited to the upper limit of the first parameter in VPS even in the case where VPS is not used. Accordingly, even in the case where VPS is not used, decoder **200** may be capable of defining the second parameter in the range where processing is performed efficiently, as is the case where VPS is used.

Stated differently, decoder **200** may be capable of adjusting the upper limit of the maximum number of one or more temporal sublayers in a sequence unit to the upper limit of the maximum number of one or more temporal sublayers in a layer set unit. This may enable decoder **200** to define the upper limit of the maximum number of one or more temporal sublayers in a sequence unit in the range where processing is performed efficiently in accordance with the upper limit of the maximum number of one or more temporal sublayers in a layer set unit.

For example, the first parameter may have a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in a layer set unit. The second parameter may have a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in a sequence unit. The fixed value may be 6.

In the case where the video parameter set is used for decoding the video, the second parameter may be limited to a range from 0 to the first parameter. In the case where a video parameter set is not used for decoding a video, on the other hand, the second parameter may be limited to a range from 0 to 6.

This may enable decoder **200** to limit the maximum number of temporal sublayers in a sequence unit to a range from 1 to the maximum number of temporal sublayers in a layer set unit in the case where VPS is used. In the case where VPS is not used, decoder **200** may be capable of

limiting the maximum number of temporal sublayers in a sequence unit to a range from 1 to 7. Accordingly, decoder **200** may be capable of defining the maximum number of temporal sublayers in a sequence unit in the range where processing is performed efficiently.

For example, the circuitry in decoder **200** may read a third parameter from a sequence parameter set. The third parameter here indicates that the video parameter set is used for decoding the video and also indicates the identifier of the video parameter set, when the third parameter is greater than 0. When the third parameter is 0, the third parameter indicates that no video parameter set is used for decoding the video.

When the third parameter is greater than 0, the second parameter may be limited to less than or equal to the first parameter. When the third parameter is 0, on the other hand, the second parameter may be limited to less than or equal to a fixed value.

This may enable decoder **200** to switch between limiting the second parameter in the SPS to less than or equal to the first parameter in VPS or limiting the second parameter in the SPS to a fixed value, in accordance with the third parameter indicating whether VPS is used or not. In other words, decoder **200** may be capable of adaptively limiting the second parameter in accordance with the third parameter.

Moreover, since decoder **200** reads the third parameter from the SPS, decoder **200** may be capable of switching, on a per sequence basis, between limiting the second parameter to less than or equal to the first parameter in VPS or limiting the second parameter to less than or equal to a fixed value. In other words, decoder **200** may be capable of adaptively limiting the second parameter in accordance with the third parameter read from the SPS.

For example, entropy decoder **202** in decoder **200** may perform the operation described above as the circuitry in decoder **200**. Entropy decoder **202** may perform the operation described above in cooperation with other components.

In the above description, not using any video parameter set means not using any of video parameter sets.

In the above description, the first parameter may be `vps_max_sublayers_minus1`. Alternatively, the first parameter may be a parameter expressed by `vps_max_sublayers`. In other words, the first parameter may be a parameter having the maximum value per se as a value instead of `sps_max_sublayers_minus1` having a value obtained by subtracting 1 from the maximum number.

The second parameter may be `sps_max_sublayers_minus1`.

Alternatively, the second parameter may be a parameter expressed by `vps_max_sublayers`. In other words, the second parameter may be a parameter having the maximum number as a value instead of `sps_max_sublayers_minus1` having a value obtained by subtracting 1 from the maximum number.

The third parameter may be `sps_video_parameter_set_id`. Alternatively, the third parameter may be a parameter indicating the order of video parameter sets as an identifier. Alternatively, when there is only one video parameter set at maximum, the third parameter may be a flag that may take two values. For example, when the flag indicates 0, no video parameter set is used, and when the flag indicates 1, 1 may be indicated by the flag as the identifier of the only video parameter set available.

A layer set unit may be expressed as a stream unit or a first data unit.

A sequence unit may be expressed as a second data unit. A stream may be expressed as a bitstream or a layer set

including one or more layers. For example, a layer included in the layer set is defined by multi-layer coding such as spatially-scalable video coding or multi-view video coding, separately from temporal sublayers.

For example, a bitstream may be stored in a non-transitory medium.

The bitstream may include a video parameter set and a sequence parameter set for causing a decoder that reads out the bitstream to perform a decoding process.

The video parameter set may include a first parameter. The first parameter may be read by the decoder in the case where the video parameter set is used in the decoding process, and may be a parameter that indicates the maximum number of one or more temporal sublayers in a layer set unit. The layer set unit here may be a unit that corresponds to a bitstream and includes one or more layers.

A sequence parameter set may include a second parameter. The second parameter may be a parameter that indicates the maximum number of one or more temporal sublayers in a sequence unit for which the sequence parameter set is referred to in the bitstream.

In the case where the video parameter set is used in the decoding process, the second parameter may be limited to less than or equal to the first parameter. In the case where the video parameter set is not used in the decoding process, on the other hand, the second parameter may be limited to less than or equal to a fixed value.

This may enable reading, from SPS, the second parameter consistent with the first parameter in VPS in the case where VPS is used. Accordingly, it may be possible to inhibit inconsistency between a parameter in the VPS and a parameter in the SPS. In the case where VPS is not used, it may be possible to read, from SPS, the second parameter that is limited to a fixed value.

Accordingly, it may be possible to define the second parameter in a range where processing is performed efficiently.

By limiting the second parameter to the first parameter or a fixed value, it may be possible to inhibit the maximum number of one or more temporal sublayers from being an indefinite value. It may be thus possible to process one or more temporal sublayers in accordance with the maximum number that is an indefinite value.

[Other Examples]

Encoder **100** and decoder **200** according to the present embodiment may be used as an image encoder and an image decoder, respectively, or may be used as a video encoder and a video decoder, respectively.

Alternatively, encoder **100** and decoder **200** may be used as an entropy encoding device and an entropy decoding device, respectively. In other words, encoder **100** and decoder **200** may correspond to only entropy encoder **110** and entropy decoder **202**, respectively. The other elements may be included in another device.

Moreover, encoder **100** may include an input unit and an output unit.

For example, one or more pictures are input to the input unit of encoder **100** and output from the output unit of encoder **100**. Decoder **200** may also include an input unit and an output unit. For example, an encoded bitstream is input to the input unit of decoder **200** and one or more pictures are output from the output unit of decoder **200**. The encoded bitstream may include control information and quantized coefficients to which variable length coding has been applied.

The term “encode” may be replaced with a term such as store, include, write into, describe, signal, send, signal, or

store. For example, encoding information may mean including information in a bitstream. Moreover, the term “decode” may be replaced with a term such as read out, parse, read, read in, derive, obtain, receive, extract, or restore. For example, decoding information may mean obtaining information from a bitstream.

At least part of each of the examples described above may be utilized as an encoding method, a decoding method, an entropy encoding method, an entropy decoding method, or any other method.

Each of the elements may be configured of dedicated hardware or may be implemented by executing a software program suitable for the element. Each of the elements may be implemented by a program executor such as a CPU or a processor reading and executing a software program recorded on a recording medium such as a hard disc or a semiconductor memory.

Specifically, encoder **100** and decoder **200** may each include processing circuitry, and storage electrically coupled to the processing circuitry and accessible from the processing circuitry. For example, the processing circuitry corresponds to processor a1 or b1 and the storage corresponds to memory a2 or b2.

The processing circuitry includes at least one of the dedicated hardware and the program executor, and performs processing using the storage. If the processing circuitry includes a program executor, the storage stores a software program to be executed by the program executor.

The software which implements encoder **100** or decoder **200** according to each of the aforementioned embodiments, for instance, is a program as follows.

For example, the program may cause a computer to execute an encoding method that includes: in the case where a video parameter set is used for encoding a video, writing a first parameter into the video parameter set, where the first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit, and the layer set unit is a unit which corresponds to a stream and includes one or more layers; and writing a second parameter into a sequence parameter set, where the second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit, and the sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for encoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for encoding the video, the second parameter is limited to less than or equal to a fixed value.

For example, the program may cause a computer to execute a decoding method that includes: in the case where a video parameter set is used for decoding a video, reading a first parameter from the video parameter set, where the first parameter indicates the maximum number of one or more temporal sublayers in a layer set unit, and the layer set unit is a unit which corresponds to a stream and includes one or more layers; and reading a second parameter from a sequence parameter set, where the second parameter indicates the maximum number of one or more temporal sublayers in a sequence unit, and the sequence unit is a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to. In the case where the video parameter set is used for decoding the video, the second parameter is limited to less than or equal to the first parameter. In the case where the video parameter set is not used for decoding the video, the second parameter is limited to less than or equal to a fixed value.

The elements may be circuits, as described above. These circuits may constitute one circuitry as a whole, or may be separate circuits. Each element may be implemented by a general-purpose processor or by a dedicated processor.

Processing performed by a specific element may be performed by a different element. The order of performing processes may be changed or the processes may be performed in parallel. An encoder/decoder may include encoder 100 and decoder 200.

The ordinal numbers such as the first and the second used in the description may be switched where necessary. A new ordinal number may be provided for or any of the existing ordinal numbers may be removed from the elements.

The above has given a description of aspects of encoder 100 and decoder 200 based on the embodiments, yet the aspects of encoder 100 and decoder 200 are not limited to the embodiments. The aspects of encoder 100 and decoder 200 may also encompass various modifications that may be conceived by those skilled in the art to the embodiments, and embodiments achieved by combining elements in different embodiments, so long as they do not depart from the scope of the present disclosure.

One or more of the aspects disclosed herein may be implemented in combination with at least one or more of other aspects in the present disclosure. In addition, one or more of the aspects disclosed herein may be implemented by combining, with other aspects, part of the processes indicated in any of the flow charts, part of the configuration of any of the devices, part of any of the syntaxes described in one or more of the aspects disclosed herein, etc.

[Implementations and Applications]

As described in each of the above embodiments, each functional or operational block may typically be realized as an MPU (micro processing unit) and memory, for example. Moreover, processes performed by each of the functional blocks may be realized as a program execution unit, such as a processor which reads and executes software (a program) recorded on a medium such as ROM. The software may be distributed. The software may be recorded on a variety of media such as semiconductor memory. Note that each functional block can also be realized as hardware (dedicated circuit).

The processing described in each of the embodiments may be realized via integrated processing using a single apparatus (system), and, alternatively, may be realized via decentralized processing using a plurality of apparatuses. Moreover, the processor that executes the above-described program may be a single processor or a plurality of processors. In other words, integrated processing may be performed, and, alternatively, decentralized processing may be performed.

Embodiments of the present disclosure are not limited to the above exemplary embodiments; various modifications may be made to the exemplary embodiments, the results of which are also included within the scope of the embodiments of the present disclosure.

Next, application examples of the moving picture encoding method (image encoding method) and the moving picture decoding method (image decoding method) described in each of the above embodiments will be described, as well as various systems that implement the application examples. Such a system may be characterized as including an image encoder that employs the image encoding method, an image decoder that employs the image decoding method, or an image encoder-decoder that includes both the image encoder and the image decoder. Other configurations of such a system may be modified on a case-by-case basis.

FIG. 110 illustrates an overall configuration of content providing system ex100 suitable for implementing a content distribution service. The area in which the communication service is provided is divided into cells of desired sizes, and base stations ex106, ex107, ex108, ex109, and ex110, which are fixed wireless stations in the illustrated example, are located in respective cells.

In content providing system ex100, devices including computer ex111, gaming device ex112, camera ex113, home appliance ex114, and smartphone ex115 are connected to internet ex101 via internet service provider ex102 or communications network ex104 and base stations ex106 through ex110.

Content providing system ex100 may combine and connect any of the above devices. In various implementations, the devices may be directly or indirectly connected together via a telephone network or near field communication, rather than via base stations ex106 through ex110. Further, streaming server ex103 may be connected to devices including computer ex111, gaming device ex112, camera ex113, home appliance ex114, and smartphone ex115 via, for example, internet ex101. Streaming server ex103 may also be connected to, for example, a terminal in a hotspot in airplane ex117 via satellite ex116.

Note that instead of base stations ex106 through ex110, wireless access points or hotspots may be used. Streaming server ex103 may be connected to communications network ex104 directly instead of via internet ex101 or internet service provider ex102, and may be connected to airplane ex117 directly instead of via satellite ex116.

Camera ex113 is a device capable of capturing still images and video, such as a digital camera. Smartphone ex115 is a smartphone device, cellular phone, or personal handyphone system (PHS) phone that can operate under the mobile communications system standards of the 2G, 3G, 3.90, and 4G systems, as well as the next-generation 5G system.

Home appliance ex114 is, for example, a refrigerator or a device included in a home fuel cell cogeneration system.

In content providing system ex100, a terminal including an image and/or video capturing function is capable of, for example, live streaming by connecting to streaming server ex103 via, for example, base station ex106. When live streaming, a terminal (e.g., computer ex111, gaming device ex112, camera ex113, home appliance ex114, smartphone ex115, or a terminal in airplane ex117) may perform the encoding processing described in the above embodiments on still-image or video content captured by a user via the terminal, may multiplex video data obtained via the encoding and audio data obtained by encoding audio corresponding to the video, and may transmit the obtained data to streaming server ex103. In other words, the terminal functions as the image encoder according to one aspect of the present disclosure.

Streaming server ex103 streams transmitted content data to clients that request the stream. Client examples include computer ex111, gaming device ex112, camera ex113, home appliance ex114, smartphone ex115, and terminals inside airplane ex117, which are capable of decoding the above-described encoded data. Devices that receive the streamed data decode and reproduce the received data. In other words, the devices may each function as the image decoder, according to one aspect of the present disclosure.

[Decentralized Processing]

Streaming server ex103 may be realized as a plurality of servers or computers between which tasks such as the processing, recording, and streaming of data are divided. For example, streaming server ex103 may be realized as a content delivery network (CDN) that streams content via a network connecting multiple edge servers located throughout the world. In a CDN, an edge server physically near a client is dynamically assigned to the client. Content is cached and streamed to the edge server to reduce load times. In the event of, for example, some type of error or change in connectivity due, for example, to a spike in traffic, it is possible to stream data stably at high speeds, since it is possible to avoid affected parts of the network by, for example, dividing the processing between a plurality of edge servers, or switching the streaming duties to a different edge server and continuing streaming.

Decentralization is not limited to just the division of processing for streaming; the encoding of the captured data may be divided between and performed by the terminals, on the server side, or both. In one example, in typical encoding, the processing is performed in two loops. The first loop is for detecting how complicated the image is on a frame-by-frame or scene-by-scene basis, or detecting the encoding load. The second loop is for processing that maintains image quality and improves encoding efficiency. For example, it is possible to reduce the processing load of the terminals and improve the quality and encoding efficiency of the content by having the terminals perform the first loop of the encoding and having the server side that received the content perform the second loop of the encoding. In such a case, upon receipt of a decoding request, it is possible for the encoded data resulting from the first loop performed by one terminal to be received and reproduced on another terminal in approximately real time. This makes it possible to realize smooth, real-time streaming.

In another example, camera ex113 or the like extracts a feature amount from an image, compresses data related to the feature amount as metadata, and transmits the compressed metadata to a server. For example, the server determines the significance of an object based on the feature amount and changes the quantization accuracy accordingly to perform compression suitable for the meaning (or content significance) of the image. Feature amount data is particularly effective in improving the precision and efficiency of motion vector prediction during the second compression pass performed by the server.

Moreover, encoding that has a relatively low processing load, such as variable length coding (VLC), may be handled by the terminal, and encoding that has a relatively high processing load, such as context-adaptive binary arithmetic coding (CABAC), may be handled by the server.

In yet another example, there are instances in which a plurality of videos of approximately the same scene are captured by a plurality of terminals in, for example, a stadium, shopping mall, or factory. In such a case, for example, the encoding may be decentralized by dividing processing tasks between the plurality of terminals that captured the videos and, if necessary, other terminals that did not capture the videos, and the server, on a per-unit basis. The units may be, for example, groups of pictures (GOP), pictures, or tiles resulting from dividing a picture. This makes it possible to reduce load times and achieve streaming that is closer to real time.

Since the videos are of approximately the same scene, management and/or instructions may be carried out by the server so that the videos captured by the terminals can be

cross-referenced. Moreover, the server may receive encoded data from the terminals, change the reference relationship between items of data, or correct or replace pictures themselves, and then perform the encoding. This makes it possible to generate a stream with increased quality and efficiency for the individual items of data.

Furthermore, the server may stream video data after performing transcoding to convert the encoding format of the video data. For example, the server may convert the encoding format from MPEG to VP (e.g., VP9), and may convert H.264 to H.265.

In this way, encoding can be performed by a terminal or one or more servers. Accordingly, although the device that performs the encoding is referred to as a “server” or “terminal” in the following description, some or all of the processes performed by the server may be performed by the terminal, and likewise some or all of the processes performed by the terminal may be performed by the server. This also applies to decoding processes.

[3D, Multi-angle]

There has been an increase in usage of images or videos combined from images or videos of different scenes concurrently captured, or of the same scene captured from different angles, by a plurality of terminals such as camera ex113 and/or smartphone ex115. Videos captured by the terminals are combined based on, for example, the separately obtained relative positional relationship between the terminals, or regions in a video having matching feature points.

In addition to the encoding of two-dimensional moving pictures, the server may encode a still image based on scene analysis of a moving picture, either automatically or at a point in time specified by the user, and transmit the encoded still image to a reception terminal. Furthermore, when the server can obtain the relative positional relationship between the video capturing terminals, in addition to two-dimensional moving pictures, the server can generate three-dimensional geometry of a scene based on video of the same scene captured from different angles. The server may separately encode three-dimensional data generated from, for example, a point cloud and, based on a result of recognizing or tracking a person or object using three-dimensional data, may select or reconstruct and generate a video to be transmitted to a reception terminal, from videos captured by a plurality of terminals.

This allows the user to enjoy a scene by freely selecting videos corresponding to the video capturing terminals, and allows the user to enjoy the content obtained by extracting a video at a selected viewpoint from three-dimensional data reconstructed from a plurality of images or videos.

Furthermore, as with video, sound may be recorded from relatively different angles, and the server may multiplex audio from a specific angle or space with the corresponding video, and transmit the multiplexed video and audio.

In recent years, content that is a composite of the real world and a virtual world, such as virtual reality (VR) and augmented reality (AR) content, has also become popular. In the case of VR images, the server may create images from the viewpoints of both the left and right eyes, and perform encoding that tolerates reference between the two viewpoint images, such as multi-view coding (MVC), and, alternatively, may encode the images as separate streams without referencing. When the images are decoded as separate streams, the streams may be synchronized when reproduced, so as to recreate a virtual three-dimensional space in accordance with the viewpoint of the user.

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In the case of AR images, the server superimposes virtual object information existing in a virtual space onto camera information representing a real-world space, based on a three-dimensional position or movement from the perspective of the user. The decoder may obtain or store virtual object information and three-dimensional data, generate two-dimensional images based on movement from the perspective of the user, and then generate superimposed data by seamlessly connecting the images. Alternatively, the decoder may transmit, to the server, motion from the perspective of the user in addition to a request for virtual object information. The server may generate superimposed data based on three-dimensional data stored in the server, in accordance with the received motion, and encode and stream the generated superimposed data to the decoder. Note that superimposed data includes, in addition to RGB values, an a value indicating transparency, and the server sets the a value for sections other than the object generated from three-dimensional data to, for example, 0, and may perform the encoding while those sections are transparent. Alternatively, the server may set the background to a determined RGB value, such as a chroma key, and generate data in which areas other than the object are set as the background.

Decoding of similarly streamed data may be performed by the client (i.e., the terminals), on the server side, or divided therebetween. In one example, one terminal may transmit a reception request to a server, the requested content may be received and decoded by another terminal, and a decoded signal may be transmitted to a device having a display. It is possible to reproduce high image quality data by decentralizing processing and appropriately selecting content regardless of the processing ability of the communications terminal itself. In yet another example, while a TV, for example, is receiving image data that is large in size, a region of a picture, such as a tile obtained by dividing the picture, may be decoded and displayed on a personal terminal or terminals of a viewer or viewers of the TV. This makes it possible for the viewers to share a big-picture view as well as for each viewer to check his or her assigned area, or inspect a region in further detail up close.

In situations in which a plurality of wireless connections are possible over near, mid, and far distances, indoors or outdoors, it may be possible to seamlessly receive content using a streaming system standard such as MPEG-Dynamic Adaptive Streaming over HTTP (MPEG-DASH). The user may switch between data in real time while freely selecting a decoder or display apparatus including the user's terminal, displays arranged indoors or outdoors, etc. Moreover, using, for example, information on the position of the user, decoding can be performed while switching which terminal handles decoding and which terminal handles the displaying of content. This makes it possible to map and display information, while the user is on the move in route to a destination, on the wall of a nearby building in which a device capable of displaying content is embedded, or on part of the ground. Moreover, it is also possible to switch the bit rate of the received data based on the accessibility to the encoded data on a network, such as when encoded data is cached on a server quickly accessible from the reception terminal, or when encoded data is copied to an edge server in a content delivery service.

[Web Page Optimization]

FIG. 111 illustrates an example of a display screen of a web page on computer ex111, for example. FIG. 112 illustrates an example of a display screen of a web page on smartphone ex115, for example. As illustrated in FIG. 111 and FIG. 112, a web page may include a plurality of image

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links that are links to image content, and the appearance of the web page differs depending on the device used to view the web page. When a plurality of image links are viewable on the screen, until the user explicitly selects an image link, or until the image link is in the approximate center of the screen or the entire image link fits in the screen, the display apparatus (decoder) may display, as the image links, still images included in the content or I pictures; may display video such as an animated gif using a plurality of still images or I pictures; or may receive only the base layer, and decode and display the video.

When an image link is selected by the user, the display apparatus performs decoding while giving the highest priority to the base layer. Note that if there is information in the HyperText Markup Language (HTML) code of the web page indicating that the content is scalable, the display apparatus may decode up to the enhancement layer. Further, in order to guarantee real-time reproduction, before a selection is made or when the bandwidth is severely limited, the display apparatus can reduce delay between the point in time at which the leading picture is decoded and the point in time at which the decoded picture is displayed (that is, the delay between the start of the decoding of the content to the displaying of the content) by decoding and displaying only forward reference pictures (I picture, P picture, forward reference B picture). Still further, the display apparatus may purposely ignore the reference relationship between pictures, and coarsely decode all B and P pictures as forward reference pictures, and then perform normal decoding as the number of pictures received over time increases.

[Autonomous Driving]

When transmitting and receiving still image or video data such as two- or three-dimensional map information for autonomous driving or assisted driving of an automobile, the reception terminal may receive, in addition to image data belonging to one or more layers, information on, for example, the weather or road construction as metadata, and associate the metadata with the image data upon decoding. Note that metadata may be assigned per layer and, alternatively may simply be multiplexed with the image data.

In such a case, since the automobile, drone, airplane, etc., containing the reception terminal is mobile, the reception terminal may seamlessly receive and perform decoding while switching between base stations among base stations ex106 through ex110 by transmitting information indicating the position of the reception terminal. Moreover, in accordance with the selection made by the user, the situation of the user, and/or the bandwidth of the connection, the reception terminal may dynamically select to what extent the metadata is received, or to what extent the map information, for example, is updated.

In content providing system ex100, the client may receive, decode, and reproduce, in real time, encoded information transmitted by the user.

[Streaming of Individual Content]

In content providing system ex100, in addition to high image quality, long content distributed by a video distribution entity, unicast or multicast streaming of low image quality, and short content from an individual are also possible. Such content from individuals is likely to further increase in popularity. The server may first perform editing processing on the content before the encoding processing, in order to refine the individual content. This may be achieved using the following configuration, for example.

In real time while capturing video or image content, or after the content has been captured and accumulated, the server performs recognition processing based on the raw

data or encoded data, such as capture error processing, scene search processing, meaning analysis, and/or object detection processing. Then, based on the result of the recognition processing, the server—either when prompted or automatically—edits the content, examples of which include: correction such as focus and/or motion blur correction; removing low-priority scenes such as scenes that are low in brightness compared to other pictures, or out of focus; object edge adjustment; and color tone adjustment. The server encodes the edited data based on the result of the editing. It is known that excessively long videos tend to receive fewer views. Accordingly, in order to keep the content within a specific length that scales with the length of the original video, the server may, in addition to the low-priority scenes described above, automatically clip out scenes with low movement, based on an image processing result. Alternatively, the server may generate and encode a video digest based on a result of an analysis of the meaning of a scene.

There may be instances in which individual content may include content that infringes a copyright, moral right, portrait rights, etc. Such instance may lead to an unfavorable situation for the creator, such as when content is shared beyond the scope intended by the creator. Accordingly, before encoding, the server may, for example, edit images so as to blur faces of people in the periphery of the screen or blur the inside of a house, for example. Further, the server may be configured to recognize the faces of people other than a registered person in images to be encoded, and when such faces appear in an image, may apply a mosaic filter, for example, to the face of the person. Alternatively, as pre- or post-processing for encoding, the user may specify, for copyright reasons, a region of an image including a person or a region of the background to be processed. The server may process the specified region by for example, replacing the region with a different image, or blurring the region. If the region includes a person, the person may be tracked in the moving picture, and the person's head region may be replaced with another image as the person moves.

Since there is a demand for real-time viewing of content produced by individuals, which tends to be small in data size, the decoder first receives the base layer as the highest priority, and performs decoding and reproduction, although this may differ depending on bandwidth. When the content is reproduced two or more times, such as when the decoder receives the enhancement layer during decoding and reproduction of the base layer, and loops the reproduction, the decoder may reproduce a high image quality video including the enhancement layer. If the stream is encoded using such scalable encoding, the video may be low quality when in an unselected state or at the start of the video, but it can offer an experience in which the image quality of the stream progressively increases in an intelligent manner. This is not limited to just scalable encoding; the same experience can be offered by configuring a single stream from a low quality stream reproduced for the first time and a second stream encoded using the first stream as a reference.

[Other Implementation and Application Examples]

The encoding and decoding may be performed by LSI (large scale integration circuitry) ex500 (see FIG. 110), which is typically included in each terminal. LSI ex500 may be configured of a single chip or a plurality of chips. Software for encoding and decoding moving pictures may be integrated into some type of a medium (such as a CD-ROM, a flexible disk, or a hard disk) that is readable by, for example, computer ex111, and the encoding and decoding may be performed using the software. Furthermore, when smartphone ex115 is equipped with a camera, video

data obtained by the camera may be transmitted. In this case, the video data is coded by LSI ex500 included in smartphone ex115.

Note that LSI ex500 may be configured to download and activate an application. In such a case, the terminal first determines whether it is compatible with the scheme used to encode the content, or whether it is capable of executing a specific service. When the terminal is not compatible with the encoding scheme of the content, or when the terminal is not capable of executing a specific service, the terminal first downloads a codec or application software and then obtains and reproduces the content.

Aside from the example of content providing system ex100 that uses internet ex101, at least the moving picture encoder (image encoder) or the moving picture decoder (image decoder) described in the above embodiments may be implemented in a digital broadcasting system. The same encoding processing and decoding processing may be applied to transmit and receive broadcast radio waves superimposed with multiplexed audio and video data using, for example, a satellite, even though this is geared toward multicast, whereas unicast is easier with content providing system ex100.

[Hardware Configuration] FIG. 113 illustrates further details of smartphone ex115 shown in FIG. 110. FIG. 114 illustrates a configuration example of smartphone ex115.

Smartphone ex115 includes antenna ex450 for transmitting and receiving radio waves to and from base station ex110, camera ex465 capable of capturing video and still images, and display ex458 that displays decoded data, such as video captured by camera ex465 and video received by antenna ex450. Smartphone ex115 further includes user interface ex466 such as a touch panel, audio output unit ex457 such as a speaker for outputting speech or other audio, audio input unit ex456 such as a microphone for audio input, memory ex467 capable of storing decoded data such as captured video or still images, recorded audio, received video or still images, and mail, as well as decoded data, and slot ex464 which is an interface for Subscriber Identity Module (SIM) ex468 for authorizing access to a network and various data. Note that external memory may be used instead of memory ex467.

Main controller ex460, which comprehensively controls display ex458 and user interface ex466, power supply circuit ex461, user interface input controller ex462, video signal processor ex455, camera interface ex463, display controller ex459, modulator/demodulator ex452, multiplexer/demultiplexer ex453, audio signal processor ex454, slot ex464, and memory ex467 are connected via bus ex470.

When the user turns on the power button of power supply circuit ex461, smartphone ex115 is powered on into an operable state, and each component is supplied with power from a battery pack.

Smartphone ex115 performs processing for, for example, calling and data transmission, based on control performed by main controller ex460, which includes a CPU, ROM, and RAM. When making calls, an audio signal recorded by audio input unit ex456 is converted into a digital audio signal by audio signal processor ex454, to which spread spectrum processing is applied by modulator/demodulator ex452 and digital-analog conversion and frequency conversion processing are applied by transmitter/receiver ex451, and the resulting signal is transmitted via antenna ex450. The received data is amplified, frequency converted, and analog-digital converted, inverse spread spectrum processed by modulator/demodulator ex452, converted into an analog audio signal by audio signal processor ex454, and then

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output from audio output unit ex457. In data transmission mode, text, still-image, or video data is transmitted by main controller ex460 via user interface input controller ex462 based on operation of user interface ex466 of the main body, for example.

Similar transmission and reception processing is performed. In data transmission mode, when sending a video, still image, or video and audio, video signal processor ex455 compression encodes, by the moving picture encoding method described in the above embodiments, a video signal stored in memory ex467 or a video signal input from camera ex465, and transmits the encoded video data to multiplexer/demultiplexer ex453. Audio signal processor ex454 encodes an audio signal recorded by audio input unit ex456 while camera ex465 is capturing a video or still image, and transmits the encoded audio data to multiplexer/demultiplexer ex453. Multiplexer/demultiplexer ex453 multiplexes the encoded video data and encoded audio data using a determined scheme, modulates and converts the data using modulator/demodulator (modulator/demodulator circuit) ex452 and transmitter/receiver ex451, and transmits the result via antenna ex450.

When a video appended in an email or a chat, or a video linked from a web page, is received, for example, in order to decode the multiplexed data received via antenna ex450, multiplexer/demultiplexer ex453 demultiplexes the multiplexed data to divide the multiplexed data into a bitstream of video data and a bitstream of audio data, supplies the encoded video data to video signal processor ex455 via synchronous bus ex470, and supplies the encoded audio data to audio signal processor ex454 via synchronous bus ex470. Video signal processor ex455 decodes the video signal using a moving picture decoding method corresponding to the moving picture encoding method described in the above embodiments, and video or a still image included in the linked moving picture file is displayed on display ex458 via display controller ex459. Audio signal processor ex454 decodes the audio signal and outputs audio from audio output unit ex457. Since real-time streaming is becoming increasingly popular, there may be instances in which reproduction of the audio may be socially inappropriate, depending on the user's environment. Accordingly, as an initial value, a configuration in which only video data is reproduced, i.e., the audio signal is not reproduced, may be preferable; and audio may be synchronized and reproduced only when an input is received from the user clicking video data, for instance.

Although smartphone ex115 was used in the above example, three other implementations are conceivable: a transceiver terminal including both an encoder and a decoder; a transmitter terminal including only an encoder; and a receiver terminal including only a decoder. In the description of the digital broadcasting system, an example is given in which multiplexed data obtained as a result of video data being multiplexed with audio data is received or transmitted. The multiplexed data, however, may be video data multiplexed with data other than audio data, such as text data related to the video. Further, the video data itself rather than multiplexed data may be received or transmitted.

Although main controller ex460 including a CPU is described as controlling the encoding or decoding processes, various terminals often include Graphics Processing Units (GPUs). Accordingly, a configuration is acceptable in which a large area is processed at once by making use of the performance ability of the GPU via memory shared by the CPU and GPU, or memory including an address that is managed so as to allow common usage by the CPU and

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GPU. This makes it possible to shorten encoding time, maintain the real-time nature of streaming, and reduce delay. In particular, processing relating to motion estimation, deblocking filtering, sample adaptive offset (SAO), and transformation/quantization can be effectively carried out by the GPU, instead of the CPU, in units of pictures, for example, all at once.

Although only some exemplary embodiments of the present disclosure have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the present disclosure.

INDUSTRIAL APPLICABILITY

The present disclosure is applicable to, for example, television receivers, digital video recorders, car navigation systems, mobile phones, digital cameras, digital video cameras, video conference systems, and electron mirrors.

What is claimed is:

1. An encoder comprising:

circuitry; and

memory coupled to the circuitry, wherein

in operation, the circuitry:

writes a first parameter into a video parameter set, the first parameter indicating a maximum number of one or more temporal sublayers in a layer set unit, the layer set unit being a unit which corresponds to a stream and includes one or more layers; and

writes a second parameter into a sequence parameter set, the second parameter indicating a maximum number of one or more temporal sublayers in a sequence unit, the sequence unit being a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to,

in a case where a reference is made to the video parameter set, the second parameter is limited to less than or equal to the first parameter, and

in a case where no reference is made to the video parameter set, the second parameter is limited to less than or equal to a fixed value.

2. The encoder according to claim 1, wherein the fixed value is an upper limit of the first parameter.

3. The encoder according to claim 1, wherein the first parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the layer set unit,

the second parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the sequence unit,

the fixed value is 6,

in the case where the reference is made to the video parameter set, the second parameter is limited to a range from 0 to the first parameter, and

in the case where no reference is made to the video parameter set, the second parameter is limited to a range from 0 to 6.

4. The encoder according to claim 1, wherein the circuitry writes a third parameter into the sequence parameter set, the third parameter (i) indicating that the reference is made to the video parameter set and indicating an identifier of the video parameter set, when the third parameter is greater than 0, and (ii) indicating

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that no reference is made to the video parameter set, when the third parameter is 0,
 when the third parameter is greater than 0, the second parameter is limited to less than or equal to the first parameter, and
 when the third parameter is 0, the second parameter is limited to less than or equal to the fixed value.
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 5. A decoder comprising:
 circuitry; and
 memory coupled to the circuitry, wherein
 in operation, the circuitry:
 reads a first parameter from a video parameter set, the first parameter indicating a maximum number of one or more temporal sublayers in a layer set unit, the layer set unit being a unit which corresponds to a stream and includes one or more layers, and
 reads a second parameter from a sequence parameter set, the second parameter indicating a maximum number of one or more temporal sublayers in a sequence unit, the sequence unit being a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to,
 in a case where a reference is made to the video parameter set, the second parameter is limited to less than or equal to the first parameter, and
 in a case where no reference is made to the video parameter set, the second parameter is limited to less than or equal to a fixed value.
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 6. The decoder according to claim 5, wherein the fixed value is an upper limit of the first parameter.
 7. The decoder according to claim 5, wherein the first parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the layer set unit,
 the second parameter has a value obtained by subtracting 1 from the maximum number of one or more temporal sublayers in the sequence unit,
 the fixed value is 6,
 in the case where the reference is made to the video parameter set, the second parameter is limited to a range from 0 to the first parameter, and
 in the case where no reference is made to the video parameter set, the second parameter is limited to a range from 0 to 6.
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 8. The decoder according to claim 5, wherein the circuitry reads a third parameter from the sequence parameter set, the third parameter (i) indicating that the reference is made to the video parameter set and indicating an identifier of the video parameter set, when the third parameter is greater than 0, and (ii) indicating that no reference is made to the video parameter set, when the third parameter is 0,
 when the third parameter is greater than 0, the second parameter is limited to less than or equal to the first parameter, and
 when the third parameter is 0, the second parameter is limited to less than or equal to the fixed value.
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 9. A decoder comprising:
 circuitry; and
 memory coupled to the circuitry, wherein
 in operation, the circuitry:
 reads a first parameter from a first parameter set, the first parameter indicating a maximum number of one or more temporal sublayers in a layer set unit, the layer set unit being a unit which corresponds to a
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stream and includes one or more layers, the first parameter set being a parameter set for the layer set unit; and
 reads a second parameter from a second parameter set, the second parameter indicating a maximum number of one or more temporal sublayers in a random access unit, the random access unit being a unit which corresponds to at least a portion of the stream, the second parameter set being a parameter set for the random access unit,
 in a case where a reference is made to the first parameter set, the second parameter is limited to less than or equal to the first parameter, and
 in a case where no reference is made to the first parameter set, the second parameter is limited to less than or equal to a fixed value.
 10. An encoding method comprising:
 writing a first parameter into a video parameter set, the first parameter indicating a maximum number of one or more temporal sublayers in a layer set unit, the layer set unit being a unit which corresponds to a stream and includes one or more layers; and
 writing a second parameter into a sequence parameter set, the second parameter indicating a maximum number of one or more temporal sublayers in a sequence unit, the sequence unit being a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to, wherein
 in a case where a reference is made to the video parameter set, the second parameter is limited to less than or equal to the first parameter, and
 in a case where no reference is made to the video parameter set, the second parameter is limited to less than or equal to a fixed value.
 11. A decoding method comprising:
 reading a first parameter from a video parameter set, the first parameter indicating a maximum number of one or more temporal sublayers in a layer set unit, the layer set unit being a unit which corresponds to a stream and includes one or more layers; and
 reading a second parameter from a sequence parameter set, the second parameter indicating a maximum number of one or more temporal sublayers in a sequence unit, the sequence unit being a unit which corresponds to at least a portion of the stream and for which the sequence parameter set is referred to, wherein
 in a case where a reference is made to the video parameter set, the second parameter is limited to less than or equal to the first parameter, and
 in a case where no reference is made to the video parameter set, the second parameter is limited to less than or equal to a fixed value.
 12. A non-transitory medium storing a bitstream, the bitstream including a video parameter set and a sequence parameter set for causing a decoder that reads out the bitstream to perform a decoding process, wherein
 the video parameter set includes a first parameter indicating a maximum number of one or more temporal sublayers in a layer set unit, the layer set unit being a unit which corresponds to the bitstream and includes one or more layers,
 the sequence parameter set includes a second parameter indicating a maximum number of one or more temporal sublayers in a sequence unit, the sequence unit being a unit for which the sequence parameter set is referred to in the bitstream,

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in a case where a reference is made to the video parameter set, the second parameter is limited to less than or equal to the first parameter, and

in a case where no reference is made to the video parameter set, the second parameter is limited to less than or equal to a fixed value.

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